

HWT-NVR800 9.0.0 API Reference

2022-06-25

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Keywords: *Provide keywords in both Chinese and English to facilitate English speaking users' search.*

Abstract: *Provide abstract in both Chinese and English to facilitate English speaking users' search.*

List of terms and abbreviations

English	Abbreviations	Definition/Description

About This Document

URL prefix: <https://IP:443>

Note: When invoking an API, you need to combine the URL prefix with the API address. In the URL prefix, *IP address* indicates the IP address of the HWT-NVR800, and **443** is the default port number for accessing the HWT-NVR800 on the OMU portal. Example:
<https://3.2.20.25:443/API/SystemConfig/General/Get>

API invoking description:

1. List all the OMU portal-related APIs, parameters (including visible parameters on the portal and invisible parameters), API description, and parameter description of the HWT-NVR800.
2. Invisible parameters refer to those submitted in API URLs.
3. List the description about success codes and failure codes that may be returned by each API.
4. HWT-NVR800 APIs are classified into Range, Get, and Set APIs. A Range or Get API can be invoked to obtain a maximum of 64 items at a time, while a Set API can be invoked to set a maximum of 10 items. Otherwise, the API will report an error indicating too frequent operations.

1 API Description

1.1 Common Error Codes

General	param_error	Incorrect input parameter.
	no_permission	No permission.
	first_login	Forcible password change at the first login.
	part_failed	Failed to save some channels.
	no_support	Not supported.
	frequent_operation	Too frequent operations.
	passwd_expired_login	Your password has expired. You must change it before logging in.
	default_failed	Failed to restore the default value.
	token_generation_failed	Failed to generate a token.
	token_invalid	Invalid token.
	device_busy	The number of APIs being processed by the device reaches the maximum.
	user_expired_login	The user account expires.
	user_locked_login	The user account is locked.
	date_param_error	The date parameter is incorrect.
Session	no_login	You have not logged in.
	expired	The login expired.
	one_IE	One browser can be used by only one user to log in.
	logout	You have logged out.
	login_at_other	Your account has logged in elsewhere.
	device_reboot	The device restarts.
	passwd_expired	The password has expired.
	param_changed	The user parameter has been modified.
	network_changed	The network port has been changed.
	ssl_error	Certificate re-signing.
	netip_limited	The login IP address is blocklisted.
	forced_offline	Forcible logout.

	user_expired	The user account has expired.
Secondary authentication	current_pwd_error_ntime	If the number of authentication attempts exceeds a specified value, the value of data contains remain_locked_time , indicating the remaining locking time.
Parameter saving	save_failed	Saving failed.
	pwd_weak	The new camera password must meet complexity requirements.
	modify_failed	Modification failed.
	modify_failed_pwd_err	The user name or password for changing the IP address of the camera is incorrect.
	modify_failed_syntax_err	The syntax (mask or gateway) for modifying the IP address of the camera is incorrect.
Search	search_failed	Search failed.
Recording playback	device_play_locked	The board is playing back the video and rejects the web request.
Operation	operation_failed	Operation failed.
User name, password, and permission change	session_invalid	
	illegal_request	Invalid request.
	overreach	Child users can only set their own information.
	short_modify_time	The password change interval is too short.
	current_pwd_error	Incorrect password.
	username_empty	The user name is empty.
	username_repeat	Duplicate user name.
	username_invalid	The user name consists of only letters, digits, and underscores.
	unmatched_pwd	The two entered passwords are inconsistent.
	pwd_empty	The password is empty.
	pwd_length_err	Incorrect password length.
	pwd_equal_name	The password must differ from the user name spelled forward and backward.
	pwd_equal_old	The new password must be different from previous <i>N</i> (specified by number

		in the value of data) passwords.
	pwd_repeated	A new password must contain a minimum of two characters that are not present in the old password.
	pwd_weak_rule	The password must contain at least two types of the following: uppercase letters, lowercase letters, digits, special characters, and spaces.
	pwd_invalid	Invalid password.
	pwd_risk	The password is in the weak password dictionary file.
Email test	user_auth_failed	Authentication failed.
	data_error	Data sending error.
	net_unreachable_or_dns_wrong	Unreachable network or DNS server error.
	connect_server_err	Failed to connect to the server.
	check_smtp_port	The connection fails. Check whether the port is correct.
	tls_ssl_handshake_err	TLS/SSL handshake failed.
	email_connect_err	Connection error. Check the recipient account.
Upgrade	in_user_interface	User interface.
	updating	The board is being upgraded.
	lack_memory	Insufficient memory.
	file_error	File error.
	no_need_upgrade	The current version is the latest and does not need to be updated.
HTTP	not_modified	No web page modification (304).
	http_redirect_https	Redirection to HTTPS from HTTP (307).
	not_found	The web page cannot be found (404).
	method_not_allowed	The request method is not allowed (405).
	payload_too_large	Overloaded (413).
	uri_too_long	Long URI (414).
	internal_server_error	Internal server error (500).
	service_unavailable_error	The service is unavailable (503,

		exceeding the maximum number of TCP connections).
Camera parameter setting		For details, see Additional Error Codes.
DDNS test	connect_error	Network connection error.
	send_error	Sending error.
	receive_error	Receiving error.
	network_info_error	Network information error.
	unknown_error	Unknown error.

1.2 Additional Error Codes

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_noerror	No error.
pu_error_code_common_error	Minor error.
pu_error_code_password_error	Incorrect user name or password.
pu_error_code_no_priority	You are not authorized to perform the operation.
pu_error_code_no_init	The SDK is not initialized.
pu_error_code_has_init	The SDK has been initialized.
pu_error_code_version_no_match	The SDK does not match the device version.
pu_error_code_startup_network_error	Network startup error.
pu_error_code_socket_error	Network socket error.
pu_error_code_connect_error	Network connection error.
pu_error_code_send_timeout	Message sending timed out.
pu_error_code_recv_timeout	Message receiving timed out.
pu_error_code_operation_timeout	Device operation timed out.
pu_error_code_out_of_memory	Failed to allocate the memory due to insufficient memory space.
pu_error_code_server_is_shutdown	The link is disconnected on the server.
pu_error_code_too_many_realplay	The number of live video channels exceeded the maximum.
pu_error_code_log_error	Log error.
pu_error_code_no_free_port	Insufficient port resources.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_has_reg_event_callback	The event callback function has been registered.
pu_error_code_register_fail	Proactive camera registration with the platform failed.
pu_error_code_register_redirect	The camera was redirected when registering with the platform.
pu_error_code_input_null_ptr	The input parameter is a null pointer.
pu_error_code_opr_local_file_fail	Failed to operate the local file.
pu_error_code_file_format_invalid	Invalid file format.
pu_error_code_vlpr_watermark_invalid	Failed to verify the digital watermark of the snapshot.
pu_error_code_rest_pkg_type_error	Incorrect RESTful packet format.
pu_error_code_xml_type_error	If the size of the data structure converted into an XML file exceeds 60 KB, the SDK can send a maximum of 64 KB data.
pu_error_code_sdkinit_link_mode_error	A connection mode error occurred during SDK initialization.
pu_error_code_sdkinit_localip_error	A local IP address error occurred during SDK initialization.
pu_error_code_sdklogin_para_error	Incorrect login parameter.
pu_error_code_code_is_invalid	Invalid error code.
pu_error_code_sdkgetversion_para_error	An error occurred when obtaining the version.
pu_error_code_identifyid_error	Incorrect user ID.
pu_error_code_sdkgetdevice_para_error	An error occurred when obtaining device parameter settings.
pu_error_code_alarmreport_cb_is_null	The alarm reporting callback function is empty.
pu_error_code_voicetb_cb_is_null	The voice intercom callback function is empty.
pu_error_code_realplay_cb_is_null	The live video viewing callback function is empty.
pu_error_code_load_playdll_error	An error occurred when loading the video library.
pu_error_code_cfgfile_error	The configuration file is incorrect.
pu_error_code_unsupport_alarm_type	Unknown alarm type.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_sdkupgrade_para_error	Upgrade parameter error.
pu_error_code_sdkupgrade_error	Upgrade error.
pu_error_code_sdkupgrade_file_error	Incorrect upgrade file.
pu_error_code_unsupport_platform_type	Unknown platform type.
pu_error_code_sdksetplatform_para_error	An error occurred when setting platform parameters.
pu_error_code_unsupport_encode_type	Unknown audio and video encoding type.
pu_error_code_unsupport_stream_type	Unknown data stream type.
pu_error_code_unsupport_bitrate_type	Unknown bit rate type.
pu_error_code_unsupport_ptz_cmd	Unknown PTZ command.
pu_error_code_cruise_para_error	Incorrect tour parameter.
pu_error_code_alarm_tmp_para_error	Incorrect temperature alarm parameter.
pu_error_code_alarm_motion_detec_para_error	Incorrect motion detection alarm parameter.
pu_error_code_alarm_input_para_error	Incorrect input alarm parameter.
pu_error_code_alarm_hide_para_error	Incorrect lens blocking alarm parameter.
pu_error_code_alarm_invade_para_error	Incorrect intrusion detection alarm parameter.
pu_error_code_alarm_count_para_error	Incorrect object counting alarm parameter.
pu_error_code_alarm_disturb_para_error	Incorrect interference alarm parameter.
pu_error_code_privacy_para_error	Incorrect privacy mask parameter.
pu_error_code_osd_para_error	Incorrect OSD parameter.
pu_error_code_ptz_para_error	Incorrect PTZ control parameter.
pu_error_code_sdkguard_para_error	Incorrect alert enabling or disabling parameter.
pu_error_code_sdkgetalarm_para_error	An error occurred when obtaining alarm parameters.
pu_error_code_para_error	Parameter error.
pu_error_code_player_func_error	An error occurred when invoking the video library function.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_sdksavealldata_error	An error occurred when saving live video data.
pu_error_code_sdkgetguardstate_para_error	Incorrect parameter for obtaining the alert status (enabled or disabled).
pu_error_code_mutuallyexclusive_relationship	The setting failed because of mutual exclusion.
pu_error_code_device_relogin	You have logged in to the device.
pu_error_code_invalid_config_file	The configuration file is invalid.
pu_error_code_wrong_rsp_type	The response message type is incorrect.
pu_error_code_rsp_meg_parse_error	Failed to parse the response message.
pu_error_code_mismatch_pwd_complexity	The password does not meet complexity requirements.
pu_error_code_alarm_audio_up_para_error	Sudden volume increase alarm parameter error.
pu_error_code_alarm_audio_down_para_error	Sudden volume decrease alarm parameter error.
pu_error_code_transparent_channel_data_cb_is_null	The data callback function of the transparent channel is empty.
pu_error_code_no_enough_continuous_mem	Insufficient continuous memory.
pu_error_code_ircmode_para_error	Incorrect IR Cut type.
pu_error_code_cert_para_error	Incorrect certificate path.
pu_error_code_import_facelibrary_error	An error occurred when importing target records into the target list.
pu_error_code_importing_facelib_error	The target records are being imported into a target list.
pu_error_code_unknown_sv_type	Unknown service type.
pu_error_code_not_open_tls	Operations related to the target list cannot be performed because TLS is not enabled.
pu_error_invalid_updatefile_num	The number of upgrade files exceeded the maximum.
pu_error_user_cancel_upgrade	Upgrade canceled.
pu_error_not_enough_memory_allocated	JSON data cannot be loaded because the input receive cache is too small.
pu_error_mutualauth_error	Reverse authentication failed.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_getconfigfile	Obtaining the configuration file...
pu_error_code_getfacelib	Obtaining target list information...
pu_error_code_getfaceinfo	Obtaining target information...
pu_error_code_manualsnapshot	A snapshot is being manually taken.
pu_error_code_onekeycollection	One-click collection is in progress.
pu_error_code_getfacepic	Obtaining target images...
pu_error_code_getimportcsv	Obtaining the target image import result...
pu_error_code_getqualityjudge	Obtaining the image quality evaluation result...
pu_error_auto_reconnecting_error	Reconnecting... Please wait.
pu_error_code_getkeyfile	Obtaining the key file...
pu_error_code_lackwpcap	There is no WPCAP dependency library. Download it.
pu_error_rsa_key_create_error	Failed to create the RSA key.
pu_error_rsa_key_encrypt_error	RSA encryption failed.
pu_error_manual_snatshot_buf_len_error	An error occurred when obtaining the buffer for storing manually captured images.
pu_error_json_error	JSON-related issue.
pu_error_code_gethotmap	Obtaining the heat map...
pu_error_http_bad_request	Invalid HTTP command.
pu_error_http_conflict	HTTP command conflict.
pu_error_code_dec_initdec_error	Decoding initialization failed.
pu_error_code_dec_startdec_error	Failed to start decoding.
pu_error_code_dec_stopdec_error	Failed to stop decoding.
pu_error_code_dec_too_many_channel	There are too many decoding channels.
pu_error_code_dec_error_port_num	Incorrect port quantity.
pu_error_code_dec_error_dec_handel	Incorrect decoding handle.
pu_error_code_dec_error_unsupport_decode_type	The decoding type is not supported.
pu_error_code_dec_error_invalid_channel_id	Invalid channel ID.
pu_error_code_server_common_error	Minor error.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_syntax_error	Incorrect configuration syntax.
pu_error_code_unsupport_error	The request is not supported.
pu_error_code_parameter_error	Incorrect parameter.
pu_error_invalid_vcu_id	Incorrect device ID.
pu_error_code_invalid_channel	Incorrect channel ID.
pu_error_code_invalid_serial_no	Incorrect serial port number.
pu_error_code_invalid_alarm_no	Incorrect alarm port.
pu_error_code_unsupported_media_type	Unsupported media type.
pu_error_code_partly_support	Only some parameters are set successfully.
pu_error_code_too_many_plans	The number of plans exceeded the maximum.
pu_error_code_no_disk	The device has no hard disk.
pu_error_code_open_sound_fail	Failed to enable the audio device.
pu_error_code_voice_opend	The server's voice intercom is in use.
pu_error_code_device_busy	The device is busy.
pu_error_code_allocate_resource	Resource allocation error.
pu_error_code_operate_flash_error	Flash memory error.
pu_error_code_unsupported_video_format	Unsupported video standard.
pu_error_code_ip_address_conflict	IP address conflict.
pu_error_code_lte_dhcp_conflict	The LTH setting conflicts with the DHCP setting.
pu_error_code_sip_stack_error	SIP protocol stack error.
pu_error_code_result_unsupported_dest	The service or app is being started.
pu_error_code_result_passwd_weak	Weak password.
pu_error_code_config_password	Incorrect configuration file password.
pu_result_check_all_alg	Failed to create the algorithm. Check whether all algorithms are upgraded.
pu_result_server_initcomplete	ITGT initialization is incomplete.
pu_result_net_pkt_type_need_close	Both multicast and unicast functions need to be disabled.
pu_result_net_pkt_type_need_open_one	Either the multicast or unicast function needs to be enabled.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_null_pointer	Null pointer.
pu_error_code_null_element	The XML file does not have the corresponding node.
pu_error_code_invalid_verification	Incorrect user name or password.
pu_error_code_invalid_version_file	Failed to verify the version file.
pu_error_code_invalid_kernel_file	Failed to verify the kernel upgrade file.
pu_error_code_invalid_rootfs_file	Failed to verify the file system upgrade file.
pu_error_code_first_login	First login.
pu_error_code_client_already_login	You have logged in to the client.
pu_error_code_user_locked	The user account is locked.
pu_error_code_many_user	The number of users exceeded the maximum.
pu_error_code_user_name_conflict	Duplicate user name.
pu_error_code_no_user_name	The user does not exist.
pu_error_code_user_nopower	The user does not have the permission.
pu_error_code_not_need_update	The version numbers are the same. No upgrade is required.
pu_error_code_not_support_ability	The capability parameter is not supported.
pu_error_code_preset_in_guard_position	The preset position has been set as the home position.
pu_error_code_preset_in_cruise_track	The preset position has been added to the tour.
pu_error_code_the_cruise_is_running	The tour is being executed.
pu_error_code_update	The system is being upgraded.
pu_error_code_reboot	The system is restarting.
pu_error_code_restore	The device is being restored to default settings.
pu_error_code_recover_oprate_forbidden	No operation such as configuration is allowed during default setting restoration.
pu_error_code_roi_error_by_mjpeg	Failed to set the ROI in MJPEG format.
pu_error_code_roi_error_by_nosubstream	Failed to set the ROI for the secondary stream because only the primary

Error Code Returned When Camera Parameters Are Set	Description
	stream is available.
pu_error_code_tz_error_by_conflict	A conflict occurred when tripwire crossing detection is enabled.
pu_error_code_oc_error_by_conflict	A conflict occurred when object counting is enabled.
pu_error_code_md_error_by_conflict	A conflict occurred when motion detection is enabled.
pu_error_code_tw_error_by_conflict	A conflict occurred when tripwire crossing detection is enabled.
pu_error_code_area_error_by_conflict	A conflict occurred when area detection is enabled.
pu_error_code_vd_error_by_conflict	A conflict occurred when video interference detection is enabled.
pu_error_code_igt_enable_conflict	Failed to enable illumination enhancement or lens blocking detection because intelligent analysis has been enabled.
pu_error_code_illumination_enable_conflict	Failed to enable intelligent analysis or lens blocking detection because illumination enhancement has been enabled.
pu_error_code_videoblock_enable_conflict	Failed to enable intelligent analysis or illumination enhancement because lens blocking detection has been enabled.
pu_error_code_igt_fail_by_switchoff	Failed to set the intelligent service because intelligent analysis is not enabled.
pu_error_code_tw_error_by_over_num	The number of tripwires exceeded the maximum.
pu_error_code_area_error_by_over_num	The number of areas exceeded the maximum (4).
pu_error_code_area_error_by_over_total_num	The number of areas exceeded the maximum (10).
pu_error_code_area_error_by_zero_num	No area is configured.
pu_error_code_area_point_error_by_over_num	The number of area vertexes exceeded the maximum.
pu_error_code_area_abn_error_by_conflict	Failed to enable removed object detection because abandoned object detection has been enabled.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_area_rmv_error_by_conflict	Failed to enable abandoned object detection because removed object detection has been enabled.
pu_error_code_vim_global_para_error	Failed to set global parameters for the intelligent analysis algorithm.
pu_error_code_area_notsurport_error	Invalid intelligent analysis area.
pu_error_code_privacy_mask_area_beyond	The privacy mask area exceeded the total area.
pu_error_code_media_adapt_bw_mutex	The adaptive bandwidth function conflicts with the encoding type or bit rate type.
pu_error_code_ip_filter_rule_same	Duplicate IP address filtering rule.
pu_error_code_wd_frame_conflict	When the WDR mode is set to manual or automatic, a conflict will occur if the full frame rate is set to 50 or 60 fps.
pu_error_code_frame_wd_conflict	When the full frame rate is 50 or 60 fps, a conflict will occur if the WDR mode is set to manual or automatic.
pu_error_code_conflict_videoblock_or_igt_is_enable	Failed to enable illumination enhancement because lens blocking detection or intelligent analysis has been enabled.
pu_error_code_busy_curise	Failed to set tour parameters during a tour.
pu_error_code_device_close	The device is not started.
pu_error_code_busy_preset	The preset position is in use and cannot be deleted.
pu_error_code_invalid_igt_area_id	The intelligent analysis area does not exist.
pu_error_code_invalid_igt_line_id	The intelligent analysis line does not exist.
pu_error_code_invalid_privacy_id	The privacy mask area does not exist.
pu_error_code_in_user_define_scan	Pattern scan is in progress (many functions are unavailable).
pu_error_code_set_drc_backlight_conflict	The backlight compensation and DRC functions cannot be enabled simultaneously.
pu_error_code_curise_in_guard_position	A preset position in the tour has been configured as the home position.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_disk_abnormal	SD card error.
pu_error_code_privacymask_region_intersect	Privacy mask areas overlapped.
pu_error_code_invalid_samepasswd	The new password must be different from the old password.
pu_error_code_not_supported_update	The upgrade file is outdated. Select an upgrade file of a later version.
pu_error_code_many_record_files	Overwhelming number of recordings or images. (The camera supports query of 500 recordings at most. You are advised to query an appropriate number of recordings, shortening the query time.)
pu_error_code_conflict_with_mosaic	The operation conflicts with the mosaic setting.
pu_error_code_conflict_with_osd	The operation conflicts with the OSD setting.
pu_error_code_conflict_with_pic	The operation conflicts with the image overlay setting.
pu_error_code_video_enc_close	The video encoder is not started.
pu_error_code_isp_para_conflict	ISP parameter conflict.
pu_error_code_wb_high_temp_lower	In AWB automatic tracking mode, the lower threshold of the color temperature is mistakenly greater than or equal to the upper threshold of the color temperature.
pu_error_code_mjpeg_unsupport_watermark	The MJPEG does not support the digital watermark function.
pu_error_code_isp_conflict_with_framerate	WDR or backlight compensation and the full frame rate cannot be enabled at the same time.
pu_error_code_framerate_conflict_with_isp	// The 50 or 60 fps conflicts with WDR or BLC.
pu_error_code_osd_string_len_err	The OSD character length is incorrect.
pu_error_code_disk_mount_error	The disk is not mounted.
pu_error_code_importing_face_lib	Importing data into the target list... Please wait.
pu_error_code_adding_one_face	Adding targets... Please wait.
pu_error_code_no_record_files	No recording file is found.
pu_error_code_extra_device_open_fail	Failed to enable the external device.

Error Code Returned When Camera Parameters Are Set	Description
puerror_code_mode_para_with_pic	The OSD parameter for switching to the custom or GB mode conflicts with image overlay.
pu_error_code_web_sdk_key_file	The key file is being transferred.
pu_error_code_zoom_is_restricted	The zoom/focus function is restricted. Reset the lens.
pu_error_code_audio_detect_mic_not_support	Audio exception detection is not supported in the case of MIC connection.
pu_error_code_cif_conflict	The CIF resolution and corridor mode cannot be enabled at the same time.
pu_error_code_corridor_conflict	The CIF resolution and corridor mode cannot be enabled at the same time.
pu_result_irc_mode_not_support_link	The day/night linkage is not supported.
pu_result_month_plan_has_start	The ISP package plan has been enabled.
pu_result_irc_link_has_start	The day/night linkage has been enabled.
pu_result_err_point_outof_range	The point coordinate is out of range.
pu_result_err_line_become_point	The start point and end point of a line segment cannot overlap.
pu_result_err_line_cross	Line crossing is not allowed.
pu_error_isp_ae_manual_conflict_wd	Manual exposure and WDR cannot be enabled at the same time.
pu_error_isp_ae_manual_conflict_flicker	Manual exposure and background frequency cannot be enabled at the same time.
pu_error_isp_ae_manual_conflict_hlc	Manual exposure and highlight compensation cannot be enabled at the same time.
pu_error_isp_ae_shutpri_conflict_wd	The shutter priority exposure mode and WDR cannot be enabled at the same time.
pu_error_isp_ae_shutpri_conflict_flicker	The shutter priority exposure mode and background frequency cannot be enabled at the same time.
pu_error_isp_ae_shutpri_conflict_hlc	The shutter priority exposure mode and HLC cannot be enabled at the same time.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_isp_backlight_conflict_wd	BLC and WDR cannot be enabled at the same time.
pu_error_isp_backlight_conflict_hlc	BLC and HLC cannot be enabled at the same time.
pu_error_isp_wd_conflict_hlc	WDR and HLC cannot be enabled at the same time.
pu_error_isp_defog_conflict_wd	The defogging mode and WDR cannot be enabled at the same time.
pu_error_isp_hlc_conflict_framerate	HLC and frame rate 50 or 60 fps cannot be used at the same time.
pu_error_isp_wd_conflict_framerate	WDR and frame rate 50 or 60 fps cannot be used at the same time.
pu_error_isp_ae_manual_conflict_backlight	Manual exposure and BLC cannot be enabled at the same time.
pu_error_isp_hlc_conflict_slow_shutter	HLC and slow shutter cannot be enabled at the same time.
pu_error_isp_wd_conflict_slow_shutter	WDR and slow shutter cannot be enabled at the same time.
pu_error_multi_stream_conflict_face_dt	The multi-stream function and the target detection function cannot be enabled at the same time. Currently, only M2120-EFI supports target detection and dual streams (secondary stream resolution D1 or above).
pu_error_isp_defog_conflict_fog	Defogging and fog detection cannot be enabled at the same time.
pu_error_code_download_file_error	Failed to download the file.
pu_error_code_connection_server_interrupted	Disconnected from server.
pu_error_code_request_overtime	The request timed out.
pu_error_code_connection_server_fail	Failed to connect to the server.
pu_error_code_connection_not_found	The connection is not found.
pu_error_code_reconnecting_to_server	Reconnecting to the server...
pu_error_code_file_not_exist	The file does not exist.
pu_error_code_invalid_updatefile	Invalid upgrade file.
pu_error_code_updatefile_bad	The file is damaged.
pu_error_code_invalid_file_format	Incorrect file format.
pu_error_code_upgrade_type_err	The upgrade file type does not match

Error Code Returned When Camera Parameters Are Set	Description
	the request type.
pu_error_code_upgrade_param_null	The upgrade parameter is empty.
pu_error_code_upgrade_not_upload	The upgrade file is not uploaded.
pu_error_code_upgrade_verify_fail	Failed to verify the integrity of the upgrade file.
pu_error_code_upgrade_sign_fail	Failed to verify the digital signature of the upgrade file.
pu_error_code_upgrade_space_full	The upgrade storage space is full.
pu_error_code_upgrade_device_fail	Failed to adapt to the list of supported devices.
pu_error_result_alg_ver_err_u	The algorithm package version is incorrect. The algorithm package version must be consistent with the VRC version of the software package.
pu_error_result_alg_ver_need_upgrade_u	Upgrade the algorithm package after upgrading the version package.
pu_error_result_need_face_alg_upgrade	The target-related algorithm package needs to be upgraded.
pu_error_result_need_vhd_alg_upgrade	The object classification algorithm package needs to be upgraded.
pu_error_result_omni_fail_u1	Failed to upgrade the rotatable lens (2001) of the compound-eye camera.
pu_error_result_omni_fail_u2	Failed to upgrade the rotatable lens (2002) of the compound-eye camera.
pu_error_result_not_support	Version rollback is not supported after device power-off in dual-system mode.
pu_error_result_upgrade_reboot	The device is being upgraded, restarted, or restored.
pu_error_result_pdd_send_tar_err_u	The primary camera failed to send packets to the secondary camera.
pu_error_result_pdd_upg_slave_overtime_u	The upgrade of the secondary camera timed out.
pu_error_result_pdd_backhaul_u	Failed to return the primary camera upgrade package.
pu_error_sd_not_format	The SD card is not formatted.
pu_error_code_custom_package_undefined	The stream profile and ISP profile are not defined.
pu_error_code_ptz_busy	The PTZ is not stopped.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_osdi_area_not_exist	The area does not exist. The index value is incorrect.
pu_error_code_osdi_area_too_large	The horizontal width of the area exceeded 180 pixels.
pu_error_code_osdi_area_invalid_verdegree	The area coordinate does not comply with the lower-left and upper-right rules.
pu_error_code_osdi_index_already_exist	The index number of the new area already exists.
pu_error_code_osdi_hor_or_ver_equal	The horizontal coordinate and vertical coordinate of the area are the same.
pu_error_code_osdi_index_invalid	The index number of the area must be greater than or equal to 1.
pu_error_code_osdi_index_equal	Duplicate index number of the area.
pu_error_code_dst_already_open	The DST has been enabled.
pu_error_code_set_time_out_of_range	The configured time is out of the system time range.
pu_stabilizer_conflict_area_crop	Electronic image stabilization and area cropping cannot be enabled at the same time.
pu_area_crop_conflict_stabilizer	Area cropping and electronic image stabilization cannot be enabled at the same time.
pu_result_stabilizer_conflict_vlpr	Electronic image stabilization and license plate detection cannot be enabled at the same time.
pu_error_code_itgt_imrs_err_point_null	Null pointer.
pu_error_code_itgt_imrs_err_read_file	Failed to read the file.
pu_error_code_itgt_imrs_err_xml_parser	Failed to parse the XML file.
pu_error_code_itgt_imrs_err_img_tpye	The image type is not supported.
pu_error_code_itgt_imrs_err_resolution_diff	The resolution is different from the initial setting.
pu_error_code_itgt_imrs_err_vertex_illegal	Improper vertex coordinates.
pu_error_code_itgt_imrs_err_resolution_unsupport	The resolution is not supported.
pu_error_code_itgt_imrs_err_param_err	Parameter error.
pu_error_code_itgt_imrs_err_malloc_failed	Failed to allocate the memory.
pu_error_code_itgt_imrs_err_not_config	Configuration error.

Error Code Returned When Camera Parameters Are Set	Description
pu_error_code_itgt_imrs_err_param_set	Parameter setting error.
pu_error_code_itgt_imrs_err_lane_overlap	Incorrect lane line.
pu_error_code_itgt_imrs_err_campus_lane_num_limit	Only one lane is supported in entrance/exit mode.
pu_error_result_vlpr_err_line_outof_range	Setting failed. The detection line is out of the lane line.
pu_error_code_vlpr_imrs_err_campus_srv_not_support_manual_snap	Manual snapshot taking is not supported in entrance/exit mode.
pu_error_code_vlpr_para_not_enable	The motor vehicle detection function is not enabled for the micro checkpoint camera.
pu_error_code_cert_not_rsa_key	The certificate is not an RSA public key for authentication.
pu_error_code_short_key	The certificate key is too short.
pu_error_code_weak_cert	The certificate key is too weak.
pu_error_code_cert_overdue	The certificate has expired.
pu_error_code_cert_pri_key_pwd_not_match	The private key of the certificate does not match the password.
pu_error_code_get_temperature_vw_ret_fail	Failed to obtain the temperature from the driver.
pu_result_its_err_invalid_lane_type	The lane line type is out of range.
pu_result_its_err_invalid_lane_num	The number of lanes is out of range.
pu_result_its_err_invalid_lane_dir	Invalid lane direction.
pu_result_its_err_invalid_lane_drv_dir	Invalid regulated driving direction of the lane.
pu_result_its_err_invalid_lane_purpose	Invalid lane usage.
pu_result_its_err_invalid_lane_attr	Invalid lane attribute.
pu_result_its_err_invalid_car_drv_dir	Invalid driving direction.
pu_result_its_err_invalid_sign_dir	Invalid marked direction.
pu_result_its_err_invalid_normal_sign_speed	Invalid marked speed for small vehicles.
pu_result_its_err_invalid_normal_low_speed	Invalid lower speed limit for small vehicles.
pu_result_its_err_invalid_normal_high_speed	Invalid upper speed limit for small vehicles.
pu_result_its_err_invalid_bigvehicle_sign_speed	Invalid marked speed for large

Error Code Returned When Camera Parameters Are Set	Description
	vehicles.
pu_result_its_err_invalid_bigvehicle_low_speed	Invalid lower speed limit for large vehicles.
pu_result_its_err_invalid_bigvehicle_high_speed	Invalid upper speed limit for large vehicles.
pu_result_its_err_invalid_extern_lamp_bitmap	Invalid external illuminator association.
pu_result_its_err_invalid_snap_interval	The capture interval is out of range.
pu_result_its_err_invalid_dt_type	The detection mode is out of range.
pu_result_its_err_invalid_regulation_num	The number of violations exceeded the maximum.
pu_result_its_err_invalid_regulation_type	Invalid violation type.
pu_result_its_err_invalid_snap_num	The number of traffic violation snapshots is out of range.
pu_result_its_err_invalid_runningred_priority	Invalid red-light-running detection priority.
pu_result_its_err_invalid_over_lane_sensitivity	The sensitivity for detecting marked lane violations is invalid.
pu_result_its_err_invalid_vinbikelane_staytime	The dwell duration for detecting motor vehicles in non-motorized vehicle lanes is invalid.
pu_result_its_err_invalid_over_buslane_staytime	Invalid bus lane occupation time.
pu_result_its_err_invalid_parkingtime	Invalid dwell duration for detecting parking in yellow zones.
pu_result_its_err_invalid_light_dt_type	The traffic light access mode is out of range.
pu_result_its_err_invalid_light_area_num	Invalid number of traffic light areas.
pu_result_its_err_invalid_light_num	Invalid number of traffic lights.
pu_result_its_err_invalid_light_arrange_type	Invalid traffic light direction.
pu_result_its_err_invalid_yellow_time	Invalid duration of the yellow light.
pu_result_its_err_invalid_light_area	Invalid traffic light area.
pu_result_its_err_line_outof_range	The detection line or stop line is out of the lane line or right boundary line range.
pu_result_its_err_invalid_normal_speed_range_config	The lower speed limit for small vehicles is greater than the upper speed limit.
pu_result_its_err_invalid_normal_sign_speed_config	The marked speed for small vehicles is

Error Code Returned When Camera Parameters Are Set	Description
	out of range.
pu_result_its_err_invalid_bigvehicle_speed_range_config	The lower speed limit for large vehicles is greater than the upper speed limit.
pu_result_its_err_invalid_bigvehicle_sign_speed_config	The marked speed for large vehicles is out of range.
pu_result_its_err_invalid_returndata	Invalid return value.
pu_result_its_err_invalid_congestion_threshold	Invalid traffic congestion threshold.
pu_result_its_err_invalid_snap_interval_type	Invalid capture interval type.
pu_result_its_err_invalid_adapt_snap_lowspeed	The maximum low speed that can be detected during the adaptive capture interval is invalid.
pu_result_its_err_invalid_adapt_snap_midspeed	The maximum medium speed that can be detected during the adaptive capture interval is invalid.
pu_result_its_err_invalid_adapt_lowspeed_interval	The adaptive low-speed capture interval is invalid.
pu_result_its_err_invalid_adapt_midspeed_interval	The adaptive medium-speed capture interval is invalid.
pu_result_its_err_invalid_adapt_hispeed_interval	The adaptive high-speed capture interval is invalid.
pu_result_its_err_invalid_lane_order	Setting failed. Draw lane lines and the right lane boundary from left to right in sequence.
pu_result_its_err_invalid_light_configduplication	Duplicate traffic light direction.
pu_result_its_err_invalid_pretestline_location	The front detection line should be below the stop line.
pu_result_its_err_portid_repeat	Setting failed due to duplicate port numbers.
pu_error_result_hbtp_err_bad_request	Request error (HBTP error code).
pu_error_result_hbtp_err_unauthorized	Not authorized (HBTP error code).
pu_error_result_hbtp_err_forbidden	Disabled (HBTP error code).
pu_error_result_hbtp_err_not_found	Not found (HBTP error code).
pu_error_result_hbtp_err_internal_server_error	Internal server error (HBTP error code).
pu_result_tgt_function_mutual_lrpr	Disable license plate recognition.
pu_result_tgt_function_mutual_fd	Disable target detection.
pu_result_tgt_function_mutual_cd	Disable parking detection.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_tgt_function_mutual_va	Disable behavior analysis.
pu_result_tgt_function_mutual_hc	Disable crossing-line head counting.
pu_result_tgt_function_mutual_at	Disable automatic tracking.
pu_result_tgt_function_mutual_vhd	Disable object classification.
pu_result_tgt_function_mutual_its	Disable ITS.
pu_result_tgt_function_mutual_qd	Disable queue length detection.
pu_result_tgt_function_mutual_cdd	Disable crowd density detection.
pu_result_tgt_function_mutual_iball	Disable traffic violation detection for the PTZ dome camera.
pu_result_tgt_function_mutual_vhdface	Disable the multi-purpose function for the camera.
pu_result_itgt_mode_notsupport_alg	This algorithm is not supported in the current mode.
pu_result_tgt_vhd_detection_err_wrong_video_mode	Object classification does not support the corridor mode. Disable the corridor mode.
pu_result_tgt_function_mutual_beh	Disable intelligent alarm reporting.
pu_result_tgt_function_mutual_campus_trafficevent	Disable vehicle detection at entrances and exits.
pu_result_tgt_function_mutual_campus_alg_type	Vehicle detection at entrances and exits supports only algorithms for recognizing Chinese license plates.
pu_result_tgt_function_mutual_md	Disable motion detection.
pu_result_tgt_function_mutual_od	Disable lens blocking detection.
pu_result_tgt_function_mutual_sc	Disable scene change detection.
pu_result_tgt_function_mutual_ood	Disable defocus detection.
pu_result_tgt_function_mutual_ca	Disable crowd gathering detection.
pu_result_tgt_function_mutual_hm	Disable the heat map function.
pu_result_tgt_function_mutual_trafficevent_campus	Disable all vehicle event detection functions.
pu_result_tgt_function_mutual_traffic_statistic_campus	Disable traffic flow statistics.
pu_result_tgt_function_mutual_vd	Disable video quality diagnosis.
pu_result_tgt_function_mutual_stabilizer	Disable electronic image stabilization.
pu_result_tgt_function_mutual_img_quality_judge	Disable all other services during image quality evaluation.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_tgt_function_mutual_face_rec	Disable target recognition.
pu_result_tgt_function_mutual_fc	Disable high-density target capture.
pu_result_tgt_function_mutual_traffic_flow_campus	Disable the video-based vehicle detector.
pu_result_tgt_function_mutual_traffic_flow_campus	Disable fight detection.
pu_result_tgt_function_mutual_climb	Disable climbing detection.
pu_result_tgt_function_mutual_fall_down	Disable fall detection.
pu_result_tgt_function_mutual_run	Disable running detection.
pu_result_tgt_function_mutual_ft	Disable the scheduled capture mode for target recognition.
pu_result_tgt_function_mutual_firedetect	Disable fire detection.
pu_result_tgt_function_mutual_temperature_control	Disable temperature measurement.
pu_result_tgt_function_mutual_visible_beh	Disable intelligent alarm reporting for the visible light channel.
pu_result_tgt_function_mutual_thermal_beh	Disable intelligent alarm reporting for the thermal imaging channel.
pu_result_tgt_function_mutual_distortion	Disable distortion correction.
pu_result_tgt_function_mutual_intelligent_encoding	Disable SuperCoding.
pu_result_tgt_function_mutual_face_attri	Disable target attribute recognition.
pu_result_tgt_function_mutual_continuous_imq	Disable continuous image quality diagnosis.
pu_result_tgt_function_mutual_fl	Disable the low-bandwidth capture mode for target detection.
pu_result_tgt_function_mutual_vhdfacemix	Disable the full intelligence mode.
pu_result_tgt_function_mutual_track	Disable linkage between the radar detector and PTZ dome camera.
pu_result_tgt_function_mutual_ytn	Disable the 1 + N mode.
pu_result_tgt_function_mutual_irt	Disable radar- and PTZ dome camera-based intrusion detection.
pu_result_itgt_err_face_success	Target images are imported to the target list successfully.
pu_result_itgt_err_facerec_disable	Target recognition is not enabled.
pu_result_itgt_err_data_update	Failed to update data.
pu_result_itgt_err_opreation_notsupport	The operation is not supported.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_itgt_err_opreation_unfinished	The batch import operation is not complete.
pu_result_itgt_err_facelib_oversize	The number of target lists exceeded the upper limit.
pu_result_itgt_err_facenum_oversize	The number of targets exceeded the upper limit.
pu_result_itgt_err_db_error	The database operation failed.
pu_result_itgt_err_facelist_repat	Duplicate target list.
pu_result_itgt_err_facepic_oversize	The image width or height is out of range.
pu_result_itgt_err_face_decode	Image decoding failed.
pu_result_itgt_err_face_detection	Failed to detect targets.
pu_result_itgt_err_face_feature	Failed to extract target features.
pu_result_itgt_err_face_dao	Failed to write target data into the database.
pu_result_itgt_err_face_file	Failed to write target data into the file.
pu_result_itgt_err_face_algdisable	The target algorithm is disabled.
pu_result_itgt_err_face_nofeature	There is no feature in the target list.
pu_result_itgt_err_face_uzip	Failed to decompress the ZIP package.
pu_result_itgt_err_face_csv	Failed to parse the CSV file.
pu_result_itgt_err_face_encrypt	Failed to encrypt or decrypt target data.
pu_result_itgt_err_face_taskunknow	The task is not processed.
pu_result_itgt_err_feature_extract_success	Features are extracted successfully.
pu_result_itgt_err_feature_extracted	Features have been extracted.
pu_result_itgt_err_facerec_enable_fail	Failed to enable target recognition because the algorithm package does not exist or its version does not match.
pu_result_itgt_err_face_feature_load_processing	Target recognition has historical alerts. Features need to be loaded upon system startup.
pu_result_itgt_err_face_feature_pac_import_processing	The target feature package is being imported.
pu_result_itgt_err_face_pic_pac_import_processing	The target image package is being imported.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_itgt_err_face_feature_get_processing	Target features are being exported.
pu_result_itgt_err_face_feature_extract_processing	Target features are being extracted.
pu_result_itgt_err_emmc_status_abnormal	The eMMC card status is abnormal.
pu_result_itgt_err_face_quickcap_notsupport_black	Ultra-fast capture does not support blocklist-based alert.
pu_result_itgt_err_facerec_disk_space_not_enough	The disk space is insufficient.
pu_result_itgt_err_face_record_check	Failed to verify target records.
pu_result_itgt_err_facerec_multi_frame	Multi-frame integration for target recognition is not supported.
pu_result_itgt_err_close_multi_frame_info	Disable multi-frame integration for target recognition.
pu_result_itgt_err_feature_file	Failed to read the feature structure from the feature.bin file.
pu_result_itgt_err_feature_notifyone	Failed to encapsulate data in the NotifyOne.
pu_result_itgt_err_database_encrypt_key	Incorrect database key.
pu_result_itgt_err_json_file_parse_fail	Failed to parse the JSON file.
pu_result_itgt_err_face_pic_extract_processing	A single image extraction task is in progress.
pu_result_itgt_err_face_pic_extract_timeout	The single image extraction task timed out.
pu_result_itgt_err_face_lib_control_oversize	The number of target alert reference lists exceeded the upper limit.
pu_result_itgt_err_face_access	The image does not exist.
pu_result_itgt_err_file_oversize	The file size exceeded the threshold.
pu_result_itgt_err_feature_synergy_notsupport	Disable feature-based collaboration.
pu_result_itgt_err_face_defaultcap	Enable prioritized capture.
pu_result_itgt_err_facedb_encrypting	The key is being updated for the target list. Try again later.
pu_result_itgt_err_record_inzip_oversizeu	The number of target records in the target list exceeded the upper limit
pu_result_tgt_car_datection_err_wrong_video_mode	When parking detection is enabled, the corridor mode and mirroring mode must be disabled.
pu_result_itgt_vhd_err_wrong_video_mode	Object classification and the corridor mode cannot be enabled at the same

Error Code Returned When Camera Parameters Are Set	Description
	time.
pu_result_itgt_vlpr_err_wrong_video_mode	Vehicle recognition and the corridor mode cannot be enabled at the same time.
pu_result_itgt_not_face_detetion_mode_err_wrong_video_mode	The corridor mode must be disabled for -H camera models in non-target checkpoint mode (not supported in full intelligence, micro checkpoint, or behavior analysis mode).
pu_result_lic_invalid	Failed to obtain the license content due to the invalid license.
pu_result_lic_repeated_esn_ver_mismatch	You are not allowed to repeatedly load the ESN or load the license file that does not match the device version.
pu_result_lic_format_error	Invalid license file.
pu_result_lic_no_prd_features_present	The license file does not contain the product feature segment.
pu_result_lic_product_invalid	Invalid product name.
pu_result_lic_expired	The license file has expired.
pu_result_lic_get_config_error	Failed to obtain the control item configuration.
pu_result_lic_lsn_revoked	The license file has expired.
pu_result_lic_esnany_and_six_mon_vldity_not_allowed	You are not allowed to repeatedly load a commissioning license file whose ESN is ANY and validity period is six months.
pu_result_lic_max_features_reached	The number of features in the license file exceeded the maximum.
pu_result_lic_invalid_feature_type	The feature type in the license file is incorrect.
pu_result_lic_lm_not_enabled	The license component is not started.
pu_result_lic_no_features_in_normal_state	All features in the license file have expired or are in the trial period.
pu_result_lic_esn_mismatch	The ESN in the license file is different from the actual ESN.
pu_result_lic_version_mismatch	The device version in the license file is different from the actual version.
pu_result_lic_no_license_activited	No activated license.
pu_result_smtp_picnum_err	Incorrect image quantity.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_smtp_server_port_err	Incorrect server port.
pu_result_smtp_server_name_err	Incorrect server name.
pu_result_smtp_server_pwd_err	Incorrect server password.
pu_result_smtp_server_ip_err	Incorrect server IP address.
pu_result_smtp_sender_add_err	Incorrect sender address.
pu_result_smtp_revpara_err	Incorrect receiver information.
pu_result_mtu_err_with_ipv6_open	The MTU is incorrect in the IPv6 scenario.
pu_result_ip_not_support	You are not allowed to set network segments to special ones, for example, 127.0.0.1/8 and 169.254.0.0/16.
pu_result_tgt_humancount_err_wrong_mode	The mirror mode or corridor mode and head counting cannot be enabled at the same time.
pu_result_dotx_user_err = 12560	Incorrect 802.1X user name.
pu_result_itgt_err_invalid	Invalid intelligent analysis license.
pu_result_media_utgt_not_enabled	An intelligent analysis channel needs to be created for media streams, but the intelligent analysis module is not enabled.
pu_result_tgt_switch_mode_err_wrong_video_mode	The switched mode does not support the corridor mode.
pu_result_tgt_switch_mode_err_enable_alg	Failed to enable the algorithm for the switched mode. Check the model.
pu_result_tgt_switch_mode_ftp_need_reconfig	The mode is switched successfully. FTP needs to be reconfigured.
pu_result_hwapi_notification_reach_max	The number of notifications has reached the maximum.
pu_result_hwapi_ssh_username_passwd_wrong	Incorrect user name or password.
pu_result_hwapi_ssh_sys_error	System error.
pu_result_hwapi_need_ssl_error	Only security protocols are supported.
pu_result_ods_move_device_not_init	The rotatable lens is not initialized.
pu_result_ods_move_device_abnormal	An exception occurred on the rotatable lens.
pu_result_get_ods_move_device_log_info_fail	Failed to obtain rotatable lens logs.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_get_ods_move_device_config_fail	Failed to obtain the rotatable lens configuration file.
pu_result_set_ods_move_device_config_fail	Failed to set the rotatable lens configuration file.
pu_result_set_ods_move_device_reboot_fail	Failed to restart the rotatable lens.
pu_result_set_ods_move_device_restore_fail	Failed to restore the rotatable lens to default settings.
pu_result_ods_sdk_fail	SDK internal error.
pu_result_need_stop_track_err	The trajectory analysis function is enabled. Therefore, calibration cannot be performed, and positioning cannot be linked. Disable the trajectory analysis function first.
pu_result_in_calibrating_err	The calibration is in progress.
pu_result_not_calibrating_err	The calibration is not complete. Therefore, positioning cannot be linked, and the trajectory analysis function cannot be enabled. Complete the calibration first.
pu_result_not_in_calibrating_err	The calibration is not in progress, so you do not need to cancel the calibration or query the progress.
pu_result_close_ytn_info	1 + N has been enabled for the fixed lens. Disable it first.
pu_result_forced_vhd_face_info	The fixed lens needs to be forcibly switched to the full intelligence mode.
pu_result_not_support_face_selcted_in_track	Linked trajectory analysis is enabled for the rotatable lens, so the target capture mode cannot be set to optimal.
pu_result_calib_not_support_mirror	Calibration cannot be performed in mirror mode.
pu_result_set_fixed_facerec_fail	Failed to configure the fixed lens of the compound-eye camera for target recognition.
pu_result_set_mov1_facerec_fail	Failed to configure rotatable lens 1 of the compound-eye camera for target recognition.
pu_result_set_mov2_facerec_fail	Failed to configure rotatable lens 2 of the compound-eye camera for target recognition.
pu_result_prase_xml_fail	Failed to parse the XML server

Error Code Returned When Camera Parameters Are Set	Description
	address.
pu_result_itgt_err_db_success	db_success
pu_result_itgt_err_db_norecord	No record in the database.
pu_result_itgt_err_db_unknown_type	Incorrect database type.
pu_result_itgt_err_1tn_disable	The 1 + N function is disabled.
pu_result_itgt_err_1tn_metadata_beyond_limit	The number of displayed target recognition items in 1 + N mode exceeded the upper limit.
pu_result_itgt_err_1tn_delete_datetime_period_over_one_day	The interval for deleting historical records exceeded one day.
pu_result_itgt_err_1tn_delete_starttime_later_than_endtime	The start time for deleting historical records cannot be later than the end time.
pu_result_itgt_err_1tn_delete_endtime_later_than_curtime	The end time for deleting historical records cannot be later than the current time.
pu_result_onvif_28181_conflict	The ONVIF protocol and the GB/T 28181 protocol cannot be enabled at the same time.
pu_result_process_num_conflict	Only two of GB/T 28181, ONVIF, SDK, and GA/T 1400 can be enabled at the same time.
pu_result_onvif_not_enabled	Failed to set parameters because ONVIF is not enabled.
pu_result_itgt_no_disk	No SD card.
pu_result_itgt_err_need_eval_first	Perform manual capture for evaluation.
pu_result_itgt_save_config_error	Failed to save configurations.
pu_result_no_key_file_err	No key file.
pu_result_set_check_key_file_fail	Failed to verify the key file.
pu_result_exit_3app_config	Failed to configure the container engine.
pu_result_exit_3app_stop_container	Failed to stop the container.
pu_result_exit_3app_start_container	Failed to start the container.
pu_result_exit_3app_space_not_enough	Insufficient space.
pu_result_exit_3app_para_cnt	Invalid parameter.
pu_result_exit_3app_create_container	Failed to create the container.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_exit_3app_name_invalid	Invalid app name.
pu_result_exit_3app_file_not_exists	The file does not exist.
pu_result_exit_3app_base_version_not_installed	The basic version is not installed.
pu_result_exit_3app_invalid_position	Invalid installation location.
pu_result_exit_3app_data_disk_size_unknown	The data disk size is not specified.
pu_result_exit_3app_start_lcrd	Failed to start the container engine.
pu_result_exit_3app_entry_not_specified	No entry program is specified for the self-developed app.
pu_result_exit_3app_vendor_invalid	Invalid vendor name.
pu_result_exit_3app_arch_not_match	The installation package is inconsistent with the SDC architecture.
pu_result_exit_running_can_not_format_sd	Failed to format the SD card because an app is running.
pu_result_exit_3app_tar_file_extract_fail	Failed to decompress the installation package.
pu_result_exit_3app_cert_invalid	Failed to verify the certificate.
pu_result_exit_3app_sign_verify_fail	Failed to verify the signature file.
pu_result_exit_3app_license_fail	Invalid license file.
pu_result_exit_3app_license_trial_time_out	The license trial period has ended.
pu_result_exit_3app_cert_file_not_exist	The certificate file does not exist.
pu_result_appsdk_28181_conflict_error	The GB/T 28181 protocol and the cloud service interconnection protocol cannot be enabled at the same time.
pu_result_player_ret_error	Failure. The control is defined as 1. Mapping is required.
pu_result_player_ret_param_invalid	Invalid input parameter. The control is defined as 2. Mapping is required.
pu_result_player_ret_save_pic_file_error	Failed to save the image file.
pu_result_player_ret_alloc_mem_error	Failed to allocate the cache.
pu_result_player_ret_order_error	Incorrect invoking sequence.
pu_result_player_ret_channel_not_exist_error	The channel does not exist.
pu_result_player_ret_jpeg_compress_error	JPEG compression failed.
pu_result_player_ret_create_file_error	Failed to create the file.
pu_result_player_ret_capture_no_picture	No decoded snapshots can be taken.

Error Code Returned When Camera Parameters Are Set	Description
pu_result_itgt_mode_notsupport_alg_behavior_analysis	This algorithm is not supported in the current mode. Switch the camera algorithm mode to behavior analysis and try again.
pu_result_itgt_mode_notsupport_alg_human_card_mode	This algorithm is not supported in the current mode. Switch the camera algorithm mode to target checkpoint and try again.
pu_result_itgt_mode_notsupport_alg_micro_card_mode	This algorithm is not supported in the current mode. Switch the camera algorithm mode to micro checkpoint and try again.
pu_result_tgt_function_mutual_ld	Disable off-duty detection.
pu_result_tgt_function_mutual_face_mask	Disable mask alarm.
PU_RESULT_TGT_FUNCTION_MUTUAL_FACE_ATTRI_OPEN	Enable target attribute recognition.
pu_result_tgt_function_mutual_ped_attri	Disable personal attribute recognition.
pu_error_code_open_alg_first	This function is not supported in the current mode. Enable the proper mode under Settings > Network > Device Management .

1.3 Request Description

For all APIs (except those invoked before login), the HTTP header needs to contain the values of **Cookie** and **X-csrfToken** returned by the login API for authentication. For details about login, see section 11.1 "Login." API requests are in the following format. The requested data and returned data are in the value of the **data** parameter. If an error occurs, the server returns the error code and error cause with the data.

POST /API/RecordConfig/Set?2020-05-07%2014:03:25 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "version": "1.0",
  "data": {
```

```
    }  
  }  
}
```

Success responses are in the following format:

```
HTTP/1.1 200 OK  
Content-Type: application/json  
Content-Length: 40  
  
{  
  "version": "x.x",  
  "result": "success",  
  "data": {  
    }  
}
```

Failure response format:

```
HTTP/1.1 400 OK  
Content-Type: application/json  
Content-Length: 40  
  
{  
  "version": "x.x",  
  "result": "success",  
  "reason": "*****",  
  "error_code": "user_blocked",  
  "data": {}  
}
```

For some APIs related to channels, if the operation is successful for some channels and fails for others, the server returns **part_failed** as the value of **error_code** and returns the failed channels and error causes. The format is as follows:

```
HTTP/1.1 400 OK  
Content-Type: application/json  
Content-Length: 40  
  
{  
  "version": "x.x",  
  "error_code": "part_failed",  
  "reason": "x.x",
```

```

"ch_error_code": [{"error_code": "no_support", "channel": "CH1"}],
"data": {}
}

```

2 System Parameter Configuration

2.1 General Parameters

2.1.1 Obtaining Values of Basic System Parameters

API Description					
Function		This API is used to obtain values of basic system parameters.			
API URL		/API/SystemConfig/General/Get			
Request Method		POST			
Message ID					
Description		API used to obtain the parameter value range: /API/SystemConfig/General/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
device_name	Device name.	String	Max length: 31 bytes		Device name.
menu_timeouts	LDU timeout.	Int	0, 30, 60, 120, 300, 600		Menu auto lock time. Unit: second
session_timeout	Web session timeout.	Int	5–1440		Web session timeout. Unit: minute
preview_session_timeout	Indicates whether preview or playback session timeout is enabled.	Bool			Indicates whether preview or playback session timeout is enabled.
cert_warn_time	Certificate warning time.	Int	0–90		Certificate warning time. Unit: day
root_cert_warn	Indicates	Bool			Indicates whether

	whether warning is enabled for the device root certificate.			warning is enabled for the device root certificate.
ai_switch	Indicates whether AI is enabled.	Bool		Indicates whether AI is enabled (supported only for HWT-NVR800s with one or two disk trays).

2.1.2 Setting Basic System Parameters

API Description					
Function	This API is used to set basic system parameters.				
API URL	/API/SystemConfig/General/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
device_name	Device name.	String	N	Maximum length: 31 bytes	In the range.
session_timeout	Web session timeout.	Int	N	5–1440	In the range.
preview_session_timeout	Preview or playback session timeout.	Bool	N	Bool	Bool
cert_warn_time	Certificate warning time.	Int	N	0–90	In the range.
root_cert_warn	Indicates whether to enable the warning for the device root certificate.	Bool	N	Bool	Bool
ai_switch	Indicates whether to enable AI.	Bool	N	Bool	Bool

menu_timeouts	LDU timeout.	Int	N	0, 30, 60, 120, 300, 600	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.2 Date and Time Parameters

2.2.1 Obtaining Values of System Time Parameters

API Description					
Function	This API is used to obtain values of system time parameters.				
API URL	/API/SystemConfig/DateTime/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/SystemConfig/DateTime/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
date	System date.	String	yyyy/mm/dd	System date. Format: yyyy/mm/dd	
time	System time.	String	hh/mm/ss	System time. Format: hh:mm:ss	
date_format	Date format.	String	"MM/DD/YYYY", "YYYY-MM-DD".	Date format.	

			"DD/MM/YYYY"	
time_zone	Time zone.	String	"GMT-12:00", "GMT-11:00", "GMT-10:00", "GMT-9:30", "GMT-9:00", "GMT-8:00", "GMT-7:00", "GMT-6:00", "GMT-5:00", "GMT-4:30", "GMT-4:00", "GMT-3:30", "GMT-3:00", "GMT-2:00", "GMT-1:00", "GMT", "GMT+1:00", "GMT+2:00", "GMT+3:00", "GMT+3:30", "GMT+4:00", "GMT+4:30", "GMT+5:00", "GMT+5:30", "GMT+5:45", "GMT+6:00", "GMT+6:30", "GMT+7:00", "GMT+8:00", "GMT+9:00", "GMT+9:30", "GMT+10:00", "GMT+10:30", "GMT+11:00", "GMT+11:30", "GMT+12:00", "GMT+12:45", "GMT+13:00", "GMT+14:00"	39 time zones are used, which are aligned with the time zones supported by HOLOWITS box cameras.

2.2.2 Setting System Time Parameters

API Description					
Function	This API is used to set system time parameters.				
API URL	/API/SystemConfig/DateTime/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
date	System time.	String	N	yyyy/mm/dd	Fixed format.
time	System time.	String	N	hh:mm:dd	Fixed format.
date_format	Date format.	String	N	"MM/DD/YYYY", "YYYY-MM-DD", "DD/MM/YYYY"	Fixed options.
time_zone	Time zone.	String	N	"GMT-12:00", "GMT-11:00", "GMT-10:00", "GMT-9:30", "GMT-9:00", "GMT-8:00", "GMT-7:00", "GMT-6:00", "GMT-5:00", "GMT-4:30", "GMT-4:00", "GMT-3:30", "GMT-3:00", "GMT-2:00", "GMT-1:00", "GMT",	Fixed options.

				"GMT+1:00", "GMT+2:00", "GMT+3:00", "GMT+3:30", "GMT+4:00", "GMT+4:30", "GMT+5:00", "GMT+5:30", "GMT+5:45", "GMT+6:00", "GMT+6:30", "GMT+7:00", "GMT+8:00", "GMT+9:00", "GMT+9:30", "GMT+10:00", "GMT+10:30", "GMT+11:00", "GMT+11:30", "GMT+12:00", "GMT+12:45", "GMT+13:00", "GMT+14:00"	
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.3 NTP Parameters

2.3.1 Obtaining Values of NTP Parameters

API Description	
Function	This API is used to obtain values of NTP parameters.
API URL	/API/SystemConfig/NTP/Get
Request Method	POST

Message ID					
Remarks		API used to obtain the parameter value range: /API/SystemConfig/NTP/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
ntp_enable	Indicates whether NTP is enabled.	Bool	Bool		NTP enable flag
custom_server	Customized time calibration server.	String	Maximum length: 63 bytes		Custom server address. Note: Use this variable when Server is set to User-Defined .

2.3.2 Setting NTP Parameters

API Description					
Function	This API is used to set NTP parameters.				
API URL	/API/SystemConfig/NTP/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
ntp_enable	Indicates whether to enable NTP.	Bool	N	Bool	Bool
custom_server	Customized time calibration	String	N	Maximum length: 63	Use this variable when Server is set to User-Defined . The value can

	server.			bytes	contain only letters, digits, hyphens (-), and periods (.). The hyphen cannot be at the beginning or end of the value.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.4 DST Parameters

2.4.1 Obtaining Values of DST Parameters

API Description					
Function		This API is used to obtain values of daylight saving time (DST) parameters.			
API URL		/API/SystemConfig/DST/Get			
Request Method		POST			
Message ID					
Remarks		API used to obtain the parameter value range: /API/SystemConfig/DST/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
dst_enable	Indicates whether DST is enabled.	Bool	Bool		Indicates whether DST is enabled.
time_offset	Offset.	Int	1, 2		DST offset value. Unit: hour
dst_mode	DST mode.	String	"Week", "Date"		DST mode.
start_date	Start date.	String	MM/DD/YYYY		DST start date. Format: MM/DD/YYYY

end_date	End date.	String		DST end date. Format: MM/DD/YYYY
start_hour	Start time.	String	hh:mm:ss	DST start time. Format: hh:mm:ss
end_hour	End time.	String		DST end time. Example: 00:20:00 Format: hh:mm:ss
start_month	Start month.	String	"Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"	Start month.
end_month	End month.	String		End month.
start_week	Start week of the month.	String	"1st", "2nd", "3rd", "4th", "Last"	Start week of the month.
end_week	End week of the month.	String		End week of the month.
start_weekday	Start day.	String	"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"	Which day start from.
end_weekday	End day.	String		Which day end of.

2.4.2 Setting DST Parameters

API Description	
Function	This API is used to set DST parameters.
API URL	/API/SystemConfig/DST/Set

Request Method	POST				
Message ID					
Remarks					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
dst_enable	Indicates whether to enable DST.	Bool	N	Bool	Boolean value.
time_offset	Offset.	Int	N	1, 2	Fixed options.
dst_mode	DST mode.		N	"Week", "Date"	Fixed options.
start_date	Start date.	String	N	MM/DD/YYYY	Fixed format.
end_date	End date.	String	N		
start_hour	Start time.	String	N	hh:mm:ss	Fixed format.
end_hour	End time.	String	N		
start_month	Start month.	String	N	byte "Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"	Fixed options.
end_month	End month.	String	N		
start_week	Start weekday.	String	N	"1st", "2nd", "3rd", "4th", "Last"	Fixed options.
end_week	End weekday.	String	N		
start_weekday	Start day.	String	N	byte	Fixed options.

end_weekday	End day.	String	N	"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"	
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result	Return value.	String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.5 Output Configuration Parameters

2.5.1 Obtaining Values of Local Output Parameters

API Description					
Function	This API is used to obtain values of local output parameters.				
API URL	/API/SystemConfig/Output/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/SystemConfig/Output/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
output		JSON object			JSON information about the elements in the object is as

				follows.
Elements in the output object				
LIVE-OUT, SPOT-OUT	Video output mode.	JSON object		JSON information about the elements in the object is as follows. SPOT-OUT (not used)
JSON information about the elements in the video output mode object				
video_output	Video output.	String		Video output.
output_resolution	Output resolution.	String	1024x768, 1280x1024, 1440x900, 720P(1280x720), 1080P(1920x1080), 1680x1050, 1600x1200, 1920x1200, 2K(2560x1440), 4K(3840x2160)	The resolution 1680 x 1050 is unavailable for HWT-NVR800s with 16 disk trays.

2.5.2 Setting Local Output Parameters

API Description					
Function	This API is used to set local output parameters.				
API URL	/API/SystemConfig/Output/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
output		JSON object	Y		JSON information about the elements in the output object is as follows.
Elements in the output object					
LIVE-OUT,	Video output	JSON	Y	""	Fixed options.

SPOT-OUT	mode.	object			SPOT-OUT (not used)
Elements in the video output mode object					
video_output	Video output.	String	N	LIVE-OUT,	Fixed options.
output_resolution	Output resolution.	String	N	1024x768, 1280x1024, 1440x900, 720P(1280x720), 1080P(1920x1080) , 1680x1050, 1600x1200, 1920x1200, 2K(2560x1440), 4K(3840x2160)	The resolution 1680 x 1050 is unavailable for HWT-NVR800s with 16 disk trays.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result	Return value.	String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.6 User Parameters

2.6.1 Obtaining Values of User Parameters

API Description					
Function	This API is used to obtain values of user parameters.				
API URL	/API/SystemConfig/User/Get (to be deprecated) (/API/SystemConfig/User/V2/Get) (general)				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/SystemConfig/User/Range (API for internal use only: /API/SystemConfig/User/V2/Range)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
user_level		String	N	"ADMIN" "API_ADMIN" (for internal use only) "USER1" "USER2" ... "USER6"	This parameter needs to be transferred only when you want to obtain user permission parameters.
type		String	N	"SavePasswor d" "SavePermissi on" "SaveTime" "GetPermissio n" "LockUser"	This parameter needs to be transferred only when you want to obtain user permission parameters.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
ADMIN, API_ADMIN (for internal use only) USER1, USER2, ... USER6		JSON object			Elements in the user object are as follows.
Elements in the user object					
user_state		String	"None" "Normal", "NotYetStart", "Expires", "ExpiresClearInfo"		User status. None Normal NotYetStart: The user information has not expired. Expires: The user information has expired and will be cleared three days after the expiration time.

				ExpiresClearInfo : The user information has expired and has been cleared.
pwd_unequal_to_old_number		String	"1", "2"	The new password must be different from the previous <i>N</i> passwords.
min_password_length		Int	Value range: 8–15	Minimum password length.
min_password_expires		Int	Value range: 0–997	Minimum password validity period.
advice_password_expires		Int	Value range: 0–998	Recommended password validity period.
max_password_expires		Int	Value range: 0–999	Maximum password validity period.
user_enable		Bool		Indicates whether a user is enabled.
expires_switch		Bool		Indicates whether the validity period is enabled.
start_date		String		Start date of the validity period. Format: MM/DD/YYYY
start_time		String		Start time of the validity period. Format: hh:mm:ss
end_date		String		End date of the validity period. Format: MM/DD/YYYY
end_time		String		End time of the validity period. Format: hh:mm:ss
username		String	Maximum length: 16 bytes	User name.

password_empty		Bool		Indicates whether the password is empty.
modify_password_time		String		Last date and time of password change.
user_locked		Bool		Indicates whether a user is locked.
permission		JSON object		Elements in the permission object are as follows.
Elements in the permission object				
log_search	Log search permission.	Bool	Bool	Log search permission.
parameter	Parameter modification permission.	Bool	Bool	Parameter modification permission.
auto_reboot	Automatic restart permission.	Bool	Bool	Automatic restart permission.
remote_login	Remote login permission.	Bool	Bool	Remote login permission.
ip_camera_maintain	Camera maintenance permission.	Bool	Bool	Camera maintenance permission.
ipc_channel_manage	Camera management permission.	Bool	Bool	Camera management permission.
alarm_config	Alarm setting permission.	Bool	Bool	Alarm setting permission.
record_config	Recording setting permission.	Bool	Bool	Recording setting permission.
channel_config	Channel setting permission.	Bool	Bool	Channel setting permission.
face_search	Target library search permission.	Bool	Bool	Target list search permission.
backup_enable	Backup permission.	Bool	Bool	Backup permission.
live_enable	Preview	Bool	Bool	Preview

	permission.			permission.
playback_enable	Playback permission.	Bool	Bool	Video playback permission.
ptz_enable	PTZ control permission.	Bool	Bool	PTZ control permission.
backup_channel	Backup channel permission.	Array	CH1 to CHn	Backup channel permission. Each array bit represents a channel with a string.
live_channel	Preview channel permission.	Array		Preview channel permission. Each array bit represents a channel with a string.
playback_channel	Playback channel permission.	Array		Video playback channel permission. Each array bit represents a channel with a string.
ptz_channel	PTZ control channel permission.	Array		PTZ control channel permission. Each array bit represents a channel with a string.

2.6.2 Setting User Parameters

API Description					
Function		This API is used to set user parameters.			
API URL		/API/SystemConfig/User/Set (to be deprecated) (/API/SystemConfig/User/V2/Set) (general)			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data	Mandatory	Value Range	Server

		Type	(Y/N)		Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
type		String	Y	"SavePassword" "SavePermission" "SaveTime" "LockUser"	Set the user name and password. Use SavePassword to enable and disable a user. Use SavePermission to set permission information. Use SaveTime to set the password validity period. Use LockUser to lock or unlock a user.
pwd_unequal_to_old_number		String	N	"1", "2"	Fixed range.
min_password_length		Int	N	8–15	Fixed range.
min_password_expires		Int	N	0–997	Fixed range.
advice_password_expires		Int	N	0–998	Fixed range.
max_password_expires		Int	N	0–999	Fixed range.
user_info		JSON object	Y	Note: When type is set to SaveTime , this parameter cannot be transferred. In other cases, this parameter is mandatory.	JSON information about the elements in the user_info object is as follows.
Elements in the user_info object					
ADMIN, API_ADMIN (for internal use only)	User ID.	JSON object	N		JSON information about the elements in the

USER1, USER2, ... USER3					object is as follows.
Elements in the user object					
user_enable	Indicates whether to enable a user.	Bool	N	Bool	Indicates whether to enable a user. The admin user does not need to be enabled. Sub-users can be enabled or disabled by the admin or apiadmin user only. (To disable a user, use only the user_enable and username parameters.)
expires_switch		Bool	N	Bool	Indicates whether to enable the user validity period.
start_date	Start date of the user validity period.	String	N	MM/DD/YYYY	Fixed format.
start_time	Start time of the user validity period.	String	N	hh:mm:ss	Fixed format.
end_date	End date of the user validity period.	String	N	MM/DD/YYYY	Fixed format.
end_time	End time of the user validity period.	String	N	hh:mm:ss	Fixed format.
username	User name.	String	N	Maximum length: 16 bytes	In the range. (Only letters, digits, and underscores are allowed.)
enc_password	Encryption password.	JSON object	N	This parameter needs to be transferred only when you want to set	For details, see Table 2-1.

				a user password.	
secondary_authentication	Authentication password.	JSON object	N	This parameter needs to be transferred for authentication only when you want to set a user password or lock or unlock a user.	For details, see Table 2-2. 1. When the admin (or apiadmin) user modifies the admin (apiadmin) parameter, secondary authentication is required. 2. User authentication is required when a user changes the password. 3. Authentication is required when a user is locked.
user_locked	Indicates whether to lock a user.	Bool	N	Bool	Indicates whether to lock a user.
permission	Permission.	JSON object	N		JSON information about the elements in the permission object is as follows. (Note: For general users only.)
Elements in the permission object					
log_search	Log search permission.	Bool		Bool	Bool
parameter	Parameter modification permission.	Bool		Bool	Bool
auto_reboot	Automatic restart permission.	Bool		Bool	Bool
remote_login	Remote login permission.	Bool		Bool	Bool

ip_camera_maintain	Camera maintenance permission.	Bool		Bool	
ipc_channel_manage	Camera management permission.	Bool		Bool	
alarm_config	Alarm setting permission.	Bool		Bool	
record_config	Recording setting permission.	Bool		Bool	
channel_config	Channel setting permission.	Bool		Bool	
face_search	Target list search permission.	Bool		Bool	
backup_enable	Backup permission.	Bool		Bool	Bool
live_enable	Preview permission.	Bool		Bool	Bool
playback_enable	Playback permission.	Bool		Bool	Bool
ptz_enable	PTZ control permission.	Bool		Bool	Bool
backup_channel	Backup channel permission.	Array		CH1 to CHn	The value must be in the format of CHn. The value of n cannot exceed the maximum number of channels of a device.
live_channel	Preview channel permission.	Array			
playback_channel	Playback channel permission.	Array			
ptz_channel	PTZ channel permission.	Array			
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

Table 2-1 Elements in the enc_password object

Elements in the enc_password object				
cipher	String		Maximum length: 1024 bytes	Encrypted password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."
seq	int		0-1000000	Sequence number, which is the same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."

Table 2-2 Elements in the secondary_authentication object

Elements in the secondary_authentication object				
cipher	String		Maximum length: 1024 bytes	Encrypted password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."
seq	int		0-1000000	Sequence number, which is the same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."

2.7 Backup Password Configuration

This section consists of two parts: page configuration and obtaining the backup password during backup.

Page configuration: The real password is not transferred in the Get operation, and only the password status is transferred. If the value is empty, the value is displayed empty. If the value is not empty, the value must be anonymized on the client. During the Set operation, the password is

encrypted using the RSA algorithm and then transmitted to the server. For details about the algorithm, see section 11.5 "Public Key and Random Number for PBKDF2."

Obtaining the backup password: In this scenario, the backup password needs to be transferred to the client due to service requirements. The RSA public key is used for encryption, and the private key is used for decryption. The data is encrypted on the server and decrypted on the client. The encrypted data is converted into a hex string for transmission.

2.7.1 Obtaining Values of Backup Password Parameters

API Description					
Function	This API is used to obtain values of backup password parameters.				
API URL	/API/SystemConfig/BackupEncrypt/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/SystemConfig/BackupEncrypt/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
password_empty		Bool	Bool	Indicates whether the password is empty.	
password		String	Empty	Backup password. The returned value is an empty string. This parameter needs to be displayed on the client. If password_empty is set to true , the value of this parameter is empty. If password_empty is set to false , the value of this parameter is an anonymous character string.	

2.7.2 Setting Backup Password Parameters

API Description					
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Function	This API is used to set backup password parameters.				
API URL	/API/SystemConfig/BackupEncrypt/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
enc_password	Encryption password.	JSON Object	N	This parameter needs to be transferred only when a password is set.	(If this parameter is not transferred, the backup password is not set.) For details, see Table 2-1.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

2.7.3 Obtaining the Backup Password

API Description					
Function	This API is used to obtain the backup password.				
API URL	/API/SystemConfig/BackupEncrypt/TransKey/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					

Pubkey	Public key.	String	Y	Maximum length: 512 bytes	The client generates an RSA public/private key pair and sends the public key to the server.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
key		String	Maximum length: 1024 bytes	The server uses the public key to perform RSA-OAEP encryption and returns the encrypted data.	

2.8 System Information Parameters

API Description					
Function	This API is used to obtain values of system information parameters.				
API URL	/API/SystemInfo/Base/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
device_name		String	Maximum length: 32 bytes		Device name.
device_type		String	Maximum length: 32 bytes		Device type.
hardware_version		String	Maximum length: 32 bytes		Hardware version.
software_version		String	Maximum length: 32 bytes		Software version.
build_time		String	Maximum length: 32 bytes		Build time.

video_format		String	Maximum length: 32 bytes	Video format.
hdd_volume		String	Maximum length: 32 bytes	HDD capacity.
ip_address		String	Maximum length: 32 bytes	IP address.
web		String	Maximum length: 32 bytes	Internal and external web ports.
mac_address		String	Maximum length: 32 bytes	MAC address.
network_state		String	Maximum length: 32 bytes	Network status.
public_cloud_state		String	Maximum length: 32 bytes	Public cloud status.
channel_num		String	Maximum length: 32 bytes	Channel number.

2.9 Channel Information Parameters

API Description					
Function	This API is used to obtain values of channel information parameters.				
API URL	/API/SystemInfo/Channel/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object	Number of channels	JSON information about the elements in the object is as follows.	
channel_max		Int		Total number of channels on the	

				current device.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
alias		String	Maximum length: 32 bytes	Camera alias.
camera_type	Camera type.	String	"fixed_bullet", "bullet", "ball", "dome"	The values represent the box camera, bullet camera, fixed dome camera, and PTZ dome camera respectively.
state		String	Maximum length: 8 bytes "Offline", "Online"	Camera connection status.
mainstream		String	Maximum length: 32 bytes 1920 x 1080, 15 fps, 8192 kbit/s	Primary stream.
substream		String		Secondary stream.
mobile_stream		String		Mobile stream.
motion_detection		String	Maximum length: 10 bytes "Support", "Nonsupport"	Motion detection status.
privacy_zone		String		Privacy mask status.

2.10 Recording Information Parameters

API Description					
Function	This API is used to obtain values of recording information parameters.				
API URL	/API/SystemInfo/Record/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data	Object	N		

	structure.				
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object	Number of channels	JSON information about the elements in the object is as follows.	
channel_max		Int		Total number of channels on the current device.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
record_state		String	Maximum length: 3 bytes "On", "Off"	Recording status.	
record_switch		Bool	Bool	Indicates whether recording is enabled.	
stream_type		String	Maximum length: 12 bytes "Mainstream", "Substream", "DualStream"	Video stream type.	
resolution		String	Maximum length: 32 bytes "Primary stream resolution Secondary stream resolution"	Resolution. Format: Primary stream resolution Secondary stream resolution	
fps		String	Maximum length: 32 bytes "Primary Stream FPS Secondary Stream FPS"	Camera's frame rate. (Unit: fps) Format: Primary Stream FPS Secondary Stream FPS	
bitrate		String	Maximum length: 32 bytes "Primary stream bit rate Secondary stream bit rate"	Camera's bit rate. (Unit: kbit/s) Format: Primary stream bit rate Secondary stream bit rate	

2.11 Network State Parameters

API Description	
Function	This API is used to obtain values of network state parameters.

API URL		/API/SystemInfo/Network/Get			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
wan		JSON object		JSON information about the elements in the wan object is as follows.	
lan		JSON object		JSON information about the elements in the lan object is as follows.	
pppoe		String	Maximum length: 7 bytes	Indicates whether PPPoE is enabled. This parameter is reserved.	
toe		String	"Enable", "Disable"	Indicates whether TOE acceleration is enabled.	
port		JSON object		JSON information about the elements in the port object is as follows.	
bandwidth		JSON object		JSON information about the elements in the bandwidth object is as follows.	
Elements in the wan object					
dhcp		String	Maximum length: 7 bytes "Enable", "Disable"	Indicates whether DHCP is enabled.	
ip_address		String	Maximum length: 16 bytes	IP address, for example, 192.168.1.24.	
subnet_mask		String	Maximum length: 16 bytes	Subnet mask, for example, 255.255.255.0.	
gateway		String	Maximum length: 16 bytes	Default gateway address, for example, 192.168.1.1.	

mac_address		String	Maximum length: 17 bytes	Physical address, for example, 88-60-50-4E-87-29.
ipv6_address		String	Maximum length: 39 bytes	IPv6 address. (This parameter is available only when the HWT-NVR800 supports IPv6.)
ipv6_gateway		String	Maximum length: 39 bytes	IPv6 gateway address. (This parameter is available only when the HWT-NVR800 supports IPv6.)
dns1		String	Maximum length: 16 bytes	Primary DNS server address, for example, 8.8.8.8.
dns2		String	Maximum length: 16 bytes	Secondary DNS server address, for example, 8.8.8.8.
Elements in the lan object				
ip_address		String	Maximum length: 16 bytes	IP address of the switch.
subnet_mask		String	Maximum length: 16 bytes	Subnet mask of the switch.
gateway		String	Maximum length: 16 bytes	Gateway address. This parameter is available only when the HWT-NVR800 supports dual network adapters.
Elements in the port object				
web		String	Internal port: 80, 28080, 38080, 48080, and 58080 External ports are not limited. Indicates whether UPnP is enabled. The value is Inactive or Active . UPnP status. The value is Enable or Disable .	Format: Internal port ID, external port ID, UPnP enabled on the port, UPnP status of the port
client		String	Internal ports: 9000, 9001, 9002, 9003, and 9004 External ports are not limited. Indicates whether UPnP is enabled. The value is Inactive or	

			Active. UPnP status: Enable or Disable (Deprecated)	
rtsp		String	Internal ports: 554, 2554, 3554, 4554, and 5554 External ports are not limited. Indicates whether UPnP is enabled. The value is Inactive or Active . UPnP status: Enable or Disable (Deprecated)	
https		String	Internal ports: 443, 2443, 3443, 4443, and 5443 External ports are not limited. Indicates whether UPnP is enabled. The value is Inactive or Active . UPnP status: Enable or Disable	
Elements in the bandwidth object				
total_bandwidth		String	Maximum length: 32 bytes	Total bandwidth.
used_bandwidth		String	Maximum length: 32 bytes	Used bandwidth.

2.12 Obtaining Values of Page Configuration Parameters

API Description	
Function	This API is used to obtain values of page configuration parameters.
API URL	/API/Login/DevicePage/Get
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
main		JSON array			The following tables describe the elements in the main array.
Elements in the main array					
		JSON object	The array contains six objects, which correspond to the channel management, recording management, alarm configuration, network configuration (available only for the admin or apiadmin user), storage management, and system configuration menus, respectively.		For JSON information about the elements in the objects, see the following tables: Table 2-3 Table 2-4 Table 2-5 Table 2-6 Table 2-7 Table 2-8

Table 2-3 Elements in the object corresponding to the channel management menu

Elements in the object corresponding to the channel management menu					
title		String		"Channel"	Fixed string, identifying the channel management menu.
sub		JSON array		The array contains five objects, which correspond to the channel configuration, preview, image control, privacy mask, and motion detection submenus respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the channel configuration submenu					

title	String		"Channel"	Fixed string, identifying the channel configuration submenu.
pages	JSON array		There are two objects in the array, corresponding to digital channel and protocol management pages respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the digital channel page				
title	String		"IP Channels"	Fixed string, identifying the digital channel page.
page	String		"chn_ipChn"	Fixed string, which is used to locate the digital channel page.
Elements in the object corresponding to the protocol configuration page				
title	String		"Protocol Manage"	Fixed string, identifying the protocol configuration page.
page	String		"chn_protocolMng"	Fixed string, which is used to locate the protocol configuration page.
Elements in the object corresponding to the preview submenu				
title	String		"Live"	Fixed string, identifying the preview submenu.
pages	JSON array		The array contains one object, which corresponds to the preview page.	JSON information about each element is as follows.
Elements in the object corresponding to the preview page				
title	String		"Live"	Fixed string, identifying the preview page.
page	String		"chn_osd"	Fixed string,

				which is used to locate the preview page.
Elements in the object corresponding to the image control submenu				
title	String		"Image Control"	Fixed string, identifying the image control submenu.
pages	JSON array		The array contains one object, which corresponds to the image control page.	JSON information about each element is as follows.
Elements in the object corresponding to the image control page				
title	String		"Image Control"	Fixed string, identifying the image control page.
page	String		"chn_imgCtrl"	Fixed string, which is used to locate the image control page.
Elements in the object corresponding to the privacy mask submenu				
title	String		"Video Cover"	Fixed string, identifying the privacy mask submenu.
pages	JSON array		The array contains one object, which corresponds to the privacy mask page.	JSON information about each element is as follows.
Elements in the object corresponding to the privacy mask page				
title	String		"Video Cover"	Fixed string, identifying the privacy mask page.
page	String		"chn_videoCover"	Fixed string, which is used to locate the privacy mask page.
Elements in the object corresponding to the motion detection submenu				
title	String		"Motion"	Fixed string, identifying the motion detection submenu.

pages	JSON array		The array contains one object, which corresponds to the motion detection page.	JSON information about each element is as follows.
Elements in the object corresponding to the motion detection page				
title	String		"Motion"	Fixed string, identifying the motion detection page.
page	String		"chn_motion"	Fixed string, which is used to locate the motion detection page.

Table 2-4 Elements in the object corresponding to the recording management menu

Elements in the object corresponding to the recording management menu				
title	String		"Record"	Fixed string, identifying the recording management menu.
sub	JSON array		The array contains two objects, which correspond to the encoding parameter setting and recording setting submenus respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the encoding parameter setting submenu				
title	String		"Encode"	Fixed string, identifying the code parameter setting submenu.
pages	JSON array		The array contains two objects, which correspond to the primary stream page and secondary stream page, respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the primary stream page				
title	String		"Mainstream"	Fixed string, identifying the primary stream page.
page	String		"stm_main"	Fixed string, which is used to

				locate the primary stream page.
Elements in the object corresponding to the secondary stream page				
title	String		"Substream"	Fixed string, identifying the secondary stream page.
page	String		"stm_sub"	Fixed string, which is used to locate the secondary stream page.
Elements in the object corresponding to the recording setting submenu				
title	String		"Record"	Fixed string, identifying the recording setting submenu.
pages	JSON array		The array contains two objects, which correspond to the recording setting page and recording plan page respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the recording setting page				
title	String		"Record"	Fixed string, identifying the recording setting page.
page	String		"rec_param"	Fixed string, which is used to locate the recording setting page.
Elements in the object corresponding to the recording plan page				
title	String		"Record Schedule"	Fixed string, identifying the recording plan page.
page	String		"schedules_rec"	Fixed string, which is used to locate the recording plan page.

Table 2-5 Elements in the object corresponding to the alarm configuration menu

Elements in the object corresponding to the alarm configuration menu				
title	String		"Alarm"	Fixed string, identifying the alarm configuration menu.
sub	JSON array		The array contains six objects, which correspond to the motion detection alarm, I/O alarm, alarm linkage, system exception alarm, alarm plan, and email plan submenus.	JSON information about each element is as follows.
Elements in the object corresponding to the motion detection alarm submenu.				
title	String		"AlarmMotion"	Fixed string, identifying the motion detection alarm submenu.
pages	JSON array		The array contains one object, which corresponds to the motion detection alarm page.	JSON information about each element is as follows.
Elements in the object corresponding to the motion detection alarm page				
title	String		"Motion"	Fixed string, identifying the motion detection alarm page.
page	String		"alarm_motion"	Fixed string, which is used to locate the motion detection alarm page.
Elements in the object corresponding to the I/O alarm submenu				
title	String		"IO"	Fixed string, identifying the I/O alarm submenu.
pages	JSON array		The array contains one object, which corresponds to the I/O alarm setting page.	JSON information about each element is as follows.
Elements in the object corresponding to the I/O alarm setting page				
title	String		"IO"	Fixed string, identifying the I/O alarm setting

				page.
page	String		"alarm_io"	Fixed string, which is used to locate the I/O alarm setting page.
Elements in the object corresponding to the alarm linkage submenu				
title	String		"PTZ Linkage"	Fixed string, identifying the alarm linkage submenu.
pages	JSON array		The array contains one object, which corresponds to the alarm linkage setting page.	JSON information about each element is as follows.
Elements in the object corresponding to the alarm linkage setting page				
title	String		"PTZ Linkage"	Fixed string, identifying the alarm linkage setting page.
page	String		"alarm_ptzLinkage"	Fixed string, which is used to locate the alarm linkage setting page.
Elements in the object corresponding to the system exception alarm submenu				
title	String		"Exception"	Fixed string, identifying the system exception alarm submenu.
pages	JSON array		The array contains one object, which corresponds to the exception alarm page.	JSON information about each element is as follows.
Elements in the object corresponding to the exception alarm page				
title	String		"Exception"	Fixed string, identifying the exception alarm page.
page	String		"alarm_exception"	Fixed string, which is used to locate the exception alarm page.

Elements in the object corresponding to the alarm plan submenu.				
title	String		"Alarm Schedule"	Fixed string, identifying the alarm plan submenu.
pages	JSON array		The array contains one object, which corresponds to the alarm plan page.	JSON information about each element is as follows.
Elements in the object corresponding to the alarm plan page				
title	String		"Alarm Schedule"	Fixed string, identifying the alarm plan page.
page	String		"schedules_alarm"	Fixed string, which is used to locate the alarm plan page.
Elements in the object corresponding to the email plan submenu.				
title	String		"Email Schedule"	Fixed string, identifying the email plan submenu.
pages	JSON array		The array contains one object, which corresponds to the email plan page.	JSON information about each element is as follows.
Elements in the object corresponding to the email plan page				
title	String		"Email Schedule"	Fixed string, identifying the email plan page.
page	String		"schedules_email"	Fixed string, which is used to locate the email plan page.

Table 2-6 Elements in the object corresponding to the network configuration menu

Elements in the object corresponding to the network configuration menu				
title	String		"Network"	Fixed string, identifying the network configuration menu.

sub	JSON array		The array contains seven objects, which correspond to the basic network settings, email settings, HTTPS, certificate management, network restriction, cloud service, and platform access submenus respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the basic network settings submenu				
title	String		"Net General"	Fixed string, identifying the basic network settings submenu.
pages	JSON array		The array contains two objects, which correspond to the basic configuration and port configuration pages respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the basic configuration page				
title	String		"General"	Fixed string, identifying the basic configuration page.
page	String		"net_general"	Fixed string, which is used to locate the basic network configuration page.
Elements in the object corresponding to the port configuration page				
title	String		"Port Configuration"	Fixed string, identifying the port configuration page.
page	String		"net_port_conf"	Fixed string, which is used to locate the port configuration page.
Elements in the object corresponding to the email submenu				
title	String		"Email"	Fixed string, identifying the

				email submenu.
pages	JSON array		The array contains one object, which corresponds to the email page.	JSON information about each element is as follows.
Elements in the object corresponding to the email page				
title	String		"Email Configuration"	Fixed string, identifying the email page.
page	String		"net_email"	Fixed string, which is used to locate the email page.
Elements in the object corresponding to the HTTPS submenu				
title	String		"HTTPS"	Fixed string, identifying the HTTPS submenu.
pages	JSON array		The array contains one object, which corresponds to the HTTPS page.	JSON information about each element is as follows.
Elements in the object corresponding to the HTTPS page				
title	String		"HTTPS"	Fixed string, identifying the HTTPS page.
page	String		"net_https"	Fixed string, which is used to locate the HTTPS page.
Elements in the object corresponding to the certificate management submenu				
title	String		"CERT"	Fixed string, identifying the certificate management submenu.
pages	JSON array		The array contains one object, which corresponds to the certificate management page.	JSON information about each element is as follows.
Elements in the object corresponding to the certificate management page				

title	String		"CERT"	Fixed string, identifying the certificate management page.
page	String		"net_cert"	Fixed string, which is used to locate the certificate management page.
Elements in the object corresponding to the network restriction submenu				
title	String		"IP FILTER"	Fixed string, identifying the network restriction submenu.
pages	JSON array		The array contains one object, which corresponds to the network restriction page.	JSON information about each element is as follows.
Elements in the object corresponding to the network restriction page				
title	String		"IP FILTER"	Fixed string, identifying the network restriction page.
page	String		"net_ipFilter"	Fixed string, which is used to locate the network restriction page.
Elements in the object corresponding to the cloud service submenu				
title	String		"Shared Cloud"	Fixed string, identifying the cloud service submenu.
pages	JSON array		The array contains one object, which corresponds to the cloud service page.	JSON information about each element is as follows.
Elements in the object corresponding to the cloud service page				
title	String		"Shared Cloud"	Fixed string,

				identifying the cloud service page.
page	String		"storage_shared_cloud"	Fixed string, which is used to locate the cloud service page.
Elements in the object corresponding to the platform access submenu				
title	String		"Platform"	Fixed string, identifying the platform access submenu.
pages	JSON array		The array contains two objects, which correspond to the ONVIF service and GB/T 28181 pages respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the ONVIF service page				
title	String		"Onvif Server"	Fixed string, identifying the ONVIF service page.
page	String		"net_onvif"	Fixed string, which is used to locate the ONVIF service page.
Elements in the object corresponding to the GB/T 28181 page				
title	String		"T28181 Config"	Fixed string, identifying the GB/T 28181 page.
page	String		"net_gb28181"	Fixed string, which is used to locate the GB/T 28181 page.

Table 2-7 Elements in the object corresponding to the storage management menu

Elements in the object corresponding to the storage management menu				
title	String		"Device"	Fixed string, identifying the storage management

				menu.
sub	JSON array		The array contains one object, which corresponds to the hard disk management submenu.	JSON information about each element is as follows.
Elements in the object corresponding to the hard disk management submenu				
title	String		"Disk"	Fixed string, identifying the hard disk management submenu.
pages	JSON array		The array contains one object, which corresponds to the hard disk management page.	JSON information about each element is as follows.
Elements in the object corresponding to the hard disk management page				
title	String		"Disk"	Fixed string, identifying the hard disk management page.
page	String		"storage_disk"	Fixed string, which is used to locate the hard disk management page.

Table 2-8 Elements in the object corresponding to the system configuration menu

Elements in the object corresponding to the system configuration menu				
title	String		"System"	Fixed string, identifying the system configuration menu.
sub	JSON array		The array contains five objects, which correspond to the basic configuration, multi-user management, backup password, system maintenance, and system information submenus respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the basic configuration submenu				

title	String		"Sys General"	Fixed string, identifying the basic configuration submenu.
pages	JSON array		The array contains three objects, which correspond to the basic configuration, date and time, and output pages respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the basic configuration page				
title	String		"General"	Fixed string, identifying the basic configuration page.
page	String		"sys_general"	Fixed string, which is used to locate the basic system configuration page.
Elements in the object corresponding to the date and time page				
title	String		"Date and Time"	Fixed string, identifying the date and time page.
page	String		"sys_date_time"	Fixed string, which is used to locate the date and time page.
Elements in the object corresponding to the output page				
title	String		"Output Configuration"	Fixed string, identifying the output page.
page	String		"sys_output_conf"	Fixed string, which is used to locate the output page.
Elements in the object corresponding to the multi-user or sub-user management submenu				
title	String		"Multi-User" and "Sub-User": distinguish the admin (or apiadmin) user from other users.	Fixed string, identifying the multi-user or sub-user

				management submenu.
pages	JSON array		The array contains one object, which corresponds to the multi-user or sub-user management page.	JSON information about each element is as follows.
Elements in the object corresponding to the multi-user or sub-user management page				
title	String		"Multi-User"/"Sub-User"	Fixed string, identifying the multi-user or sub-user management page.
page	String		"sys_user"/"sys_sub_user"	Fixed string, which is used to locate the multi-user or sub-user management page.
Elements in the object corresponding to the backup password submenu				
title	String		"Encrypt Backup"	Fixed string, identifying the backup password submenu.
pages	JSON array		The array contains one object, which corresponds to the backup password page.	JSON information about each element is as follows.
Elements in the object corresponding to the backup password page				
title	String		"Encrypt Backup"	Fixed string, identifying the backup password page.
page	String		"sys_encrypt_backup"	Fixed string, which is used to locate the backup password page.
Elements in the object corresponding to the system maintenance submenu				
title	String		"Maintenance"	Fixed string, identifying the system

				maintenance submenu.
pages	JSON array		The array contains four objects, which correspond to the factory setting restoration page, system upgrade page, automatic maintenance page, and developer mode page respectively. When a sub-user logs in, only the automatic maintenance page is available.	JSON information about each element is as follows.
Elements in the object corresponding to the factory setting restoration page				
title	String		"Sys Load Default"	Fixed string, identifying the factory setting restoration page.
page	String		"sys_load_default"	Fixed string, which is used to locate the factory setting restoration page.
Elements in the object corresponding to the system upgrade page				
title	String		"Sys Upgrade"	Fixed string, identifying the system upgrade page.
page	String		"sys_upgrade"	The value is a fixed string, which is used to locate the system upgrade page.
Elements in the object corresponding to the automatic maintenance page				
title	String		"Sys Auto Reboot"	Fixed string, identifying the automatic maintenance page.
page	String		"sys_auto_reboot"	Fixed string, which is used to locate the automatic maintenance page.
Elements in the object corresponding to the developer mode page				
title	String		"Sys Develop"	Fixed string,

				identifying the developer mode page.
page	String		"sys_develop"	Fixed string, which is used to locate the developer mode page.
Elements in the object corresponding to the system information submenu				
title	String		"Information"	Fixed string, identifying the system information submenu.
pages	JSON array		The array contains four objects, which correspond to the device information, channel information, recording information, and network information pages respectively.	JSON information about each element is as follows.
Elements in the object corresponding to the device information page				
title	String		"Information"	Fixed string, identifying the device information page.
page	String		"sysInfo_base"	Fixed string, which is used to locate the device information page.
Elements in the object corresponding to the channel information page				
title	String		"Channel Information"	Fixed string, identifying the channel information page.
page	String		"sysInfo_chn"	Fixed string, which is used to locate the channel information page.
Elements in the object corresponding to the recording information page				
title	String		"Record Info"	Fixed string, identifying the recording

				information page.
page	String		"sysInfo_rec"	Fixed string, which is used to locate the recording information page.
Elements in the object corresponding to the network information page				
title	String		"Network State"	Fixed string, identifying the network information page.
page	String		"sysInfo_net"	Fixed string, which is used to locate the network information page.

3 Network Parameter Configuration

3.1 Basic Network Parameters

3.1.1 Obtaining Values of Basic Network Parameters

API Description					
Function	This API is used to obtain values of basic network parameters.				
API URL	/API/NetworkConfig/NetBase/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/NetBase/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks

wan		JSON object		JSON information about the elements in the object is as follows.
lan		JSON object		JSON information about the elements in the object is as follows.
pppoe		JSON Object		JSON information about the elements in the object is as follows.
port		JSON array		JSON information about the elements in the array is as follows.
extern_ip		String	Maximum length: 16 bytes	External IP address. This parameter is valid only when the external network is enabled. Note: This parameter can be obtained but cannot be set.
toe		Bool	Bool	Indicates whether TOE acceleration is enabled. This parameter is not supported currently.
Elements in the wan object				
dhcp		Bool	Bool	Indicates whether DHCP is enabled.
ip_address		String	Maximum length: 16 bytes	IP address, for example, 192.168.1.24.
subnet_mask		String	Maximum length: 16 bytes	Subnet mask, for example, 255.255.255.0.
gateway		String	Maximum length: 16 bytes	Default gateway address, for example, 192.168.1.1.
ipv6_address		String	Maximum length: 39 bytes	IPv6 address. (This parameter is available only when the HWT-NVR800 supports IPv6.)
ipv6_prefixlen		String	Maximum length: 3 bytes	(This parameter is available only when the HWT-NVR800 supports IPv6.)
ipv6_gateway		String	Maximum length: 39 bytes	IPv6 gateway address.

				(This parameter is available only when the HWT-NVR800 supports IPv6.)
dns1		String	Maximum length: 16 bytes	Primary DNS server address, for example, 8.8.8.8.
dns2		String	Maximum length: 16 bytes	Secondary DNS server address, for example, 8.8.8.8.
ipc_dhcp		Bool	Bool	Indicates whether camera DHCP is enabled.
Elements in the lan object				
ip_address		String	Maximum length: 16 bytes	IP address of the switch. If the network segment of the IP address is changed, the system will restart.
subnet_mask		String	Maximum length: 16 bytes	Subnet mask of the switch (parameter change or system restart).
gateway		String	Maximum length: 16 bytes	Gateway address. This parameter is available only when the HWT-NVR800 supports dual network adapters.
configure_segment		String	Maximum length: 200 bytes	Route network segment. This parameter is available only when the HWT-NVR800 supports dual network adapters.
Elements in the pppoe array (not used)				
enable		Bool	Bool	Indicates whether PPPoE is enabled.
user		String	Maximum length: 35 bytes	PPPoE user name.
password		String	Maximum length: 35 bytes	PPPoE password.
Elements in the port array				
		JSON object		JSON information about the elements in the object is as follows.

Elements in the object				
service		String	"Web", "Https",	Port type.
protocol		String	"TCP"	Protocol type. Note: This parameter can be obtained but cannot be set.
internal_port		Int	[0,80] [443,2443,3443,4443,5443]	The internal port number range corresponds to "Web" and "Https" in sequence.
external_port		Int	1–65535	External port number.
upnp_status		String	"Inactive", "Active"	UPnP status. Note: This parameter can be obtained but cannot be set.
upnp		Bool	Bool	Indicates whether UPnP is enabled on the port.
mapping_strategy		String	"Manual" "Auto"	Mapping policy.

3.1.2 Setting Basic Network Parameters

API Description					
Function	This API is used to set basic network parameters.				
API URL	/API/NetworkConfig/NetBase/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandator y (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
wan		JSON object	N		JSON information about the elements is

					as follows.
lan		JSON object	N		JSON information about the elements is as follows.
pppoe		JSON array	N		JSON information about the elements in the array is as follows.
port		JSON array	N		JSON information about the elements in the array is as follows.
toe		Bool	N	Bool	Indicates whether to enable TOE acceleration. This parameter is not supported currently.
Elements in the wan object					
dhcp		Bool	N	Bool	Bool
ip_address		String	N	Maximum length: 16 bytes	IP address format.
subnet_mask		String	N	Maximum length: 16 bytes	IP address format.
gateway		String	N	Maximum length: 16 bytes	Gateway address.
ipv6_address		String	N	Maximum length: 39 bytes	IPv6 address. (This parameter is available only when the HWT-NVR800 supports IPv6.)
ipv6_prefixlen		String	N	Maximum length: 3 bytes	(This parameter is

					available only when the HWT-NVR800 supports IPv6.)
ipv6_gateway		String	N	Maximum length: 39 bytes	IPv6 gateway address. (This parameter is available only when the HWT-NVR800 supports IPv6.)
dns1		String	N	Maximum length: 16 bytes	Primary DNS server IP address.
dns2		String	N	Maximum length: 16 bytes	Secondary DNS server IP address.
ipc_dhcp		Bool	N	Bool	Indicates whether to enable camera DHCP.
Elements in the lan object					
ip_address		String	N	Maximum length: 16 bytes	IP address format. If the value is changed, the system will restart.
subnet_mask		String	N	Maximum length: 16 bytes	Subnet mask. If the value is changed, the system will restart.
gateway		String	N	Maximum length: 16 bytes	Gateway address. This parameter is available only when the HWT-NVR800 supports dual network adapters.

configure_segment		String	N	Maximum length: 200 bytes	Route network segment. This parameter is available only when the HWT-NVR800 supports dual network adapters.
Elements in the pppoe array (not used)					
enable		Bool	N	Bool	Bool
user		String	N	Maximum length: 35 bytes	In the range. Special characters will be checked.
password		String	N	Maximum length: 35 bytes	In the range.
Elements in the port array					
		JSON object			JSON information about the elements in the objects of the port array is as follows.
Elements in the objects of the port array					
service		String	Y	Maximum length: 6 bytes "Web", "Https",	Fixed options in the range.
internal_port		Int	N	[0,80] [443,2443,3443,4443,5443]	The internal port number range corresponds to "Web" and "Https" in sequence.
external_port		Int	N	1–65535	In the range.
upnp_status		String	N	Maximum length: 8 bytes "Inactive", "Active"	Fixed options in the range.
upnp		Bool	N	Bool	Bool
mapping_strategy		String	N	"Manual" "Auto"	Fixed options.

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)

3.1.3 Setting Network Test Parameters

API Description					
Function	This API is used to set network test parameters.				
API URL	/API/NetworkConfig/NetBase/Test				
Request Method	POST				
Message ID					
Remarks					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
net_test		JSON object	Y		JSON information about the elements in the net_test object is as follows.
Elements in the net_test object					
nic_choice	Network adapter.	String	Y	"WAN", "LAN"	WAN : local connection; LAN : internal connection Note: LAN can be transmitted only when the LAN information is available in section 3.1.1 "Obtaining Values of Basic Network Parameters."
destination	Destination address.	String	Y	Maximum length: 127 bytes	Network address to be tested.
Output Parameters					

Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success for success and failed for failure
reason		String		This parameter is returned when a request error occurs.
error_code		String		incorrect address : The destination address is incorrect.
data	Data structure.	Object		If the operation is successful, elements in the data object are returned. If the operation fails, an empty string is returned.
Elements in the data object				
avgtime		float	[0.000–5000.000]	Network delay, in milliseconds.
lost		float	[0.000–100.000]	Packet loss rate, in percentage.

3.2 Email Parameters

3.2.1 Obtaining Values of Email Parameters

API Description					
Function	This API is used to obtain values of email parameters.				
API URL	/API/NetworkConfig/Email/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/Email/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
data	Data structure.	Object	Y		

Elements in the data object				
email_enable		Bool	Bool	Indicates whether the email function is enabled.
encryption		String	Maximum length: 7 bytes "Disable", "SSL", "TLS", "Auto"	Encryption type.
smtp_port		Int	1–65535	SMTP port.
smtp_server		String	Maximum length: 127 bytes	SMTP server.
username				Email user name.
password				Email login password.
sender				Sender address.
password_empty		Bool	Bool	Indicates whether the password is empty.
interval_time		Int	1, 3, 5, and 10	Sending interval. Unit: minute
recvemail		JSON object		Recipient email address. JSON information about the elements in the object is as follows.
Elements in the recvemail object				
recvemail_1		String	Maximum length: 127 bytes	Email address of the recipient.
recvemail_2		String	Maximum length: 127 bytes	Email address of the recipient.
recvemail_3		String	Maximum length: 127 bytes	Email address of the recipient.

3.2.2 Setting Email Parameters

API Description	
Function	This API is used to set email parameters.
API URL	/API/NetworkConfig/Email/Set
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
email_enable		Bool	N	Bool	Bool
encryption		String	N	Maximum length: 7 bytes "Disable", "SSL", "TLS", "Auto"	Fixed options in the range.
smtp_port		Int	N	1–65535	In the range.
smtp_server		String	N	Maximum length: 127 bytes	In the range.
username			N		In the range.
sender			N		In the range.
enc_password	Encryption password.	JSON object	N		If the password is not changed, this parameter does not need to be transferred. For details, see Table 2-1.
interval_time		Int	N	1, 3, 5, and 10	Fixed options in the range.
recvemail		JSON object	N		Recipient email address. JSON information about the elements in the object is as follows.
Elements in the recvemail object					
recvemail_1		String	N	Maximum length: 127 bytes	In the range.
recvemail_2		String	N	Maximum length: 127 bytes	In the range.
recvemail_3		String	N	Maximum length: 127 bytes	In the range.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)

3.3 Email Test Parameters

API Description

Function	This API is used to set email test parameters.
API URL	/API/NetworkConfig/Email/Test
Request Method	POST
Message ID	
Remarks	During the test, the email_test_flag parameter is mandatory. 1. If the value of email_test_flag is Start, the system starts to test whether the email function is normal. In this mode, test_state and test_id cannot be transferred, and the following parameters are optional: email_enable, encryption, sender, interval_time, smtp_port, and enc_password. When the email function is disabled, the test is invalid and other parameters are also optional. When the email function is enabled, smtp_server and username must be transferred. 2. If the value of email_test_flag is Stop, the email test is stopped. Only the email_test_flag parameter needs to be transferred. 3. If the value of email_test_flag is Query, the email test result is obtained. That is, the current API is invoked to test the email, and test_id returned by the test is used as the input parameter to invoke the current API again to obtain the test result. The only difference is that the value of email_test_flag is changed to Query. In the current mode, only the test_id parameter needs to be transferred.

Input Parameters

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
email_enable		Bool	N	Bool	Bool

encryption		String	N	Maximum length: 7 bytes "Disable", "SSL", "TLS", "AUTO"	Fixed options in the range.
smtp_port		Int	N	1-65535	In the range.
smtp_server		String	N	Maximum length: 127 bytes	In the range.
username			N		In the range.
sender			N		In the range.
recvemail_1			N		In the range. The email format is checked.
recvemail_2			N		
recvemail_3			N		
enc_password	Encryption password.	JSON object	N		For details, see Table 2-1.
interval_time		Int	N	1,3,5,10	Fixed options in the range.
test_id		Int	N	1-65535	In the range.
email_test_flag		String	Y	"Start", "Stop", "Query"	Fixed options.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
test_state		String	"testing": testing "ok": test completed	Email test status. (If the test fails, an error is returned.)	

3.4 IP Filter Parameters

3.4.1 Obtaining Values of Network Restriction Parameters

API Description	
Function	This API is used to obtain values of network restriction parameters.
API URL	/API/NetworkConfig/IPFilter/Get
Request Method	POST

Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/IPFilter/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
enable		Bool	Bool	Indicates whether the IP filter function is enabled.	
choose		String	Maximum length: 9 bytes "Whitelist", "Blacklist"	Restriction mode.	
restricted_type		String	Maximum length: 9 bytes "Whitelist", "Blacklist"	Blocklist or trustlist.	
whitelist		JSON array		JSON information about the elements in the array is as follows.	
blacklist		JSON array		JSON information about the elements in the array is as follows.	
Elements in the whitelist/blacklist array					
start_address		String	Maximum length: 16 bytes	Start filtered IP address.	
end_address		String	Maximum length: 16 bytes	End filtered IP address.	
ip_type		String	"Ipv4", "Ipv6"	IP address type.	

3.4.2 Setting Network Restriction Parameters

API Description	
Function	This API is used to set network restriction parameters.

API URL		/API/NetworkConfig/IPFilter/Set			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
enable		Bool	N	Bool	Bool
choose		String	N	Maximum length: 9 bytes "Whitelist", "Blacklist"	Fixed options in the range.
restricted_type		String	N	Maximum length: 9 bytes "Whitelist", "Blacklist"	Fixed options in the range.
whitelist		JSON array	N		
blacklist		JSON array	N		
Elements in the whitelist/blacklist array					
start_address		String	N	Maximum length: 16 bytes	In the range. The IP address format is checked.
end_address		String	N	Maximum length: 16 bytes	In the range. The IP address format is checked.
ip_type		String	N	Maximum length: 4 bytes "Ipv4", "Ipv6"	Fixed options in the range.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

3.5 HTTPS Parameters

3.5.1 Obtaining Values of the HTTPS Parameters

API Description					
Function	This API is used to obtain values of HTTPS parameters.				
API URL	/API/NetworkConfig/https/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/https/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
file_type	Certificate installation type.	String	"Default", "Custom"		Default: default installation Custom: custom installation
file_exist	Indicates whether the certificate file exists.	Int	0 or 1		The options include 0 (no) and 1 (yes).
subject		String			Issuing object (required for custom installation).
issuer		String			Issuer (required for custom installation).
not_before		String			Start time (required for custom installation).
not_after		String			End time (required for custom installation).

3.5.2 Setting HTTPS Parameters

API Description					
Function	This API is used to set HTTPS parameters.				
API URL	/API/NetworkConfig/https/Set				
Request Method	POST				
Message ID					
Remarks	When operate is set to Install, ca_file and key_file must be transferred, and root_ca_file and enc_password are optional. When operate is set to Uninstall, other parameters are optional. When operate is set to Switch, file_type must be transferred.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
operate	Operation type.	String	Y	"Install", "Uninstall", "Switch"	Fixed options, indicating certificate installation, uninstallation, and switchover respectively.
file_type	Certificate installation type.	String	N	"Default", "Custom"	Fixed options.
ca_file	Certificate.	String	N	0-10240	In the range.
key_file	Key.	String	N	0-10240	In the range.
enc_password	Private key password.	JSON object	N		For details, see Table 2-1.
root_ca_file	Root certificate.	String	N	0-10240	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

Table 3-1 Error codes

install_failed	Installation failed.
uninstall_failed	Uninstallation failed.
uninstall_not_allow	Uninstallation is not allowed because the device is in use.
switch_failed	Switchover failed.
upload_failed	Installation failed.
upload_success	Installation success.
length_too_long	Oversized data.
unsafe_siganature	Insecure certificate.
cert_key_not_match	Certificate mismatch.
invalid_cert_time	Invalid certificate time.
invalid_private_key	Invalid private key file.
invalid_cert	Invalid certificate file.
invalid_key_usage	Invalid certificate key usage.
invalid_cert_chain	Invalid certificate chain.
invalid_cacert	Invalid root certificate.
invalid_cacert_time	Invalid root certificate time.

3.6 Certificate Management Parameters

API Description					
Function	This API is used to set certificate management parameters.				
API URL	/API/NetworkConfig/CertManage/Set				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/CertManage/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
cert_info	Certificate.	Json object	Y		JSON information about the elements in the object is as follows.
Elements in the cert_info object					
cloud		JSON object	N		JSON information about the elements in the object is as follows.
sdk		JSON object	N		
device		JSON object	N		
Elements in the cloud , sdk , and device objects					
operate		String	N	"Upload", "Restore"	Fixed options, indicating uploading the certificate and restoring the default settings respectively.
root_cert		String	N	0–4096	In the range (Uninstall not needed).
local_cert		String	N	0–2048	In the range (Uninstall not needed, Device type not needed).
local_private_key		String	N	0–2048	
enc_password		JSON object	N		For details, see Table 2-1.
install_unsafe_cert		Bool	N	true, false	Uninstall not needed. If the root certificate contains an invalid certificate or the certificate is insecure, you can choose whether to install the certificate.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

3.7 ONVIF Service

3.7.1 Obtaining Values of ONVIF Service Parameters

API Description	
Function	This API is used to obtain values of ONVIF service parameters.

API URL	/API/NetworkConfig/OnvifServer/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/OnvifServer/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
switch		Bool	Bool		Enable the function
auth_type		String	"Digest_sha256", "Digest", "Digest/WSSE", "WSSE", "None"		Authentication mode.
transmit_type		String	"HTTP/HTTPS", "HTTPS", "HTTP"		Transmission protocol.
username		String	0-16		User name.
password_empty		Bool	Bool		Indicates whether the password is empty.
client_transmit_type		String	"HTTPS", "HTTP"		Transmission protocol of the client.
md5switch		Bool	Bool		Indicates whether MD5 is disabled.
link_status		JSON array			Connection status information. Elements in the array are as follows.
Elements in the link_status array					
		JSON object			Status of a single connection. Elements in a single object are as follows.
Elements in a single object					
remote_ip		String	Maximum length: 16 bytes		Client IP address.

login_state		Int	0: connecting; 1: connected; 2: not connected	Connection status.
status_code		String	"Param_err", "Ipcstatus_unauther", "Session_not_found", "Not_implemented", "Network_error", "Onvif_switch_close"	Reason for being offline. The values represent the following respectively: incorrect parameter, incorrect user name or password, session not found, function not implemented, network connection failed, and ONVIF service being disabled.

3.7.2 Setting ONVIF Service Parameters

API Description					
Function	This API is used to set ONVIF service parameters.				
API URL	/API/NetworkConfig/OnvifServer/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
switch		Bool	N	Bool	Bool
auth_type		String	N	"Digest_sha256", "Digest", "Digest/WSSE", "WSSE", "None"	Fixed range.
transmit_type		String	N	"HTTP/HTTPS", "HTTPS", "HTTP"	Fixed range.
username		String	N	0-16	Limited length

					and characters.
client_transmit_type		String	N	"HTTPS", "HTTP"	Fixed range.
md5switch		Bool	N	Bool	Bool
enc_password		JSON object	N		
Elements in the enc_password object					
seq		Int	Y	0-1000000	Fixed range (same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2").
cipher		String	Y	Maximum length: 1024 bytes	Fixed length. (The value of cipher is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2".)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

3.8 GB/T 28181-compliant Platform Parameters

3.8.1 Obtaining Values of GB/T 28181-compliant Platform Parameters

API Description					
Function	This API is used to obtain values of GB/T 28181-compliant platform parameters.				
API URL	/API/NetworkConfig/T28181/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/T28181/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)
server_port		Int	1-65535		Server port.
local_port		Int	5060, 5061, 5062, 5063, 5064		Local port.
stream_port		Int	55550, 55551, 55552, 55553, 55554		Stream port.
heart_beat_time		Int	5-255(s)		Heartbeat interval.
expires		Int	30-100000(s)		Registration validity period.
enable_flag		Bool			Indicates whether to the platform is enabled.
max_timeouts		Int	3-255		Maximum timeout count.
stream_type		String	"Primary", "Secondary"		Stream type, which can be primary or secondary streams.

link_status		String	"Not_Connect", "Register_success", "Network_error", "Password_deviceld_error", "GB28181_close", "GB28181_open"	Connection status. The values represent the following respectively: disconnected, registration succeeded (online), network error, incorrect password or device ID, GB/T 28181 disabled, and unregistered.
server_ip		String	Maximum length: 32 bytes	Platform IP address.
server_id		String	Value range: 1–20 bytes	Device ID (a numeric string).
device_id		String	Value range: 1–20 bytes	Device ID (a numeric string).
password		String	Value range: 1–32 bytes	Password. Leave this parameter empty when obtaining parameters.
password_empty		Bool		Indicates whether the actual password is empty.
device_name		String	Maximum length: 32 bytes	
server_domain		String	Maximum length: 12 bytes	Server domain.
channel_nvr_id		JSON array		HWT-NVR800 channel ID. The elements in the array are as follows.
Elements in the channel_nvr_id array				
		JSON object		ID of a single channel. The elements are as follows.
Elements in a single object				
channel_id		String	Maximum length: 20 bytes	Channel ID (a numeric string).
md5_enable		Bool		Indicates whether MD5 is disabled.

3.8.2 Setting GB/T 28181-compliant Platform Parameters

API Description					
Function	This API is used to set GB/T 28181-compliant platform parameters.				
API URL	/API/NetworkConfig/T28181/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
server_port		Int	N	1-65535	Fixed range.
local_port		Int	N	5060, 5061, 5062, 5063, 5064	Fixed range.
stream_port		Int	N	55550, 55551, 55552, 55553, 55554	Fixed range.
heart_beat_time		Int	N	5-255(s)	Fixed range.
expires		Int	N	30-100000(s)	Fixed range.
enable_flag		Bool	N	This parameter is optional. If the value of this parameter is true , the server_ip parameter must be valid. When enable_flag is set to false , the enc_password parameter cannot be transferred.	
max_timeouts		Int	N	3-255	Fixed range.
stream_type		String	N	"Primary", "Secondary"	Fixed options.
server_ip		String	N	Maximum length: 32 bytes	Fixed range.
server_id		String	N	Value range: 1–20 bytes	Fixed range.
device_id		String	N	Value range: 1–20 bytes	Fixed range.
device_name		String	N	Maximum length: 32 bytes	Fixed range.

server_domain		String	N	Maximum length: 32 bytes	Fixed range.
channel_nvr_id		JSON array	N		For details, see Table 3-2.
enc_password		JSON object	N		For details, see Table 2-1.
md5_enable		Bool	N	Bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

Table 3-2 Elements in the channel_nvr_id array

Elements in the channel_nvr_id array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
channel_id		String	Maximum length: 20 bytes	Channel ID (a numeric string).

3.9 RTSP Parameters

3.9.1 Obtaining Values of RTSP Parameters

API Description	
Function	This API is used to obtain values of RTSP parameters.
API URL	/API/NetworkConfig/RtspServer/Get
Request Method	POST
Message ID	
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/RtspServer/Range

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
auth_type		String	"sha256_auth", "md5_auth"		Authentication mode.
rtspmd5switch		Bool	Bool		Indicates whether MD5 is disabled.

3.9.2 Setting RTSP Parameters

API Description					
Function	This API is used to set RTSP parameters.				
API URL	/API/NetworkConfig/RtspServer/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
auth_type		String	N	"sha256_auth", "md5_auth"	Fixed options.
rtspmd5switch		Bool	N	Bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

3.10 FTP Parameters

3.10.1 Obtaining Values of FTP Parameters

API Description					
Function	This API is used to obtain values of FTP parameters.				
API URL	/API/NetworkConfig/FtpServer/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/NetworkConfig/FtpServer/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)
ftpport		Int	1–65535		Server port.
ftploginname		String	Maximum length: 63 bytes		User name.
ftploginpwd		String	Maximum length: 63 bytes		Password.
picquality		String	"high_definition", "standard_definition"		Image quality, which can be HD or SD.
ftpipaddr		String	Maximum length: 63 bytes		Platform IP address.
protocol		String	"FTP", "SFTP"		Protocol, which can be FTP or SFTP.
dirtytype		String	"root_directory", "one_directory", "second_directory", "three_directory", "four_directory"		Directory structure, which can be level-1 directory, level-2

				directory, level-3 directory, or level-4 directory.
subdirtype1		String	"equipment_ip", "time_ym", time_ymd", "channel_number", "user_defined"	Level-1 directory. The options include the device IP address, time (YYYY-MM), time (YYYY-MM-D D), channel ID, and user-defined value.
subdirtype2		String	"equipment_ip", "time_ym", time_ymd", "channel_number", "user_defined"	Level-2 directory. The options include the device IP address, time (YYYY-MM), time (YYYY-MM-D D), channel ID, user-defined value.
subdirtype3		String	"equipment_ip", "time_ym", time_ymd", "channel_number", "user_defined"	Level-3 directory. The options include the device IP address, time (YYYY-MM), time (YYYY-MM-D D), channel ID, user-defined value.
subdirtype4		String	"equipment_ip", "time_ym", time_ymd", "channel_number", "user_defined"	Level-4 directory. The options include the device IP address, time (YYYY-MM), time

				(YYYY-MM-DD), channel ID, user-defined value.
customname1		String	Maximum length: 63 bytes	User-defined path of the level-1 directory.
customname2		String	Maximum length: 63 bytes	User-defined path of the level-2 directory.
customname3		String	Maximum length: 63 bytes	User-defined path of the level-3 directory.
customname4		String	Maximum length: 63 bytes	User-defined path of the level-4 directory.
picfilename1		String	"channel", "time"	Image naming element 1: channel ID and time.
picfilename2		String	"channel", "time"	Image naming element 2: channel ID and time.
separator		String	":", "+", "-", "_", "="	Separator of image naming elements.
password_empty		Bool	true, false	Indicates whether the password is empty.
channel_parameter		object array		Channel parameters. For details, see Table 3-3.

Table 3-3 Elements in the channel_parameter array

Elements in the **channel_parameter** array

		JSON object		The following part describes elements in a single object, indicating the capture interval of a channel and whether FTP is enabled.
Elements in a single object				
channel		String	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	If the parameter of a channel needs to be modified, this parameter must be transferred.
snapinterval		int		Capture interval of the channel specified by the channel parameter, in seconds.
enableftp		Bool		Indicates whether FTP is enabled for the channel specified by the channel parameter. The options include true (yes) and false (no).

3.10.2 Setting FTP parameters

API Description					
Function	This API is used to set FTP parameters.				
API URL	/API/NetworkConfig/FtpServer/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
ftpport		Int	N	1–65535	Fixed range.

ftploginname		String	N	Maximum length: 63 bytes	Fixed range.
picquality		String	N	"high_definition", "standard_definition"	Fixed options.
ftpipaddr		String	N	Maximum length: 63 bytes	Fixed range.
protocol		String	N	"FTP", "SFTP"	Fixed options.
dirtytype		String	N	"root_directory", "one_directory", "second_directory", "third_directory", "four_directory"	Fixed options.
subdirtytype1		String	N	"equipment_ip", "time_ym", "time_ymd", "channel_number", "user_defined"	Fixed options.
subdirtytype2		String	N	"equipment_ip", "time_ym", "time_ymd", "channel_number", "user_defined"	Fixed options.
subdirtytype3		String	N	"equipment_ip", "time_ym", "time_ymd", "channel_number", "user_defined"	Fixed options.
subdirtytype4		String	N	"equipment_ip", "time_ym", "time_ymd", "channel_number", "user_defined"	Fixed options.
customname1		String	N	Maximum length: 63 bytes	Fixed range.
customname2		String	N	Maximum length: 63 bytes	Fixed range.
customname3		String	N	Maximum length: 63 bytes	Fixed range.
customname4		String	N	Maximum length: 63 bytes	Fixed range.
picfilename1		String	N	"channel", "time"	Fixed options.
picfilename		String	N	"channel", "time"	Fixed options.

2					
separator		String	N	".", "+", "-", "_", "="	Fixed options.
enc_password		JSON object	N		For details, see Table 2-1.
channel_parameter		object array	N		Channel parameters. For details, see Table 3-3.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)

3.10.3 Setting FTP Test Parameters

API Description

Function	This API is used to set FTP test parameters.
API URL	/API/NetworkConfig/FtpServer/Test
Request Method	POST
Message ID	
Remarks	

Input Parameters

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
ftpport		Int	N	1–65535	Bool
ftpploginname		String	N	Maximum length: 63 bytes	Fixed range.
ftpipaddr		String	N	Maximum length: 63 bytes	Fixed range.

protocol		String	N	"FTP", "SFTP"	Fixed options.
enc_password		JSON object	N		For details, see Table 2-1.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

4 Channel Parameters

4.1 Channel Configuration Parameters

4.1.1 Searching for Cameras

API Description					
Function	This API is used to search for cameras.				
API URL	/API/ChannelConfig/RemoteDev/Search				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range (for the API for searching for cameras and the API for setting camera parameters): /API/ChannelConfig/RemoteDev/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
device_info		JSON array			JSON information about the elements in the array is as follows.

Elements in the device_info array				
		JSON object		Information about a single camera. The elements are as follows.
Elements in a single object				
ip_address		String	Maximum length: 63 bytes	IP address or domain name.
subnet_mask		String	Maximum length: 15 bytes	Subnet mask.
gateway		String	Maximum length: 15 bytes	Gateway.
dns1		String	Length: 7–15 bytes	DNS.
dns2		String	Length: 7–15 bytes	DNS.
port		Int	1–65535	Port number.
web_port		Int	1–65535	Media port number. (If no media port number is available, this parameter is not transferred.)
channel_num		Int	The maximum value cannot exceed the maximum number of HWT-NVR800 channels.	Number of channels supported by each device (read-only)
protocol		String	Maximum length: 32 bytes "HWTSDK", "Onvif", "Custom 1", "Custom 2", "Custom 3", "Custom 4", "Custom 5", "Custom 6", "Custom 7", "Custom 8", "Custom 9", "Custom 10",	Camera access protocol.

			"Custom 11", "Custom 12", "Custom 13", "Custom 14", "Custom 15", "Custom 16"	
manufacturer		String	Maximum length: 35 bytes	Vendor.
device_type		String	Maximum length: 35 bytes	Device type.
device_type_flag		String	Maximum length: 32 bytes	Device type flag.
mac_address		String	Maximum length: 35 bytes	MAC address.
software_version		String	Maximum length: 35 bytes	Firmware version.
version_flag		Int	0–255	Version flag.
activesign		Int	0–2	Activation flag.
serial_number		String	Maximum length: 32 bytes	Serial number.
camera_type		String	"fixed_bullet", "bullet", "ball", "dome"	The values represent the box camera, bullet camera, fixed dome camera, and PTZ dome camera respectively.

4.1.2 Setting Camera Parameters

API Description	
Function	This API is used to set camera parameters.
API URL	/API/ChannelConfig/RemoteDev/Set
Request Method	POST
Message ID	
Remarks	This API is used to change IP addresses of cameras whose information has been found on the OMU portal.
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
device_info		JSON array	Y		JSON information about the elements in the array is as follows.
seq		Int	Y		Sequence number.
Elements in the device_info array					
		JSON object			Information about a single camera. The elements are as follows.
Elements in a single object					
old_ip_address		String	Maximum length: 63 bytes	Old IP address of a camera.	
ip_address		String	Maximum length: 63 bytes	New IP address of a camera.	
subnet_mask		String	Maximum length: 15 bytes	Subnet mask.	
gateway		String	Maximum length: 15 bytes	Gateway.	
port		Int	1–65535	Port number.	
web_port		Int	1–65535	Media port number. (If the parameter is obtained, set it to the original value. If the parameter is not obtained, this parameter is not transferred.)	
protocol		String	Maximum length: 32 bytes " HWTSDK ", "Onvif",	Camera access protocol. This parameter is mandatory when the camera IP address is changed, indicating that the camera IP address is changed through the HOLOWITS or ONVIF protocol.	
mac_address		String	Maximum length: 35 bytes	MAC address.	
username		String	Maximum length: 31 bytes	User name.	
cipher		String	Maximum length: 1024 bytes	Encrypted password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for	

				PBKDF2."
Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)

4.1.3 Obtaining Values of Protocol Management Parameters

API Description					
Function	This API is used to obtain values of protocol management parameters.				
API URL	/API/ChannelConfig/ProtocolManage/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range (for the API for obtaining values of protocol management parameters and the API for setting these parameters): /API/ChannelConfig/ProtocolManage/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
protocol_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the protocol_info object					
protocol1-protocol16		Json object		Elements in the protocol object are as follows.	
Elements in the protocol object					
protocol_name		String	Maximum length: 15 bytes	Protocol name.	
custom_stream		Json Array		Elements in the array are as follows.	

Elements in the custom_stream array				
stream_name		String	"Mainstream", "Substream"	
enable		Bool	true, false	Indicates whether streams are enabled. This parameter is available only for secondary streams.
type		String	"Rtsp"	
port		Int	1–65535	
source_path		String	Maximum length: 119 bytes	

4.1.4 Setting Protocol Management Parameters

API Description					
Function	This API is used to set protocol management parameters.				
API URL	/API/ChannelConfig/ProtocolManage/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
protocol_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the protocol_info object					
protocol1-protocoln		JSON object	N		JSON information about the elements in the object is as follows.
Elements in the protocol object					

protocol_name		String	N	Maximum length: 15 bytes	Fixed options.
custom_stream		JSON array	N		Elements in the array are as follows.
Elements in the custom_stream array					
stream_name		String	N	"Mainstream", "Substream"	Fixed options.
enable		Bool	N	true, false	Bool (available only for secondary streams)
type		String	N	"Rtsp"	Fixed options.
port		Int	N	1–65535	In the range.
source_path		String	N	Maximum length: 119 bytes	In the range (Chinese characters are not allowed).
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

4.1.5 Obtaining Values of Parameters on the Channel Management Page

API Description	
Function	This API is used to obtain values of parameters on the channel management page.
API URL	/API/ChannelConfig/IPChannel/Get
Request Method	POST
Message ID	
Remarks	API used to obtain the parameter value range (for the API for obtaining values of parameters on the channel management page and the API for setting these parameters): /API/ChannelConfig/IPChannel/Range
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Channel array. The maximum number of elements in the array cannot exceed the number of channels.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object	Maximum length: 10 bytes	Elements in the channel object are as follows.	
Elements in the channel object					
state		String	Maximum length: 14 bytes "NotConfigured", "Online", "Unauther", "Netwrong", "Connect" "Bandlimit", "Locked", "MD5NotSupport"	Channel status. The values are as follows: NotConfigured : No camera is added to the channel; Online : The channel is online; Unauther : Incorrect user name or password; Netwrong : Network connection error; Connect : The channel is being connected; Bandlimit : Limited bandwidth; Locked : The user is locked; MD5NotSupport : MD5 is disabled for connecting to the platform.	
switch_mode		String	Maximum length: 11	Switch mode, which can be automatic or manual. This	

			bytes "ManualMode", "AutoMode"	parameter is available only for PoE channels.
alias		String	Maximum length: 35 bytes	Alias of a camera.
alias_pos		String	Maximum length: 12 bytes "LeftTop", "RightTop", "LeftBottom", "RightBottom", "Hide"	Position of the camera alias.
ip_address		String	Maximum length: 63 bytes	IP address or domain name.
subnet_mask		String	Maximum length: 15 bytes	Subnet mask.
gateway		String	Maximum length: 15 bytes	Gateway.
port		Int	1–65535	Media port.
protocol		String	Maximum length: 32 bytes "HWTSDK", "Onvif", "Custom 1", "Custom 2", "Custom 3", "Custom 4", "Custom 5", "Custom 6", "Custom 7", "Custom 8", "Custom 9", "Custom 10", "Custom 11", "Custom 12", "Custom 13", "Custom 14", "Custom 15", "Custom 16"	Camera access protocol.
username		String	Maximum length: 31	User name.

			bytes	
password		String	Maximum length: 31 bytes	User password. This parameter is left empty.
password_empty		Bool		Indicates whether the password is empty.
manufacturer		String	Maximum length: 35 bytes	Device type.
device_type		String	Maximum length: 35 bytes	MAC address.
mac_address		String	Maximum length: 35 bytes	Vendor.
software_version		String	Maximum length: 35 bytes	Firmware version.
switch_mode_flag		Bool		
security		String	Maximum length: 12 bytes "Risk", "Weak", "Medium", "Strength", "Default"	Password security.
can_modify_pwd		Bool		Indicates whether the password can be changed.
use_default_pwd		Bool	False (The value is false when the value is obtained.)	Indicates whether the default password is used (for HOLOWITS box cameras).
modify_all_chn_pwd		Bool	False (The value is false when the value is obtained.)	Indicates whether the passwords of all channels are changed (for HOLOWITS box cameras).
network_mode		String	"Dhcp", "Static"	Network mode (for cameras that go online through the ONVIF protocol).
can_set_netmode		Bool		Indicates whether the network mode can be selected.
channel_num		Int		Total number of connected channels.
camera_type	Camera type.	String	"fixed_bullet", "bullet", "ball", "dome"	The values represent the box camera, bullet camera, fixed dome camera, and PTZ dome camera respectively.
serial_number		String	Maximum length: 31 bytes	Camera serial number.

4.1.6 Setting Parameters on the Channel Management Page

API Description					
Function	This API is used to set parameters on the channel management page.				
API URL	/API/ChannelConfig/IPChannel/Set				
Request Method	POST				
Message ID					
Remarks	When operation_type is set to AddOrEditChannel, the input parameters are as follows: ip_address, subnet_mask, gateway, port, username, cipher, seq, alias, alias_pos, protocol, manufacturer, device_type, mac_address, software_version, and camera_type. (When an inactivated camera is added, the password configured during camera addition is used as the activation password for login.) When operation_type is set to EditIPCPParam, the input parameters are as follows: ip_address, subnet_mask, gateway, and network_mode. When operation_type is set to EditIPCPwd, the input parameters are as follows: username, enc_password, use_default_pwd, and modify_all_chn_pwd.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
operation_type		String	N	"AddOrEditChannel", "EditIPCPParam", "EditIPCPwd"	Operation type. You can add or modify channel parameters, and edit camera parameters or password of a connection.

remove_ipc		string array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	When a camera is deleted, only this parameter needs to be transferred.
Elements in the channel_info object					
CH1-CHn		JSON object	N		JSON information about the elements in the object is as follows.
Elements in the channel object					
switch_mode		String	N	Maximum length: 11 bytes "ManualMode", "AutoMode"	Fixed options in the range. Switch mode, which can be automatic or manual. This parameter is available only for PoE channels.
alias		String	N	Maximum length: 35 bytes	In the range (Chinese characters are allowed).
alias_pos		String	N	Maximum length: 12 bytes "LeftTop", "RightTop", "LeftBottom", "RightBottom", "Hide"	Fixed options in the range.
ip_address		String	N	Maximum length: 63 bytes	Domain name format in the range.
subnet_mask		String	N	Maximum length: 15 bytes	Domain name format in the range.
gateway		String	N	Maximum length: 15	Domain name format in the

				bytes	range.
port		Int	N	1–65535	In the range.
protocol		String	N	Maximum length: 32 bytes "HWTSDK", "Onvif", "Custom 1", "Custom 2", "Custom 3", "Custom 4", "Custom 5", "Custom 6", "Custom 7", "Custom 8", "Custom 9", "Custom 10", "Custom 11", "Custom 12", "Custom 13", "Custom 14", "Custom 15", "Custom 16"	In the range.
username		String	N	Maximum length: 35 bytes	In the range.
cipher	Channel password.	String	N	Maximum length: 1024 bytes (The key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2" is used for encryption.)	Encrypted password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2." (This parameter is used when a camera is added or the connection password is changed.)

seq		Int	N	0-1000000	Sequence number, which is the same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2." (This parameter is used when a camera is added or the connection password is changed.)
enc_password	Password.	JSON object	N		For details, see Table 2-1 (used for changing the camera password).
manufacturer		String	N	Maximum length: 35 bytes	In the range.
device_type		String	N	Maximum length: 35 bytes	In the range.
mac_address		String	N	Maximum length: 35 bytes	In the range.
software_version		String	N	Maximum length: 35 bytes	In the range.
switch_mode_flag		Bool			
modify_all_chn_pwd		Bool			Indicates whether to change passwords of all channels.
network_mode		String		"Dhcp", "Static"	Network mode (for cameras that go online through the ONVIF protocol).

use_default_pwd		Bool			Indicates whether to use the default password.
camera_type		String		"fixed_bullet", "bullet", "ball", "dome"	The values represent the box camera, bullet camera, fixed dome camera, and PTZ dome camera respectively.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

Table 4-1 Elements in the enc_password object

Elements in the enc_password object					
new_cipher		String	Maximum length: 1024 bytes	Encrypted new password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."	
seq		Int	0-1000000	Sequence number, which is the same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."	
cipher		String	Maximum length: 1024 bytes	Encrypted old password, which is generated based on the value of key returned by the API in section 11.5 "Public Key and Random	

				Number for PBKDF2."
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4.2 OSD Parameters

4.2.1 Obtaining Values of OSD parameters

API Description					
Function	This API is used to obtain value of OSD parameters.				
API URL	/API/ChannelConfig/OSD/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/OSD/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CH n		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for	

				digital channels. Note: This parameter is unavailable when the channel is online.
name		JSON object		JSON information about the elements in the object is as follows.
osd1-osdx		JSON object	x indicates the number of OSDs.	JSON information about the elements in the object is as follows.
datetime		JSON object		JSON information about the elements in the object is as follows.
covert		Bool		Preview status.
alpha		Int	0-128	OSD transparency. This parameter is available only when transparency is supported.
Elements in the name and osd1-osdx objects				
show		Bool	Bool	Indicates whether the channel name is superimposed on images.
text		String	Maximum length: 63 bytes	OSD text.
pos		JSON object		JSON information about the elements in the object is as follows.
Elements in the datetime object				
show		Bool		Indicates whether the date is displayed in the video window.
date_format		String	"MM/DD/YYYY", "YYYY-MM-DD", "DD/MM/YYYY"	Date format. Note: This parameter is unavailable for analog channels.
time_format		Int	24, 12	Time mode. The unit is hour. Note: HOLOWITS cameras support only the 24-hour format.
pos		JSON object		JSON information about the elements in the object is as follows.
show_week		Bool	Bool	Indicates whether the weekday is displayed.
Elements in the pos object				
x		Int	0-704	X coordinate.
y		Int	0-576	Y coordinate.

4.2.2 Setting OSD Parameters

API Description					
Function	This API is used to set OSD parameters.				
API URL	/API/ChannelConfig/OSD/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON Object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info array					
CH1-CHn		JSON Object	N		Elements in the channel object are as follows.
Elements in the channel object					
name		JSON object	N		JSON information about the elements in the object is as follows.
osd1-osdx		JSON object	N	x indicates the number of OSDs.	JSON information about the elements is as follows.
datetime		JSON object	N		JSON information about the elements in the object is as follows.
covert		Bool	N	Bool	Bool
Elements in the name and osd1-osdx objects					
show		Bool	N	Bool	Bool
text		String	N	Maximum length: 63 bytes	In the range (Chinese characters are allowed).

pos		JSON object	N		JSON information about the elements in the object is as follows.
Elements in the datetime object					
show		Bool	N	Bool	Bool
date_format		String	N	"MM/DD/YYYY", "YYYY-MM-DD", "DD/MM/YYYY"	Fixed options.
time_format		Int	N	24, 12	Fixed options. (Note: HOLOWITS cameras support only the 24-hour format.
pos		JSON object	N		JSON information about the elements in the object is as follows.
show_week		Bool	N	Bool	Bool
Elements in the pos object					
x		Int	N	0–704	In the range.
y		Int	N	0–576	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

4.3 Image Control Parameters

4.3.1 Obtaining Values of Image Control Parameters

API Description	
Function	This API is used to obtain values of image control parameters.
API URL	/API/ChannelConfig/ImageControl/Get
Request Method	POST
Message ID	
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/ImageControl/Range

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.	
mirror_mode		String	"Disable", "VerticalMirroring",	Mirroring mode.	

			"HorizontalMirroring", "All"	
corridor_mode		String	"Enable", "Disable"	Corridor mode (available only for supported cameras).
ir-cut_mode		String	"ColorMode", "BlackWhiteMode", "AutoMode"	IR-CUT mode.
day/night_mode		String	"SignalTriggered", "ImageTriggered"	Day/Night mode.
sensitivity		String	"Low", "Middle", "High"	Detection sensitivity.
exposure_mode		String	"Auto", "Manual"	Exposure mode.
shutter_speed		String	"1/3", "1/4", "1/8", "1/12", "1/15", "1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000", "1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	Shutter speed.
gain		Int	0–100	Gain.
coefficient		Int	0–100	Exposure compensation coefficient.
background_frequency		String	"Disable", "50Hz", "60Hz"	Background frequency.
auto_slow_shutter		String	"Enable", "Disable"	Slow shutter.
shutter_limit(s)		String	1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000",	Shutter limit. (Slow shutter is disabled.)

			"1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	
shutter_limit(s)		String	"1/3", "1/4", "1/8", "1/12", "1/15",	Shutter limit (when slow shutter is enabled).
back_light		String	"WDR", "HLC", "BacklightCompensation", "Disable"	Backlight.
wdr_coefficient		Int	0–100	WDR coefficient.
hlc_coefficient		String	"Low", "Middle", "High"	HLC coefficient.
white_balance		String	"Auto", "Manual" "AutoTracking", "OnePush", "Indoor", "Outdoor", "FixedSodiumLamp", "FluorescentLamp1", "FluorescentLamp2", "IncandescentLamp", "Sunlight"	White balance.
red_fine-tuning		Int	0–100	Red fine-tuning.
blue_fine-tuning		Int	0–100	Blue fine-tuning.
red_tuning		Int	0–100	Red tuning.
blue_tuning		Int	0–100	Blue tuning.
ct_upper_limit		Int	1500–15000	Upper limit of the color temperature.
ct_lower_limit		Int	1500–15000	Lower limit of the color temperature (the lower limit must be less than the upper limit).
defog_mode		String	"Disable", "Auto", "Manual"	Defogging mode.
defogging_level		Int	0–100	Defogging level.
elect-img-stabilization		String	"Enable", "Disable"	Electronic image stabilization.
denoising		String	"Disable", "CommonMode", "ExpertMode"	Noise reduction.
denoising_level		Int	0–100	Noise reduction level.

spatial_denoising_level		Int	0–100	Spatial noise reduction level.
temporal_denoising_level		int	0-100	Temporal noise reduction level.

4.3.2 Setting Image Control Parameters

API Description					
Function	This API is used to set image control parameters.				
API URL	/API/ChannelConfig/ImageControl/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandato ry (Y/N)	Value Range	Server Verificati on Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
mirror_mode		String	N	"Disable", "VerticalMirroring", "HorizontalMirroring", "All"	Fixed options.
corridor_mode		String	N	"Enable", "Disable"	Fixed options (available only for supported cameras).

ir-cut_mode		String	N	"ColorMode", "BlackWhiteMode", "AutoMode"	Fixed options.
day/night_mode		String	N	"SignalTriggered", "ImageTriggered"	Fixed options.
sensitivity		String	N	"Low", "Middle", "High"	Fixed options.
exposure_mode		String	N	"Auto", "Manual"	Fixed options.
shutter_speed		String	N	"1/3", "1/4", "1/8", "1/12", "1/15", "1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000", "1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	In the range.
gain		Int	N	0–100	In the range.
coefficient		Int	N	0–100	In the range.
background_frequency		String	N	"Disable", "50Hz", "60Hz"	Fixed options.
auto_slow_shutter		String	N	"Enable", "Disable"	Fixed options.
shutter_limit(s)		String	N	1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000", "1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	Fixed options. (Slow shutter is disabled.)

shutter_limit(s)		String	N	"1/3", "1/4", "1/8", "1/12", "1/15",	Fixed options. (Slow shutter is enabled.)
back_light		String	N	"WDR", "HLC", "BacklightCompensation", "Disable"	Fixed options.
wdr_coefficient		Int	N	0–100	In the range.
hlc_coefficient		String	N	"Low", "Middle", "High"	Fixed options.
white_balance		String	N	"Auto", "Manual" "AutoTracking", "OnePush", "Indoor", "Outdoor", "FixedSodiumLamp", "FluorescentLamp1", "FluorescentLamp2", "IncandescentLamp", "Sunlight"	Fixed options.
red_fine-tuning		Int	N	0–100	In the range.
blue_fine-tuning		Int	N	0–100	In the range.
red_tuning		Int	N	0–100	In the range.
blue_tuning		Int	N	0–100	In the range.
ct_upper_limit		Int	N	1500–15000	In the range.
ct_lower_limit		Int	N	1500–15000	In the range.
defog_mode		String	N	"Disable", "Auto", "Manual"	Fixed options.
defogging_level		Int	N	0–100	In the range.
elect-img-stabilization		String	N	"Enable", "Disable"	Fixed options.
denoising		String	N	"Disable", "CommonMode",	Fixed options.

				"ExpertMode"	
denoising_level		Int	N	0–100	In the range.
spatial_denoising_level		Int	N	0–100	In the range.
temporal_denoising_level		int	N	0-100	In the range.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)

Pay attention to the following when setting parameters:

1. When the white balance is in automatic tracking mode, the lower limit of the color temperature cannot be greater than the upper limit of the color temperature.
2. Manual exposure cannot be enabled with WDR, highlight compensation, background frequency, or backlight compensation at the same time.
3. In automatic exposure mode, slow shutter, WDR, and highlight compensation cannot be enabled at the same time.
4. WDR and the defogging mode cannot be enabled at the same time.

4.3.3 Restoring Default Values of Image Control Parameters

API Description	
Function	This API is used to restore default values of image control parameters and return parameters.
API URL	/API/ChannelConfig/ImageControl/Default
Request Method	POST
Message ID	
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the array is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
mirror_mode		String	"Disable", "VerticalMirroring", "HorizontalMirroring", "All"		Mirroring mode.
corridor_mode		String	"Enable", "Disable"		Corridor mode (available only for

				supported cameras).
ir-cut_mode		String	"ColorMode", "BlackWhiteMode", "AutoMode"	IR-CUT mode.
day/night_mode		String	"SignalTriggered", "ImageTriggered"	Day/Night mode.
sensitivity		String	"Low", "Middle", "High"	Detection sensitivity.
exposure_mode		String	"Auto", "Manual"	Exposure mode.
shutter_speed		String	"1/3", "1/4", "1/8", "1/12", "1/15", "1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000", "1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	Shutter speed.
gain		Int	0–100	Gain.
coefficient		Int	0–100	Exposure compensation coefficient.
background_frequency		String	"Disable", "50Hz", "60Hz"	Background frequency.
auto_slow_shutter		String	"Enable", "Disable"	Slow shutter.
shutter_limit(s)		String	1/25", "1/30", "1/50", "1/60", "1/90", "1/100", "1/120", "1/125", "1/180", "1/250", "1/350", "1/500", "1/725", "1/1000", "1/1500", "1/2000", "1/3000", "1/4000", "1/6000", "1/10000", "1/30000", "1/100000"	Shutter limit (when slow shutter is disabled).
shutter_limit(s)		String	"1/3", "1/4", "1/8", "1/12",	Shutter limit (when

			"1/15",	slow shutter is enabled).
back_light		String	"WDR", "HLC", "BacklightCompensation", "Disable"	Backlight.
wdr_coefficient		Int	0–100	WDR coefficient.
hlc_coefficient		String	"Low", "Middle", "High"	HLC coefficient.
white_balance		String	"Auto", "Manual", "AutoTracking", "OnePush", "Indoor", "Outdoor", "FixedSodiumLamp", "FluorescentLamp1", "FluorescentLamp2", "IncandescentLamp", "Sunlight"	White balance.
red_fine-tuning		Int	0–100	Red fine-tuning.
blue_fine-tuning		Int	0–100	Blue fine-tuning.
red_tuning		Int	0–100	Red tuning.
blue_tuning		Int	0–100	Blue tuning.
ct_upper_limit		Int	1500–15000	Upper limit of the color temperature.
ct_lower_limit		Int	1500–15000	Lower limit of the color temperature (the lower limit must be less than the upper limit).
defog_mode		String	"Disable", "Auto", "Manual"	Defogging mode.
defogging_level		Int	0–100	Defogging level.
elect-img-stabilization		String	"Enable", "Disable"	Electronic image stabilization.
denoising		String	"Disable", "CommonMode", "ExpertMode"	Noise reduction.
denoising_level		Int	0–100	Noise reduction level.
spatial_denoising_level		Int	0–100	Spatial noise reduction level.
temporal_denoising_level		int	0-100	Temporal noise

I				reduction level.
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4.4 Video Color Parameters

4.4.1 Obtaining Values of Video Color Parameters

API Description					
Function	This API is used to obtain values of video color parameters.				
API URL	/API/ChannelConfig/Color/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/Color/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CH<i>n</i> The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CH<i>n</i> . The value of <i>n</i> cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as	

				follows.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
hue		Int	0-100	Hue.
bright		Int	0-100	Brightness.
contrast		Int	0-100	Contrast.
saturation		Int	0-100	Saturation.
sharpness		Int	0-100	Sharpness.
support_default		Bool	Bool	Indicates whether the default values can be restored.

4.4.2 Setting Video Color Parameters

API Description	
Function	This API is used to set video color parameters.
API URL	/API/ChannelConfig/Color/Set
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object 11	N		Elements in the channel object are as follows.
Elements in the channel object					
hue		Int	N	0-100	In the range.
bright		Int	N	0-100	In the range.
contrast		Int	N	0-100	In the range.
saturation		Int	N	0-100	In the range.
sharpness		Int	N	0-100	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

4.4.3 Restoring Default Values of Video Color Parameters

API Description	
Function	This API is used to restore default values of video color parameters and return the parameters.
API URL	/API/ChannelConfig/Color/Default
Request Method	POST
Message ID	
Remarks	Only HOLOWITS cameras support the function of restoring default colors.

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.	
hue		Int	0-100	Hue.	
bright		Int	0-100	Brightness.	
contrast		Int	0-100	Contrast.	
saturation		Int	0-100	Saturation.	
sharpness		Int	0-100	Sharpness.	
support_default		Bool	Bool	Indicates whether the default values can be restored.	

4.5 Video Cover Parameters

4.5.1 Obtaining Values of Privacy Mask Parameters

API Description					
Function	This API is used to obtain values of privacy mask parameters.				
API URL	/API/ChannelConfig/VideoCover/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/VideoCover/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CH<i>n</i> The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CH<i>n</i> . The value of <i>n</i> cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object. For details, see Table 4-2.

Table 4-2 Elements in the channel_info object

Elements in the channel_info object				
CH1-CHn		JSON		Elements in a

		object		single object are as follows.
Elements in a single object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
privacy_zone_enable		Bool	(Obtained for HWTSDK box cameras)	Indicates whether privacy mask is enabled.
maxnum		Int	(Obtained for HWTSDK box cameras)	Maximum number of masked areas.
curnum		Int	(Obtained for HWTSDK box cameras)	Current number of masked areas.
zone_info_p		JSON array	(Obtained for HWTSDK box cameras)	The array contains a maximum of eight JSON objects as required by the HWTSDK protocol. For details, see Table 4-3.
zone_info		JSON array	Obtained for ONVIF-compliant online box cameras	The array contains a maximum of eight JSON objects as required by the ONVIF protocol. For details, see

				Table 4-5.
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Table 4-3 Elements in the zone_info_p array

Elements in the zone_info_p array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
zone_no		Int	1-8	ID of a masked area.
color		Int	0x0-0xFFFFFFFF	Color of a masked area.
point		JSON array		Area coordinates. For details, see Table 4-4.

Table 4-4 Elements in the point array

Elements in the point array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
left		Int	0–704 (HWTSDK)	X coordinate.
top		Int	0–576 (HWTSDK)	Y coordinate.

Table 4-5 Elements in the zone_info array

Elements in the zone_info array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				

zone_no		Int	1-4	ID of a masked area.
zone_enable		Bool	Bool	Indicates whether privacy mask is enabled for the current area.
rect		JSON object		Area coordinates. For details, see Table 4-6.

Table 4-6 Elements in the rect object

Elements in the rect object				
left		Int	0-704	X coordinate.
top		Int	0-576	Y coordinate.
width		Int	0-704	Width.
height		Int	0-576	Height.

4.5.2 Setting Privacy Mask Parameters

API Description					
Function	This API is used to set privacy mask parameters.				
API URL	/API/ChannelConfig/VideoCover/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object. For

					details, see Table 4-2.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

4.6 PTZ Parameters

4.6.1 Obtaining Values of PTZ Parameters

API Description					
Function	This API is used to obtain values of PTZ parameters.				
API URL	/API/ChannelConfig/PTZ/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/PTZ/Range (Note: This API is supported only when the hardware supports the RS-485 serial port.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CH<i>n</i> The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CH<i>n</i> . The value of <i>n</i> cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	

channel_info		JSON object		JSON information about the elements in the object is as follows.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
signal_type		String	"Analog", "Digital"	Signal type.
protocol		String	"Pelco-D", "Pelco-P"	Protocol.
baudrate		String	"1200", "2400", "4800", "9600"	Baud rate.
databit		String	"8", "7", "6", "5"	Data bit.
stopbit		String	"1", "2"	Stop bit.
parity		String	"None", "Odd", "Even", "Mark", "Space"	Parity bit.
address		Int	1-255	Address code.

4.6.2 Setting PTZ Parameters

API Description					
Function	This API is used to set PTZ parameters.				
API URL	/API/ChannelConfig/PTZ/Set				
Request Method	POST				
Remarks	Note: This API is supported only when the hardware supports the RS-485 serial port.				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.

Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
signal_type		String	N	"Analog", "Digital"	In the range.
protocol		String	N	"Pelco-D", "Pelco-P"	In the range.
baudrate		String	N	"1200", "2400", "4800", "9600"	In the range.
databit		String	N	"8", "7", "6", "5"	In the range.
stopbit		String	N	"1", "2"	In the range.
parity		String	N	"None", "Odd", "Even", "Mark", "Space"	In the range.
address		Int	N	1-255	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)

4.7 Transparently Transmitting Camera Parameters

API Description					
Function	This API is used to transparently transmit camera parameters.				
API URL	/API/ChannelConfig/PassThrough				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structur	Object	Y		

	e.				
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
request_url	Request signaling.	String	N	Maximum length: 1024 bytes	In the range.
request_data	Input parameter, which is in JSON format.	String	N	Maximum length: 10240 bytes	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation success) or other values (cause of failure)
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					

CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
respond_data		String	Maximum length: 128 KB	Input parameter, which is in JSON format.

4.8 1 + N Parameters

4.8.1 Obtaining Values of 1 + N Parameters

API Description					
Function	This API is used to obtain values of 1 + N parameters.				
API URL	/API/ChannelConfig/MultimachineMutual/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/MultimachineMutual/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.

Elements in the channel object				
subordination		String	"master_device", "secondary_device"	The values represent the primary device and the secondary device respectively.
device_type		String	Maximum length: 35 bytes	Device type.
pairing_state		Bool	Bool	Matching status.
on_state		Bool	Bool	Indicates whether a channel is enabled.
bonded_device		int array	0, 1, ..., n The maximum number of channels varies depending on the capabilities of the device.	Bound secondary devices. A maximum of four channels can be bound. In this version, only one channel can be bound. The value 0 is for channel 1, value 1 is for channel 2, and the rest can be deduced by analogy.

4.8.2 Setting 1 + N Parameters

API Description					
Function	This API is used to set 1 + N parameters.				
API URL	/API/ChannelConfig/MultimachineMutual/Set				
Request Method	POST				
Message ID					
Remarks	This API is used only for the primary device. The operation types include enable, bind, and unbind. The bind operation is valid only when the primary device is enabled. To unbind a channel, only this channel needs to be transferred.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
operation	Operation.	String	Y	"enable", "bind", "unbind"	Operation type. enable : Enable a

					channel; bind : Bind a channel; unbind : Unbind a channel. (Note: This parameter is used only for the primary device.)
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn	Channel.	JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
on_state	Indicates whether a channel is enabled.		Y	Bool	This parameter is available only when operation is set to enable .
bonded_device	Bound secondary devices.	int array	Y	0, 1, ..., n The maximum number of channels varies depending on the capabilities of the device.	A maximum of four channels can be bound. In this version, only one channel can be bound. The value 0 is for channel 1, value 1 is for channel 2, and the rest can be deduced by analogy. This parameter is available only when operation is set to bind .
onvif_user	ONVIF user name.	String	Y	0-32	Limited length and characters. This parameter is available only when operation is set to bind .
enc_password	ONVIF encryption password.	JSON object	N	This parameter needs to be transferred only when a user	For details, see Table 2-1. This parameter is available only

				password is set.	when operation is set to bind .
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		success (operation successful) or other values (cause of failure)

4.9 Longitude and Latitude Parameters

4.9.1 Obtaining Values of Longitude and Latitude Parameters

API Description					
Function	This API is used to obtain values of longitude and latitude parameters.				
API URL	/API/ChannelConfig/LongandLatitude/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/ChannelConfig/LongandLatitude/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	N		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a

				the capabilities of the device.	device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the channel_info object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
longitude		String	Maximum length: 20 bytes	Longitude.	
latitude		String	Maximum length: 20 bytes	Latitude.	

4.9.2 Setting Longitude and Latitude Parameters

API Description					
Function	This API is used to set longitude and latitude parameters.				
API URL	/API/ChannelConfig/LongandLatitude/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					

CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
longitude		String	Y	Maximum length: 20 bytes	In the range.
latitude		String	Y	Maximum length: 20 bytes	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	success (operation success) or other values (cause of failure)	

5 Stream Parameter Configuration

5.1 Encode Parameters

5.1.1 Obtaining Values of Stream Parameters

API Description					
Function	This API is used to obtain values of stream parameters.				
API URL	/API/StreamConfig/MainStream/Get /API/StreamConfig/SubStream/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/StreamConfig/MainStream/Range /API/StreamConfig/SubStream/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
status		String	"Offline", "Online"		Channel status. This parameter is available only for digital

				channels. Note: This parameter is unavailable when the channel is online.
max_frame_rate		Int	20, 25, 30, 50, 60	Full frame rate.
resolution		String	"1280 x 720", "1280 x 960", "1920 x 1080", "2304 x 1296", "2048 x 1520", "2592 x 1520" ...	Resolution.
fps		Int	0–60	Frame rate.
video_encode_type		String	"H.264", "H.265"	Video encoding type.
bitrate_control		String	"CBR", "VBR"	Code stream type.
video_quality		String	"Lowest", "Lower", "Low", "Medium", "Higher", "Highest"	Image quality. (Note: This parameter is available only when Type is set to VBR .)
bitrate_mode		String	"Predefined", "User-defined"	Stream mode.
bitrate		Int	Primary 32, 48, 64, 80, 96, 128, 160, 192, 224, 256, 320, 384, 448, 512, 640, 768, 896, 1024, 1280, 1536, 1792, 2048, 3072, 4096, 5120, 6144, 7168, 8192, 10240, 12288, 16384	Pre-defined stream size.

			tr e a m	
			S e c o n d a r y s t r e a m	32, 48, 64, 80, 96, 128, 160, 192, 224, 256, 320, 384, 448, 512, 640, 768, 896, 1024, 1280, 1536, 1792, 2048, 3072, 4096, 5120, 6144, 7168, 8192, 10240, 12288, 16384
custom_bit rate		Int	8–100000	Custom stream size.
audio		Bool	bool	Audio switch.
i_frame_int erval		Int	1–500	I-frame interval. (This parameter is available only for HOLOWIT S protocols.)

5.1.2 Setting Stream Parameters

API Description	
Function	This API is used to set stream parameters.
API URL	/API/StreamConfig/MainStream/Set /API/StreamConfig/SubStream/Set
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
max_frame_rate		Int	N	20, 25, 30, 50, 60	Fixed options.
resolution		String	N	"1280 x 720", "1280 x 960", "1920 x 1080", "2304 x 1296", "2048 x 1520", "2592 x 1520" ...	Fixed options.
fps		Int	N	0–60	In the range. This parameter is available only when resolution is set.
video_encode_type		String	N	"H.264", "H.265"	Fixed option. This parameter is available only when

						resolution is set.
bitrate_control		String	N	"CBR", "VBR"		Fixed option. This parameter is available only when resolution is set.
video_quality		String	N	"Lowest", "Lower", "Low", "Medium", "Higher", "Highest"		Fixed option. This parameter is available only when resolution is set.
bitrate_mode		String	N	"Predefined", "User-defined"		Fixed option. This parameter is available only when resolution is set.
bitrate		Int	N	Primary stream	32, 48, 64, 80, 96, 128, 160, 192, 224, 256, 320, 384, 448, 512, 640, 768, 896, 1024, 1280, 1536, 1792, 2048, 3072, 4096, 5120, 6144, 7168, 8192, 10240, 12288, 16384	Fixed option. This parameter is available only when resolution is set.
				Secondary stream	32, 48, 64, 80, 96, 128, 160, 192, 224, 256, 320, 384, 448, 512, 640, 768, 896, 1024, 1280, 1536, 1792, 2048, 3072, 4096, 5120, 6144, 7168, 8192, 10240, 12288, 16384	

custom_bitrate		Int	N	8–100000	In the range. This parameter is available only when resolution is set.
audio		Bool	N	bool	Bool
i_frame_interval		Int	N	1–500	In the range. (This parameter is available only for HOLOWITS protocols.)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

5.2 RTSP Preview Stream Parameters

API Description	
Function	This interface is used to obtain values of RTSP preview stream parameters.
API URL	/API/Preview/GetRtspUrl/Get
Request Method	POST
Message ID	
Description	When multiple request parameters are sent to the client, only one type of URL can be returned at a time. Therefore, the priority of the returned URL is as follows: 1. When both is_dualtalk and is_device are set to true , the HWT-NVR800 intercom URL is returned. Note that is_device cannot be set individually. Otherwise, only the URLs of the primary and secondary streams of the channel are returned. 2. If is_dualtalk is set to true and is_device does not exist or is set to false , the channel intercom URL is returned. 3. If is_dualtalk is not transferred or is set to false and is_metadata is set to true , the channel metadata URL is returned. 4. If is_dualtalk is not transferred or set to false and is_extenxion is set to true , the URL of extended channel metadata is returned. If is_device , is_dualtalk , is_metadata , and is_extenxion are not transferred or all set to false , the URLs of the primary and secondary streams of the channel are returned.

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
transport_type		String	Y	"tcp", "udp", "http", and "https"	Fixed range.
is_device	Indicates whether to request device streams.	Bool	N	bool	Bool
is_dualtalk	Indicates whether to request intercom streams.	Bool	N	bool	Bool
is_metadata	Indicates whether to request metadata streams.	Bool	N	bool	Bool
is_extension	Indicates whether to request extended metadata streams.	Bool	N	bool	When extended metadata is requested, the extended metadata URL is used preferentially and the is_metadata parameter is

				invalid. (The is_extenxion parameter is applicable only to the internal client. For other clients that require the metadata URL, use the URL returned by the is_metadata parameter.)
Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
channel_info		JSON object		JSON information about the elements in the object is as follows.
dualtalk		String	"rtsp://192.168.1.100:554/rtsp/dualtalk?channel=0&token=123"	NVR intercom stream
Elements in the channel_info object				
CH1-CHn		JSON object		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object				
channel		String	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Channel No.
mainstream		String	Maximum length: 127 bytes For details, see "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=0&token=123".	Primary stream URL or metadata URL.

			Metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=0&metadata=true&token=123" Extended metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=0&metadata=true&extensions=true&token=123"	
substream		String	Maximum length: 127 bytes For details, see "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=1&token=123". Metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=1&metadata=true&token=123" Extended metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=1&metadata=true&extensions=true&token=123"	Secondary stream URL or metadata URL.
dualtalk		String	"rtsp://192.168.1.100:554/rtsp/dualtalk?channel=4&token=123"	Intercom stream URL.

Note: If the video cannot be played in VLC media player, update the player to the latest version.

5.2.1 Reference Documents for Live Video Viewing Through RTSP

For details, see all sections in the RFC 2326 (<https://tools.ietf.org/html/rfc2326>).

5.3 Obtaining the Key for Live Video Viewing and Recording Playback Through Proprietary Protocols

API Description					
Function	This API is used to obtain the key for proprietary protocol authentication.				
API URL	/API/Web/Get_Private_Key				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
number	Number of keys obtained.	Unsigned int	N	1–32	Fixed range.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
id_keys		JSON array	JSON object		For details about the array elements, see Table 5-1.

Table 5-1 Elements in the id_keys array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Elements in a single object are as follows.
Elements in a single object				
id		String	64 bytes	Key ID.
key		String	64 bytes	Key used to verify live video viewing and recording playback through proprietary protocols.

6 Alarm Parameter Configuration

6.1 I/O Alarm Parameters

6.1.1 Obtaining Values of I/O Alarm Parameters

API Description					
Function	This API is used to obtain values of I/O alarm parameters.				
API URL	/API/AlarmConfig/IO/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/AlarmConfig/IO/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
alarm_in		JSON array	N	"Local<-1"... "Local<-n" "IP_CH1<-1"... "IP_CHn<-1" "IP_CH1<-2"... "IP_CHn<-2" The maximum number of alarm input sources varies depending on the capabilities of a device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
"Local<-1"... "Local<-n"		JSON object	The maximum number of channels varies	Alarm input information about the elements in the JSON	

"IP_CH1<-1"...			depending on the capabilities of the device. Alarm input: Alarm input can be triggered by the local system or camera (up to two channels).	object is as follows.
"IP_CHn<-1"				
"IP_CH1<-2"...				
"IP_CHn<-2"				
Elements in an alarm input object				
alarm_type		String	"NormallyOpen", "NormallyClose", "Off"	I/O alarm status.
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds.
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds.
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output can be triggered by the local system or camera (up to two channels).
post_recording		String	"30", "60", "120", "300"	Recording delay, in seconds.
snap_enable		Bool	bool	Indicates whether image capture is enabled.
send_email		Bool	bool	Indicates whether to send emails upon alarm occurrence.
show_message		Bool	bool	Indicates whether to display the alarm ID.
full_screen		Bool	bool	Indicates whether to display alarms in full screen mode.
channel		Array	CH1 to CHn	Indicates whether the alarm linkage for a channel is enabled. Each value represents a channel switch.

detect_interval		Int	1–256s	Alarm check interval (frontend I/O).
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether to display the date when the alert plan starts to be executed. The options are true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan.
alarm_plan_list		JSON array	JSON object	Alert plan list (frontend I/O parameters). For details, see Table 6-1.

Table 6-1 Alert plan array elements

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Elements in a single object are as follows.
Elements in a single object				
cycle_type		Int	0 and 1	Cycle mode. The values are as follows: 0 : none; 1 : weekly.
period		Int	0–7	Week day. The value 0 indicates every day, and the values 1 to 7 indicate Monday to Sunday.
start_date		String	Start date. The date is in YYYY-MM-DD format.	Fixed format.
start_time		String	Start time. The time is in hh:mm:ss format.	Fixed format.
end_time		String	End time.	Fixed format.

			The time is in hh:mm:ss format.	
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6.1.2 Setting I/O Alarm Parameters

API Description					
Function	This API is used to set I/O alarm parameters.				
API URL	/API/AlarmConfig/IO/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
"Local<-1"... "Local<-n" "IP_CH1<-1"... "IP_CHn<-1" "IP_CH1<-2"... "IP_CHn<-2"		JSON object	N	The maximum number of channels varies depending on the capabilities of the device. Alarm input: Alarm input can be triggered by the local system or camera (up to two channels).	Alarm input information about the elements in the JSON object is as follows.
Elements in an alarm input object					
alarm_type		String	N	"NormallyOpen", "NormallyClose", "Off"	Fixed options.
latch_time		String	N	"10", 20, "40", "60"	Fixed option (unit: second).

buzzer		String		"0", "1", "3", "5", "10", "20", "40", "60"	Fixed option (unit: second).
alarm_out		Array		"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Fixed options.
post_recording		String		"30", "60", "120", "300"	Fixed option (unit: second).
snap_enable		Bool		bool	Indicates whether to enable target capture.
send_email		Bool		bool	Indicates whether to send emails upon alarm occurrence.
show_message		Bool		bool	Indicates whether to display the alarm ID.
full_screen		Bool		bool	Indicates whether to display alarms in full screen mode.
channel		Array		CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
detect_interval		Int		1–256s	Alarm check interval (frontend I/O)
alarm_plan_list		JSON array		JSON object	Alert plan list (frontend I/O parameters). For details, see Table

					6-1.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

6.2 Motion Alarm Parameters

6.2.1 Obtaining Values of Motion Detection Parameters

API Description					
Function	This API is used to obtain values of motion detection parameters.				
API URL	/API/AlarmConfig/Motion/Get				
Request Method	POST				
Message ID					
Description	Parameters on the channel configuration page: switch, mbcol, mbrow, maxrect, rect, sensitivity, intervals, and region_setting (available only for ONVIF-compliant box cameras) Parameters on the alarm configuration page: buzzer, alarm_out, latch_time, record_enable, post_recording, record_channel, snap_enable, show_message, send_email, full_screen, alarm_plan_list, and show_start_date API used to obtain the parameter value range: /API/AlarmConfig/Motion/Range. (The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum	The value must be in the format of CHn . The value of n cannot exceed the maximum number of

				number of channels varies depending on the capabilities of the device.	channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
channel_max		Int		Total number of channels of the current device. This parameter is available only when the parameter range is obtained.	
Elements in the channel_info object					
CH1-CHn		JSON object		Channel information about the elements in the JSON object is as follows.	
Elements in the CH1-CHn object					
status		String	"Offline","Online"	Channel status. This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.	
switch		Bool	Bool	Indicates whether the motion detection alarm is enabled. (ChannelConfig)	
mbcol		Int	1–44 (ONVIF)	The motion detection area is divided into multiple blocks consisting of multiple rows and columns. This parameter is determined by the camera and does not need to be set. (ChannelConfig)	
mbrow		Int	1–30 (ONVIF)	The motion detection area is divided into multiple blocks consisting of multiple rows and columns. This parameter is determined by the camera and does not need to be set. (ChannelConfig)	

maxrect		Int	4 (HWTSDK)	Maximum number of blocks in a motion detection area. (ChannelConfig)
rect		JSON array	(HWTSDK)	JSON information about the elements in the array is as follows. (ChannelConfig)
sensitivity		Int	1–5	Motion detection sensitivity. (ChannelConfig)
intervals		Int	1–256 (HWTSDK)	Motion alarm check interval, in seconds. (ChannelConfig)
audio_enable		Bool	bool	Indicates whether audible alarms are enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Signed-in.wav", "Please Sort Waste.wav",	Audio file. (ChannelConfig)

			"Off-Duty Conduct.wav", "Deep Water Zone.wav", "Waste Sorting in Specified Period.wav"	
audio_play back_time s		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light _illuminato r_enable		Bool	bool	Indicates whether the light flashing is enabled. (ChannelConfig)
flash_dura tion		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light _illuminato r		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_en able		Bool	bool	Indicates whether to enable video recording for a channel upon alarm occurrence. (AlarmConfig)
record_ch		Array	CH1 to CHn	Alarm output channel. (AlarmConfig)

annel				Indicates whether alarm linkage is enabled for the channel. (AlarmConfig)
post_recording		String	"30","60","120","300"	Recording delay, in seconds. (AlarmConfig)
region_setting		Array		This parameter is available only for ONVIF-compliant box cameras. (ChannelConfig) The area is divided into a maximum of 36 rows, and each row contains a maximum of 44 blocks. The values of mbrow and mbcol determine the number of rows supported by the camera and the number of blocks in each row. Each element describes a block in a row. 1: Motion detection is enabled for the block. 0: Motion detection is disabled for the block.
snap_enable		Bool	bool	Indicates whether image capture is enabled. (AlarmConfig)
show_message		Bool	bool	Indicates whether to display the alarm ID. (AlarmConfig)
send_email		Bool	bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
full_screen		Bool	bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether to display the date when the alert plan starts to be executed. The options are true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan.
alarm_plan_list		JSON array	JSON object.	Alert plan list. For details, see Table 6-1. (AlarmConfig)
Elements in the rect array				
left		Int	0–704	X coordinate.
top		Int	0–576	Y coordinate.

width		Int	0–704	Width.
height		Int	0–576	Height.

6.2.2 Setting Motion Detection Parameters

API Description					
Function	This API is used to set motion detection parameters.				
API URL	/API/AlarmConfig/Motion/Set				
Request Method	POST				
Message ID					
Description	Parameters on the channel configuration page: switch, maxrect, rect, sensitivity, intervals, and region_setting (available only for ONVIF-compliant box cameras) Parameters on the alarm configuration page: buzzer, alarm_out, latch_time, record_enable, post_recording, record_channel, snap_enable, show_message, send_email, full_screen, and alarm_plan_list				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					

switch		Bool	N	bool	Indicates whether to enable the motion detection alarm. (ChannelConfig)
maxrect		Int	N	4 (HWTSDK)	In the range. (ChannelConfig)
rect		JSON array	N	(HWTSDK)	(ChannelConfig)
sensitivity		Int	N	1–5	In the range. (ChannelConfig)
intervals		Int	N	1–256 (HWTSDK)	In the range. (ChannelConfig)
audio_enable		Bool	N	bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not	Audio output. (ChannelConfig)

				Allowed.wav" "Welcome.wav", "Signed-in.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_play back_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light _illuminator _enable		Bool	N	bool	Indicates whether to enable light flashing. (ChannelConfig)
flash_durat ion		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light _illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Fixed option (unit: second). (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Fixed option. (AlarmConfig)
latch_time		String	N	"10", "20", "40",	Fixed option (unit:

				"60"	second). (AlarmConfig)
record_enable		Bool	N	bool	Indicates whether to enable video recording for a channel upon alarm occurrence. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Indicates whether to enable alarm linkage for the channel. (AlarmConfig)
post_recording		String		"30", "60", "120", "300"	Fixed option (unit: second). (AlarmConfig)
region_setting		Array			This parameter is available only for ONVIF-compliant box cameras. (AlarmConfig) The area is divided into a maximum of 36 rows, and each row contains a maximum of 44 blocks. The values of mbrow and mbcol determine the number of rows supported by the camera and the number of blocks in each row. Each element describes a block in a row. 1: Motion detection is enabled for the block. 0: Motion detection is disabled for the block.
snap_enable		Bool		bool	Indicates whether to enable snapshot. (AlarmConfig)
show_message		Bool		bool	Indicates whether to display the alarm ID. (AlarmConfig)
send_email		Bool		bool	Indicates whether to send emails upon alarm occurrence.

					(AlarmConfig)
full_screen		Bool		bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
alarm_plan_list		JSON array		JSON object	Alert plan list. For details, see Table 6-1. (AlarmConfig)
Elements in the rect array					
left		Int		0–704 (HWTSDK)	In the range.
top		Int		0–576 (HWTSDK)	In the range.
width		Int		0–704	In the range.
height		Int		0–576	In the range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

6.3 Alarm Linkage to PTZ Rotation

6.3.1 Obtaining Values of PTZ Alarm Linkage Parameters

API Description					
Function		This API is used to obtain values of PTZ alarm linkage parameters.			
API URL		/API/AlarmConfig/PTZLinkage/Get			
Request Method		POST			
Message ID					
Description		API used to obtain the parameter value range: /API/AlarmConfig/PTZLinkage/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object		Channel information about the elements in the JSON object is as follows.	
Elements in the CH1-CHn object					
switch		Bool	bool	Indicates whether the channel is enabled.	
motion		Bool	bool	Indicates whether the motion alarm is triggered.	
io		Array		Indicates whether the alarm input is enabled. If the value obtained by channel 0(1) is 1 , Local<-1 (Local<-2) is checked on the interface.	
ptz_info		JSON array		JSON information about the elements in the array is as follows.	
Elements in the ptz_info array					
ptz_chn		String	CH1 to CHn	Channel to which the alarm is linked.	

linkage_ptz_point_index		String	"None" or "1" to "255"	Point corresponding to the alarm-linked channel.
ptz_switch		Bool	bool	Indicates whether the alarm linkage groups are enabled. (4 groups)

6.3.2 Setting PTZ Alarm Linkage Parameters

API Description					
Function	This API is used to set PTZ alarm linkage parameters.				
API URL	/API/AlarmConfig/PTZLinkage/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
switch		Bool	N	bool	Bool
motion		Bool	N	bool	Bool
io		Array	N		If the value obtained by channel 0(1) is 1 , Local<-1 (Local<-2) is checked on the interface.
ptz_info		JSON array	N		JSON information about the elements in the array

					is as follows.
Elements in the ptz_info array					
ptz_chn		String	N	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
linkage_ptz_point_index		String	N	None or 1 to 255	In the range.
ptz_switch		Bool	N	bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

6.4 Exception Alarm Parameters

6.4.1 Obtaining Values of Exception Alarm Parameters

API Description					
Function	This API is used to obtain values of exception alarm parameters.				
API URL	/API/AlarmConfig/Exception/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/AlarmConfig/Exception/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks

exception_info		JSON object		JSON information about the elements in the object is as follows.
Elements in the exception_info object				
video_loss		JSON object		JSON information about the elements in the object is as follows.
disk_error		JSON object		
no_space_on_disk		JSON object		
high_temperature		JSON object		
cpu_fan_error		JSON object		
Elements in video_loss , video_loss , and no_space_on_disk objects				
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds.
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Indicates whether the alarm output channels are enabled. Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds.
send_email		Bool	bool	Indicates whether to send emails upon alarm occurrence.

6.4.2 Setting Exception Alarm Parameters

API Description	
Function	This API is used to set exception alarm parameters.

API URL		/API/AlarmConfig/Exception/Set			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
exception_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the exception_info array					
video_loss		JSON object	N		JSON information about the elements in the object is as follows.
disk_error		JSON object	N		
no_space_on_disk		JSON object	N		
high_temperature		JSON object	N		
cpu_fan_error		JSON object	N		
Elements in video_loss , disk_error , and no_space_on_disk objects					
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Fixed option (unit: second).
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2"	Fixed format.

				The maximum number of alarm output sources varies depending on the capabilities of a device.	
latch_time		String	N	"10", "20", "40", "60"	Fixed option (unit: second).
send_email		Bool	N	bool	Indicates whether to send emails upon alarm occurrence.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

6.5 Alarm Linkage to Audio Output

6.5.1 Obtaining Values of Alarm-triggered Audio Parameters

API Description					
Function	This API is used to obtain values of alarm-triggered audio parameters.				
API URL	/API/AlarmConfig/AudioLinkage/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/AlarmConfig/AudioLinkage/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON	N	CH1 to CHn	The value must be in the

		array		The maximum number of channels varies depending on the capabilities of the device.	format of CH_n . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CH _n		JSON object		Channel information about the elements in the JSON object is as follows.	
Elements in the CH1-CH_n object					
status		String	"Offline","Online"	Channel status. This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.	
volume		Int	0–100	Audio volume.	
audio_list		JSON array	JSON object	The elements in the audio_list array are as follows.	
Elements in the audio_list array					
file_name		String		Audio file name. (For warning-type SuperColor cameras, a maximum of 25 audio files are supported, including 19 pre-defined audio files. For other types of cameras, a maximum of 12 audio files are supported, including 6 pre-defined audio files.)	
internal		Bool		Indicates whether a file is a pre-defined audio file. (Pre-defined audio files cannot be deleted.)	

6.5.2 Adding an Audio File

API Description					
Function	This API is used to add an audio file.				
API URL	/API/AlarmConfig/AudioLinkage/Add				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CH<i>n</i>	Fixed format.
volume		Int	N	0–100	Fixed format.
file_name		String	N	Minimum length: 4 bytes Maximum length: 20 bytes	Audio file name, which must comply with the following rules: 1. The file name extension must be .wav 2. Supports ASCII printable characters, excluding spaces and the following characters: @ # \$ & () - ' 3. Cannot start with the following characters: + - .
file_data		String	N	Base64 (audioData)	File data encrypted using Base64. (The file is a .wav audio file in PCM encoding format and the file size cannot exceed 512 KB.)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

6.5.3 Deleting an Audio File

API Description					
Function	This API is used to delete an audio file.				
API URL	/API/AlarmConfig/AudioLinkage/Clear				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		
Elements in the channel_info object					
CH1-CHn		JSON object	Y		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
audio_list		JSON array	Y		The elements in the audio_list array are as follows.
Elements in the audio_list array					
file_name		String	Y	Maximum length: 128 bytes	Audio file name. (The pre-defined files cannot be deleted.)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

7 Storage Parameter Configuration

7.1 Disk Parameters

7.1.1 Obtaining Values of Disk Parameters

API Description					
Function	This API is used to obtain values of disk parameters.				
API URL	/API/StorageConfig/Disk/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the value range of parameters for obtaining and setting disk parameters: /API/StorageConfig/Disk/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
over_write		String	"Off", "Auto", "1Day", "3Days", "7Days", "14Days", "30Days", "90Days"	Disk overwrite type.	
esata_record		Bool	bool	Indicates whether the eSATA-based recording is enabled.	
hdd_format_type		String	"AllHddData" "OnlyHddRecord", "OnlyHddData"	Disk formatting type. /*Format the entire disk.*/ /*Format only the recording partition.*/ /*Format only the common partition.*/	

disk_info		JSON array		JSON information about the elements in the array is as follows.
Elements in the disk_info array				
id		Int		Disk ID.
model		String	Maximum length: 40 bytes	Disk model.
serial_no		String	Maximum length: 20 bytes	Disk serial number.
firmware		String	Maximum length: 8 bytes	Disk firmware model.
device_type		String	"None", "Normal", "Esata", "Sd", "Usb", "Network", "Raid"	Disk type.
active		String	"Inactive", "Active"	Disk status.
status		String	"NoHdd", "Unformat", "Normal", "Full", "ReadOnly", "HddError", "Connecting", "Offline", "Occupied", "Oversized", "Broken", "Degrade", "Rebuilding", "Backup", "RaidHdd"	Disk Status.
total_size		Int	Unit: MB	Total capacity.
free_size		Int	Unit: MB	Remaining capacity.
total_time		Int	Unit: second	Total time.
free_time		Int	Unit: second	Remaining time.

format_enable		Bool		Indicates whether to format the disk.
delete_enable		Bool		Indicates whether to delete the disk.
disk_type		String	"ReadAndWriteDisk", "ReadOnlyDisk"	Disk group type.
disk_group_id		Int	1 to maximum value	Disk group ID. For models with one or two disk trays, the maximum value is 16 . For other models, the maximum value is 20 .

7.1.2 Setting Disk Parameters

API Description					
Function	This API is used to set disk parameters.				
API URL	/API/StorageConfig/Disk/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
disk_info		JSON array	N		JSON information about the elements in the array is as follows.
over_write		String	N	"Off", "Auto", "1Day", "3Days", "7Days", "14Days", "30Days", "90Days"	Fixed options.

Elements in the disk_info array				
id		Int		Disk ID. This parameter is mandatory.
disk_type		String	"ReadAndWriteDisk", "ReadOnlyDisk"	Disk group type.
disk_group_id		Int	1 to maximum value	Disk group ID. For models with one or two disk trays, the maximum value is 16 . For other models, the maximum value is 20 . Note: This parameter can be set only when a disk group exists.
Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
disk_info		JSON array	Maximum length: 256 bytes	JSON information about the elements in the array is as follows.

7.1.3 Formatting a Hard Disk

API Description					
Function	This API is used to format a hard disk				
API URL	/API/StorageConfig/Disk/Format				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
hdd_format_type		String	N	"AllHddData" "OnlyHddRecord", "OnlyHddData"	Disk formatting type. /*Format the entire disk.*/ /*Format only the recording partition.*/ /*Format only the common partition.*/ If this parameter is not set, the entire disk is formatted by default.
hdd_info		JSON array	Y		Information about the disk to be formatted. JSON information about the elements in the array is as follows.
Elements in the hdd_info array					
		JSON object	Y		The elements in a single disk formatting object are as follows.
Elements in a single disk formatting object.					
hdd_id		Int	Y	1– MAX_HDD_GROUP_NUM	ID of the disk to be formatted.
hdd_type		String	Y	"Normal", "Esata"	Type of the disk to be formatted.
Output Parameters					

Parameter	Description	Data Type	Value Range	Remarks
result		String		The values include success (operation success) or other values (cause of failure).

7.1.4 Obtaining Disk Formatting Progress

API Description					
Function	This API is used to obtain the disk formatting progress.				
API URL	/API/StorageConfig/Disk/Format/Progress				
Request Method	POST				
Message ID					
Description	Note: This API is used to obtain the formatting progress of the disk mentioned in section 7.1.3 "Formatting a Hard Disk," not the disk formatting progress on the LDU.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String			The values include success (operation success) or other values (cause of failure).
hdd_id		Int			ID of the disk to

				be formatted. This parameter does not exist when the last result of the formatting progress is returned.
hdd_format_state		String	"Formatting" "Ok"	Formatting status.
hdd_format_percent		Int	0–100	Formatting progress, in percentage.
hdd_format_info		JSON Array		Information about the disk formatting result. The JSON information about the elements in the array is as follows.
Elements in the hdd_format_info array				
		JSON object		Information about formatting of a single hard disk. The elements are as follows.
Elements in the hdd_format_info array				
hdd_id		Int		ID of the formatted disk.
hdd_format_result		String	"Idle","Ok","InvalidParam" "NoExist","HwError" "Failed","Oversized"	Disk formatting result.

7.2 Disk Group Parameters

7.2.1 Obtaining Values of Disk Group Parameters

API Description	
Function	This API is used to obtain values of disk group parameters.
API URL	/API/StorageConfig/DiskGroup/Get

Request Method	POST				
Message ID					
Remarks	This API cannot be invoked currently.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String			The values include success (operation success) or other values (cause of failure).
disk_group_info		JSON array			JSON information about the elements in the array is as follows.
Elements in the disk_group_info array					
disk_group_type		String	"None", "RecordDiskGroup", "RedundanceDiskGroup"		Disk group type.
group_num		Int	1– MAX_HDD_GROUP_NUM		Recording disk group ID/ Redundant disk group No.
channel		String	CH1 to CH<i>n</i>		Recording channel added to the disk group. Note: A channel can be added to only one disk group.

7.3 Public Cloud Parameters

7.3.1 Obtaining Values of Public Cloud Parameters

API Description	
Function	This API is used to obtain values of public cloud parameters.
API URL	/API/StorageConfig/SharedCloud/Get

Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/StorageConfig/SharedCloud/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
enable		Bool			Indicates whether the public cloud is enabled.
sync_time		Bool			Indicates whether to synchronize system time with the public cloud.
cloud_state		Bool			Public cloud connection status.
bind_state		Bool			Device binding status.
qrcode_string		String	Maximum length: 63 bytes	Character string of the public cloud QR code.	
server_address		String	Maximum length: 127 bytes	IP address of the public cloud server.	
bind_nickname		String	Maximum length: 63 bytes	Bound user name.	
bind_phone		String	Maximum length: 63 bytes	Bound mobile phone number.	
bind_email		String	Maximum length: 63 bytes	Bound email address.	
bind_time		String	Maximum length: 31 bytes	Binding time.	
cur_country_code		String	Maximum length: 3 bytes	Country/Region code of the public cloud.	
country		JSON array			JSON information about the elements in the array is as follows.
Elements in the country array					

		JSON object		A single country element is described below.
country_name		String	Maximum length: 127 bytes	Country/Region name obtained through the public cloud API.
country_code		String	Maximum length: 3 bytes	Country/Region code obtained through the public cloud API.

7.3.2 Setting Public Cloud Parameters

API Description					
Function	This API is used to set public cloud parameters.				
API URL	/API/StorageConfig/SharedCloud/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
enable		Bool	N	bool	Indicates whether to enable the public cloud.
server_address		String	N	Maximum length: 127 bytes	IP address of the public cloud server.
sync_time		Bool	N	bool	Indicates whether to synchronize system time with the public cloud.
cur_country_code		String	N	Maximum length: 3 bytes	Country/Region code of the public cloud.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks

result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).
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7.3.3 Forcibly Removing Binding from the Public Cloud

API Description					
Function	This API is used to forcibly remove binding from the public cloud.				
API URL	/API/StorageConfig/SharedCloud/Unbind				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

8 Plan Parameter Configuration

8.1 Plan Parameters

8.1.1 Obtaining Values of Plan Parameters

API Description					
Function	This API is used to obtain values of plan parameters.				
API URL	/API/Schedules/Record/Get /API/Schedules/Email/Get /API/Schedules/Alarm/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Schedules/Record/Range, /API/Schedules/Email/Range, /API/Schedules/Alarm/Range /API/Schedules/Alarm/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					

CH1-CHn		JSON object		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object				
category		JSON array		JSON information about the elements in the array is as follows.
Elements in the category array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
schedule_type		String	"Normal", "Io", "IntelligentAnalysis", "Exception", "Buzzer", "AlarmOut"	Types of plans that can be configured. Recording plan. The options include Normal and Io . Email plan. The options include Io , Exception , and IntelligentAnalysis . Alarm plan. The options include Buzzer and AlarmOut . Note: The actual plan types that can be configured depend on the device capability.
week		JSON array		JSON information about the elements in the array is as follows.
Elements in the week array				
day		String	Sun, Mon, Tue, Wed, Thu, Fri, Sat	Day of a week.
time		Array	0 : Plan is disabled during the time segment. 1 : Plan is enabled during the time segment.	Each array bit (int) identifies half an hour.

8.1.2 Setting Plan Parameters

API Description

Function	This interface is used to set plan parameters.				
API URL	/API/Schedules/Record/Set /API/Schedules/Email/Set /API/Schedules/Alarm/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
category		JSON array	N		JSON information about the elements in the array is as follows.
Elements in the category array					
		JSON object	N		Elements in a single object are as follows.
Elements in a single object					
schedule_type		String	N	"Normal", "Io", "IntelligentAnalysis", "Exception",	Fixed options.

				"Buzzer", "AlarmOut"	
week		JSON array	N		JSON information about the elements in the array is as follows.
Elements in the week array					
day		String	N	Sun, Mon, Tue, Wed, Thu, Fri, Sat	Fixed options.
time		Array	N	0 : The plan is disabled during the time segment. 1 : The plan is enabled during the time segment.	Each array bit (int) identifies half an hour.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

9 Recording Parameter Configuration

9.1 Recording Parameters

9.1.1 Obtaining Values of Recording Parameters

API Description					
Function	This API is used to obtain values of recording parameters.				
API URL	/API/RecordConfig/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/RecordConfig/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CHn		JSON object		Channel information about the elements in the JSON object is as follows.	
Elements in the CH1-CHn object					
record_switch		Bool	bool	Indicates whether the recording is enabled.	
stream_mode		String	"Mainstream", "DualStream"	Code stream type.	
prerecord		Bool	bool	Indicates whether to the pre-recording is enabled.	

9.1.2 Setting Recording Parameters

API Description					
Function		This API is used to set recording parameters.			
API URL		/API/RecordConfig/Set			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
record_switch		Bool	N	bool	Bool
stream_mode		String	N	"Mainstream", "DualStream"	Fixed options.
prerecord		Bool	N	bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

9.1.3 Obtaining Values of Manual Server Recording Parameters

API Description					
Function	This API is used to obtain values of manual server recording parameters.				
API URL	/API/ManualRecordConfig/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
channel		JSON array	N	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
Elements in the channel_info object					
CH1-CH n		JSON object		Channel information about the elements in the JSON object is as follows.	
Elements in the CH1-CHn object					
manual_record		Bool	bool	Manual recording status.	

9.1.4 Configuring Manual Server Recording

API Description					
Function	This API is used to configure manual server recording.				
API URL	/API/ManualRecordConfig/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.

Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
manual_record		Bool	N	bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

9.2 Recording Search

9.2.1 Searching for Recordings

API Description					
Function	This API is used to search for recordings				
API URL	/API/Playback/SearchRecord/Search				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CH<i>n</i>	The value must be in the format of CH<i>n</i> . The value of <i>n</i> cannot exceed the

					maximum number of channels of a device.
start_date		String	Y	Search start date, in the format of MM/DD/YYYY.	Fixed format.
end_date		String	Y	Search end date, in the format of MM/DD/YYYY.	Fixed format.
start_time		String	Y	Search start time. Format: hh:mm:ss	Fixed format.
end_time		String	Y	Search end time. Format: hh:mm:ss	Fixed format.
stream_mode		String	N	"Mainstream","Substream"	Stream type. If this parameter is not transferred, primary stream recordings are searched by default.
record_type		Int	N	1<<0 (scheduled recordings) 1<<2 (recordings triggered by motion detection alarms) 1<<3 (recordings triggered by I/O alarms) 1<<4 (recordings triggered by tripwire crossing detection alarms) 1<<7 (recordings triggered by all intelligent alarms) 1<<9 (recordings triggered by intrusion detection alarms) 1<<17 (recordings triggered by audio alarms) 1<<18 (manual	Recording type.

				recordings) 1<<20 (recordings triggered by lens blocking alarms) 1<<10 (recordings triggered by target detection alarms) 1<<11 (recordings triggered by crossing-line head counting alarms) 1<<12 (recordings triggered by area entry detection alarms) 1<<13 (recordings triggered by area exit detection alarms) 1<<14 (recordings triggered by fast movement alarms) 1<<15 (recordings triggered by loitering detection alarms) 1<<19 (recordings triggered by queue length detection alarms) 1<<21 (recordings triggered by crowd density detection alarms) 1<<22 (recordings triggered by mask detection alarms) 1<<23 (recordings triggered by third-party algorithm-based alarms) 1<<24 (recordings triggered by parking violation detection alarms) 1<<25 (recordings triggered by appearance frequency detection alarms) 1<<26 (recordings triggered by abandoned object detection alarms) 1<<27 (recordings	
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				triggered by removed object detection alarms) 1<<28 (recordings triggered by electric bike detection alarms) 1<<29 (recordings triggered by off-duty detection alarms) 0x7FFFFFFE (recordings triggered by all alarms) 0xFFFFFFFF (all recordings)	
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
record		JSON array		JSON information about the elements in the array is as follows.	
Elements in the record array					
channel		String	CH1 to CHn	Channel No.	
stream_mode		String	"Mainstream","Substream"	Indicates whether to enable recording.	
record_type		Int	1<<0 (scheduled recordings) 1<<2 (recordings triggered by motion detection alarms) 1<<3 (recordings triggered by I/O alarms) 1<<4 (recordings triggered by tripwire crossing detection alarms) 1<<7 (recordings triggered by intelligent alarms) 1<<9 (recordings triggered by intrusion detection alarms) 1<<17 (recordings triggered by audio alarms) 1<<18 (manual	Recording type.	

			recordings) 1<<20 (recordings triggered by lens blocking alarms) 1<<10 (recordings triggered by target detection alarms) 1<<11 (recordings triggered by crossing-line head counting alarms) 1<<12 (recordings triggered by area entry detection alarms) 1<<13 (recordings triggered by area exit detection alarms) 1<<14 (recordings triggered by fast movement alarms) 1<<15 (recordings triggered by loitering detection alarms) 1<<19 (recordings triggered by queue length detection alarms) 1<<21 (recordings triggered by crowd density detection alarms) 1<<22 (recordings triggered by mask detection alarms) 1<<23 (recordings triggered by third-party algorithm-based alarms) 1<<24 (recordings triggered by parking violation detection alarms) 1<<25 (recordings triggered by appearance frequency detection alarms) 1<<26 (recordings triggered by abandoned object detection alarms) 1<<27 (recordings triggered by removed object detection alarms) 1<<28 (recordings	
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			triggered by electric bike detection alarms) 1<<29 (recordings triggered by off-duty detection alarms) 0x7FFFFFFE (recordings triggered by all alarms) 0xFFFFFFFF (all recordings)	
start_date		String		Recording file start date, in the format of MM/DD/YYYY.
end_date		String		Recording file end date, in the format of MM/DD/YYYY.
start_time		String		Recording file start time, in the format of hh:mm:ss.
end_time		String		Recording file end time, in the format of hh:mm:ss.
size		Int		File size in bytes.
record_id		Unsigned int	0–0xFFFFFFFF	
disk_event_id		Unsigned int	0–65535	
event_id		Unsigned int	0–0xFFFFFFFF	Event ID, which is an input parameter of the API for locking recordings.
locked		Bool		Indicates whether an event is locked.

9.3 Recording Search by Month

9.3.1 Searching for Recordings by Month

API Description	
Function	This API is used to search for recordings by month.
API URL	/API/Playback/SearchMonth/Get
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		JSON array	Y	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Each array bit represents a channel with a string.
start_date		String	Y	Search start date, in the format of MM/DD/YYYY.	Fixed format.
stream_type		String	Y	"Mainstream","Substream"	Fixed options.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
is_has_rec		Int array	0 and 1	Each array bit indicates whether recordings are generated in that day of the month. 0: no 1: yes	
record_type		Int array	0–3	Each array bit indicates the type of recordings generated in that day of the month. 0: none 1: alarm-triggered 2: normal 3: both	

9.4 Image Search

9.4.1 Searching for Images

API Description	
Function	This API is used to search for images.

API URL		/API/Playback/Picture/Get			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String array	Y	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Each array bit represents a channel with a string.
start_date		String	Y		Search start date, in the format of MM/DD/YYYY. Each array bit represents a channel with a string. This parameter is transferred upon the first search. If the image list exists in the web system, this parameter does not need to be transferred.
end_date		String	Y	"Mainstream","Substream"	Search end date, in the format of MM/DD/YYYY. Each array bit represents a channel with a string. This parameter

					is transferred upon the first search. If the image list exists in the web system, this parameter does not need to be transferred.
start_time		String	Y		Search start time, in the format of hh:mm:ss. Each array bit represents a channel with a string. This parameter is transferred upon the first search. If the image list exists in the web system, this parameter does not need to be transferred.
end_time		String	Y		Search end time, in the format of hh:mm:ss. Each array bit represents a channel with a string. This parameter is transferred upon the first search. If the image list exists in the web system, this parameter does not need to be transferred.
pic_info		String	N	Only the pic_info parameter needs to be transferred for you to obtain information about	During the first search, the web system does not contain the pic_info

				an image.	parameter. Therefore, this parameter is not transferred. Then, the board performs a search and returns information about all images to the web system, including the data of the first image.
pic_sort		Int	N		Image sorting order by time. The options include 0 (in sequence) and 1 (in reverse order).
record_type		Unsigned int	N	1<<0 (common) 1<<2 (snapshot triggered by motion detection alarms) 1<<3 (snapshot triggered by I/O alarms) 1<<7 (snapshot triggered by intelligent analysis) 1<<17 (snapshot triggered by audio alarms) 1<<18 (manual snapshot) 1<<20 (snapshot triggered by lens blocking detection alarms) 1<<4 (snapshot triggered by tripwire crossing detection alarms) 1<<9 (snapshot triggered by intrusion detection alarms) 1<<11 (snapshot triggered by crossing-line head counting alarms) 1<<12 (snapshot triggered by area entry detection alarms) 1<<13 (snapshot triggered	In the range.

				by area exit detection alarms) 1<<14 (snapshot triggered by fast movement detection alarms) 1<<15 (snapshot triggered by loitering detection alarms) 1<<19 (snapshot triggered by queue length detection alarms) 1<<21 (snapshot triggered by crowd density detection alarms) 1<<22 (snapshot triggered by mask detection alarms) 1<<23 (snapshot triggered by third-party algorithm-based alarms) 1<<24 (snapshot triggered by parking violation detection alarms) 1<<25 (snapshot triggered by appearance frequency detection alarms) 1<<26 (snapshot triggered by abandoned object detection alarms) 1<<27 (snapshot triggered by removed object detection alarms) 1<<28 (snapshot triggered by electric bike detection alarms) 1<<29 (snapshot triggered by off-duty detection alarms) 0x7FFFF (ALL)	
msg_type		String	N		This parameter is returned when requested. You do not need to set this parameter upon the first search. (The client can record the requested image index.)

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).
overload		Bool		The image size cannot exceed 5000 KB.
msg_type		String		This parameter is returned when requested.
picture		JSON object		Returns the information about the image and content of the image encoded using Base64. For details about elements in this object, see Table 9-1.
all_pic_info		Jsonarray		If the web system does not contain the pic_info parameter, the board searches for and returns this parameter. For details about the elements in the array, see Table 9-2.
all_pic_num		Int	0–5000	If the web system does not contain the pic_info parameter, the board searches for and returns the parameter.

Table 9-1 Elements in a single object

Parameter	Description	Data Type	Value Range	Remarks
channel		String		
alias		String	Maximum length: 35 bytes	Channel alias.
time		String		yyyy/MM/dd hh:mm:ss
size		Unsigned int		Image size, in bytes.
record_type		Unsigned int	1<<0 (common) 1<<2 (snapshot triggered by motion detection)	

			alarms) 1<<3 (snapshot triggered by I/O alarms) 1<<7 (snapshot triggered by intelligent analysis) 1<<17 (snapshot triggered by audio alarms) 1<<18 (manual snapshot) 1<<20 (snapshot triggered by lens blocking detection alarms) 1<<4 (snapshot triggered by tripwire crossing detection alarms) 1<<9 (snapshot triggered by intrusion detection alarms) 1<<11 (snapshot triggered by crossing-line head counting alarms) 1<<12 (snapshot triggered by area entry detection alarms) 1<<13 (snapshot triggered by area exit detection alarms) 1<<14 (snapshot triggered by fast movement detection alarms) 1<<15 (snapshot triggered by loitering detection alarms) 1<<19 (snapshot triggered by queue length detection alarms) 1<<21 (snapshot triggered by crowd density detection alarms) 1<<22 (snapshot triggered by mask detection alarms) 1<<23 (snapshot triggered by third-party algorithm-based alarms) 1<<24 (snapshot triggered by parking violation detection alarms) 1<<25 (snapshot triggered by appearance frequency detection alarms) 1<<26 (snapshot triggered by abandoned object detection alarms) 1<<27 (snapshot triggered by removed object detection alarms) 1<<28 (snapshot triggered by electric bike detection alarms) 1<<29 (snapshot triggered by off-duty detection alarms) 0x7FFFF (ALL)	
image		String	Base64	Image content.
third_object		JSON object		Additional information

				about the third-party algorithm alarm. This parameter is available only when image alarm type contains third-party algorithm alarms. For details, see Table 9-3.
--	--	--	--	--

Table 9-2 Array elements

Parameter	Description	Data Type	Value Range	Remarks
pic_info		String	Maximum length: 128 bytes	Unique ID of an image. After obtaining the parameter, you can invoke the current API again and use this parameter as an input parameter. Then, you can obtain data of the image encoded using Base64.
channel		String		
alias		String	Maximum length: 35 bytes	Channel alias.
time		String		yyyy/MM/dd hh:mm:ss
size		Unsigned int		Image size, in bytes.
record_type		Unsigned int	1<<0 (common) 1<<2 (snapshot triggered by motion detection alarms) 1<<3 (snapshot triggered by I/O alarms) 1<<7 (snapshot triggered by intelligent	Image type.

			analysis) 1<<17 (snapshot triggered by audio alarms) 1<<18 (manual snapshot) 1<<20 (snapshot triggered by lens blocking detection alarms) 1<<4 (snapshot triggered by tripwire crossing detection alarms) 1<<9 (snapshot triggered by intrusion detection alarms) 1<<11 (snapshot triggered by crossing-line head counting alarms) 1<<12 (snapshot triggered by area entry detection alarms) 1<<13 (snapshot triggered by area exit detection alarms) 1<<14 (snapshot triggered by fast movement detection alarms) 1<<15 (snapshot triggered by loitering detection alarms) 1<<19 (snapshot triggered by queue length detection alarms) 1<<21 (snapshot triggered by crowd density detection alarms) 1<<22 (snapshot triggered by mask detection alarms) 1<<23 (snapshot triggered by third-party algorithm-based alarms) 1<<24 (snapshot triggered by parking violation detection alarms) 1<<25 (snapshot triggered by appearance frequency detection alarms) 1<<26 (snapshot triggered by abandoned object detection alarms) 1<<27 (snapshot triggered by removed object detection alarms) 1<<28 (snapshot triggered by electric bike detection alarms) 1<<29 (snapshot triggered by off-duty detection alarms) 0x7FFF (ALL)	
--	--	--	---	--

Table 9-3 Elements in third_object

Parameter	Description	Data Type	Value Range	Remarks
-----------	-------------	-----------	-------------	---------

	tion			
alog_name		String	Maximum length: 63 bytes	Name of third-party algorithms (AppName).

9.4.1.2 Example

For the first search:

The request format is as follows:

POST/API/Playback/Picture/Get?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrftoken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data":{
    "channel":[
      "CH3",
      "CH4"
    ],
    "start_date":"11/03/2020",
    "start_time":"00:00:00",
    "end_date":"11/03/2020",
    "end_time":"23:59:59",
    "record_type":4294967295,
    "pic_sort":0
  }
}
```

Response:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 729

```

{
  "result": "success",
  "data": {
    "overload": false,
    "all_pic_num": 2,
    "all_pic_info": [
      {
        "pic_info": "AAAABAAAAAAQAAEACAAAAAIAAAAA3EVv9CwSDQMLFAAAMDcD1kYAAAAAAAAAAAA",
        "channel": "CH3",
        "alias": "channel 3",
        "time": "2020/11/03 13:18:44",
        "size": 18134,
        "record_type": 262144
      },
      {
        "pic_info": "AAAAAwEAAAAQAAEACAAAAAIAAAAA3EVv9CwSDQMLFADWdjC1kYAAAEAAAAAAAA",
        "channel": "CH3",
        "alias": "channel 3",
        "time": "2020/11/03 13:18:44",
        "size": 18134,
        "record_type": 262144
      }
    ]
  }
}

```

In subsequent searches, you only need to use the **pic_info** parameter in **all_pic_info** obtained from the first search as the input parameter and invoke the current API again to obtain the image information. Information about only one image can be obtained at a time. If there are multiple images in **all_pic_info**, you need to revoke the API for multiple times to obtain all images.

The request format is as follows:

```

POST/API/Playback/Picture/Get?2020-10-17%2014:21:33 HTTP/1.1
Content-Type: application/json; charset=UTF-8
Accept: application/json, text/javascript, */*; q=0.01

```

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data":
  {
    "pic_info": "AAAABAAAAAAQAAEACAAAAAIAAAAA3EVv9CwSDQMLFAAAMDcD1kYAAAAAAAAAAAA",
    "msg_type": "send_0"
  }
}
```

Response format:

Response:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 23109

```
{
  "result": "success",
  "data": {
    "msg_type": "send_0",
    "picture": {
      "channel": "CH3",
      "alias": "channel 3",
      "time": "2020/11/03 13:18:44",
      "size": 18134,
      "record_type": 262144,
      "image": "base64_encode_data" // The content omitted here is the data of image encoded using Base64.
    }
  }
}
```

9.5 RTSP Stream Playback

API Description

Function		This API is used to obtain the URL for playing back RTSP streams.			
API URL		/API/Playback/PlaybacRtspkUrl/Get			
Request Method		POST			
Port Description		The interval between the start time and end time cannot exceed seven days.			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Each array bit represents a channel with a string. (Only a single channel is requested.)
transport_type		String	Y	"tcp", "udp", "http", and "https"	Fixed range.
is_metadata		Bool	N	bool	Indicates whether to request metadata streams.
is_extenxion	Indicates whether to request extended metadata	Bool	N	bool	When extended metadata is requested, the extended metadata URL is used preferentially and the is_metadata parameter is invalid.

	a stre ams				(The is_extensio n parameter is applicable only to the internal client. For other clients that require the metadata URL, use the URL returned by the is_metadat a parameter.)
stream_mod e		String	N	"Mainstream", "Substream"	Fixed options.
start_date		String	Y	Search start date, in the format of YYYY-MM-DD.	Fixed format. The UTC time is used.
end_date		String	Y	Search end date, in the format of YYYY-MM-DD.	Fixed format. The UTC time is used.
start_time		String	Y	Search start time, in the format of hh:mm:ss.	Fixed format. The UTC time is used.
end_time		String	Y	Search end time, in the format of hh:mm:ss.	Fixed format. The UTC time is used.
record_type		Int	Y	1<<0 (scheduled recordings) 1<<2 (recordings triggered by motion detection alarms) 1<<3 (recordings triggered by I/O alarms) 1<<4 (recordings triggered by tripwire crossing detection alarms) 1<<7 (recordings triggered by intelligent alarms) 1<<9 (recordings triggered by intrusion detection alarms) 1<<17 (recordings triggered by audio alarms)	

			1<<18 (manual recordings) 1<<20 (recordings triggered by lens blocking alarms) 1<<10 (recordings triggered by target detection alarms) 1<<11 (recordings triggered by crossing-line head counting alarms) 1<<12 (recordings triggered by area entry detection alarms) 1<<13 (recordings triggered by area exit detection alarms) 1<<14 (recordings triggered by fast movement alarms) 1<<15 (recordings triggered by loitering detection alarms) 1<<19 (recordings triggered by queue length detection alarms) 1<<21 (recordings triggered by crowd density detection alarms) 1<<22 (recordings triggered by mask detection alarms) 1<<23 (recordings triggered by third-party algorithm-based alarms) 1<<24 (recordings triggered by parking violation detection alarms) 1<<25 (recordings triggered by appearance frequency detection alarms) 1<<26 (recordings triggered by abandoned object detection alarms) 1<<27 (recordings triggered by removed object detection alarms) 1<<28 (recordings triggered by electric bike detection alarms) 1<<29 (recordings triggered by off-duty detection alarms) 0x7FFFFFFE (recordings triggered by all alarms) 0xFFFFFFFF (all recordings)	
Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
playback_url		String	Maximum length: 127 bytes Example: rtsp://192.168.1.100:554/rtsp/playback?channel=	URL for playing back RTSP

			2&subtype=0&recordtype=4294967295&starttime=2020-07-03T00:29:37Z&endtime=2020-07-03T15:33:30Z&token=123 Metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=0&metadata=true&token=123" Extended metadata URL: "rtsp://192.168.1.100:554/rtsp/streaming?channel=4&subtype=1&metadata=true&extensions=true&token=123"	streams of the requested channel. If the request fails, the cause of the error is returned.
--	--	--	---	---

Note: If RTSP stream cannot be played back in VLC media player, update the player to the latest version.

9.5.1 References

1. RFC 2326 (<https://tools.ietf.org/html/rfc2326>)
2. Section 6.10 in *ONVIF™ Streaming Specification* (<https://www.onvif.org/specs/stream/ONVIF-Streaming-Spec.pdf>).

6.10 Go To Time

As stated in [RFC 2326] section 10.5, a PLAY command received when replay is already in progress will not take effect until the existing play operation has completed. This specification adds a new RTSP header, "Immediate", which overrides this behaviour for the PLAY command that it is used with:

```
PLAY rtsp://192.168.0.1/path/to/recording RTSP/1.0
CSeq: 123
Session: 12345678
Require: onvif-replay
Range: clock=20090615T114900.440Z-
Rate-Control: no
Immediate: yes
```

You need to add the extended field framed by the red box in the above figure to overwrite the current playing operation.

3. Section 13.5.1 "End-of-Stream" in *RFC 7826* (<https://tools.ietf.org/html/rfc7826>).

If the encoding changes during playback, the server sends the following message: ANNOUNCE
 rtsp://172.16.10.142:443/rtsp/playback?channel=1&subtype=0&starttime=2021-01-06T7:07:00Z&endtime=2021-01-06T7:09:00Z RTSP/1.0

CSeq: 7

Date: Wed, 06 Jan 2021 18:50:50 GMT

v=0

o=- 67850940035 67850940035 IN IP4 172.16.8.117

s=RTSP Session

c=IN IP4 0.0.0.0

t=0 0

m=video 0 RTP/AVP 96

a=control:rtsp://172.16.10.142:443/rtsp/playback?channel=1&subtype=0&starttime=2021-01-06T7:07:00Z&endtime=2021-01-06T7:09:00Z&trackID=1

a=rtpmap:96 H265/90000

a=x-onvif-track:VideoTrackToken_0_main

a=framerate:25

a=x-dimensions:1920,1080

a=recvonly (This message is used to prompt users to update the SDP of the client.)

For details, see section 10.3 in *RFC 2326* (<https://tools.ietf.org/html/rfc2326#page-32>).

9.6 Recording Event Lock

9.6.1 Locking Recording Events

API Description					
Function	This API is used to lock recording events.				
API URL	/API/Playback/EventLock/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
event_id		Unsigned int	Y	For details about the value of the parameter ranges from 0x0 to 0xFFFFFFFF , see the event_id parameter in the return value of /API/Playback/SearchRecord/Search .	Fixed range.
lock		Bool	Y		Fixed range.

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).

9.7 Recording Bookmark

9.7.1 Searching for Recording Bookmarks

API Description					
Function	This API is used to search for recording bookmarks.				
API URL	/API/Playback/RecordLabel/Search				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
start_date		String	Y	Search start date, in the format of MM/DD/YYYY.	Fixed format.
end_date		String	Y	Search end date, in the format of MM/DD/YYYY.	Fixed format.
start_time		String	Y	Search start time, in the format of hh:mm:ss.	Fixed format.
end_time		String	Y	Search end time, in the format of hh:mm:ss.	Fixed format.
keyword	Key word.	string	N	Maximum length: 20 bytes	Fixed format.

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
label_list		JSON array		JSON information about the elements in the array is as follows.
Elements in the label_list array				
name		String	Maximum length: 20 bytes	Bookmark name.
channel		String	CH1 to CHn	Channel No.
date		String	The date format is MM/DD/YYYY	Date.
time		String	The time format is hh:mm:ss	Time.
label_id		Unsigned int	0–0xFFFFFFFF	
record_id		Unsigned int	0–0xFFFFFFFF	

9.7.2 Configuring Recording Bookmarks

API Description					
Function	This API is used to configure recording bookmarks.				
API URL	/API/Playback/RecordLabel/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
operate	Operation.	String	Y	"Add", "Delete",	Fixed format.

				"Modify"	
name		String	This parameter is mandatory when operate is set to Add or Modify .	Maximum length: 20 bytes	Fixed format.
channel		String	This parameter is mandatory when operate is set to Add .	CH1 to CHn	Fixed format.
date		String	This parameter is mandatory when operate is set to Add .	Label date, in the format of MM/DD/YYYY.	Fixed format.
time		String	This parameter is mandatory when operate is set to Add .	Label time, in the format of hh:mm:ss	Fixed format.
label_id		Unsigned int	This parameter is mandatory when operate is set to Delete or Modify .	0–0xFFFFFFFF	Fixed format.
record_id		Unsigned int	This parameter is mandatory when operate is set to Delete or Modify .	0–0xFFFFFFFF	Fixed format.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

9.8 RTSP Stream Download

API Description	
Function	This API is used to obtain the URL for downloading RTSP streams.
API URL	/API/Playback/DownloadRtspkUrl/Get
Request Method	POST

Port Description		The interval between the start time and end time cannot exceed seven days.			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Each array bit represents a channel with a string. (Only a single channel is requested.)
transport_type		String	Y	"tcp", "udp", "http", and "https"	Fixed range.
stream_mode		String	N	"Mainstream", "Substream"	Fixed options.
start_date		String	Y	Search start date, in the format of YYYY-MM-DD.	Fixed format. The UTC time is used.
end_date		String	Y	Search end date, in the format of YYYY-MM-DD.	Fixed format. The UTC time is used.
start_time		String	Y	Search start time, in the format of hh:mm:ss.	Fixed format. The UTC time is used.
end_time		String	Y	Search end time, in the format of hh:mm:ss.	Fixed format. The UTC time is used.
record_ty		Int	Y	1<<0 (scheduled recordings)	

pe				1<<2 (recordings triggered by motion detection alarms) 1<<3 (recordings triggered by I/O alarms) 1<<4 (recordings triggered by tripwire crossing detection alarms) 1<<7 (recordings triggered by intelligent alarms) 1<<9 (recordings triggered by intrusion detection alarms) 1<<17 (recordings triggered by audio alarms) 1<<18 (manual recordings) 1<<20 (recordings triggered by lens blocking alarms) 1<<10 (recordings triggered by target detection alarms) 1<<11 (recordings triggered by crossing-line head counting alarms) 1<<12 (recordings triggered by area entry detection alarms) 1<<13 (recordings triggered by area exit detection alarms) 1<<14 (recordings triggered by fast movement alarms) 1<<15 (recordings triggered by loitering detection alarms) 1<<19 (recordings triggered by queue length detection alarms) 1<<21 (recordings triggered by crowd density detection alarms) 1<<22 (recordings triggered by mask detection alarms) 1<<23 (recordings triggered by third-party algorithm-based alarms) 1<<24 (recordings triggered by parking violation detection alarms) 1<<25 (recordings triggered by appearance frequency detection alarms) 1<<26 (recordings triggered by abandoned object detection alarms) 1<<27 (recordings triggered by removed object detection alarms) 1<<28 (recordings triggered by electric bike detection alarms) 1<<29 (recordings triggered by off-duty detection alarms)	
----	--	--	--	--	--

				0x7FFFFFFE (recordings triggered by all alarms) 0xFFFFFFFF (all recordings)	
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
download_url		String	Maximum length: 127 bytes Reference: "rtsp://172.16.10.24:443/rtsp/download?channel=4&subtype=0&recordtype=1&starttime=2020-11-04T00:00:00Z&endtime=2020-11-04T11:15:24Z&token=fc4b71eab35ac0b1b77454f5b88d858c9ec9b2472b990045f812730278b0d5b8"		URL for downloading RTSP streams of the requested channel. If the request fails, the cause of the error is returned.

10 Factory Setting Restoration

10.1 Checking Whether the System Needs to Be Restarted After Each Module Is Restored to Factory Defaults

API Description					
Function	This API is used to check whether the system needs to be restarted after each module is restored to factory defaults.				
API URL	/API/Maintenance/Reset/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Maintenance/Reset/Range This API is used to check whether the system needs to be restarted after a function field is restored to factory defaults. You can query a single function field individually or multiple function fields at the same time. Whether this field needs to be queried is determined only when the value of this field is true.				
Input Parameters					
Parameter	Description	Data Type	Mandatory	Value Range	Server Verification Rule

			(Y/N)		
data	Data structure.	Object	Y		
Elements in the data object					
channel	Channel management.	Bool	N	bool	Bool
alarm	Alarm configuration.	Bool	N	bool	Bool
network	Network configuration.	Bool	N	bool	Bool
device	Storage management.	Bool	N	bool	Bool
system	System configuration.	Bool	N	bool	Bool
system_maintain	System maintenance.	Bool	N	bool	Bool
attendance	Target recognition-based attendance.	Bool	N	bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
need_reboot		Bool	bool	Indicates whether the system needs to be restarted. The options	

				include true (yes) and false (no).
--	--	--	--	--

10.2 Restoring Factory Defaults

API Description					
Function	This API is used to restore factory defaults.				
API URL	/API/Maintenance/Reset/Set				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Maintenance/Reset/Range You can restore a single module to factory defaults.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel	Channel management.	Bool	N	bool	Bool
alarm	Alarm configuration.	Bool	N	bool	Bool
network	Network configuration.	Bool	N	bool	Bool
device	Storage management.	Bool	N	bool	Bool

system	System configuration.	Bool	N	bool	Bool
system_maintain	System maintenance.	Bool	N	bool	Bool
attendance	Target recognition-based attendance.	Bool	N	bool	Bool
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2. (This parameter is needed only when you set parameters for factory setting restoration.)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.3 Automatic Reboot

10.3.1 Obtaining Values of Automatic Reboot Parameters

API Description	
Function	This API is used to obtain values of automatic reboot parameters.
API URL	/API/Maintenance/AutoReboot/Get
Request Method	POST

Message ID					
Description		API used to obtain the parameter value range: /API/Maintenance/AutoReboot/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
auto_reboot		Bool	bool		Indicates whether to enable automatic reboot.
period_mode		String	"EveryDay", "EveryWeek", "EveryMonth"		Maintenance period.
week		String	"Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"		Weekdays of the week. This parameter is valid only when period_mode is set to EveryWeek .
day		Int	1–31		Days of the month. This parameter is valid only when period_mode is set to EveryMonth .
time		String	hh:ss		Restart time, in the format of hh:ss. (Each digit must contain two bits.)
show_reboot_button		Bool	This parameter cannot be transferred during parameter setting.		Indicates whether to display the restart button. This parameter is available only for the admin (apiadmin) user.

10.3.2 Setting Automatic Reboot Parameters

API Description	
Function	This API is used to set automatic reboot parameters.
API URL	/API/Maintenance/AutoReboot/Set

Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
auto_reboot		Bool	N	bool	Bool
period_mode		String	N	"EveryDay", "EveryWeek", "EveryMonth"	Fixed options.
week		String	N	"Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"	Fixed options.
day		Int	N	1–31	In the range.
time		String	N	hours:minutes	Fixed format.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.4 Device Reboot

10.4.1 Rebooting the Device

API Description	
Function	This API is used to reboot the device.
API URL	/API/Maintenance/DeviceReboot/Set
Request Method	POST
Message ID	
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned and the system is restarted. If the operation fails, other values (cause of failure) are returned.	

10.5 System Upgrade

10.5.1 Obtaining the System Upgrade Token

API Description					
Function	This API is used to obtain the system upgrade token.				
API URL	/API/Maintenance/SystemUpgrade/Token				
Request Method	POST				
Message ID					
Description	This API is used together with the system upgrade API. After the upgrade token is generated, the system allocates memory to the system upgrade API. If the system upgrade API is not invoked within one minute, the system is restarted.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data	Object	Y		

	structure.				
Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
file_name			Y	string	Upgrade file name.
file_size			Y	int	Upgrade file size.
upgrade_head			Y	Int array	The first 1 KB data of the upgrade file.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
upgrade_token		String		Upgrade token.	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.5.2 Upgrading the System

API Description				
Function	This API is used to upgrade the system.			
API URL	/API/Maintenance/SystemUpgrade/Upgrade			
Request Method	POST			
Message ID				
Description	This API is used together with the API for obtaining the system upgrade token.			
Input Parameters				
The request header must contain the upgrade token in the following format: X-Upgrade-Token: xxxxxx. The value of xxxxxx is the upgrade token obtained through the API in section 10.5.1 "Obtaining the System Upgrade Token." The upgrade package data is transferred in form-data format.				
Output Parameters				
Parameter	Description	Data	Value Range	Remarks

		Type		
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned and the system is restarted. If the operation fails, other values (cause of failure) are returned.

Table 10-1 Upgrade error codes

in_user_interface	User interface.
updating	The board is being upgraded.
lack_memory	Insufficient memory.
file_error	File error.
no_need_upgrade	The current version is the latest.
size_error	The file size exceeded the upper limit (maximum file size returned in file_limit).
file_invalid	Invalid file.

10.6 Log Writing

10.6.1 Writing Logs

API Description					
Function	This API is used to write logs.				
API URL	/API/Maintenance/Log/Set				
Request Method	POST				
Message ID					
Description	This API is used when some operations of the client need to be written to the board, but logs need to be saved to the HWT-NVR800. For example, the client saves recordings to the local host or captures images locally, which is irrelevant to the board.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
sub_type		String	Y	"manual_capture",	Fixed options.

				"start_manual_record", "stop_manual_record", "picture_backup" "write_logstr"	
logstr		String	N	"Path config save success" "Path config save failed""User view Privacy Statement" "Download finish""playback single frame play" "preview play start" "preview play stop" "Preview digital zoom start" "Preview digital zoom stop" "playback play start" "playback play stop" "Download start""Download stop"	Fixed options. This parameter is mandatory when sub_type is set to write_logstr . This parameter is invalid when sub_type is set to other values.
channel		String array	N	CH1 to CHn	Fixed options.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.7 Cloud-based Upgrade

10.7.1 Performing Cloud-based Upgrade Detection

API Description	
Function	This API is used to perform cloud-based upgrade detection.
API URL	/API/Maintenance/CloudUpgrade/Check
Request Method	POST
Message ID	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
is_detect		Bool	Y	true: enable detection; false: disable detection and obtain data only	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
detect_result		String	Maximum length: 127 bytes	Detection result.	
need_upgrade		Bool		Indicates whether a new version needs to be upgraded.	
current_version		String		Current version.	
new_version		String		Latest version.	

10.7.2 Performing Cloud-based System Upgrade

API Description					
Function		This API is used to perform cloud-based system upgrade.			
API URL		/API/Maintenance/CloudUpgrade/Upgrade			
Request Method		POST			
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the upgrade is successful, success is returned. The system restarts for the upgrade after the upgrade package is downloaded. If the upgrade fails, other values (cause of failure) are returned.	

10.7.3 Checking the Cloud-based System Upgrade Progress

API Description					
Function	This API is used to check the progress of cloud-based system upgrade.				
API URL	/API/Maintenance/CloudUpgrade/Upgrade/Progress				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
upgrade_state		String	"upgrade_detect_fail"/*Upgrade check fails.*/ "upgrade disk partition full"/*Disk		Upgrade status and error codes.

			partition for upgrade is full.*/ "upgrade_disk_partition_invalid"/*Disk partition for upgrade is invalid.*/ "upgrade_disk_partition_umount"/*Disk partition for upgrade is unmounted.*/ "upgrade_is_detecting"/*Checking the upgrade version...*/ "upgrade_is_upgrading"/*Upgrading...*/ "upgrade_cloud_switch_off"/*Cloud-based upgrade is disabled.*/ "upgrade_file_name_error"/*File name error*/ "upgrade_usdtype_error"/*USB type error*/ upgrade_download /*Downloading...Do not disconnect the network.*/ upgrade_upgrading /*Upgrading...Do not power off the device.*/ upgrade_memory_not_enough /*Insufficient memory*/ upgrade_file_error /*File error*/ upgrade_upgrade_succeed /*Upgrade success*/ upgrade_parameter_error /*Parameter error*/ upgrade_no_u_disc /*No USB flash drive*/ upgrade_no_upgrade_software /*No upgrade software*/ upgrade_software_is_new /*The software is the latest version.*/ upgrade_software_packet_error /*Software package operation error*/ upgrade_language_version_error /*Software language error*/ upgrade_file_name_too_long upgrade_failed /*Upgrade failure*/ upgrade_exit upgrade_uncipher upgrade_download_failed /*Download failure*/ upgrade_download_network_error /*Network error*/	
--	--	--	---	--

			upgrade_environmentvar_different /*The startup parameter has been upgraded.*/ upgrade_usdtype_error /*USB flash drive format error*/ upgrade_unknown/*Unknown*/	
upgrade_percent		Unsigned int		Upgrade progress. This parameter is transferred only when upgrade_state is set to upgrade_upgrading or upgrade_downloading .

10.8 Developer Mode

10.8.1 Obtaining Values of Developer Mode Parameters

API Description					
Function	This API is used to obtain values of developer mode parameters.				
API URL	/API/Maintenance/DevelopMode/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Maintenance/DevelopMode/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
enable_sdk_log	SDK log.	Bool			Indicates whether to enable SDK logs.
enable_cloud_log	Cloud log.	Bool			Indicates whether to enable

				cloud logs.
enable_dev_log	Device log.	Bool		Indicates whether to enable device logs.

10.8.2 Setting Developer Mode Parameters

API Description					
Function	This API is used to set developer mode parameters.				
API URL	/API/Maintenance/DevelopMode/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
enable_sdk_log	SDK log.	Bool	N		Bool
enable_cloud_log	Cloud log.	Bool	N		Bool
enable_dev_log	Device log.	Bool	N		Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.8.3 Obtaining Log Token in Developer Mode

API Description	
Function	This API is used to obtain the log token in developer mode.
API URL	/API/Maintenance/DevelopMode/Token
Request Method	POST

Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
x-download-token		String	0–32		Token value.

10.8.4 Downloading Logs Generated in Developer Mode

API Description					
Function	This API is used to download logs generated in developer mode.				
API URL	/API/Maintenance/DevelopMode/Download				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
x-download-token		String	Y	0–32	In the range.
x-csrf-token		String	Y		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

10.8.5 Deleting Logs Generated in Developer Mode

API Description					
Function	This API is used to delete logs generated in developer mode.				
API URL	/API/Maintenance/DevelopMode/Clear				
Request Method	POST				
Message ID					
Description	Only the compressed log file is deleted. The log that is being written is not deleted.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
secondary_authentication		JSON object	Y		Used to verify the password of the admin (apiadmin) user for secondary authentication. For details, see Table 2-2.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

10.9 Camera Maintenance

10.9.1 Obtaining Camera Status Before Restart

API Description	
Function	This API is used to obtain camera status before the camera restarts.
API URL	/API/IPCMaintain/IPCReboot/Get
Request Method	POST

Message ID					
Description		API used to obtain the parameter value range: /API/IPCMaintaint/IPCReboot/Range			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			Information about each channel.
Elements in the channel_info object					
CH1-CHn		JSON object			Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
ip_address	IP address/Domain.	String	Maximum length: 63 bytes		
state	Camera status.	String	"Offline", "Online"		
software_version	Firmware version.	String	Maximum length: 35 bytes		

10.9.2 Restarting Cameras

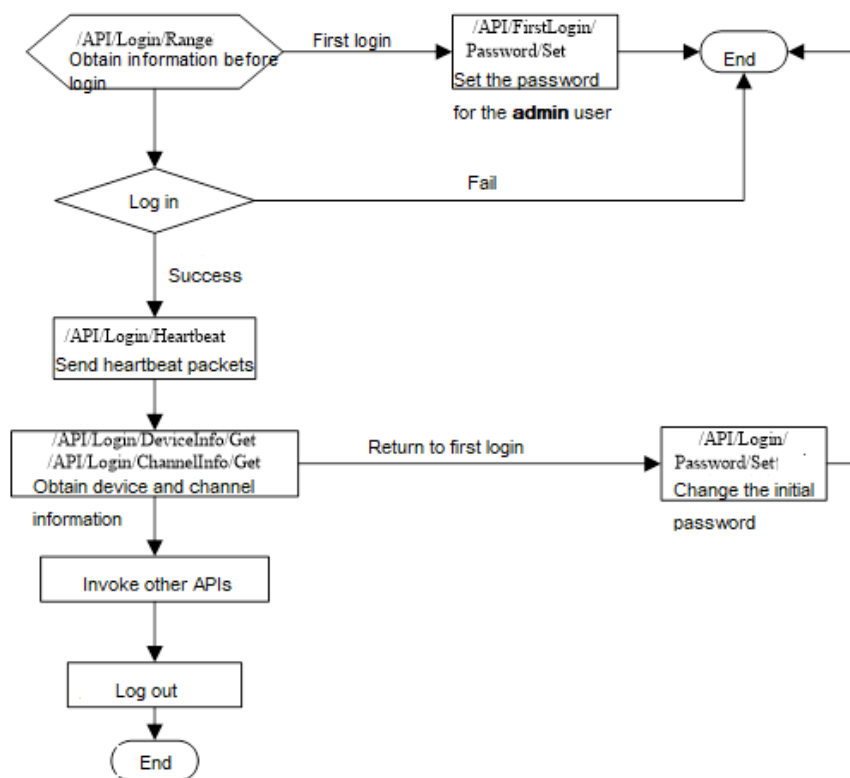
API Description	
Function	This API is used to restart cameras.
API URL	/API/IPCMaintaint/IPCReboot/Set
Request Method	POST
Message ID	
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel_info		JSON object	Y		
Elements in the channel_info object					
CH1-CHn		JSON object	N		Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
reboot_switch		Bool	Y		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			Information about each channel.
Elements in the channel_info object					
CH1-CHn		JSON object			Channel information about the elements in the JSON object is as follows.
Elements in the CH1-CHn object					
state	Camera operation status.	String	"Success", "Failed"		

11 Login and Logout

This section describes APIs for login, logout, heartbeat, preview, and for obtaining device or channel information.

For details about logging in to the client and subsequent procedure, see the figure below.



The heartbeat packets need to be sent periodically after successful login. If no heartbeat packet is received on the server within 30s, the session is terminated and the message "no_heartbeat" is returned for all requests.

In the development phase, you can set keepalive parameters to extend the heartbeat interval to 180s. Keepalive parameters can ensure that sessions are not restricted by the session duration.

11.1 Login

For details about the login process, see the figure above.

11.1.1 Obtaining Device Information Before Login

API Description					
Function	This API is used to obtain device information before login.				
API URL	/API/Login/Range				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
username		JSON object		User name type and value range.	
password		JSON object		Password type and value range.	
lang_strs	Language list.	JSON array		List of supported languages.	
default_lang		String		Default language.	
first_login_flag	Login with an empty password.	Bool	Bool	Indicates whether to log in to the device web system using an empty password.	
site_version		String		Version number.	

If the value of **first_login_flag** is **true**, the system is used for the first time and no password is set for the **admin** user. In this case, you need to invoke **/API/FirstLogin/Password/Set** to set the password.

11.1.2 Obtaining Device Information

API Description					
Function	This API is used to obtain device information.				
API URL	/API/Login/DeviceInfo/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandator y (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_num		Int		Total number of	

				channels.
stream_profile		JSON object	Stream type and value range.	Supports switch among several stream types. (Only Mainstream and Substream are used.)
support_ftp_upgrade		Bool		Indicates whether to support FTP upgrade. (This parameter is not used.)
default_stream		String	"Mainstream", "Substream",	Default stream type of the device web system.
pushinfo_type		JSON object	"Motion" "Io" "Videoloss", "Pir"	Type and value range of push parameters. Pir is not supported.
ptz_support	Indicates whether PTZ is supported.	Bool	true, false	
local_alarmin_num		Int		Number of alarm input channels supported by the board. (A maximum of 16 channels are supported.)
local_alarmout_num		Int		Number of alarm output channels supported by the board. (A maximum of four channels are supported.)
suggested_modify_pwd			bool	Indicates whether to prompt suggestions for changing the password.
support_face_config		Bool	bool	Indicates whether to support the function of querying and setting target detection parameters.
support_ie_down_snap		Bool	bool	Indicates whether to support the function of downloading snapshots from the

				device web system.
support_human_vehicle_search		Bool	bool	Indicates whether to support pedestrian and vehicle search.
preview_num		Int		Number of channels for previewing primary streams.
mac_addr		String	Maximum length: 35 bytes	MAC address.
support_io_record		Bool		Indicates whether I/O alarms exist in the program based on the total number of I/O alarms reported by the frontend and backend.
support_manual_record		Bool		Indicates whether to support search of the manual recordings.
support_sound_record		Bool		Indicates whether to support display. The options include 0 (hide) and 1 (display).
support_playback_new_rec_detail		Bool		Indicates whether to support NewRecordDetailRsp (New_GetRecFileList) search.
support_playback_new_rec_file_download		Bool		Indicates whether to support ewRecordDetailRsp (New_GetRecFileList) recording download and whether to support the function of encrypting recording files and querying the encrypted recording files.
support_more_chn_playback		Bool		Indicates whether multi-channel playback is supported on the device web page (more than four channels).
support_video_cover_record		Bool		Indicates whether to display lens blocking

				alarms.
support_smart_record		Bool		Indicates whether to support intelligent recording.
support_substream_playback		Bool		Indicates whether to support playback of secondary streams.
support_pir_record		Bool		Indicates whether to support PIR recording. (currently not supported by the device)
support_flood_light		Bool		Indicates whether to enable the light to flash when an alarm is generated. (currently not supported by the device)
support_speaker		Bool		Indicates whether voice intercom is supported.
show_certificate_remain_time		Bool		true: The validity period of the custom certificate is less than 8 days.
certificate_remain_time		Int	0–10	Validity period of the custom certificate.
media_port	Media port.	Int		For details, see <i>Communication Matrix</i> .
media_external_port	Media port mapped by the external IP address.	Int		For details, see <i>Communication Matrix</i> .
local_ip		String		
upgrade_file_max_size		Int	0-100	Maximum size of the upgrade file on the board, in MB.
support_audio_volume		Bool		Indicates whether to support volume adjustment.

sales_area		String	Max length: 64 bytes	Sales region. The options include Foreign and Local .
support_auth_login		Bool		Indicates whether to enable authorized login.

11.1.3 Obtaining Channel Information

API Description					
Function	This API is used to obtain channel information.				
API URL	/API/Login/ChannelInfo/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_param		JSON object		JSON information about the elements in the array is as follows.	
Elements in the channel_param object					
channel	Channel.	String			
channel_name	Channel name.	String			
channel_alias	Channel alias.	String			
connect_status		String	"NotConfigured", "Online", "Offline", "Sleep"	Channel connection status.	
offline_reason		String	Maximum length: 14 bytes	Channel status. The values are as follows: NotConfigured : No camera is added to the channel; Online : The	

			"NotConfigured", "Online", "Unauther", "Netwrong", "Connect" "Bandlimit", "Locked", "MD5NotSupport"	channel is online; Unauther : Incorrect user name or password; Netwrong : Network connection error; Connect : The channel is being connected; Bandlimit : Limited bandwidth; Locked : The user is locked; MD5NotSupport : MD5 is disabled for connecting to the platform.
ability	Channel capability.	String array	"FaceAlarm", "BehicleAlarm", "Fisheye", "Binoculars", "Light", "Audio", "TalkHalf", "Siren", "Ptz", "NewPtz", "Iris", "HkPtz", "Mainstream" "Substream"	
alarm_in_num		Int		Number of alarm input channels supported by a camera.
alarm_out_num		Int		Number of alarm output channels supported by the board.
camera_type		String	"fixed_bullet", "bullet", "ball" "dome", "Uknown"	Camera type. The following types are supported: box camera, bullet camera, fixed dome camera, PTZ dome camera, and unknown types. (The icon displayed for an unknown type of a camera subjects to that of the fixed_bullet type.)
ipc_init		Bool		Indicates whether the camera is initialized.

11.1.4 Setting the User Password

API Description	
Function	This API is used to set the user password.
API URL	/API/Login/Password/Set

	/API/FirstLogin/Password/Set				
Request Method	POST				
Message ID					
Description	Application scenarios: /API/Login/Password/Set is used to change the password when a user logs in for the first time or the password expires. /API/FirstLogin/Password/Set is used to set the password of the admin user when the device is powered on for the first time.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
enc_password	Password	JSON object	Y		
Elements in the enc_password object					
cipher	Encryption password, which is the same as the value of Pubkey returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."	String	Y	1024 bytes	
seq	Sequence number, which is the same as the value of seq returned by the API in section 11.5 "Public Key and Random Number for PBKDF2."	Int	Y	0–1000000	
Output Parameters					
Parameter	Description	Data Type	Value Range	Note	
result		String	Maximum	The values include success	

			length: 256 bytes	(operation success) or other values (cause of failure).
--	--	--	-------------------	---

The password must be sent to the server after being encrypted using RSA. For details, see section 11.5 "Public Key and Random Number for PBKDF2."

11.1.5 Logging In to the Web System

API Description					
Function	This API is used to log in to the web system.				
API URL	/API/Web/Login				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Note	
Login successful					
last_login_time		String	Max_length:64	Time of last successful login (timestamp).	
last_login_ip		String	Max_length: 64	IP address used in the last successful login.	
fail_login_count		Int		Number of login failures since the last successful login.	
pwd_remain_time		Int		Remaining valid days of the password.	
Login failed					
error_code		String	no_permission, login_failed_or_block, already_login, black_ip	no_permission: You do not have the remote permission. login_failed_or_block: Login failed or the system is locked. Try again after waiting for a period of time specified in block_retain_time .	

				already_login : This device has been logged in elsewhere. black_ip : The IP address is in the blacklist and cannot be connected.
block_retain_time		Int		Duration during which the system will be locked due to excessive login failures.

This API is a web login API. The client uses digest authentication for login. When the login is successful, the **Set-cookie** and **X-csrfToken** parameters are returned in the HTTP header.

For example:

Set-Cookie:

session=54f47bdcec65156e41b51d945bdcb4b6a9c77fae237b1c40d74c08db4a5eeb8f;
Secure;HttpOnly;path=/

X-csrfToken: 93e2a23d518767b646ffd9a58b24b48665923ed797dbbb6b1e9760ebb9335b14

When a client accesses a static file after login, the cookie must be contained in the HTTP header.
For example:

Cookie: session=54f47bdcec65156e41b51d945bdcb4b6a9c77fae237b1c40d74c08db4a5eeb8f;

When an API is accessed, the **X-csrfToken** parameter must be contained in the HTTP header. For example:

X-csrfToken: 93e2a23d518767b646ffd9a58b24b48665923ed797dbbb6b1e9760ebb9335b14.

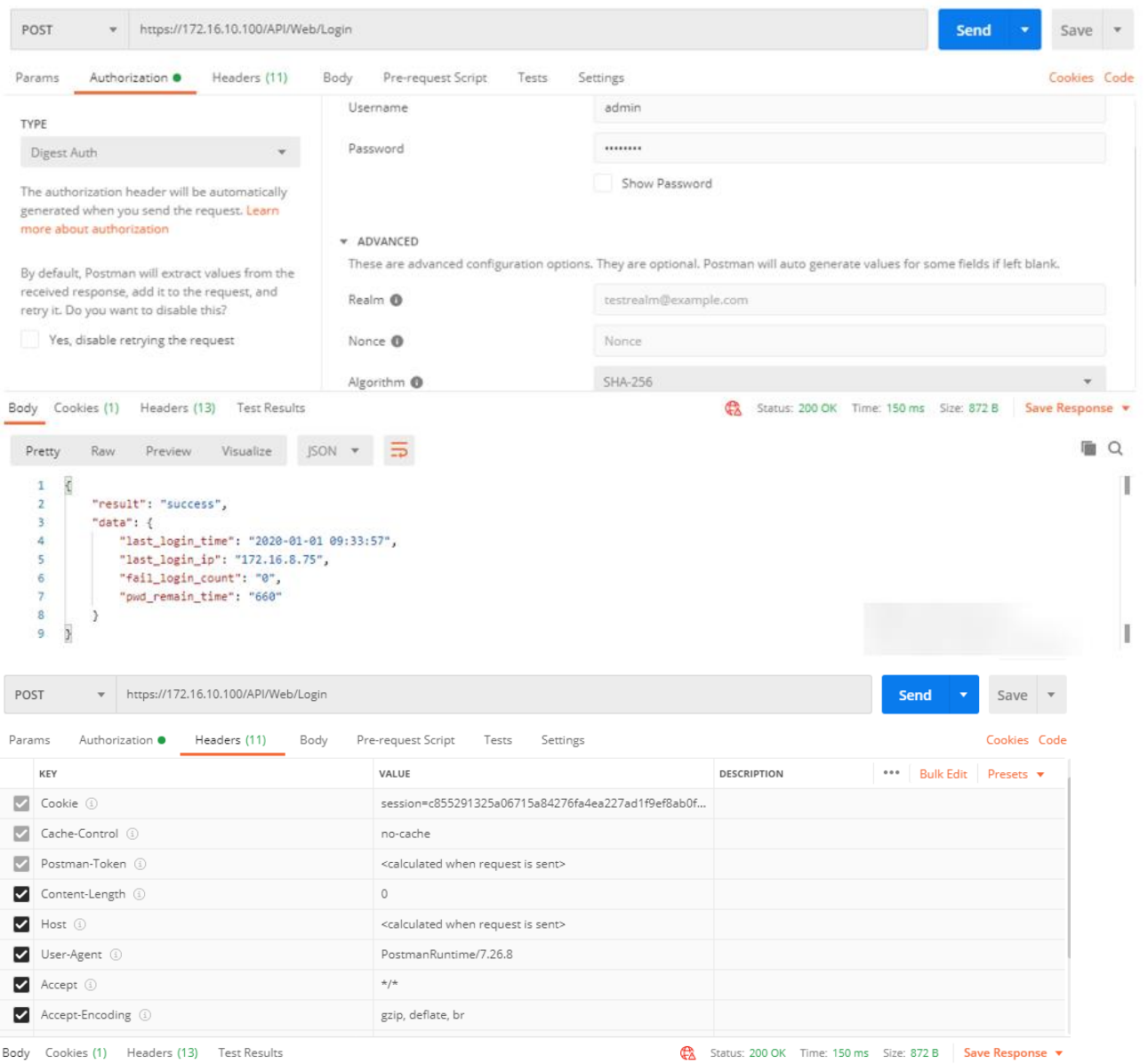
After the login is successful, the session is maintained for 5 minutes (default value), which is configurable. When the client accesses the server after login, the server updates the session expiration time.

Currently, the digest authentication algorithm is RFC 2617. It will be changed to RFC 7616 and also be compatible with RFC 2617. If you are not familiar with the digest authentication, refer to related documents on the Internet.

11.1.5.1 Instruction and Example

Use **Digest Auth** and enter the user name and password for login. The SHA-256 encryption algorithm is used by default.

Perform the following operations on Postman:



POST ▼ https://172.16.10.100/API/Web/Login Send Save ▼

Params **Authorization** ● Headers (11) Body Pre-request Script Tests Settings Cookies Code

TYPE
Digest Auth ▼

The authorization header will be automatically generated when you send the request. [Learn more about authorization](#)

By default, Postman will extract values from the received response, add it to the request, and retry it. Do you want to disable this?
☐ Yes, disable retrying the request

ADVANCED
These are advanced configuration options. They are optional. Postman will auto generate values for some fields if left blank.

Username: admin
Password: *****
☐ Show Password

Realm: testrealm@example.com
Nonce: Nonce
Algorithm: SHA-256 ▼

Body Cookies (1) Headers (13) Test Results Status: 200 OK Time: 150 ms Size: 872 B Save Response ▼

Pretty Raw Preview Visualize JSON ▼ ↕

```

1  {
2    "result": "success",
3    "data": {
4      "last_login_time": "2020-01-01 09:33:57",
5      "last_login_ip": "172.16.8.75",
6      "fail_login_count": "0",
7      "pwd_remain_time": "660"
8    }
9  }

```

POST ▼ https://172.16.10.100/API/Web/Login Send Save ▼

Params **Authorization** ● **Headers (11)** Body Pre-request Script Tests Settings Cookies Code

KEY	VALUE	DESCRIPTION	***	Bulk Edit	Presets
☒ Cookie ①	session=c855291325a06715a84276fa4ea227ad1f9ef8ab0f...				
☒ Cache-Control ①	no-cache				
☒ Postman-Token ①	<calculated when request is sent>				
☒ Content-Length ①	0				
☒ Host ①	<calculated when request is sent>				
☒ User-Agent ①	PostmanRuntime/7.26.8				
☒ Accept ①	*/*				
☒ Accept-Encoding ①	gzip, deflate, br				

Body Cookies (1) Headers (13) Test Results Status: 200 OK Time: 150 ms Size: 872 B Save Response ▼

Select **POST** and select **Digest Auth** in **TYPE**. Enter the user name and password, and select **SHA-256** in **Algorithm** under **ADVANCED**. Use the default values on the **Headers** page, and leave the **Body** page empty.

The request format is as follows:

POST https://172.16.10.24/API/Web/Login?2020-11-10%2015:58:36 HTTP/1.1

Accept: */*

Cookie: session=e0433ec0a249d63cc73c6a745cc0367aa39f0b476554c351077df1ff8a36ccf3

Authorization: Digest algorithm="SHA-256", username="admin", realm="device",
nonce="df265b319236a49d92c83928a91dd947b64aac5785ececff1aa0c35f3a35a5c8",
uri="/API/Web/Login?2020-11-10 15:58:36",
response="9be3f4599b2965f2a7fde333e630d854d9da5eb83738bbc026d177886e0931f2",
opaque="null", qop=auth, nc=00000001, cnonce="247fe9fd623c1c42"

Content-type: application/json

X-Requested-With: XMLHttpRequest

Referer: https://172.16.10.24/index.html?_1604994970029

Accept-Language: zh-Hans-CN,zh-Hans;q=0.5

Accept-Encoding: gzip, deflate

User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0; UBrowser/6.2.4098.3)
like Gecko

Host: 172.16.10.24

Content-Length: 9

Connection: Keep-Alive

Cache-Control: no-cache

undefined

The response format is as follows:

HTTP/1.1 200 OK

Cache-Control: no-store

Connection: close

Content-Length: 146

Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors 'self'

Content-Type: application/json

Set-Cookie:

session=dc484bbc4b60d7c49e75d14c6c423bdea41ea622969eaec67c433df0f8898f6a;Secure;HttpOnly;path=/

Strict-Transport-Security: max-age=315360000; includeSubDomains; preload

WWW-Authenticate:

qop="auth",cnonce="247fe9fd623c1c42",rspauth="72a596270ecdb70032c2121096307e731cb2065063b66022b4b3b9359c962cd0",nc="00000001"

X-Content-Type-Options: nosniff

X-Frame-Options: SAMEORIGIN

X-XSS-Protection: 1; mode=block

X-csrfToken: d5f0810b18e40f2f5e475b627a59b4747c76aef2c6dbd93db787d938ae8c9dea

server: server

```
{
  "result": "success",
  "data": {
    "last_login_time": "2020-11-10 15:57:18",
    "last_login_ip": "172.16.8.18",
    "fail_login_count": "0",
    "pwd_remain_time": "722"
  }
}
```

11.2 Logout

API Description					
Function	This API is used to log out of the web system.				
API URL	/API/Web/Logout				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	The values include success (operation success) or other values (cause of failure).	

11.3 Certificate

11.3.1 Obtaining the Certificate Token

API Description					
Function	This API is used to obtain the certificate token.				
API URL	/API/Login/CertFile/Token				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
token		String	0–32	Token value.	

11.3.2 Obtaining Certificates

API Description					
Function	This API is used to obtain certificates.				
API URL	/API/Login/CertFile/Get				
Request Method	POST				
Message ID					
Description	Request data format: token=xxxxxxx				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Body content of the input parameter, that is, token=xxxxxxx , in which xxxxxxx indicates the token content returned by the API for obtaining the certificate token.					
Output Parameters					

Parameter	Description	Data Type	Value Range	Remarks
		String		Certificate content.

11.4 Heartbeat Packets

API Description					
Function	This API is used to obtain heartbeat packets.				
API URL	/API/Login/Heartbeat				
Request Method	POST				
Message ID					
Description	If no heartbeat packet is sent within 30 seconds, the session is terminated.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
keep_alive	Keep-alive.	Bool	N		
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		Return code 200 indicates that the operation is successful.

It is recommended that the **keep_alive** parameter be used only during development.

Example:

POST /API/Login/Heartbeat

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcfc81b08

Content-Length: 28

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{"data":{"keep_alive":true}}
```

Response:

HTTP/1.1 200 OK

Cache-Control: no-store

Content-Length: 20

Content-Type: application/json

```
{
  "result": "success"
}
```

11.5 Public Key and Random Number for PBKDF2

API Description					
Function	This API is used to request the public key for encrypting user passwords or a random number used for PBKDF2.				
API URL	/API/Maintenance/TransKey/Get (authentication required) /API/Login/TransKey/Get (used for the first login and no authentication is required)				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
type	Request type.	String array "Salt", "Pubkey"	Y	1–5	The maximum length of the array is 5. Multiple Salt and Pubkey elements can be requested.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
key_lists		JSON			

		array		
Elements in the object				
type		String	"Salt", "Pubkey"	Only Pubkey can be used for the first login.
key		String	0–1024	The random number is a hex string, which needs to be converted. The public key is in plain text.
iter		Int	0–1000000	Number of PBKDF2 iterations.
seq		Int	0–1000000	Each key corresponds to a seq value. When the encrypted data is transmitted, the seq value must be transferred at the same time.

11.5.1 Instruction and Example

Passwords are transmitted in different scenarios. For example, entering the current password for secondary authentication when formatting a hard disk, setting the user password, and setting the camera password.

Scenario	Type of the Requested Key	Algorithm	Request API
Secondary authentication	Salt	PBKDF2	/API/Maintenance/TransKey/Get
Set the password of the admin user for the first time	Pubkey	RSA	/API/Login/TransKey/Get /API/FirstLogin/Password/Set
Change the initial password of a common user	Pubkey	RSA	/API/Maintenance/TransKey/Get /API/Login/Password/Set
Add a user Change the user password	Pubkey	RSA	/API/Maintenance/TransKey/Get
Add a camera Modify camera settings Change camera password	Pubkey	RSA	/API/Maintenance/TransKey/Get
Set the email password, certificate password, and password of the GB/T 28181-compliant platform	Pubkey	RSA	/API/Maintenance/TransKey/Get

1. Process for the client to use PBKDF2:

```

salt = hex_to_byte(key);
enc = PBKDF2(salt, pwd, iter);
cipher = byte_to_hex(enc);

```

For details about the conversion of **byte_to_hex**, see the following:

```

for (unsigned int i = 0; i < bvc.size(); i++){
    sprintf(buf.data(), buf.size(), "%02x", bvc[i]) ;
    hexstr += buf.data();
}

```

2. Process for the client to use RSA:

```

enc = rsa_encrypt (key, pwd, RSA_PKCS1_OAEP_PADDING);
cipher = byte_to_hex(enc);

```

For details about the conversion of **rsa_encrypt**, see OpenSSL.

```

int RSA_public_encrypt(int flen, const unsigned char *from, unsigned char *to, RSA *rsa, int padding);

```

3. Example for changing a user password:

POST /API/Maintenance/TransKey/Get?2020-10-17%2017:11:01 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcfc81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```

{
    "version": "1.0",
    "data": { "type": ["Pubkey"] }
}

```

Response:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 512

```

{
    "result": "success",
    "data": {

```

```

"Key_lists":{
  "type": "Pubkey",
  "key": "-----BEGIN RSA PUBLIC
KEY-----\nMIIBCgKCAQEAvaASnInvdmnwF2/8K4VQxuu2W69veMCth/r9+518XYTHquF15FbJh\nnnCmMbsljzw7kM
ei+UL66DVCvWlYHnlzPRqdW/znKk7mmmAdogwNj/5zM13rw2YRw\njXamFUUD9IW6eUa4v57/EcN58IIUEDNIQ
C4FbozyATtzQUIWKbWtELv3UfQ8xOUw\nRZqVEe1GC/s+/GdwCqseLAg412kCepM2IV3Or/LVDLZEB3DCDEqjjv
NLWh/5T9gu\nt09xXCQlN9x+aghXrHdiYMqPtz31Wz7cAHwUcqtu6RR4wPpBl8XbGu5W+nvFYkSt\nN37rLq8svyh
qoNcO9JREu2Hm3+sO2CEB1wDAQAB\n-----END RSA PUBLIC KEY-----\n",
  "seq": 0
}}

POST /API/SystemConfig/User/Set?2020-10-17%2017:11:01 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{
  "version": "1.0",
  "data": {
    "user_info": {
      "USER1": {
        "user_enable": true,
        "username": "user1",
        "start_date": "10/17/2020",
        "end_date": "10/17/2021",
        "start_time": "16:05:28",
        "end_time": "16:05:28",
        "enc_password": {
          "seq": 0,
          "cipher": "2bf74abedca2a9b3b745f997f35551cfc95c
37838130fad6eda6ac2508b266c5883b0c605a932e1671b7f42f213b29206bff08d3749c7cc51ed0f1ac117298b7c7
3b31f9a73939582056c9b8b6e97f8a0a08e9987a972a8dd5d246dff1b912f09125638fa508e25d4fdac5febd8ccb568
1b63b3c44155681d0c12e133b8ac77c6f0b69ee2787dd5e7e39165207d0252797596d381996443ae8ff27f09d6cde
842dc01e929173287681912a4ff86f3dd7defc927176576c0d7feaec38a1d667d6e97f1a0b6e334f2fa7c549ca6e6
ba56f726b19b6f6dcfd00c34bc4c858e0dda1062a185ef60615cbc7892d69d0d314080bfc2f3b7a845f10ee638629e2
e7094"}
        },
        "type": "SavePassword"
      }
    }
  }
}

```

Response:

```

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 30

{"result": "success", "data": {}}

```

11.6 Login Authorization

The HWT-NVR800 provides the following two APIs:

1. API for obtaining a temporary login token as a logged-in user.
2. API for obtaining the session ID and token using the login token.

These two APIs can be used by any logged-in user who wants to authorize some APIs to other devices without providing information such as the user name, password, session ID, or token.

11.6.1 Obtaining the Login Token

API Description					
Function	This API is used to obtain the login token.				
API URL	/API/Get_Login_Token				
Request Method	POST				
Message ID					
Description	Used to obtain the login token to authorize login.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
limit_ip	IP address restriction.	Bool	N		If this parameter is not transferred, the type is incorrect, or the value true is transferred, the IP address is restricted. That is, the IP address of the authorized user must be the same as that of the current user. If the parameter is correctly transferred, the IP address of other users can be authorized.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
login_token		String	Fixed length: 64 bytes	The token is valid within 1 minute after being obtained.	

11.6.2 Authorizing Login Using a Token

API Description

Function	This API is used to authorize login using a token.				
API URL	/API/Web/Token_Login?login_token=***				
Request Method	POST				
Message ID					
Description	If the login authorization is successful, session and X-csrfToken are returned. If the login authorization times out, the error message "fail login_token" is returned.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks

11.6.3 Usage Description

After logging in to the system, you need to obtain the login token to authorize login. Then, **session** and **X-csrfToken** are returned for the authorized user if the login authorization is successful. The authorized user has the same permissions and obligations (for example, sending heartbeat packets) as the primary user. The primary user is not affected when the authorized user logs out. However, if the primary user logs out, the authorized user is logged out accordingly. Only one authorized user can log in to the system at a time. If another authorized user logs in to the system, the previous authorized user will be logged out.

Example:

1. Obtain a token.

```
POST /API/Get_Login_Token
```

```
Content-Type: application/json; charset=UTF-8
```

```
Accept: application/json, text/javascript, */*; q=0.01
```

```
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08
```

```
Content-Length: 0
```

```
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc
```

```
{
```

```
  "data":
```

```
  {
```

```
    "limit_ip": false
```

```
    }  
  }  
Response:  
  HTTP/1.1 200 OK  
  Cache-Control: no-store  
  Content-Length: 82  
  Content-Type: application/json  
  {  
    "login_token":  
    "facda51d3466e3f33a0eba33a24bff79d3a1349005586b519ac166e0605917a8"  
  }
```

2. Authorize login using the token. (Note that **Cookie** and **X-csrfToken** are not required in the header.)

```
  POST/API/Web/Token_Login?login_token=facda51d3466e3f33a0eba33a24bff79d3a1349005586b519ac166e0605917a8  
  Content-Length: 0  
  Response:  
  HTTP/1.1 200 OK  
  Cache-Control: no-store  
  Content-Length: 0  
  Content-Type: application/json  
  Set-Cookie:session=faf8f3514f826b00bf30b5e9d6076ae7e0cc6e3885aad67aa004af1846f00cc8;HttpOnly;path=/  
  X-csrfToken: 1f6de0116da94b7ae3f84f0590ac0c6e29d5cca9d958e2cf5fd195efbdfd82e4
```

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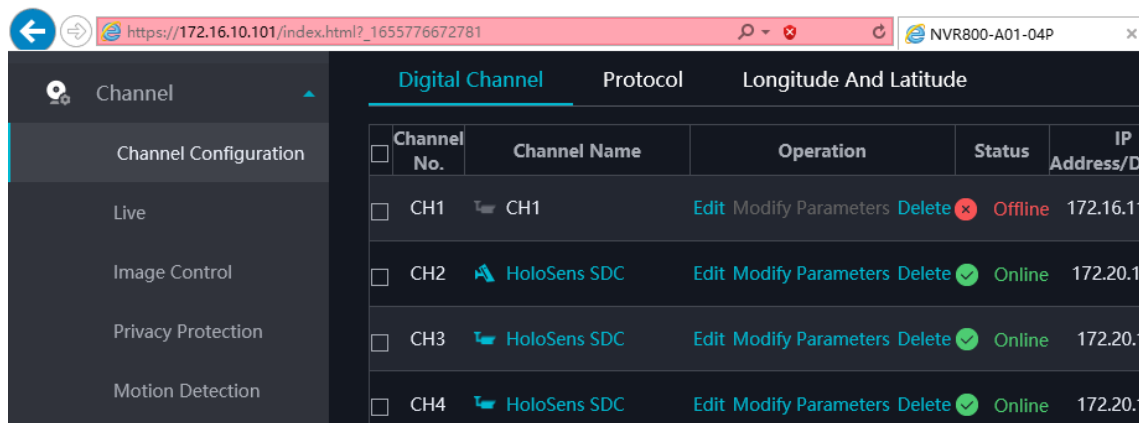
In addition to allowing you to authorize login for other users, the HWT-NVR800 also allows you to develop built-in web functions on the client.

Client:

token=/API/Get_Login_Token

open("http://192.168.1.111/?login_token=xxx&page=xxx&lg=ENU")

The display effect is as follows:



The options of **lg** can be **ENU** or **CHS**. ENU indicates English and CHS indicates Chinese. If **lg** is not transferred, the default language set on the device is used.

The options of **page** include **live**, **playback**, **config**, and **pathConfig**. If **page** is not transferred, the **config** channel configuration page is displayed by default.

12 Snapshot

12.1 Manually Taking a Snapshot

API Description					
Function	This API is used to manually take a snapshot.				
API URL	/API/Snapshot/Get				
Request Method	POST				
Message ID					
Description	Only the snapshot data is returned but not stored in the hard disk.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a

					device.
snapshot_resolution		String	N	"352 x 240", "352 x 288" ...	The snapshot resolution must meet the following requirements: 1. The maximum resolution supported by HWT-NVR800s with two disk trays is 720p. For other models, the maximum resolution is 1080p. 2. The resolution of the primary or secondary stream in the resolution list is less than or equal to the resolution of the current secondary stream. (If this parameter is not transferred, the minimum resolution is used by default.)
reset_session_timeout		Bool	N		Indicates whether to update the session timeout interval. This parameter is valid only when preview_session_timeout is enabled. If this parameter is not transferred, the session timeout interval is updated by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel		String	CH1 to CHn	Snapshot channel.	
date		String	The date format is MM/DD/YYYY	Snapshot date.	
time		String	The time format is hh:mm:ss	Snapshot time.	
img_data		String	Maximum length: 256 x 1024 bytes	Snapshot data. (JPEG format, Base64 encoding)	

12.2 Manually Taking a Snapshot from the Server

API Description

Function	This API is used to manually take a snapshot from the server.				
API URL	/API/ManualSnapshot/Capture				
Request Method	POST				
Message ID					
Description	Only snapshot result is returned and snapshot data is stored in the hard disk.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

13 PTZ Control

13.1 Obtaining Values of PTZ Parameters

API Description	
Function	This API is used to obtain values of PTZ parameters.
API URL	/API/PreviewChannel/PTZ/Get
Request Method	POST
Message ID	
Description	API used to obtain the value range of parameters for obtaining PTZ parameters and setting PTZ control: /API/PreviewChannel/PTZ/Range. (The required parameter value range can be obtained only when the channel parameter is transferred.) A new API is used in NVR 9.1 and later versions:

/API/PreviewChannel/PTZV2/Get. The API described in this section is reserved for compatibility with iClient S100 and will be deprecated in the future.					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		string	Y	CH1 to CHn	Fixed range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
preset_point		Int array	0–255	The maximum length of the array is 255 characters. Each bit in the array indicates a preset position. The value of each bit (ranges from 0 to 255) indicates the dwell duration at the preset position. If the value of a bit is 0 , no preset position is set for this bit.	
cruise_state		Bool		Tour status.	
line_scan_state		Bool		Linear scan tour status.	

13.2 Performing PTZ Control

API Description	
Function	This API is used to perform PTZ control.
API URL	/API/PreviewChannel/PTZ/Control
Request Method	POST
Message ID	
Description	When the duration for cameras to dwell at multiple preset positions is modified in batches, the value of the cmd parameter is still PTZ_CMD_CHANGECRUISETIME , but the duration needs to be transferred through preset_point_time_array . Note that the preset positions for which you want to modify the camera dwell duration must exist. Otherwise, the

modification is invalid. A new API is used in NVR 9.1 and later versions: /API/PreviewChannel/PTZV2/Control. The API described in this section is reserved for compatibility with iClient S100 and will be deprecated in the future.					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	Fixed range.
cmd		Int	Y	PTZ_CMD_UP = 1 (Move the PTZ up.) PTZ_CMD_DOWN = 2 (Move the PTZ down.) PTZ_CMD_LEFT = 3 (Move the PTZ left.) PTZ_CMD_RIGHT = 4 (Move the PTZ right.) PTZ_CMD_UPLEFT = 5 (Move the PTZ up and left.) PTZ_CMD_UPRIGHT = 6 (Move the PTZ up and right.) PTZ_CMD_DOWNLEFT = 7 (Move the PTZ down and left.) PTZ_CMD_DOWNRIGHT = 8 (Move the PTZ down and right.) PTZ_CMD_ZOOMIN = 9 (Zoom in.) PTZ_CMD_ZOOMOUT = 10 (Zoom out.) PTZ_CMD_FOCUSNEAR = 11 (Focus near.) PTZ_CMD_FOCUSFAR = 12 (Focus far.)	

				PTZ_CMD_IRISOPEN = 13 (Open up the aperture.) PTZ_CMD_IRISCLOSE = 14 (Close down the aperture.) PTZ_CMD_AUTOSCAN = 15 (Horizontal scan.) PTZ_CMD_CRUISE = 16 (Perform tour along preset positions.) PTZ_CMD_CHANGECRUISETIME = 17 (Modify dwell duration at a preset position.) PTZ_CMD_SETPRESET = 18 (Add a preset position.) PTZ_CMD_CLEARPRESET = 19 (Delete a preset position.) PTZ_CMD_CALLPRESET = 20 (Go to a preset position.) PTZ_FAST_LOCATE = 52 (3D positioning.)	
speed		Int	N	0–10	
preset_index		Int	N	1–255	Icon of a single preset position.
ctl_stop		Bool	N	true, false	true: Stops the PTZ operation.
point_cruise_time		Int	N		Dwell duration at a single preset position.
point		JSON array	N	Two elements in the object . The first one is the coordinates of the upper left corner, and the second one is the coordinates of the lower right corner.	3D positioning coordinates are contained. For details, see Table 13-1.
custom_preset_array		JSON array	N	The element is of the int type. The value of the preset position ranges from 1 to 255. The array supports cycling of a maximum of 32 preset positions. If this parameter is not transferred, all preset positions are cycled by default.	User-defined array of preset positions at which the tour is performed. That is, a user selects the preset positions from the preset positions supported by the current channel

					and transfers them to the current array. (The preset positions supported by the current channel can be obtained through the API in section 13.2.1 "Instruction and Example.") In addition, the cmd parameter needs to be set to the enumeration corresponding to PTZ_CMD_CRUISE . This parameter is invalid when other commands are used.
preset_point_time_array		Object array	N	Elements in the object . Each element represents a preset position.	Time array of the preset position. For details, see Table 13-2.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

Table 13-1 Elements in the point array

Elements in the point array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				

x		Int	0–352	X value of the coordinate.
y		Int	0–288	Y value of the coordinate.

Table 13-2 Elements in preset_point_time_array

Elements in preset_point_time_array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
preset_index		Int	1–255	Preset position.
point_cruise_time		Int	1–255	Dwell duration at the preset position.

13.2.1 Instruction and Example

PTZ direction control and lens (zoom, aperture, and focus) control are performed in two steps: control start and control end. The following uses direction control as an example. The request content is as follows:

Step 1: Start the control.

```
POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1
```

```
Content-Type: application/json; charset=UTF-8
```

```
Accept: application/json, text/javascript, */*; q=0.01
```

```
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08
```

```
Content-Length: 2438
```

```
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc
```

```
{
  "version": "1.0",
  "data": {
    "cmd": 4,
    "channel": "CH1",
    "speed": 5,
    "preset_index": 1,
    "ctl_stop": false
```

```
}
```

```
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{  
  "result": "success",  
  "data": {}  
}
```

Step 2: Stop the control.

POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{  
  "version": "1.0",  
  "data": {  
    "cmd": 4,  
    "channel": "CH1",  
    "speed": 5,  
    "preset_index": 1,  
    "ctl_stop": true  
  }  
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{  
  "result": "success",
```

```
"data": {}  
}
```

To set a preset position, perform the following steps:

1. Perform a direction or lens control, that is, rotate the PTZ to a required direction or adjust the lens.
2. Save the control as a preset position.

If you repeat the preceding operations for multiple times, you will obtain multiple preset positions.

Then, you can run the tour command to start the tour for the preset positions you just saved.

The procedure for saving a preset position is as follows:

```
POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1  
Content-Type: application/json; charset=UTF-8  
Accept: application/json, text/javascript, */*; q=0.01  
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08  
Content-Length: 2438  
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc  
  
{  
  "version": "1.0",  
  "data": {  
    "channel": "CH3",  
    "point_cruise_time": 3,  
    "preset_index": 3,  
    "cmd": 18  
  }  
}
```

The response format is as follows:

```
HTTP/1.1 200 OK  
Content-Type: application/json  
Content-Length: 230  
  
{  
  "result": "success",  
  "data": {}  
}
```

13.3 Obtaining Current PTZ Position Information

API Description					
Function	This API is used to obtain current PTZ position information.				
API URL	/API/PreviewChannel/PTZCurrentLocation/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	Fixed range.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
PTZ_hor_degree		Int		Pan angle (0° to 359°).	
PTZ_ver_degree		Int		Tilt angle (−10° to +90°).	
len_multiple		Int		Integral part of the lens zoom ratio (1–20). If the value of digital zoom is not 1 , the lens zoom ratio is the maximum value by default.)	
dot_len_multiple		Int		Decimal part of the lens zoom ratio (0–99).	
dot_len_multiple		Int		Integral part of the digital zoom (0-12). The parameter is set when the lens zoom ratio reaches the maximum value. The value 0 indicates that this parameter is not set. The total zoom ratio is the multiplication of zoom ratios.	

dot_zoom_ratio		Int		Decimal part of the digital zoom (0–99).
PTZ_dot_hor_degree		Int		Decimal part of the pan angle (0–99).
PTZ_dot_ver_degree		Int		Decimal part of the tilt angle (–99 to 99).
fabs_PTZ_hor_degree		String		Absolute pan angle.
fabs_PTZ_ver_degree		String		Absolute tilt angle.
zoom_position		Int		Zoom position.
focus_position		Int		Focus position.
zoom_flag		Bool		The values include true (zoom ratio) or false (motor position).

13.4 Obtaining Values of PTZ Home Position Parameters

API Description					
Function	This API is used to obtain values of PTZ home position parameters.				
API URL	/API/PreviewChannel/GuardPoint/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the value range of parameters for obtaining PTZ home position parameters: /API/PreviewChannel/GuardPoint/Range. (The required parameter value range can be obtained only when the channel parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn . Parameters for one channel can be obtained at a time.	Fixed range.

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
enable		Bool		Indicates whether to enable the home position.
wait_time		Int	5–3600s	Waiting time.
status		String	"Offline", "Online"	Channel status. This parameter is transferred for online channels.
guard_mode		String	"Preset", "Cruise", "Defined", "Horizontal", "Frame", "Random", "Panorama", "Vertical"	Home position mode, including the preset position, tour, pattern scan, horizontal scan, frame scan, random scan, panoramic scan, and vertical scan.
preset_index		Int	0–256	Preset position selection. The preset position parameters must be set first. This parameter corresponds to the preset position mode. For other modes, this parameter is not transferred.
cruise_index		Int	0–8	Tour position. Tour parameters must be set first. This parameter corresponds to the tour mode. For other modes, this parameter is not transferred.
defined_index		Int	0-5	Pattern scan path. Pattern scan parameters must be set first. This parameter corresponds to the pattern scan mode. For other modes, this parameter is not transferred.
interval_time		Int	5–3600s	Scan interval. This parameter corresponds to the frame scan or random scan mode. For other modes, this parameter is not transferred.

scan_speed		Int	1–10	Scan speed. This parameter corresponds to the vertical scan mode. For other modes, this parameter is not transferred.
horizontal_param		JSON object	Parameter elements corresponding to the horizontal scan mode. For other modes, this parameter is not transferred.	For details, see Table 13-3.
panorama_param		JSON object	Parameter elements corresponding to the panorama scan mode. For other modes, this parameter is not transferred.	For details, see Table 13-3.

Table 13-3 Elements in horizontal_param and panorama_param

Elements in horizontal_param and panorama_param				
scan_speed		Int	1–10	Scan speed.
horizontal_reverse		Bool		Horizontal reverse.

13.5 Setting the PTZ Home Position Parameters

API Description					
Function	This API is used to set the PTZ home position parameters.				
API URL	/API/PreviewChannel/GuardPoint/Set				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	Fixed range.
enable		Bool	N		Indicates whether to enable the home position.
wait_time		Int	N	5–3600s	Waiting time.
guard_mode		String	N	"Preset", "Cruise", "Defined", "Horizontal", "Frame", "Random", "Panorama", "Vertical"	Home position mode, including the preset position, tour, pattern scan, horizontal scan, frame scan, random scan, panoramic scan, and vertical scan.
preset_index		Int	N	0–256	Preset position selection. The preset position parameters must be set first. This parameter corresponds to the preset position mode. For other modes, this parameter is not transferred.
cruise_index		Int	N	0–8	Tour position. Tour parameters must be set first. This parameter corresponds to the tour mode. For other modes, this parameter is not

					transferred.
defined_index		Int	N	0–5	Pattern scan path. Pattern scan parameters must be set first. This parameter corresponds to the pattern scan mode. For other modes, this parameter is not transferred.
interval_time		Int	N	5–3600s	Scan interval. This parameter corresponds to the frame scan or random scan mode. For other modes, this parameter is not transferred.
scan_speed		Int	N	1–10	Scan speed. This parameter corresponds to the vertical scan mode. For other modes, this parameter is not transferred.
horizontal_param			N	Parameter elements corresponding to the horizontal scan mode. For other modes, this parameter is not transferred.	For details, see Table 13-3.
panorama_param			N	Parameter elements corresponding to the panorama scan mode. For other modes, this parameter is not transferred.	For details, see Table 13-3.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values

			(cause of failure).
--	--	--	---------------------

13.6 Obtaining Values of PTZ Parameters (V2)

API Description					
Function	This API is used to obtain values of PTZ parameters.				
API URL	/API/PreviewChannel/PTZV2/Get				
Request Method	POST				
Message ID					
Remarks	API used to obtain the parameter value range: /API/PreviewChannel/PTZV2/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	Fixed range.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
preset_point		Object array			Preset position information. For details, see Table 13-4.
cruise_state		Bool			Tour status.
line_scan_state		Bool			Linear scan tour status.
curise_track		Object array			Tour information. For details, see Table 13-5.

Table 13-4 Elements in the preset_point array

Elements in the preset_point array				
		Object array		Elements in a single object are as follows.

Elements in a single object				
preset_index		Int	1–255	Preset position index.
preset_name		String	0–31	Preset position name.
focus_enable		Bool		Indicates whether to enable focus memory.

Table 13-5 Elements in the `cruise_track` array

Elements in the cruise_track array				
		Object array		Elements in a single object are as follows.
Elements in a single object				
track_index		Int	1–8	Tour index.
track_name		String	1–31	Tour name.
track_preset		Object array		Information about preset positions in a tour. For details, see Table 13-6.

Table 13-6 Elements in the `track_preset` array

Elements in the track_preset array				
		Object array		Elements in a single object are as follows.
Elements in a single object				
preset_index		Int	1–255	Preset position index.
cruise_time		Int	30–3600	Tour duration.
cruise_speed		Int	1–10	Speed for rotating to a preset position. (This parameter is not supported for ONVIF-compliant cameras and analog PTZ devices.)
support_onekeyfocus		Bool		Indicates whether OnePush focus is supported.

13.7 Performing PTZ Control (V2)

API Description					
Function	This API is used to perform PTZ control.				
API URL	/API/PreviewChannel/PTZV2/Control				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
channel		String	Y	CH1 to CHn	Fixed range.
cmd		Int	Y	PTZ_CMD_UP = 1 (Move the PTZ up.) PTZ_CMD_DOWN = 2 (Move the PTZ down.) PTZ_CMD_LEFT = 3 (Move the PTZ left.) PTZ_CMD_RIGHT = 4 (Move the PTZ right.) PTZ_CMD_UPLEFT = 5 (Move the PTZ up and left.) PTZ_CMD_UPRIGHT = 6 (Move the PTZ up and right.) PTZ_CMD_DOWNLEFT = 7 (Move the PTZ down and left.) PTZ_CMD_DOWNRIGHT = 8 (Move the PTZ down and right.) PTZ_CMD_ZOOMIN = 9 (Zoom in.) PTZ_CMD_ZOOMOUT = 10 (Zoom out.) PTZ_CMD_FOCUSNEAR = 11 (Focus near.)	

				PTZ_CMD_FOCUSFAR = 12 (Focus far.) PTZ_CMD_IRISOPEN = 13 (Open up the aperture.) PTZ_CMD_IRISCLOSE = 14 (Close down the aperture.) PTZ_CMD_AUTOSCAN = 15 (Horizontal scan.) PTZ_CMD_CRUISE = 16 (Perform tour along preset positions, which is not supported in V2.) PTZ_CMD_CHANGECRUIS ETIME = 17 (Modify dwell duration at a preset position, which is not supported in V2.) PTZ_CMD_SETPRESET = 18 (Add a preset position.) PTZ_CMD_CLEARPRESET = 19 (Delete a preset position.) PTZ_CMD_CALLPRESET = 20 (Go to a preset position.) PTZ_FAST_LOCATE = 52 (3D positioning.) PTZ_A_KEY_FOCUS=53 (OnePush focus.) PTZ_CRUISE_TRACH_SET =54 (Set a tour.) PTZ_CRUISE_TRACH_CLR =55 (Delete a tour.) PTZ_CRUISE_TRACH_GOT O=56 (Start a tour.) PTZ_CRUISE_TRACH_STO P=57 (Stop a tour.)	
speed		Int	N	0–10	
preset_index		Int	N	1–255	Icon of a single preset position, which is requested when the preset position is invoked or deleted.

ctl_stop		Bool	N	true, false	true : Stops the PTZ operation.
point		JSON array	N	Two elements in the object . The first one is the coordinates of the upper left corner, and the second one is the coordinates of the lower right corner.	3D positioning coordinates are contained. For details, see Table 13-7.
curise_track		Object	N		For details about how to add, modify, and delete a tour, see Table 13-8.
preset_point		Object	N		For details about how to add and modify a preset position, see Table 13-9.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		The values include success (operation success) or other values (cause of failure).

Table 13-7 Elements in the point array

Elements in the point array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
x		Int	0–352	X value of the coordinate.
y		Int	0–288	Y value of the coordinate.

Table 13-8 Elements in the `curise_track` object

Elements in the curise_track object				
<code>track_index</code>	Y	Int	1–8	Tour index.
<code>track_name</code>	N	String	1–31	Tour name. (This parameter is used only when setting and adding a tour.)
<code>track_preset</code>	N	Object array		Information about preset positions in a tour. (This parameter is used only when setting and adding a tour). For details, see Table 13-6.

Table 13-9 Elements in the `preset_point` object

Elements in the preset_point object				
<code>preset_index</code>		Int	1–255	Preset position index.
<code>preset_name</code>		String	0–31	Preset position name.
<code>focus_enable</code>		Bool		Indicates whether to enable focus memory.

13.7.2 Instruction and Example

PTZ direction control and lens (zoom, aperture, and focus) control are performed in two steps: control start and control end. The following uses direction control as an example. The request content is as follows:

Step 1: Start the control.

```
POST/API/PreviewChannel/PTZV2/Control?2020-10-17%2014:21:33 HTTP/1.1
```

```
Content-Type: application/json; charset=UTF-8
```

```
Accept: application/json, text/javascript, */*; q=0.01
```

```
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08
```

```
Content-Length: 2438
```

```
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc
```

```
{
  "version": "1.0",
  "data": {
```

```
"cmd": 4,  
"channel": "CH1",  
"speed": 5,  
"preset_index": 1,  
"ctl_stop": false  
}  
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{  
  "result": "success",  
  "data": {}  
}
```

Step 2: Stop the control.

POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{  
  "version": "1.0",  
  "data": {  
    "cmd": 4,  
    "channel": "CH1",  
    "speed": 5,  
    "preset_index": 1,  
    "ctl_stop": true  
  }  
}
```

The response format is as follows:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 2976

{
  "result": "success",
  "data": {}
}
```

To set a preset position, perform the following steps:

1. Perform a direction or lens control, that is, rotate the PTZ to a required direction or adjust the lens.
2. Save the control as a preset position.

If you repeat the preceding operations for multiple times, you will obtain multiple preset positions.

Then, you can customize a tour by setting preset positions.

The procedure for saving a preset position is as follows:

```
POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1
Content-Type: application/json; charset=UTF-8
Accept: application/json, text/javascript, */*; q=0.01
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08
Content-Length: 2438
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{
  "version": "1.0",
  "data": {
    "cmd": 18,
    "channel": "CH5",
    "preset_point": {
      "preset_index": 5,
      "preset_name": "preset_5",
      "focus_enable": true
    }
  }
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 230

```
{
  "result": "success",
  "data": {}
}
```

The following is an example of setting a tour:

POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "version": "1.0",
  "data": {
    "cmd": 54,
    "channel": "CH5",
    "cruise_track": {
      "track_index": 8,
      "track_name": "API_SET",
      "track_preset": [
        {
          "preset_index": 3,
          "cruise_time": 321,
          "cruise_speed": 1
        },
        {
          "preset_index": 2,
          "cruise_time": 123,
          "cruise_speed": 1
        }
      ]
    }
  }
}
```

```
    }  
  }  
}
```

```
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 230

```
{  
  "result": "success",  
  "data": {}  
}
```

The following is an example of starting a tour:

POST/API/PreviewChannel/PTZ/Control?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{  
  "version": "1.0",  
  "data": {  
    "cmd": 56,  
    "channel": "CH5",  
    "curise_track": {  
      "track_index": 8,  
    }  
  }  
}
```

The response format is as follows:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 230

```
{
```

```

    "result": "success",
    "data": {}
}

```

14 Event

14.1 Event Push Message

14.1.1 Obtaining Event Push Messages from the Server

API Description					
Function	This API is used to obtain event push messages from the server.				
API URL	/API/Event/Check				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
reader_id		Unsigned int	N	If this parameter is not transferred, all alarm information is checked on the board and reader_id is output.	
sequence		Unsigned int	N	If this parameter is not transferred, all alarm information is checked on the board and reader_id is output.	
lap_number		Unsigned int	N	If this parameter is not transferred, all alarm information is checked on the board and reader_id is output.	

subscribe_ai_metadata		Bool	N	If this parameter is not transferred, snapshot metadata updates as well as real-time data updates collected through head counting, queue length detection, crowd density detection, and off-duty detection are not sent.	
subscribe_intelligence		Bool	N	If this parameter is not transferred, intelligent alarm events are not sent.	
need_background_img		Bool	N	If this parameter is not transferred, alarm thumbnails are not sent.	
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
reader_id		Int		Optional parameter. If the request does not contain reader_id , reader_id is created and returned in the output.	
sequence		Int		Location of the latest sequence.	
lap_number		Unsigned int		Cell to which the current sequence position belongs.	
alarm_list		JSON array		JSON information about the elements in the	

				array is as follows.
heat_alarm	"HeatAlarm"	String		Optional parameter. If the request does not contain alarm_list , heat_alarm is output.
Elements in the alarm_list array				
system_alarm		JSON array	System alarm array	Optional JSON information about the elements in the array is as follows.
channel_alarm		JSON array	Channel alarm array	Optional JSON information about the elements in the array is as follows.
face_alarm		JSON array	Target alarm array	Optional JSON information about the elements in the array is as follows.
realtime_data_report		JSON array	Real-time data reporting array	Optional JSON information about the elements in the

				array is as follows.
object_alarm		JSON array	Object classification alarm array.	Optional JSON information about the elements in the array is as follows.
time		String	"MM/DD/YYYYhh:mm:ss"	Time when an event (including alarms and real-time data) is reported.
Elements in the system_alarm array				
device_name		String	Maximum length: 31 bytes	Device name (optional).
hdd		JSON		JSON information about the object is as follows.
Elements in the hdd object				
hdd_alarm_type	"Ok", "Error"	String		Hard disk alarm type.
hdd_alarm_info	"Full", "Bad", "Unformat", "Readonly", "Warning", "NoDisk"	String		Hard disk error alarm information.
Elements in the channel_alarm array				
channel		String		
motion_alarm		Bool		(Optional)
record_flag		JSON		Recording flag

				(optional). JSON informatio n about the object is as follows.
camera_connect_status		JSON		(Optional) JSON informatio n about the object is as follows.
io_alarm		Bool		(Optional)
pir_alarm		Bool		(Optional)
low_power		Bool		Indicates whether the battery level is low (optional).
charge_power		Bool		Indicates whether the battery is being charged (optional).
channel_name		String		(Optional)
camera_type	Camera type.	String	"fixed_bullet", "bullet", "ball", "dome", "Uknown"	Camera type. The following types are supported: box camera, bullet camera, fixed dome camera, PTZ dome camera, and unknown types. (The icon displayed for an unknown type of a

				camera subjects to that of the fixed_bullet type.)
ptz_alarm		JSON		(Optional) JSON information about the object is as follows.
videoloss		Bool		(Optional)
int_alarm	Intelligent alarm reporting.	JSON		(Optional)
SnapBgImg	Alarm thumbnail.	String	Base64-encoded image data. When need_background_img is set to true , this parameter is pushed only for motion detection, I/O alarm, and intelligent alarm.	(Optional)
Elements in the record_flag object				
m		String	"R", "G"	Motion Alarm The values are R (alarm-triggered recording) or G (no alarm-triggered recording).
i		String	"R", "G"	IO Alarm The values are R (alarm-triggered recording) or G (no alarm-triggered recording).
r		String	"MR", "SR"	Recording The values

				are SR (scheduled recording status) or MR (status of manual recording on the board).
p		String	"R", "G"	PIR Alarm The values are R (alarm-triggered recording) or G (no alarm-triggered recording). PIR is not supported.
s		String	"R", "G"	Intelligent Alarm The values are R (alarm-triggered recording) or G (no alarm-triggered recording).
Elements in the camera_connect_status object				
connect_status		String	"NotConfigured", "Online", "Offline", "Sleep"	Channel connection status.
offline_reason	Channel offline reason.	String	Maximum length: 14 bytes "NotConfigured", "Online", "Unauther", "Netwrong", "Connect" "Bandlimit", "Locked" "MD5NotSupport"	Channel status. The values are as follows: NotConfigured : No camera is added to the channel;

				Online: The channel is online; Unauther: Incorrect user name or password; Netwrong: Network connection error; Connect: The channel is being connected; Bandlimit: Limited bandwidth; Locked: The user is locked; MD5NotSupport: MD5 is disabled for connecting to the platform.
ability		String array	"FaceAlarm", "VehicleAlarm", "Fisheye", "Binoculars", "Light", "Audio", "DualTalk", "Siren", "Ptz", "Newptz", "Iris", "Hkptz", "Mainstream", "Substream", "IO"	

protocol		String	"Onvif", "HWTSDK" "Custom 1", "Custom 2", "Custom 3", "Custom 4", "Custom 5", "Custom 6", "Custom 7", "Custom 8", "Custom 9", "Custom 10", "Custom 11", "Custom 12", "Custom 13", "Custom 14", "Custom 15", "Custom 16"	Camera access protocol.
input_num		Int		Alarm input.
output_num		Int		Alarm output.
Elements in the ptz_alarm object				
cur_zoom_value		Int		Current zoom position (not used currently).
zoom_step		Int	0–2	Zoom step. The values are 0 (1 step), 1 (5 steps) or 2 (20 steps) (not used currently).
cur_focus_value		Int		Current focus position (not used currently).

focus_step		Int	0–2	Focus step. The values are 0 (1 step), 1 (5 steps), or 2 (20 steps) (not used currently).
auto_focus_state		Int		Focus status. The values are 0 (complete) or 1 (focusing) (not used currently).
shift_range		Int		Not used currently.
ptz_cruise_state		Bool		Tour status.
ptz_line_scan_state		Bool		Linear scan tour status.
Elements in the int_alarm object				
alarm_val		Bool	Alarm status.	
alog_name		String	Maximum length: 63 bytes	Third-party algorithm alarm name. This parameter is returned when int_subtype is set to a third-party algorithm type.
int_subtype	Intelligent alarm subtype.	String	"Lens", "Sound", "Mask", "Queue", "People Density", "People Count", "Over Line", "Region Invasion", "Region In", "Region Out",	Lens: lens blocking alarm Sound: audio alarm Mask: mask detection alarm Queue: queue length detection alarm "People Density": crowd density detection alarm People Count: crossing-line head counting alarm Over Line: tripwire crossing alarm

			"Fast Move", "Linger", "Third Part Alog", "Item Leave", "Item Move", "Electromobile", "Illegalp Parking" "Leave_detect"	Region Intrusion: area intrusion alarm Region In: area entry alarm Region Out: area exit alarm Fast Move: fast movement alarm Linger: loitering detection alarm Third Part Alog: third-party algorithm alarm Item Leave: abandoned object alarm Item Move: removed object alarm Electromobile: electric bike detection alarm Illegalp Parking: parking violation detection alarm Leave_detect: off-duty detection alarm
Elements in a single face_alarm array				
Id	ID of the matched target in the target list.	Long		The value -1 indicates a non-listed person. Other values indicate a unique ID of a target in the target list.
GrpId	Group to which the target belongs.	Long		The value -1 indicates targets that are not matched.
Sex	Gender.	Int	1: male; 2: female	Gender of the matched target. This parameter is valid only when Id is not -1 .
Age	Age.	Int	0-254	Age of the matched target. This parameter is valid only when Id is not -1 .
Name	Name of the matched target.	String	"Mike"	
GrpName	Name of the group to which the matched target belongs.	String	"Student"	
Country	Nationality.	String	"China"	

NativePlace	Birthplace.	String	"Guangdong,Zhuhai"	
IdCode	ID card.	String	"415025199203050916"	
Job	Occupation.	String	"Software"	
Phone	Phone number.	String	"12345678902"	
Email	Email.	String	"abcd@163.com"	
Domicile	Current place of residence.	String	"Guangdong,Zhuhai,Xiangzhou ..."	
Remark	Comment.	String	"Detail of this person ..."	
MatchedImage	Matched image in the target list.	String	"base64(imgData)"	
MatchedFeature	Feature value of the matched image in the target list.	String	"base64(feature)"	
MatchedFtVersion	Version of the feature value.	Unsigned int		
AppearCnt	Number of appearance times of a captured target.	Int		The maximum value is 20 . That is, if a target appears for more than 20 times, the value of this parameter is still 20 , and the value of OverMaxCount is true .
OverMaxCount	Indicates whether the number of appearance times of a captured target exceeds the maximum value that can be displayed.	Bool		If the value of AppearCnt exceeds the maximum value, the value true is returned. Otherwise, the value false is returned.
StartTime	Snapshot time. The value is a Unix timestamp, for example, 2018-10-25 13:08:36.	Unsigned long long		

EndTime	Snapshot time. The value is a Unix timestamp, for example, 2018-10-25 13:08:36.	Unsigned long long		
Similarity	Similarity between the captured image and the matched target image in the target list.	Float		
SnapId	ID of the snapshot taken by the camera. The value is a 32-bit unsigned integer.	Unsigned int		
SnapChn	Snapshot channel.	Int	1, 2, 3...n	n indicates the maximum number of channels.
SnapFaceImg	Target snapshot.	String	"base64(imgData)"	
SnapBgImg	Background snapshot.	String	"base64(imgData)"	
SnapBodyImg	Person snapshot.	String	"base64(imgData)"	
SnapFeature	Feature value extracted from the snapshot.	String	"base64(feature)"	
SnapFtVersion	Feature value version.	Unsigned int		
Gender	Gender detected by the camera.	Int	The values are -1 (not supported), 0 (unknown), 1 (male), or 2 (female).	Gender information detected by the camera.
SnapAge	Age information detected by the camera.	Int	0–100	The value ranges from 1 to 100. Value 0 indicates that the age is unknown.
Expression	Facial	Int	FE_Unknown = 0,	The value -1 indicates

	expression.		FE_Smile, FE_Angry, FE_Sad, FE_Normal, FE_Panic,	that this parameter is not supported.
Hair	Hairstyle.	Int	HT_Unknown, HT_LongHair, HT_ShortHair	The value -1 indicates that this parameter is not supported.
Hat	Indicates whether a person is wearing a hat.	Int	HS_Unknown, HS_None, HS_HatExist,	The value -1 indicates that this parameter is not supported.
GlassType	Indicates whether a person is wearing glasses.	Int	GT_Unknown, GT_None, GT_NormalGlasses, GT_Sunglasses	The value -1 indicates that this parameter is not supported.
MouthMask	Indicates whether a person is wearing a mask.	Int	MM_Unkonwn, MM_None, MM_MaskExist	The value -1 indicates that this parameter is not supported.
Mustache	Indicates whether a person has facial hair.	Int	MT_Unknown, MT_None, MT_MustacheExist	The value -1 indicates that this parameter is not supported.
Policy	Policy of the group to which the matched target belongs.	Int	For details, see the enumerated values of DLDetectPolicy .	The value -1 indicates that this parameter is not supported.
GrpEnableAlarm	Indicates whether target alarm is enabled for a list.	Bool		
AlarmFlags	Alarm type.	Int	4-bit value, with each bit indicating the following: Bit 0: common snapshot Bit 1: alarm-triggered snapshot Bit 2: snapshot triggered by mask	The options are constants in the AIAlarmType_e enum. Multiple alarm types can be specified. The UI processing logic of the HWT-NVR800 is as follows: When multiple alarm types are specified, these alarm types are displayed in

			alarms Bit 3: snapshot triggered by alarms indicating that a non-listed person appears frequently Bit 4: snapshot triggered by alarms indicating that a listed person appears frequently	parallel.
FaceRect	Target position.	JSON object		Position of a target in the background image. For details, see Table 14-1.
Elements in a single realtime_data_report object				
channel	Channel	String		Channel that triggers data reporting.
realtime_data_type		String	"cross_counting", "queue_length", "crowd_density", "leave_detect"	cross_counting: real-time data of crossing-line head counting detection; queue_length: real-time queue length detection data; crowd_density: real-time crowd density detection data; leave_detect: real-time off-duty detection data
total_in_count		Unsigned int	0–65536	Total number of persons entering a specified area. This parameter is returned when realtime_data_type is cross_counting .
total_out_count		Unsigned int	0–65536	Total number of persons leaving a specified area. This parameter is returned when realtime_data_type is cross_counting .
now_in_count		Unsigned int	0–65536	Current number of persons entering a specified area. This parameter is returned when realtime_data_type is cross_counting .

now_out_count		Unsigned int	0–65536	Current number of persons leaving a specified area. This parameter is returned when realtime_data_type is cross_counting .
shoulder_count		Unsigned int	0–0xFFFFFFFF	Number of queuers. This parameter is returned when realtime_data_type is queue_length .
queue_time		unsigned int	0–0xFFFFFFFF	Estimated waiting time required for the current queue, in seconds. This parameter is returned when realtime_data_type is queue_length .
people_count		unsigned int	0–0xFFFFFFFF	Number of persons in the detection area. This parameter is returned when realtime_data_type is crowd_density .
area_ratio		unsigned int	0–0xFFFFFFFF	Percentage of all persons in the detection area to the entire video image. (The value needs to be divided by 10 to accurate to one decimal place.) This parameter is returned when realtime_data_type is set to crowd_density .
leavedetect_alarm_num		unsigned int	0–0xFFFFFFFF	Total number of persons who should be on duty in the detection area. This parameter is returned when realtime_data_type is leave_detect .
leavedetect_now_num		unsigned int	0–0xFFFFFFFF	Current number of persons on duty in the detection area. This parameter is returned when realtime_data_type is leave_detect .

Elements in a single object_alarm array				
SnapObjectImg	Object image.	String	"base64(imgData)"	
SnapBgImg	Object background image.	String	"base64(imgData)"	
ObjectType	Object classification alarm type.	String	"HumanBody", "Vehicle", "NoneVehicle"	HumanBody: person alarm Vehicle: motor vehicle alarm NoneVehicle: non-motorized vehicle alarm
SnapId	ID of the snapshot taken by the camera. The value is a 32-bit unsigned integer.	Unsigned int		
StartTime	Snapshot time. The value is a Unix timestamp, for example, 2018-10-25 13:08:36.	Unsigned long long		
EndTime	Snapshot time. The value is a Unix timestamp, for example, 2018-10-25 13:08:36.	Unsigned long long		
SnapChn	Snapshot channel.	Int	1, 2, 3...n	n indicates the maximum number of channels.
CarRect	Position of a captured vehicle in the background image.	JSON object		This parameter is available only when ObjectType is Vehicle . For details, see Table 14-1.
PlateRect	Position of a captured license plate in the background image.	JSON object		This parameter is available only when ObjectType is Vehicle . For details, see Table 14-1.

LicensePlate	License plate characters.	String	Maximum length: 63 bytes	This parameter is available only when ObjectType is Vehicle . If the value is an empty string, the information is unknown.
CarPreBrand	Vehicle make.	String	Maximum length: 63 bytes	This parameter is available only when ObjectType is Vehicle . If the value is an empty string, the information is unknown.
CarSubBrand	Vehicle model.	String	Maximum length: 63 bytes	This parameter is available only when ObjectType is Vehicle . If the value is an empty string, the information is unknown.
CarYearBrand	Vehicle model year.	String	Maximum length: 63 bytes	This parameter is available only when ObjectType is Vehicle . If the value is an empty string, the information is unknown.
CarPreBrandId	Vehicle make ID.	Unsigned int	0-0xffffffff	Vehicle make ID, which is returned by the camera. For details, see the supported vehicle makes provided by the camera.
CarSubBrandId	Vehicle model ID.	Unsigned int	0-0xffffffff	Vehicle model ID, which is returned by the camera. For details, see the supported vehicle models provided by the camera.
CarType	Vehicle type	Int		This parameter is available only when ObjectType is Vehicle . The value corresponds to constants in the PU_VEHICLE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_VT_MOTOR .
CarColor	Vehicle color.	Int		This parameter is

				available only when ObjectType is Vehicle. The value corresponds to constants in the PU_VEHICLE_COLOR_E enum.
PlateType	License plate type.	Int		This parameter is available only when ObjectType is Vehicle. The value corresponds to constants in the PU_PLATE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_LPT_AIRPORT currently.
PlateColor	License plate color.	Int		This parameter is available only when ObjectType is Vehicle. The value corresponds to constants in the PU_PLATE_COLOR_E enum.
NonVehicleBgRect	Position of a non-motorized vehicle in the background image.	JSON object		This parameter is available only when ObjectType is NoneVehicle. For details, see Table 14-1.
HumanBgRect	Position of a person in the background image.	JSON object		This parameter is available only when ObjectType is HumanBody. For details, see Table 14-1.
HumanImgRect	Position of a person in the captured image.	JSON object		This parameter is available only when ObjectType is HumanBody. For details, see Table 14-1.

Table 14-1 Elements in FaceRect, CarRect, PlateRect, NonVehicleBgRect, HumanBgRect, and HumanImgRect

Elements in FaceRect, CarRect, PlateRect, NonVehicleBgRect, HumanBgRect, and HumanImgRect				
Note: Use the permyriad coordinate (10000 x 10000) in the background image. If X, Y, Width, and Height are all 0, the license plate position is unknown.				
X		Int	0–10000	X coordinate of the upper left corner of the target position.
Y		Int	0–10000	Y coordinate of the upper left corner of the target position.
Width		Int	0–10000	Target position width.
Height		Int	0–10000	Target position height.

14.1.2 Instruction and Example

Events are reported in cyclic query mode. That is, the client sends a query request to the server, and the server replies to the client. Then, the client immediately sends a query request to the server again. The client can continuously query data or send a request after a period of time. However, the interval cannot exceed 30 seconds.

The server has a **reader_id** for each alarm link, and **sequence** and **lap_number** indicate indexes of the system alarm list. For the first request, no parameter needs to be transferred. The server returns the status of the device as well as a **reader_id**. The client uses the **reader_id**, **sequence**, and **lap_number** returned by the server last time to request the alarm status. If no alarm is generated, the server is blocked. If an alarm is generated during the blocking, the server returns a response immediately. If no alarm is generated, the server is blocked for about 10 seconds and returns a **heat_alarm** message. To ensure the timeliness of event reporting, the client sends the next request immediately after receiving the response. For the first request, the server returns current status for the client to parse the status as required. For subsequent requests, the alarm event is reported.

If the server does not receive the next request within 30s, **reader_id** becomes invalid and the server returns an error message. In this case, the client needs to send the first request again. The HWT-NVR800 returns the latest alarm location (**reader_id**, **sequence**, and **lap_number**) each time when a request is sent for the first time.

If the client is disconnected from the HWT-NVR800, the client needs to save the **sequence** and **lap_number** returned by the HWT-NVR800 to receive alarms from the position where the connection is interrupted after the client is reconnected to the HWT-NVR800. To receive alarms from the position where the connection is interrupted, after the first connection, use the new **reader_id** and the saved **sequence** and **lap_number** to send a request again. Note that the transferred values of **sequence** and **lap_number** must be less than or equal to the new values of **sequence** and **lap_number**. If you do not need to handle the alarms generated during the disconnection, skip this operation.

If the client needs to receive intelligent alarms, the request must contain **subscribe_ai_metadata** and **subscribe_intelligence** whose value is set to **true**. Otherwise, intelligent alarms are not returned.

For alarms such as **motion_alarm**, the value can be **true** or **false**. If the value is **true**, an alarm is generated. If the value is **false**, no alarm is generated and the client needs to process as required.

1. Example of the first request:

```
POST /API/Event/Check?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{
  "data": {}
}
```

Response:

```
HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

{
  "result": "success",
  "data": {
    "alarm_list": [
      {
        "time": "10/17/2020 14:16:31",
        "system_alarm": [
          {
            "device_name": " HWT-NVR800",
            "hdd": {
              "hdd_alarm_type": "Ok",
              "hdd_alarm_info": ""
            }
          }
        ],
        "channel_alarm": [
          {
```

```
"channel": "CH1",
"motion_alarm": false,
"io_alarm": false,
"ptz_alarm": {
  "ptz_cruise_state": false,
  "ptz_line_scan_state": false
},
"videoloss": false,
"camera_connect_status": {
  "connect_status": "Online",
  "offline_reason": "Online",
  "ability": [
    "Ptz",
    "Mainstream",
    "Substream"
  ],
  "protocol": "Onvif",
  "alarm_in_num": 0,
  "alarm_out_num": 0
},
"channel_name": "ok",
"camera_type": "dome",
"record_flag": {
  "r": "SR"
},
{
  "channel": "CH2",
  "motion_alarm": false,
  "io_alarm": false,
  "ptz_alarm": {
    "ptz_cruise_state": false,
    "ptz_line_scan_state": false
```

```
        },
        "videoloss": true,
        "camera_connect_status": {
        "connect_status": "NotConfigured",
        "offline_reason": "NotConfigured",
        "ability": [

        ],
        "alarm_in_num": 0,
        "alarm_out_num": 0

        },
        "channel_name": "",
        "camera_type": "Uknown",
        "record_flag": {

        }

        },
        {
        "channel": "CH3",
        "motion_alarm": false,
        "io_alarm": false,
        "ptz_alarm": {
        "ptz_cruise_state": false,
        "ptz_line_scan_state": false

        },
        "videoloss": true,
        "camera_connect_status": {
        "connect_status": "NotConfigured",
        "offline_reason": "NotConfigured",
        "ability": [

        ],
        "alarm_in_num": 0,
```

```
"alarm_out_num": 0
    },
    "channel_name": "",
    "camera_type": "Uknown",
    "record_flag": {
        }
    },
    {
        "channel": "CH4",
        "motion_alarm": false,
        "io_alarm": false,
        "ptz_alarm": {
            "ptz_cruise_state": false,
            "ptz_line_scan_state": false
        },
        "videoloss": true,
        "camera_connect_status": {
            "connect_status": "NotConfigured",
            "offline_reason": "NotConfigured",
            "ability": [
                ],
            "alarm_in_num": 0,
            "alarm_out_num": 0
        },
        "channel_name": "",
        "camera_type": "Uknown",
        "record_flag": {
            }
        },
        {
```

```
"channel": "CH5",
"motion_alarm": false,
"io_alarm": false,
"ptz_alarm": {
  "ptz_cruise_state": false,
  "ptz_line_scan_state": false
},
"videoloss": true,
"camera_connect_status": {
  "connect_status": "NotConfigured",
  "offline_reason": "NotConfigured",
  "ability": [
    ],
    ],
"alarm_in_num": 0,
"alarm_out_num": 0
  },
"channel_name": "",
"camera_type": "Uknown",
"record_flag": {
  }
},
{
  "channel": "CH6",
  "motion_alarm": false,
  "io_alarm": false,
  "ptz_alarm": {
    "ptz_cruise_state": false,
    "ptz_line_scan_state": false
  },
  "videoloss": true,
  "camera_connect_status": {
```

```
"connect_status": "NotConfigured",
"offline_reason": "NotConfigured",
"ability": [
    ],
"alarm_in_num": 0,
"alarm_out_num": 0
},
"channel_name": "",
"camera_type": "Uknown",
"record_flag": {
    }
},
{
    "channel": "CH7",
    "motion_alarm": false,
    "io_alarm": false,
    "ptz_alarm": {
        "ptz_cruise_state": false,
        "ptz_line_scan_state": false
    },
    "videoloss": true,
    "camera_connect_status": {
        "connect_status": "NotConfigured",
        "offline_reason": "NotConfigured",
        "ability": [
            ],
        "alarm_in_num": 0,
        "alarm_out_num": 0
    },
    "channel_name": "",
```

```
"camera_type": "Uknown",
"record_flag": {
    }
},
{
"channel": "CH8",
"motion_alarm": false,
"io_alarm": false,
"ptz_alarm": {
"ptz_cruise_state": false,
"ptz_line_scan_state": false
},
"videoloss": true,
"camera_connect_status": {
"connect_status": "NotConfigured",
"offline_reason": "NotConfigured",
"ability": [
    ],
"alarm_in_num": 0,
"alarm_out_num": 0
},
"channel_name": "",
"camera_type": "Uknown",
"record_flag": {
    }
}
]
}
],
"reader_id": 3,
```



```
"sequence": 58,  
"lap_number": 0  
}  
}
```

2. Example of a second request:

```
POST /API/Event/Check?2020-10-17%2014:21:33 HTTP/1.1  
Content-Type: application/json; charset=UTF-8  
Accept: application/json, text/javascript, */*; q=0.01  
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08  
Content-Length: 2438  
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc  
{  
  "data": {  
    "reader_id": 3,  
    "sequence": 58,  
    "lap_number": 0  
  }  
}
```

Response:

```
HTTP/1.1 200 OK  
Content-Type: application/json  
Content-Length: 97  
{  
  "result": "success",  
  "data": {  
    "alarm_list": [  
      {  
        "time": "11/04/2020 19:48:24",  
        "channel_alarm": [  
          {  
            "channel": "CH4",  
            "int_alarm": {  
              "alarm_val": true,  

```

```
"int_subtype": "Mask"
    }
    }
  ]
},
{
  "time": "11/04/2020 19:48:24",
  "channel_alarm": [
    {
      "channel": "CH1",
      "record_flag": {
        "r": "SR"
      }
    },
    {
      "channel": "CH2",
      "record_flag": {
        "r": "SR"
      }
    },
    {
      "channel": "CH3",
      "record_flag": {
        "r": "SR"
      }
    },
    {
      "channel": "CH4",
      "record_flag": {
        "s": "R",
        "r": "SR"
      }
    }
  ],
```

```
        {  
    "channel": "CH5",  
    "record_flag": {  
        }  
    },  
    {  
    "channel": "CH6",  
    "record_flag": {  
        }  
    },  
    {  
    "channel": "CH7",  
    "record_flag": {  
        }  
    },  
    {  
    "channel": "CH8",  
    "record_flag": {  
        }  
    },  
    {  
    "channel": "CH9",  
    "record_flag": {  
        }  
    },  
    {  
    "channel": "CH10",  
    "record_flag": {
```

```
        }
    },
    {
        "channel": "CH11",
        "record_flag": {

            }
        },
        {
            "channel": "CH12",
            "record_flag": {

                }
            },
            {
                "channel": "CH13",
                "record_flag": {

                    }
                },
                {
                    "channel": "CH14",
                    "record_flag": {

                        }
                    },
                    {
                        "channel": "CH15",
                        "record_flag": {

                            }
                        },
                    },
                },
            },
        },
    },
}
```

```
        {  
    "channel": "CH16",  
    "record_flag": {  
  
        }  
    },  
    {  
    "channel": "CH17",  
    "record_flag": {  
  
        }  
    },  
    {  
    "channel": "CH18",  
    "record_flag": {  
  
        }  
    },  
    {  
    "channel": "CH19",  
    "record_flag": {  
  
        }  
    },  
    {  
    "channel": "CH20",  
    "record_flag": {  
  
        }  
    },  
    {  
    "channel": "CH21",  
    "record_flag": {
```

```
        }
    },
    {
        "channel": "CH22",
        "record_flag": {

        }
    },
    {
        "channel": "CH23",
        "record_flag": {

        }
    },
    {
        "channel": "CH24",
        "record_flag": {

        }
    },
    {
        "channel": "CH25",
        "record_flag": {

        }
    },
    {
        "channel": "CH26",
        "record_flag": {

        }
    },
    },
```

```
        {
    "channel": "CH27",
    "record_flag": {

        }
    },
    {
    "channel": "CH28",
    "record_flag": {

        }
    },
    {
    "channel": "CH29",
    "record_flag": {

        }
    },
    {
    "channel": "CH30",
    "record_flag": {

        }
    },
    {
    "channel": "CH31",
    "record_flag": {

        }
    },
    {
    "channel": "CH32",
    "record_flag": {
```

```
        }
    }
]
}
],
"reader_id": 1,
"sequence": 63,
"lap_number": 0
}
```

15 AI

15.1 Operation Results

Result Type	Comment
AORT_SUCCESS = 0	Success.
AORT_NO_DB = -1	No database.
AORT_DB_EXEC_FAILED = -2	Operations on the database failed.
AORT_CALC_FEATURE_FAILED = -3	Failed to extract features.
AORT_CANCELED = -4	The operation is canceled.
AORT_NO_DISK = -5	No hard disk.
AORT_DISK_ERROR = -6	Hard disk error.
AORT_EXIST = -7	The operation object already exists.
AORT_GROUP_INVALID = -8	Invalid list.
AORT_NOT_EXIST = -9	The target does not exist.
AORT_MORE_FILE_EXIST = -10	The file already exists.
AORT_SEARCH_ERROR = -11	A search error occurred.
AORT_OVER_MAX_COUNT = -12	The number of targets exceeds the upper limit.
AORT_UPDATING_FEATURE = -13	Updating features...
AORT_NO_USABLE_IPC = -14	No IPC for feature calculation.

AORT_INVALID_PARAM = -15	Invalid parameter.
AORT_INVALID_FORMAT = -16	Incorrect format.
AORT_INVALID_RES = -17	Incorrect resolution.
AORT_INVALID_MEM = -18	Excessive file size.
AORT_CREAT_FAILED = -19	Creation failed.
AORT_MD5_NOT_MATCH = -20	MD5 mismatch.
AORT_POS_ERROR = -21	Incorrect position.
AORT_SIZE_ERROR = -22	Incorrect size.
AORT_NOT_READY = -23	Not ready.
AORT_INVALID_DB = -24	Invalid database.
AORT_GRP_INVALID_OR_INTERNAL=-25	Invalid list or internal preset list that cannot be deleted.
AORT_GRP_FACE_OVER_MAX=-26	The number of targets in the list exceeds the upper limit.
AORT_NO_PERMISSION=-27	No permission.
AORT_INVALID_STRING = -28	Invalid string.
AORT_STRING_TOO_LONG = -29	The string is too long.
AORT_FACE_INVALID = -30	The target does not exist.
AORT_FACE_NAME_NULL = -31	The target name cannot be empty.
AORT_FACE_EXISTED = -32	The target image already exists in the list.

15.2 Enums

////////////////////////////////////

```
enum AIDBOptType {
    ADOT_Get,
    ADOT_Add,
    ADOT_Modify,
    ADOT_Delete,
    ADOT_Move,
    ADOT_Max
}
```

```
};
```

Target list alarm policy. Only **DLDP_Allow** and **DLDP_Deny** are valid for customized lists.

```
enum DLDetectPolicy {
    DLDP_Allow,    // 0: allowed
    DLDP_Deny,     // 1: rejected
    DLDP_Unknown, // 2: unknown (deprecated)
    DLDP_Internal, // 3: internal list
    DLDP_Advance  // 4: advanced (deprecated)
};
```

Operation results:

```
enum AI_OPT_RESULT_TYPE {
    AORT_SUCCESS = 0,           /*<Success. */
    AORT_NO_DB = -1,            /*<No database. */
    AORT_DB_EXEC_FAILED = -2,   /*<Operations in the database failed. */
    AORT_CALC_FEATURE_FAILED = -3, /*<Failed to extract features. */
    AORT_CANCELED = -4,         /*<The operation is canceled. */
    AORT_NO_DISK = -5,          /*<No hard disk. */
    AORT_DISK_ERROR = -6,       /*<Hard disk error. */
    AORT_EXIST = -7,            /*<The target already exists. */
    AORT_GROUP_INVALID = -8,     /*<Invalid list. */
    AORT_NOT_EXIST = -9,        /*<The target does not exist. */
    AORT_MORE_FILE_EXIST = -10,
    AORT_SEARCH_ERROR = -11,     /*<Search error. */
    AORT_OVER_MAX_COUNT = -12,   /*<The number of targets exceeds the upper limit. */
    AORT_UPDATING_FEATURE = -13, /*<Updating features... */
    AORT_NO_USABLE_IPC = -14,     /*<No IPC for feature calculation. */
    AORT_INVALID_PARAM = -15,     /*<Invalid parameter. */
    AORT_INVALID_FORMAT = -16,    /*<Incorrect format. */
    AORT_INVALID_RES = -17,       /*<Incorrect resolution. */
    AORT_INVALID_MEM = -18,       /*<Excessive file size. */
    AORT_CREAT_FAILED = -19,      /*<Creation failed. */
    AORT_MD5_NOT_MATCH = -20,     /*<MD5 mismatch. */
    AORT_POS_ERROR = -21,        /*<Incorrect position. */
};
```

```
AORT_SIZE_ERROR = -22,    /*!<Incorrect size. */
AORT_NOT_READY = -23,    /*!<Not ready. */
AORT_INVALID_DB = -24 ,   /*!<Invalid database. */
AORT_GRP_INVALID_OR_INTERNAL=-25, /*!<Invalid list or internal preset list that
cannot be deleted. */
AORT_GRP_FACE_OVER_MAX=-26,/*!<The number of targets in the list has reached
upper limit. */
AORT_NO_PERMISSION=-27,/*!<No permission. */
AORT_INVALID_STRING = -28,    /*!<Invalid string. */
    AORT_STRING_TOO_LONG = -29,    /*!<The string is too long. */
    AORT_FACE_INVALID = -30,    /*!<The target does not exist. */
    AORT_FACE_NAME_NULL = -31,    /*!<The target name cannot be empty. */};
AORT_FACE_EXISTED = -32    /*!<The target image already exists in the list. */

// Target attribute enum.
enum FaceExpression_e {
    FE_Unknown,
    FE_Smile,
    FE_Angry,
    FE_Sad,
    FE_Normal,
    FE_Panic,
    FE_Max
};

enum GlassType_e {
    GT_Unknown,
    GT_None,
    GT_NormalGlasses,
    GT_Sunglasses,
    GT_Max
};

enum HairType_e {
```

```
    HT_Unknown,  
    HT_LongHair,  
    HT_ShortHair,  
    HT_Max  
};
```

```
enum HatState_e {  
    HS_Unknown,  
    HS_None,  
    HS_HatExist,  
    HS_Max  
};
```

```
enum Gender_e {  
    GD_Unknown,  
    GD_Male,  
    GD_Female,  
    GD_Max  
};
```

```
enum MouthMask_e {  
    MM_Unknown,  
    MM_None,  
    MM_MaskExist,  
    MM_Max  
};
```

```
enum Mustache_e {  
    MT_Unknown,  
    MT_None,  
    MT_MustacheExist,  
    MT_Max  
};
```

```
enum AIResultSortType {
    ARST_Ascending,
    ARST_Descending,
    ARST_None,
    ARST_Max
};

// Alarm type.
enum AIAlarmType_e {
    AAT_NormalSnap, // Normal capture upon alarm occurrence.
    AAT_FaceAlarm, // Target capture upon alarm occurrence.
    AAT_MonthMaskAlarm, // Target capture upon the occurrence of mask detection alarms.
    AAT_RepeatersAlarm, // High-frequency capture of unauthorized persons upon alarm
occurrence.
    AAT_DBRepeatersAlarm, // High-frequency capture of listed persons upon alarm occurrence.
    AAT_Max
};

////////// Enums related to person- and target-related services.

// Obtain the capture type.
enum AICamPicType {
    ACPT_Face = 0,
    ACPT_HumanBody = 1, // Person.
    ACPT_Vehicle = 2, // Motor vehicle.
    ACPT_PIDHumanBody = 3,
    ACPT_PIDVehicle = 4,
    ACPT_LCDHumanBody = 5,
    ACPT_LCDVehicle = 6,
    ACPT_NonVehicle = 7, // Non-motorized vehicle.
    ACPT_Count
};

// Vehicle make.
typedef enum PU_VEHICLE_BRAND {
    PU_LPR_VB_NOT, // Not limited.
```

PU_LPR_VB_GMC,	//GMC
PU_LPR_VB_JEEP,	//Jeep
PU_LPR_VB_MG,	//MG
PU_LPR_VB_MINI,	//MINI
PU_LPR_VB_SMART,	//smart
PU_LPR_VB_FAW,	// FAW.
PU_LPR_VB_MITSUBISHIS,	// MITSUBISHI (SOUTH EAST MOTOR).
PU_LPR_VB_MITSUBISHI,	// MITSUBISHI MOTORS.
PU_LPR_VB_MAXUS,	// SAIC Maxus.
PU_LPR_VB_SOUEASTMOTOR, //10	// SOUTH EAST MOTOR.
PU_LPR_VB_DFAC,	// Dongfeng.
PU_LPR_VB_DFXK,	// Dongfeng Sokon.
PU_LPR_VB_DFFD,	// Dongfeng Fengdu.
PU_LPR_VB_DFFS,	// Dongfeng Fengshen.
PU_LPR_VB_DFFX,	// Dongfeng Forthing.
PU_LPR_VB_ZXAUTO,	// ZX Auto.
PU_LPR_VB_ZHONGHUA,	// Brilliance Auto (Zhonghua)
PU_LPR_VB_POLARSUN,	// Polarsun Automobile.
PU_LPR_VB_TOYOTA,	// Toyota.
PU_LPR_VB_ISUZU, //20	// ISUZU.
PU_LPR_VB_WULINGCAR,	// Wuling Motors.
PU_LPR_VB_ZOTYE,	// Zotye Auto.
PU_LPR_VB_IVECO,	// Iveco.
PU_LPR_VB_PORSCHE,	// Porsche.
PU_LPR_VB_CALIFORNIACRUISER,	// Chrysler.
PU_LPR_VB_LAMBORGHINI,	// Lamborghini.
PU_LPR_VB_CADILLAC,	// Cadillac.
PU_LPR_VB_BUICK,	// Buick.
PU_LPR_VB_LIFAN,	// Lifan.
PU_LPR_VB_BAICMOTORBJ, //30	// BAIC.
PU_LPR_VB_BAW,	// BAW.
PU_LPR_VB_BEIQIWEIWANG,	// Beijing Automobile Weiwang.
PU_LPR_VB_SMA,	// Shanghai Maple Automobile.

PU_LPR_VB_HAWTAI,	// Hawtai.
PU_LPR_VB_HAWTAIXD,	// Hawtai (Hyundai).
PU_LPR_VB_SHUANGHUAN,	// Shuanghuan.
PU_LPR_VB_SSANGYONG,	// SsangYong Motor.
PU_LPR_VB_GEELY,	// Geely.
PU_LPR_VB_VENUCIA,	// Venucia.
PU_LPR_VB_HAFEI, //40	// Hafei.
PU_LPR_VB_VOLKSWAGEN,	// Volkswagen.
PU_LPR_VB_DAIHATSU,	// Daihatsu.
PU_LPR_VB_CHERY,	// Chery.
PU_LPR_VB_KERRY,	// Kerry.
PU_LPR_VB_BENZ,	// Mercedes-Benz.
PU_LPR_VB_AUDI,	// Audi.
PU_LPR_VB_RELAY,	// Rely.
PU_LPR_VB_BMW,	// BMW.
PU_LPR_VB_BAOJUN,	// Baojun.
PU_LPR_VB_ZHONGTONG, //50	// Zhongtong.
PU_LPR_VB_BJNORTH,	// Beijing North.
PU_LPR_VB_HUAXI,	// Huaxi.
PU_LPR_VB_YOUI,	// Youyi.
PU_LPR_VB_SICHUNXD,	// Sichuan Hyundai.
PU_LPR_VB_YUTONG,	// Yutong.
PU_LPR_VB_ANKAI,	// Ankai.
PU_LPR_VB_SHAOLIN,	// Shaolin.
PU_LPR_VB_YANGTSE,	// Yangtse.
PU_LPR_VB_JAC,	// JAC.
PU_LPR_VB_HIGER, //60	// Higer.
PU_LPR_VB_FOTON,	// Foton.
PU_LPR_VB_HUITONG,	// Huitong.
PU_LPR_VB_GOLDENDRAGON,	// Golden Dragon.
PU_LPR_VB_KINGLONG,	// King Long.
PU_LPR_VB_CHANGAN,	// Changan.
PU_LPR_VB_YONGMAN,	// Youngman.

PU_LPR_VB_HUANGHAI,	// Huanghai.
PU_LPR_VB_BENTLEY,	// Bentley.
PU_LPR_VB_GACTRUMPCHI,	// GAC Trumpchi.
PU_LPR_VB_GACGONOW, //70	// GAC Gonow.
PU_LPR_VB_HUMMER,	// Hummer.
PU_LPR_VB_JAGUAR,	// Jaguar.
PU_LPR_VB_SUBARU,	// Subaru.
PU_LPR_VB_SKODA,	// Škoda.
PU_LPR_VB_NISSAN,	// Nissan.
PU_LPR_VB_CHANGHE,	// Changhe.
PU_LPR_VB_HONDA,	// Honda.
PU_LPR_VB_LINCOLN,	// Lincoln.
PU_LPR_VB_PEUGEOT,	// Peugeot.
PU_LPR_VB_OPEL, //80	// Opel.
PU_LPR_VB_BYD,	// BYD.
PU_LPR_VB_JONWAY,	// Jonway.
PU_LPR_VB_HUIZHONG,	// Huizhong.
PU_LPR_VB_JMC,	// JMC.
PU_LPR_VB_VOLVO,	// Volvo.
PU_LPR_VB_FERRARI,	// Ferrari.
PU_LPR_VB_HAIMA,	// Haima.
PU_LPR_VB_TESLA,	// Tesla.
PU_LPR_VB_LEOPAARDCAR,	// Leopaard.
PU_LPR_VB_MARSALADI, //90	// Maserati.
PU_LPR_VB_HYUNDAI,	// Hyundai.
PU_LPR_VB_HYUNDAIHUAHAI,	// Hyundai or Huanghai.
PU_LPR_VB_LINIAN,	// Everus (Li Nian).
PU_LPR_VB_RIICH,	// Riich.
PU_LPR_VB_FORD,	// Ford.
PU_LPR_VB_FORDI,	// Foday.
PU_LPR_VB_LUXGEN,	// Luxgen.
PU_LPR_VB_INFINITI,	// Infiniti.
PU_LPR_VB_ROEWE,	// Roewe.

PU_LPR_VB_LOUTS, //100	// Lotus.
PU_LPR_VB_FIAT,	// Fiat.
PU_LPR_VB_ACURA,	// Acura.
PU_LPR_VB_SANY,	// Sany Heavy Industry.
PU_LPR_VB_DFACNJDY,	// Dongfeng or Nanjun.
PU_LPR_VB_DFLIUQI,	// Dongfeng Liuzhou Motor.
PU_LPR_VB_CNHTC,	// CNHTC.
PU_LPR_VB_WZFD,	// Wuzheng or Feidie.
PU_LPR_VB_KAMA,	// Kama.
PU_LPR_VB_BEIJING,	// Beijing.
PU_LPR_VB_BAICMOTORWL, //110	// BAIC Weilong.
PU_LPR_VB_BEIBEN,	// BeiBen.
PU_LPR_VB_BEIBENZK,	// BeiBen Trucks.
PU_LPR_VB_BAICMOTOR,	// BAIC.
PU_LPR_VB_HUALING,	// Hualing.
PU_LPR_VB_NANJUNDFAC,	// Nanjun or Dongfeng.
PU_LPR_VB_TANGJUN,	// Tangjun.
PU_LPR_VB_DAYUN,	// Dayun.
PU_LPR_VB_GACTIANYE,	// GAC Tianye.
PU_LPR_VB_JINGGONG,	// Jinggong.
PU_LPR_VB_HONGYAN, //120	// Hongyan.
PU_LPR_VB_JIEFANG,	// Jiefang.
PU_LPR_VB_YUEJIN,	// Yuejin.
PU_LPR_VB_JINBEI,	// Jinbei.
PU_LPR_VB_SHANQI,	// Shaanxi Automobile.
PU_LPR_VB_BALONG,	// Balong.
PU_LPR_VB_FEIDIE,	// Feidie.
PU_LPR_VB_KIA,	// Kia.
PU_LPR_VB_LANDROVER,	// Land Rover.
PU_LPR_VB_DODGE,	// Dodge.
PU_LPR_VB_SUZUKI, //130	// Suzuki.
PU_LPR_VB_GREATWALL,	// Great Wall.
PU_LPR_VB_CHANGANSHANGYONG,	// Changan SEONG YEONG.

PU_LPR_VB_LANDWIND,	// Landwind.
PU_LPR_VB_CHEVROLET,	// Chevrolet.
PU_LPR_VB_CITROEN,	// Citroën.
PU_LPR_VB_LEXUS,	// Lexus.
PU_LPR_VB_RENAULT,	// Renault.
PU_LPR_VB_FENGCHI,	// Fengchi.
PU_LPR_VB_MAZDA,	// Mazda.
PU_LPR_VB_HUANGHAIBJ, //140	// Huanghai (Beijing).
PU_LPR_VB_TRICYCLE,	// Tricycle.
PU_LPR_VB_WULING,	// Wuling.
PU_LPR_VB_HAWTAIXDB,	// Hawtai (Hyundai).
PU_LPR_VB_GEELYQQY,	// Geely Gleagle.
PU_LPR_VB_GEELYDH,	// Geely Emgrand.
PU_LPR_VB_FAWBESTURN,	// Bestune.
PU_LPR_VB_GAC,	// GAC.
PU_LPR_VB_LEOPAARD,	// Leopaard.
PU_LPR_VB_HONGQI,	// Hongqi.
PU_LPR_VB_JINBEIFOTON, //150	// Jinbei or Foton.
PU_LPR_VB_ZHONGQIHOWO,	// CNHTC HOWO.
PU_LPR_VB_BAWSHENGBAO,	// BAIC Senova.
PU_LPR_VB_GACRIYE,	// GAC Riye.
PU_LPR_VB_HEIBAO,	// Heibao.
PU_LPR_VB_DFBUS,	// Dongfeng Bus.
PU_LPR_VB_HAVAL,	// Haval.
PU_LPR_VB_HUANGHAIBUS,	// Huanghai Bus.
PU_LPR_VB_SHIDAI AUTO,	// Shidai.
PU_LPR_VB_DS,	//DS
PU_LPR_VB_AUMAN, //160	// Auman.
PU_LPR_VB_YUESHENG,	// Yusheng.
PU_LPR_VB_CHENGGONG AUTO,	// Chenggong.
PU_LPR_VB_BAWHUANSU,	// BAIC Huansu.
PU_LPR_VB_VENUCIAQC,	// Venucia.
PU_LPR_VB_LUTOS AUTO,	// Lotus.

PU_LPR_VB_DFFG,	// Dongfeng Fengguang.
PU_LPR_VB_CDW,	// CDW.
PU_LPR_VB_CCTRUCKS,	// CcTrucks.
PU_LPR_VB_HENGSHAN,	// Hengshan.
PU_LPR_VB_FUQIQITENG, //170	// Keyton.
PU_LPR_VB_SHANQIZHONGKA,	// Shaanxi Heavy-duty Motor.
PU_LPR_VB_SIMING,	// Siming.
PU_LPR_VB_ANYUANBUS,	// Anyuan Bus.
PU_LPR_VB_SHENLONG,	// Shenlong Bus.
PU_LPR_VB_ZHONGCHESHIDAI,	// CRRC Electric Vehicle.
PU_LPR_VB_BEIQINEWENERGY,	// BAIC New Energy.
PU_LPR_VB_QILUKECHE,	// Qilu Bus.
PU_LPR_VB_NANJUNQICHE,	// Nanjun.
PU_LPR_VB_DFCLKC,	// Dongfeng Chaolong Bus.
PU_LPR_VB_AOCHIAUTO, //180	// Aochi Auto.
PU_LPR_VB_ISUZUQL,	// Isuzu.
PU_LPR_VB_SCANIA,	// Scania.
PU_LPR_VB_HOWO,	// CNHTC HOWO Light Truck.
PU_LPR_VB_JIULONG,	// Jiulong.
PU_LPR_VB_MIXINGVEHICLE,	// Agitating truck.
PU_LPR_VB_VAN,	// Van.
PU_LPR_VB_LORRIES,	// Lorry.
PU_LPR_VB_OTHERS,	// Others.
PU_LPR_VB_OILTANKER,	// Tanker.
PU_LPR_VB_DAYUNZK, //190	// Dayun Heavy Truck.
PU_LPR_VB_SAICIVECOHY,	// SAIC-IVECO HONGYAN.
PU_LPR_VB_FAWGM,	// FAW-GM.
PU_LPR_VB_WUZHOU LONG,	// Wuzhoulong.
PU_LPR_VB_ZHIDOU,	// ZhiDou.
PU_LPR_VB_YANGZIRIVER,	// YANGTSE.
PU_LPR_VB_SHENGWO,	// Shenwo.
PU_LPR_VB_BORGWARD,	// Borgward.
PU_LPR_VB_HANTENG,	// Hanteng.

PU_LPR_VB_ENRANGER,	// Weichai Enranger.
PU_LPR_VB_COWIN, //200	// Cowin.
PU_LPR_VB_SWMMOTORS,	// SWM.
PU_LPR_VB_YEMAAUTO,	// Mustang.
PU_LPR_VB_HUASONG,	// Huasong.
PU_LPR_VB_QOROS,	// Qoros.
PU_LPR_VB_DAYUNQK,	// Dayun Light Truck.
PU_LPR_VB_FAWLIUTE,	// FAW Liute.
PU_LPR_VB_SINOTRUCKHH,	// Sinotruk Haohan.
PU_LPR_VB_SITRAK,	// Sinotruk Sitrak.
PU_LPR_VB_SINOTRUCKSTEYR,	// Sinotruk Steyr.
PU_LPR_VB_WEY, //210	//WEY
PU_LPR_VB_DENZA,	// Denza.
PU_LPR_VB_BISUAUTO,	// Bisu.
PU_LPR_VB_ASTONMARTIN,	// Aston Martin.
PU_LPR_VB_SAAB,	// Saab.
PU_LPR_VB_SHANNQITJ,	// Shaanxi Tongjia.
PU_LPR_VB_KANGDIGLEAGLE,	// Kandi GLEagle.
PU_LPR_VB_CHANGANKY,	// Changan Kuayue.
PU_LPR_VB_KAWEI,	// Kawei.
PU_LPR_VB_NJKL,	// Nanjing Golden Dragon Skywell.
PU_LPR_VB_GZBUS, //220	// Guangzhou Bus.
PU_LPR_VB_PICKUP,	// Pickup truck.
PU_LPR_VB_MCLAREN,	// McLaren.
PU_LPR_VB_SCHOOLBUS,	// School bus.
PU_LPR_VB_LYNKCO,	// LYNK & CO
PU_LPR_VB_DITU,	// Dito.
PU_LPR_VB_ROLLSROYCE,	// Rolls-Royce.
PU_LPR_VB_PUBLICBUS,	// Public bus.
PU_LPR_VB_UNKNOWN,	// Unknown.
PU_LPR_VB_CRANE,	// Crane.
PU_LPR_VB_ALFAROMEIO, //230	// Alfa Romeo.
PU_LPR_VB_ALL = 512	

```
} PU_VEHICLE_BRAND_E;
```

```
// Vehicle type.
```

```
typedef enum PU_VEHICLE_TYPE {
```

```
    PU_LPR_VT_NOT,           // Unknown.
```

```
    PU_LPR_VT_CAR,           // Car.
```

```
    PU_LPR_VT_TRUCK,         // Truck.
```

```
    PU_LPR_VT_VAN,           // Van.
```

```
    PU_LPR_VT_PAS,           // Bus.
```

```
    PU_LPR_VT_BUGGY,         // Buggy.
```

```
    PU_LPR_VT_SUV,           // SUV.
```

```
    PU_LPR_VT_MEDIUM_BUS,    // Medium-sized bus.
```

```
    PU_LPR_VT_MOT,           // Motorcycle (non-motorized vehicle).
```

```
    PU_LPR_VT_PEDESTRAIN,    // Pedestrian.
```

```
    PU_LPR_VT_SCHOOL_BUS, //10 // School bus.
```

```
    PU_LPR_VT_HEAVY_TRUCK,    // Dump truck.
```

```
    PU_LPR_VT_TANKER,         // High-risk vehicle.
```

```
    PU_LPR_VT_RIDEMAN,        // Cyclist.
```

```
// =====The preceding types are supported in earlier versions.
```

```
    PU_VLPR_VT_CAR_M,         // Mini sedan.
```

```
    PU_VLPR_VT_CAR_L,         // Small car.
```

```
    PU_VLPR_VT_CAR_S,         // Compact car.
```

```
    PU_VLPR_VT_CAR_TWO,       // Two-box car.
```

```
    PU_VLPR_VT_CAR_THR,       // Three-box car.
```

```
    PU_VLPR_VT_SUV_QINGKE,    // Light bus.
```

```
    PU_VLPR_VT_SUV_L, //20 // Small SUV.
```

```
    PU_VLPR_VT_SUV_S,         // Compact SUV.
```

```
    PU_VLPR_VT_SUV_Z,         // Medium SUV.
```

```
    PU_VLPR_VT_SUV_M,         // Medium and large SUV.
```

```
    PU_VLPR_VT_SUV_B,         // Large SUV.
```

```
    PU_VLPR_VT_WEIMIAN,       // Mini van.
```

```
    PU_VLPR_VT_MPV,           // MPV
```

```
    PU_VLPR_VT_JP,           // Coupe.
```

```
    PU_VLPR_VT_WEIKA,      // Mini truck.
    PU_VLPR_VT_SUV_PIKA,   // Pickup truck.
    PU_VLPR_VT_TRUCK_Z, //30 // Medium truck.
    PU_VLPR_VT_TRUCK_KEI,  // Light truck.
    PU_VLPR_VT_ZHONGKA,    // Heavy truck.
    PU_VLPR_VT_TAXI,       // Taxi.
    PU_VLPR_VT_TANK,       // Tanker.
    PU_VLPR_VT_CRANE,      // Mobile crane.
    PU_VLPR_VT_MOTOR = 120, // Motorcycle outside China
} PU_VEHICLE_TYPE_E;
```

// Vehicle color.

```
typedef enum PU_VEHICLE_COLOR {
    PU_LPR_VC_NOT, // Unknown.
    PU_LPR_VC_WHITE, // White.
    PU_LPR_VC_GRAY, // Gray (silver).
    PU_LPR_VC_YELLOW, // Yellow.
    PU_LPR_VC_PINK, // Pink.
    PU_LPR_VC_RED, // Red.
    PU_LPR_VC_GREEN, // Green.
    PU_LPR_VC_BLUE, // Blue.
    PU_LPR_VC_BROWN, // Brown.
    PU_LPR_VC_BLACK, // Black.
    PU_LPR_VC_PURPLE, // Purple.
    PU_LPR_VC_ORANGE, // Orange.
    PU_LPR_VC_CYAN, // Cyan.
    PU_LPR_VC_GOLDEN, // Golden.
    PU_LPR_VC_SILVER, // Silver.
} PU_VEHICLE_COLOR_E;
```

// License plate type.

```
typedef enum PU_LPR_PLATE_TYPE {
    PU_LPR_PLT_UNKNOWN = 0, // Unknown.
```

PU_LPR_PLT_NORMBLUE = 1, // Blue license plate.
PU_LPR_PLT_NORMBLACK = 2, // Black license plate.
PU_LPR_PLT_NORMYELLOW = 3, // Common one-row yellow license plate.
PU_LPR_PLT_DBLYELLOW = 4, // Two-row yellow license plate.
PU_LPR_PLT_POLICE = 5, // White license plate of police vehicles.
PU_LPR_PLT_WJ = 6, // White license plate of armed police vehicles.
PU_LPR_PLT_DBLWJ = 7, // Two-row white license plate of armed police vehicles.
PU_LPR_PLT_ARMY = 8, // One-row license plate of military vehicles.
PU_LPR_PLT_DBLARMY = 9, // Two-row license plate of military vehicles.
PU_LPR_PLT_EMBASSY = 10, // License plate of diplomatic vehicles.
PU_LPR_PLT_HK = 12, // Hong Kong (China) license plate.
PU_LPR_PLT_MACAU = 13, // Macao (China) license plate.
PU_LPR_PLT_AGRICULTURE = 14, // License plate of agricultural conveyance vehicles (tractors).
PU_LPR_PLT_CHANGNEI = 15, // Factory license plate.
PU_LPR_PLT_INDIVI = 16, // Personalized license plate.
PU_LPR_PLT_NEWNENRGY = 17, // License plate of new-energy vehicles.
PU_LPR_PLT_OTHER = 18, // Other types.
PU_VLPR_LPT_COACH = 19, // License plate of learner-driven vehicles.
PU_VLPR_LPT_AIRPORT = 20, // License plate of civil aviation vehicles.

PU_VLPR_LPT_INA_PRIVATE = 100, // License plate of private vehicles in Indonesia.
PU_VLPR_LPT_INA_PUBLIC = 101, // License plate of public transportation vehicles in Indonesia.
PU_VLPR_LPT_INA_EMBASSY = 102, // License plate of diplomatic vehicles in Indonesia.

PU_VLPR_LPT_KSA_PRIVATE = 110, // License plate of private vehicles in Saudi Arabia.
PU_VLPR_LPT_SAU_OTHER = 112, // License plate for other types of vehicles in Saudi Arabia.

PU_VLPR_LPT_HK_PRIVATE = 120, // License plate for private vehicles in Hong Kong (China).

PU_VLPR_LPR_BRA_PRIVATE = 130, // License plate for private vehicles in Brazil.

PU_VLPR_LPR_BRA_COMMERCIAL = 131, // License plate for business vehicles in Brazil.

PU_VLPR_LPR_BRA_OFFICIAL = 132, // License plate for government vehicles in Brazil.

PU_VLPR_LPR_BRA_DIPLOMATIC = 133, // License plate for diplomatic vehicles in Brazil.

PU_VLPR_LPR_BRA_MOTO = 134, // License plate for motorcycles in Brazil.

PU_VLPR_LPR_BRA_OTHER = 135, // License plate for other types of vehicles in Brazil.

PU_VLPR_LPR_MEX_PRIVATE = 140, // License plate for private vehicles in Mexico.

PU_VLPR_LPR_MEX_BUS = 141, // License plate for buses in Mexico.

PU_VLPR_LPR_MEX_TAXI = 142, // License plate for taxis in Mexico.

PU_VLPR_LPR_MEX_TRUCK = 143, // License plate for trucks in Mexico.

PU_VLPR_LPR_MEX_MOTO = 144, // License plate for motorcycles in Mexico.

PU_VLPR_LPR_MEX = 145, // Mexican license plate.

PU_VLPR_LPR_ARG_PRIVATE = 150, // License plate for private vehicles in Argentina.

PU_VLPR_LPR_ARG_OFFICIAL = 151, // License plate for government vehicles in Argentina.

PU_VLPR_LPR_ARG_MOTO = 152, // License plate for motorcycles in Argentina.

PU_VLPR_LPR_ARG_OTHER = 153, // License plate for other types of vehicles in Argentina.

PU_VLPR_LPR_ZAF_PRIVATE = 160, // License plate for private motor vehicles in South Africa.

PU_VLPR_LPR_ZAF_OFFICIAL = 161, // License plate for government vehicles in South Africa.

PU_VLPR_LPR_ZAF_DIPLOMATIC = 162, // License plate for diplomatic vehicles in South Africa.

PU_VLPR_LPR_ZAF_POLICE = 163, // License plate for police vehicles in South Africa.

PU_VLPR_LPR_ZAF_MILITARY = 164, // License plate for military vehicles in South Africa.

PU_VLPR_LPR_ZAF_MOTO = 165, // License plate for motorcycles in South Africa.

PU_VLPR_LPR_ZAF_OTHER = 166, // License plate for other types of vehicles in South Africa.

PU_VLPR_LPR_GHA_PRIVATE = 170, // License plate for private vehicles in Ghana.

PU_VLPR_LPR_GHA_OFFICIAL = 171, // License plate for government vehicles in Ghana.

PU_VLPR_LPR_GHA_DIPLOMATIC = 172, // License plate for diplomatic vehicles in Ghana.

```
    PU_VLPR_LPR_GHA_COMMERCIAL = 173, // License plate for business vehicles in
    Ghana.
    PU_VLPR_LPR_GHA_MOTO = 174,        // License plate for motorcycles in Ghana.
    PU_VLPR_LPR_GHA_OTHER = 175,       // License plate for other types of vehicles in
    Ghana.
} PU_PLATE_TYPE_E;

// License plate color.
typedef enum PU_LPR_PLATE_COLOR {
    PU_LPR_LC_UNKNOWN = 0,             // Unknown.
    PU_LPR_LC_BLUE = 1,                // Blue.
    PU_LPR_LC_YELLOW = 2,              // Yellow.
    PU_LPR_LC_WHITE = 3,               // White.
    PU_LPR_LC_BLACK = 4,               // Black.
    PU_LPR_LC_GREEN = 5,               // Green.
    PU_LPR_LC_GRADIENT_GREEN = 6,      // Black text on gradient green background.
    PU_LPR_LC_YELLOW_GREEN = 7,        // Black text on yellow-green background.
    PU_LPR_LC_RED_WHITE = 8,           // White text on red background.
    PU_LPR_LC_WHITE_RED = 9,           // Red text on white background.
    PU_LPR_LC_BLUE_WHITE = 10,         // White text on blue background.
    PU_LPR_LC_WHITE_BLUE = 11,         // Blue text on white background.
    PU_LPR_LC_WHITE_YELLOW = 12,       // Yellow text on white background.
    PU_LPR_LC_WHITE_GREEN = 13,        // Green text on white background.
    PU_LPR_LC_WHITE_SILVER = 14,       // Gray text on white background.
    PU_LPR_LC_SILVER_BLACK = 15,       // Black text on gray background.
} PU_PLATE_COLOR_E;
```

15.3 Target List Management

15.3.1 Adding a Target List

API Description	
Function	This API is used to add a target list.
API URL	/API/AI/FDGroup/Add

Request Method	POST				
Message ID					
Description	A maximum of 61 customized lists can be added.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Group		JSON array	Y		Elements in the Group array are as follows.
Elements the Group array					
		JSON object	Y		Elements in a single object are as follows.
Elements in a single object					
Name		String	Y	Maximum length: 128 bytes	Target list name. The following characters are not allowed: * ? \ : < > / " The value cannot start with any of the following characters: + - = @
Policy		Unsigned int	N		List alarm display policy. For details, see constants in the DLDetectPolicy enum. If this parameter is not transferred, list alarms are displayed by default.

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
Result	Operation result.	JSON array	[0,0...N] (Each element represents the operation result of a target group.)	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Group		JSON array	If the operation fails, an empty value is returned.	For details, see Table 15-1.

Table 15-1 Elements in the Group array

Elements in the Group array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Id		Int		Unique ID of a target list. This parameter is mandatory when the list information is modified. The value cannot be changed.
Name		String	Maximum length: 128 bytes	Target list name. The following characters are not allowed in the name: * ? \ : < > / " The value cannot start with any of the

				following characters: + - = @
Policy		Unsigned int		List alarm display policy. For details, see constants in the DLDetectPolicy enum.
Enabled		Unsigned int		Indicates whether to enable a list. The options include 0 (no) and 1 (yes).
AlarmEnabled		Unsigned int		Indicates whether to enable the alarm function for a list. The options include 0 (no) and 1 (yes).
CanDel		Int		Indicates whether a list can be deleted. Default lists (including the trustlist, blocklist, and unauthorized person list) are system-provided and cannot be deleted, and their names and policies cannot be edited on the UI). This parameter is read-only.
Similarity		Unsigned int	0-100	Similarity threshold for a list, which is an unsigned integer. For unauthorized person lists, the match is successful if the similarity between two images is less than the threshold; for other lists, the match is successful if the similarity between two images is larger than the threshold.
ChnAlarmOut	Alarm output.	JSON array		Each element indicates alarm output associated with a channel. For details, see Table 15-2.

ChnBuzzerOpt		JSON array	0–7, 0–7, 0–7 ...	Buzzer configuration. Each element in the array indicates the buzzer configuration of a channel. The options include 0 (disabled), 1 (1s), 2 (3s), 3 (5s), 4 (10s), 5 (20s), 6 (40s), and 7 (60s).
LatchTimeOpt		JSON array	0–8, 0–8, ...	Alarm output duration. Each element in the array indicates the alarm output duration of a channel. The options include 0 (50 ms), 1 (1s), 2 (2s), 3 (3s), 4 (5s), 5 (10s), 6 (20s), and 7 (40s), and 8 (60s).
SendEmail		JSON array	0–255, 0–255, ...	Each element is of the uint8_t type and is used in binary mode. Each bit in the binary value indicates whether to send an email for alarms of a certain type in the channel represented by the element.
ShowThumbnail		JSON array	0–255, 0–255, ...	Each element is of the uint8_t type and is used in binary mode. Each bit in the binary value indicates whether to display a thumbnail for alarms of a certain type for a channel.
AlarmSchedule		JSON array		Alarm schedule. For details, see Table 15-3.
EnableChnAlarm		JSON array	0–255, 0–255, ...	Channels where list alarms can be reported and alarm types supported for the channels. Each element (used in binary mode) in the array is of the uint8_t type and represents a

				channel, and each bit in the element represents an alarm type. The value of a bit indicates whether to report alarms of the corresponding type in the channel.
--	--	--	--	--

Table 15-2 A single element in ChnAlarmOut

A single element in ChnAlarmOut				
Example for a single channel: "ChnAlarmOut": [[0, 0, 0, 0], // Local alarm output, which depends on the number of local alarm channels. Each bit in an element (used in binary mode) indicates whether a local alarm output device is associated with the channel. [255, 255, 0, 255], // Camera alarm output 1, which depends on the number of camera's alarm channels. If the camera can connect to a maximum one alarm output device, this element specifies the first alarm output device. Each bit in this element (used in binary mode) indicates whether the camera alarm output device 1 is associated with the channel. [255, 255, 0, 255] // Camera alarm output 2, which depends on the number of camera's alarm channels. If the camera can connect to a maximum of two alarm output device, this element specifies the second alarm output device. Each bit in this element (used in binary mode) indicates whether the camera alarm output device 2 is associated with the channel.				
		JSON array	This array consists of three elements. Element 1 is for local alarm output, and elements 2 and 3 are for camera alarm output.	Elements in the array are as follows.
Elements in the array (local alarm output, camera alarm output 1, and camera alarm output 2)				
		JSON array	0–255, 0–255, 0–255	Each element is of the uint8_t type, and each bit in the element indicates whether the alarm output is associated with the channel.

Table 15-3 Elements in the AlarmSchedule array

Elements in the AlarmSchedule array				
		JSON array		Each element indicates the alarm schedule of a channel. Elements in the array are as follows.
Elements in the array				
		JSON array		Alarm schedule, with each element in the array indicating the schedule of a day (from Monday to Sunday) in the week. Elements in the array are as follows.
Elements in the array				
		JSON array	0–255, 0–255,	Each element is of the uint8_t type. Each bit in the element indicates half an hour, so there are 48 bits in the element. If the value of a bit is 1 , video recording is performed; otherwise, video recording is not performed.

15.3.2 Modifying List Alarm Parameters

API Description	
Function	This API is used to modify list alarm parameters.
API URL	/API/AI/FDGroup/Modify
Request Method	POST
Message ID	
Description	When modifying list information, you can transfer only the parameters that need to be modified.
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Group		JSON array	Y		For details, see Table 15-4.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N] (Each element represents the operation result of a target group.)	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	

Table 15-4 Elements in the Group array

Elements in the Group array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Id		Int		Unique ID of a target list. This parameter is mandatory when the list information is modified. The value cannot be changed.
Name		String	Maximum length: 128	Name of a target list.

			bytes	
Policy		Unsigned int		List alarm display policy. For details, see constants in the DLDetectPolicy enum.
Enabled		Unsigned int		Indicates whether to enable a list. The options include 0 (no) and 1 (yes).
AlarmEnabled		Unsigned int		Indicates whether to enable the alarm function for a list. The options include 0 (no) and 1 (yes).
Similarity		Unsigned int	0-100	Similarity threshold for a list, which is an unsigned integer. For unauthorized person lists, the match is successful if the similarity between two images is less than the threshold; for other lists, the match is successful if the similarity between two images is larger than the threshold.
ChnAlarmOut	Alarm output.	JSON array		Each element indicates alarm output associated with a channel. For details, see Table 15-2.
ChnBuzzerOpt		JSON array	0-7, 0-7, 0-7 ...	Buzzer configuration. Each element in the array indicates the buzzer configuration of a channel. The options include 0 (disabled), 1 (1s), 2 (3s), 3 (5s), 4 (10s), 5 (20s), 6 (40s), and 7 (60s).
LatchTimeOpt		JSON array	0-8, 0-8, ...	Alarm output duration. Each element in the array indicates the alarm output duration of a channel. The

				options include 0 (50 ms), 1 (1s), 2 (2s), 3 (3s), 4 (5s), 5 (10s), 6 (20s), and 7 (40s), and 8 (60s).
SendEmail		JSON array	0–255, 0–255, ...	Each element is of the uint8_t type and is used in binary mode. Each bit in the binary value indicates whether to send an email for alarms of a certain type in the channel represented by the element.
ShowThumbnail		JSON array	0–255, 0–255, ...	Each element is of the uint8_t type and is used in binary mode. Each bit in the binary value indicates whether to display a thumbnail for alarms of a certain type for a channel.
AlarmSchedule		JSON array		Alarm schedule. For details, see Table 15-3.
EnableChnAlarm		JSON array	0–255, 0–255, ...	Channels where list alarms can be reported and alarm types supported for the channels. Each element (used in binary mode) in the array is of the uint8_t type and represents a channel, and each bit in the element represents an alarm type. The value of a bit indicates whether to report alarms of the corresponding type in the channel.

15.3.3 Deleting a Target List

API Description

Function	This API is used to delete a target list.				
API URL	/API/AI/FDGroup/Remove				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Group		JSON array	Y		Elements in the Group array are as follows.
Elements in the Group array					
		JSON object	Y		Elements in a single object are as follows.
Id		Int	Y		ID of a target to be deleted. (Note that preset lists cannot be deleted. The CanDel parameter indicates whether a list can be deleted.)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N] (Each element represents the operation result of a target group.)	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	

ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.

15.3.4 Querying Target List Parameters

API Description					
Function	This API is used to query target list parameters.				
API URL	/API/AI/FDGroup/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
DefaultVal		Unsigned int	N		Indicates whether to obtain default alarm parameters or actual alarm parameters. The options include 1 (default alarm parameters, which can be displayed on the UI) and 0 (actual alarm parameters).
WithInternal		Unsigned int	N	0 or 1	Indicates whether the internal list is included. The internal list is

					not available for you and is only used for storing common targets. To obtain the common target list, a client needs to transfer the ID of the internal list. If this parameter and the Name parameter are not transferred, the internal list is returned by default. The options include 1 (yes) and 0 (no).
SimpleInfo		Unsigned int	N		Indicates whether to obtain simplified or complete information. Simplified information contains only the following parameters in Group: Name, Id, DetectType, Policy, Enabled, CanDel, and Similarity. Complete information contains all the parameters of a list. The options include 1 (simplified information) and 0 (complete information).
Name		String	N		Name of a list for query. If this parameter is not transferred, the

					parameters of all lists are queried. The following characters are not allowed: * ? \ : < > / " The value cannot start with any of the following characters: + - = @
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.		Int	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int	If the operation fails, 0 is returned.	Number of lists, that is, the size of the Group array.	
Group		JSON array	If the operation fails, an empty value is returned.	For details, see Table 15-1.	

15.3.5 Example and Comment

You can use the aforementioned APIs to add, delete, modify, and query lists. The parameters in Table 15-1 vary depending on the API.

1. Request example for adding a list:

```
POST/API/AI/FDGroup/Add?2020-10-17%2014:21:33 HTTP/1.1
Content-Type: application/json; charset=UTF-8
Accept: application/json, text/javascript, */*; q=0.01
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08
Content-Length: 2438
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{
  "data": {
    "Group": [
      {
        "Name": "Student_grp"
      }
    ]
  }
}
```

Response:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 2976

{
  "data": {
    "Result": [0], // Operation result. Each element corresponds to a list added. For details, see 15.1 .
    "Group": [
      {
        "Id": 5, // Unique ID of a list.
        "Name": "Student_grp", // List name.
        "Policy": 0, // List policy. For details, see constants in the DLDetectPolicy enum.
        "Enabled": 1, // Indicates whether to enable a list. The options include 0 (no) and 1 (yes).
        "AlarmEnabled": 1, // Indicates whether to enable the alarm function for a list. The options include 0 (no) and 1 (yes).
        "CanDel": 1, // Indicates whether a list can be deleted. Default lists cannot be deleted (the trustlist, blocklist, and stranger list are preset and their names and policies cannot be edited on the UI).
        "Similarity": 85, // Similarity threshold of a list, which is an unsigned integer. For stranger lists, the match is successful if the similarity between two images is less than the threshold; for other lists, the match is successful if the similarity between two images is larger than the threshold.
        "ChnAlarmOut": [ // Each element indicates alarm output associated with a channel.
          [255, 255, 0, 255], [255, 255, 0, 255], [255, 255, 0, 255] ], // This array consists of three elements. Element 1 is for local alarm output, and elements 2 and 3 are for camera alarm output 0 and 1. Each element is of the uint8_t type, and each bit in the element indicates whether the alarm output is associated with the channel.
        ]
      }
    ]
  }
}
```

```

    [ 255, 255, 0, 255], [255, 255, 0, 255], [255, 255, 0, 255] ],
    [ 255, 255, 0, 255], [255, 255, 0, 255], [255, 255, 0, 255] ],
    ...
    [ 255, 255, 0, 255], [255, 255, 0, 255], [255, 255, 0, 255] ]
    ],

    "ChnBuzzerOpt": [0, 1, 1, 2, 3, 4, 0, 1, 1, 3, 2, 2 ],    // Each element in the array indicates the buzzer
configuration of a channel.

    "LatchTimeOpt": [0, 1, 1, 2, 3, 4, 0, 1, 1, 3, 2, 2 ],    // Each element in the array indicates the alarm
output duration of a channel.

    "SendEmail": [0, 0, 0, 0], // Each element is of the uint8_t type and is used in binary mode. Each bit in
the binary value indicates whether to send an email for alarms of a certain type in the channel represented by the
element.

    "ShowThumbnail": [255, 255, 255, 255], // Each element is of the uint8_t type and is used in binary
mode.

    "AlarmSchedule": [      // Alarm schedule.

        [      // Channel 0.

            [255, 255, 255, 255, 255, 255], // Monday. Each element is of the uint8_t type. Each bit in the
element indicates half an hour, so there are 48 bits in the element. If the value of a bit is 1, video recording is
performed; otherwise, video recording is not performed.

            ...

        ],

        [      // Channel 1.

            [255, 255, 255, 255, 255, 255],

            ...

        ],

        ... // Channel n.

    ]

    "EnableChnAlarm": [255, 255, 255, 255] // Channels where list alarms can be reported and alarm
types supported for the channels. Each element (used in binary mode) in the array is of the uint8_t type and
represents a channel, and each bit in the element represents an alarm type. The value of a bit indicates whether
to report alarms of the corresponding type in the channel.

    }

    ]

    }

    }

```


2. Request example for modifying a list: (You only need to transfer the parameters that need to be modified. Read-only parameters cannot be modified.)

POST/API/AI/FDGroup/Modify?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "Group": [
      {
        "Id": 3,
        "Name": "Blocklist",
        "Policy": 1,
        "Similarity": 85,
        "Enabled": 1,
        "AlarmEnabled": 1
      },
      {
        "Id": 6,
        "Name": "List 4",
        "Policy": 0,
        "Similarity": 85,
        "Enabled": 1,
        "AlarmEnabled": 1
      }
    ]
  }
}
```

Response:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 320

```
{
  "data": {
    "Result": [0,0] // Result code. For details, see constants in the AI_OPT_RESULT_TYPE enum.
  }
}
```

3. Parameters in the body of a list deletion request are the same as those in a list modification request, except that only **Id** needs to be transferred in **Group** for deleting a list. The response body is also the same for the two kinds of requests.

4. Request for querying target list parameters:

POST/API/AI/FDGroup/Get?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "DefaultVal": 0, // Indicates whether to obtain default alarm parameters or actual alarm parameters. The
    options include 1 (default alarm parameters, which can be displayed on the UI) and 0 (actual alarm parameters).

    "WithInternal": 1, // Indicates whether the internal list is included. The internal list is not available for you
    and is only used for storing common targets. To obtain the common target list, a client needs to transfer the ID of
    the internal list.

    "SimpleInfo": 0, // Indicates whether to obtain simplified or complete information. Simplified information
    contains only the following parameters in Group: Name, Id, DetectType, Policy, Enabled, CanDel, and
    Similarity. The options include 1 (simplified information) and 0 (complete information).

    "Name" : "Trustlist" // Name of a list for query.
  }
}
```

Response: (The response body is of the same format as that of the response body returned after a target list is added.)

15.4 Target Operations

15.4.1 Adding a Target

API Description					
Function	This API is used to add a target.				
API URL	/API/AI/Faces/Add				
Request Method	POST				
Message ID					
Description	To add a target, you must specify GrpId and Image , and provide target features if there are any. You can add multiple targets at a time. Note that the Chinese phone number, email address, and employee ID are not allowed when adding a target. Verify MD5 values and do not transfer invalid MD5 values.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FaceInfo		JSON array	Y		For details, see Table 15-5.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a	

				request error occurs.
Count		Int		Number of actually added elements in FaceInfo .
Id		JSON array	[1, 2,...M]	IDs of added targets.
MD5		JSON array	Examples: "F74C70AD B0B63B00E 279E71B414 3704D", "F74C70AD B0B63B00E 279E71B414 370CC",	MD5 values of target images.
OriginalFaceInfo		JSON array		Original target information to be overwritten by new information. This parameter is returned when an existing target is added again. Since the target ID remains the same, it can be used for target match. For details, see Table 15-5.

Table 15-5 Elements in the FaceInfo or OriginalFaceInfo array

Elements in the FaceInfo or OriginalFaceInfo array				
		JSON object		(Parameters in the object vary depending on the API.) Elements in a single object are as follows.
Elements in a single object				
Id		Long long		64-bit signed target ID, which uniquely identifies a target. This parameter is not required for adding a target.

GrpId		Unsigned int		Target list ID. This parameter is invalid when you modify a target.
Sex		Unsigned int	Constants in the Gender_e enum	Gender. The options include 0 (unknown), 1 (male), 2 (female), and 3 (all)
Age		Unsigned int	(In a query scenario, if 0xff is transferred, targets at all ages are queried; if a specific age is transferred, targets at this age in the specified list are queried.)	Age.
ModifyCnt		Int		Number of times that a target has been modified, which is read-only.
Image		String	Base64-encoded image data	Main image of a target in the list, which is Base64-encoded. This parameter must be transferred for adding a target.
Feature		String	Base64-encoded feature	Feature in a target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version. Return this parameter together with the feature.
Name		String	Maximum length: 384 bytes	Name. This parameter is mandatory for adding a target but not required in the API for adding target images

				for match.
Country		String	Maximum length: 384 bytes	Nationality.
NativePlace		String	Maximum length: 384 bytes	Birthplace.
IdCode		String	Maximum length: 64 bytes	ID card number or employee ID, which cannot contain Chinese characters.
Job		String	Maximum length: 384 bytes	Occupation.
Phone		String	Maximum length: 64 bytes	Phone number, which cannot contain Chinese characters.
Email		String	Maximum length: 64 bytes	Email address, which cannot contain Chinese characters.
Domicile		String	Maximum length: 960 bytes	Residence place.
Remark		String	Maximum length: 960 bytes	Remarks.
MD5		String		MD5 value of an image. This parameter is not required for adding a target.
EnableChnAlarm		JSON array	0–255, 0–255, ...	Channels where target alarms can be reported and alarm types supported for the channels. Each element (used in binary mode) in the array of the parameter value is of the uint8_t type and represents a channel, and each bit in the element represents an alarm type. If the value in a bit is 1, the alarm type represented by this bit

				is supported in the channel represented by the element that the bit belongs to. The alarm processing logic remains unchanged.
--	--	--	--	--

15.4.1.1 Example and Comment

The values of **Image** and **Feature** can be the same as those of **FacelImage** and **Feature** in **/API/AI/SnapedFaces/GetByIndex**. You can also import external images. Such images do not have features, so the **Feature** and **FtVersion** parameters can be ignored. In this case, a target camera that supports feature extraction must be added to the HWT-NVR800. Supported image specifications are as follows:

Format: JPG (recommended), PNG, and BMP.

Dimensions: ranging from 80 x 80 to 1200 x 1200. The recommended value is 150 x 200. If the image dimensions exceed this recommended value, you are advised to scale down the image before invoking related APIs. Otherwise, the device load increases, but the target recognition effect will not be improved.

Size: no more than 500 KB. It is recommended that the image size be less than or equal to 50 KB. If the image quality is acceptable, a smaller image size is preferred. Generally, the size of a 150 x 200 JPG ID photo is about 7 KB. If the size exceeds 50 KB, scale down the image to within 150 x 200 and encode the image into a JPG image with 80% quality. Then, you can invoke related APIs for target recognition. Otherwise, the network transmission efficiency deteriorates and the device consumes more memory, but the target recognition effect will not be improved. In addition, the image data needs to be encoded using Base64.

The **GrpId** parameter specifies the list into which an image is to be imported. Other parameters in the request indicate personal information about targets, which needs to be set by yourself. Note that only images that do not exist in the list can be imported.

Request example:

```
POST/API/AI/Faces/Add?2020-10-17%2014:21:33 HTTP/1.1
Content-Type: application/json; charset=UTF-8
Accept: application/json, text/javascript, */*; q=0.01
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08
Content-Length: 2438
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{"data": {
  "FaceInfo": [
    {
      "GrpId": 1, // List.
```

```

    "Sex": 1, // Gender. The options include 1 (male) and 2 (female).
    "Age": 26,
    "Image": "base64(imgData)", // Main image of a target in the list. This parameter must
    be transferred for adding a target.
    "Feature": "base64(feature)", // Feature in a target image.
    "FtVersion": 12591105 // Feature version. Feature match is valid only when it is
    performed on features of the same version. Return this parameter together with the feature.
    "Name": "Mike", // Name.
    "Country": "China", // Nationality.

    "NativePlace": "Guangdong,Zhuhai", // Birthplace.
    "IdCode": "415025199203050916", // ID card number or employee ID.
    "Job": "Software", // Occupation.
    "Phone": "12345678902", // Phone number.
    "Email": "abcd@163.com", // Email address.
    "Domicile": "Guangdong,Zhuhai,Xiangzhou ...", // Residence place.
    "Remark": "Detail of this person ...", // Remarks.

    "EnableChnAlarm": [255, 255, 255, 255] // Channels where target alarms can be reported
    and alarm types supported for the channels. Each element (used in binary mode) in the array of
    the parameter value is of the uint8_t type and represents a channel, and each bit in the element
    represents an alarm type. If the value in a bit is 1, the alarm type represented by this bit is
    supported in the channel represented by the element that the bit belongs to. The alarm processing
    logic remains unchanged.
    },
    ...
  ]
}
}

```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 320

```

{
  "data": {
    "Count": 2,
    "Result": [0, 0], // Result code. For details, see constants in the AI_OPT_RESULT_TYPE enum.
  }
}

```



```

    "Id": [2, 3],    // IDs of added targets.

    "MD5": ["F74C70ADB0B63B00E279E71B4143704D", "0194F781438F2DE8FBE5B0469895036D"] //
MD5 values of target images.
}
}

```

15.4.2 Modifying a Target

API Description					
Function	This API is used to modify a target.				
API URL	/API/AI/Faces/Modify				
Request Method	POST				
Message ID					
Description	A target needs to be modified based on its Id or MD5. Both parameters must be valid, and MD5 takes priority. In addition, the parameter chosen for the modification must be the same for all targets. That is, you cannot modify one target based on Id but another target based on MD5. If Image and Feature (with the feature version) are specified, use them for image modification. If only Image is specified, Feature will be recalculated. Note that Image must specify the images added to lists. If Image is not specified, only other parameters (except Image and Feature) are modified. The request and response body is in the same format as that for adding a target. You can transfer only the parameters to be modified.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FaceInfo		JSON array	Y		For details, see Table 15-6.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	

Result	Operation result.	JSON array	[0,0...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of modified targets.
Id		JSON array	[1, 2,...N]	IDs of modified targets. (If the modification is based on Id , original target IDs are returned; if it is based on MD5 , -1 is returned for each target.)
MD5		JSON array	Examples: "F74C70AD B0B63B00E 279E71B414 3704D", "F74C70AD B0B63B00E 279E71B414 370CC",	MD5 values of target images. For MD5 -based modification, if a target image is not modified, the original MD5 value is returned; otherwise, a new MD5 value is returned. For Id -based modification, an empty string is returned.

Table 15-6 Elements in the FacelInfo array

Elements in the FacelInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Id		Long long		64-bit signed target ID, which uniquely

				identifies a target.
Sex		Unsigned int	Constants in the Gender_enum	Gender. The options include 0 (unknown), 1 (male), 2 (female), and 3 (all.)
Age		Unsigned int	(In a query scenario, if 0xff is transferred, targets at all ages are queried; if a specific age is transferred, targets at this age in the specified list are queried.)	Age.
Image		String	Base64-encoded image data	Main image of a target in the list, which is Base64-encoded. This parameter must be transferred for adding a target.
Feature		String	Base64-encoded feature	Feature in a target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version. Return this parameter together with the feature.
Name		String	Maximum length: 384 bytes	Name. This parameter is mandatory for adding a target but not required in the API for adding target images for match.
Country		String	Maximum length: 384 bytes	Nationality.
NativePlace		String	Maximum length: 384	Birthplace.

			bytes	
IdCode		String	Maximum length: 64 bytes	ID card number or employee ID, which cannot contain Chinese characters.
Job		String	Maximum length: 384 bytes	Occupation.
Phone		String	Maximum length: 64 bytes	Phone number, which cannot contain Chinese characters.
Email		String	Maximum length: 64 bytes	Email address, which cannot contain Chinese characters.
Domicile		String	Maximum length: 960 bytes	Residence place.
Remark		String	Maximum length: 960 bytes	Remarks.
MD5		String		MD5 value of an image. This parameter is not required for adding a target.
EnableChnAlarm		JSON array	0–255, 0–255, ...	Channels where target alarms can be reported and alarm types supported for the channels. Each element (used in binary mode) in the array of the parameter value is of the uint8_t type and represents a channel, and each bit in the element represents an alarm type. If the value in a bit is 1, the alarm type represented by this bit is supported in the channel represented by the element that the bit belongs to. The alarm processing logic remains unchanged.

15.4.3 Deleting a Target

API Description					
Function	This API is used to delete a target.				
API URL	/API/AI/Faces/Remove				
Request Method	POST				
Message ID					
Description	A target needs to be deleted based on Id or MD5 , and MD5 takes priority.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FaceInfo		JSON array	Y		Elements in the array are as follows.
Elements in the FaceInfo array					
		JSON object			Either Id or MD5 . Elements in a single object are as follows.
Elements in a single object					
Id		Long long	N		64-bit signed target ID, which uniquely identifies a target.
MD5		String	N		MD5 value of a target image.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation	JSON	[0,0...N]	If the operation is	

	result.	array		successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results About This Document."
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of deleted targets.
Id		JSON array	[1, 2,...N]	IDs of deleted targets.
MD5		JSON array	Examples: "F74C70AD B0B63B00E 279E71B414 3704D", "F74C70AD B0B63B00E 279E71B414 370CC",	MD5 values of target images.

15.4.3.1 Example and Comment

A target needs to be deleted based on **Id** or **MD5**, and **MD5** takes priority. Request example:

POST/API/AI/Faces/Remove?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
```

```
"FaceInfo": [  
    {"MD5": "4236DF810F4B2FE22E17D67A1DFDBC7C"},  
    {"MD5": "4579782D07676BF4DD5C7D0FBE981B9A"}  
]  
}  
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{  
  "data": {  
    "Count": 1,  
    "Result": [0,0 ],  
    "Id": [ -1, -1],  
    "MD5": [  
      "4236DF810F4B2FE22E17D67A1DFDBC7C ",  
      "4579782D07676BF4DD5C7D0FBE981B9A"  
    ]  
  }  
}
```

15.4.4 Changing the List to Which a Target Belongs

API Description	
Function	This API is used to change the list to which a target belongs.
API URL	/API/AI/FDGroup/Change
Request Method	POST
Message ID	
Description	The list to which a target belongs needs to be changed based on Id or MD5 , and MD5 takes priority.

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Group		Int	Y		ID of the destination list to which a target is to be moved.
FaceInfo		JSON array	Y		The elements are described in this table.
Elements in the FaceInfo array					
		JSON object			Either Id or MD5 . Elements in a single object are as follows.
Elements in a single object					
Id		Long long	N		64-bit signed target ID, which uniquely identifies a target.
MD5		String	N		MD5 value of a target image.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur	

				on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		
Id		JSON array	[1, 2,...N]	IDs of modified targets.
MD5		JSON array	Examples: "F74C70AD B0B63B00E 279E71B414 3704D", "F74C70AD B0B63B00E 279E71B414 370CC",	MD5 values of target images.

15.4.4.1 Example

Request example:

POST/API/AI/FDGroup/Change?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "Group": 28,
    "FaceInfo": [{"Id": 683}]
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{
  "data": {
    "Count": 1,
    "Result": [0],
    "Id": [683],
    "MD5": [""]
  }
}
```

15.4.5 Querying Targets in a List

API Description	
Function	This API is used to query targets in a list.
API URL	/API/AI/AddedFaces/Search
Request Method	POST
Message ID	
Description	This API returns only the number of targets found and the result is cached. To obtain more information about the targets, you need to invoke the API in 15.4.6 . For later expansion and implementation convenience, specify all target information when querying targets in a list. All parameters can be left empty (indicating that default values are used). Generally, if GrpId is transferred, all targets in a list are queried. If other parameters are transferred, you can perform query by criteria or in fuzzy mode. You can perform fuzzy query by the following parameters: Name, Country, NativePlace, IdCode, Job, Phone, Email, Domicile, and Remark. You can perform exact query by the following parameters: Age and Sex. The options of Sex are constants in the Gender_e enum, including 1 (male), 2 (female), and 3 (all). As for Age, if 0xff is transferred, targets at all ages are queried; if

		a specific age is transferred, targets at this age in the specified list are queried. Only one list is allowed in each query.			
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FaceInfo		JSON array	Y		For details, see Table 15-7.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of targets found.	

Table 15-7 Elements in the FaceInfo array

Elements in the FaceInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				

GrpId		Unsigned int		Target list ID.
Sex		Unsigned int	Constants in the Gender_enum	Gender. The options include 0 (unknown), 1 (male), 2 (female), and 3 (all.)
Age		Unsigned int	(In a query scenario, if 0xff is transferred, targets at all ages are queried; if a specific age is transferred, targets at this age in the specified list are queried.)	Age.
Name		String	Maximum length: 384 bytes	Name.
Country		String	Maximum length: 384 bytes	Nationality.
NativePlace		String	Maximum length: 384 bytes	Birthplace.
IdCode		String	Maximum length: 64 bytes	ID card number or employee ID, which cannot contain Chinese characters.
Job		String	Maximum length: 384 bytes	Occupation.
Phone		String	Maximum length: 64 bytes	Phone number, which cannot contain Chinese characters.
Email		String	Maximum length: 64 bytes	Email address, which cannot contain Chinese characters.
Domicile		String	Maximum length: 960 bytes	Residence place.

Remark		String	Maximum length: 960 bytes	Remarks.
--------	--	--------	---------------------------	----------

15.4.5.1 Example

Request example:

POST/API/AI/AddedFaces/Search?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

{

"data": {

"FaceInfo": [

{

"GrpId": 2,

"Remark": "1234" // If this parameter is not transferred and no other parameters can be used for fuzzy query, all targets in the list whose ID (which is **2** in this example) is specified by **GrpId** are queried.

}

]

}

}

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2976

```
{
  "data": {
    "Result": 0,
    "Count": 1 // Indicates that one target is found in the specified list.
  }
}
```

15.4.6 Obtaining Target Information from a List in Pagination Mode

API Description					
Function	This API is used to obtain target information from a list in pagination mode.				
API URL	/API/AI/AddedFaces/GetByIndex				
Request Method	POST				
Message ID					
Description	Before invoking this API for the first time, you must invoke the API in section 15.4.5 "Querying Targets in a List", whose returned data is cached. In non-simplified mode: 1. If WithImage is set to a non-zero value, information about a maximum of 30 target images can be obtained. 2. If WithImage is set to 0 and WithFeature is set to a non-zero value, information about a maximum of 500 target images can be obtained.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
StartIndex		Unsigned int	N		Start index of a request. For example, if the index ranges from 0 to 19 for the first request and from 20 to

					39 for the second request, the values of this parameter are 0 and 20 respectively. If this parameter is not transferred, 0 is used by default.
Count		Unsigned int	N		Number of requested targets. If this parameter is not transferred, the first target image is returned by default.
SimpleInfo		Unsigned int	N		Indicates whether to obtain simplified information, which contains only Id , GrpId , and Name of a target. The options include 1 (yes) and 0 (no). If this parameter is not transferred, non-simplified information is obtained by default.
WithImage		Unsigned int	N		Indicates whether to obtain images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, images are not obtained by default.
WithFeature		Unsigned int	N		Indicates whether to obtain features.

					The options include 1 (yes) and 0 (no). If this parameter is not transferred, features are not obtained by default.
NeedMD5		Unsigned int	N		Indicates whether MD5 values are required. This parameter is transferred even in non-simplified mode. If it is not transferred, MD5 values are not required by default.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of targets. (If the value is less than that of Count in the request, the actual number of targets has decreased.)
FaceInfo		JSON array		For details, see Table 15-5.

15.4.6.1 Example

Request example:

POST/API/AI/AddedFaces/GetByIndex?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "StartIndex": 0,
    "Count": 1,
    "SimpleInfo": 1,
    "WithImage": 0,
    "WithFeature": 0,
    "NeedMD5": 0
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 739

```
{
  "data": {
```

```
"Result": 0,

"Count": 1,

"FaceInfo": [
    {
        "Id": 1,
        "GrpId": 2,
        "Name": "mmx",
        "MD5": "D538C66812C9441BEC4575411593A18A",
        "Sex": 2,
        "Age": 28,
        "ModifyCnt": 5,
        "Country": "China",

        "NativePlace": "Guangdong,Zhuhai",
        "IdCode": "11111119",
        "Job": "Software",
        "Phone": "12345678302",
        "Email": "abcd@163.com",
        "Domicile": "Guangdong,Zhuhai,Xiangzhou ...",
        "Remark": "222",
        "EnableChnAlarm": []
    }
]
}
```

15.4.7 Obtaining Target Information Based on the Target ID or MD5 Value

API Description	
Function	This API is used to obtain target information based on the target ID or MD5 value.

API URL	/API/AI/AddedFaces/GetById				
Request Method	POST				
Message ID					
Description	This API directly obtains target information (with no need of query) based on the target ID or MD5 value. Note that the MD5 value (specified by FacesMD5) is preferred in this process. In non-simplified mode: 1. If WithImage is set to a non-zero value, information about a maximum of 30 target images can be obtained. 2. If WithImage is set to 0 and WithFeature is set to a non-zero value, information about a maximum of 500 target images can be obtained.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FacesId		JSON array	N	[1,2,3...N]	IDs of requested targets.
FacesMD5		JSON array	N	Reference: "F75C70ADB0B63B00E279E71B4143704D", "F75C70ADB0B63B00E279E71B41437CCC"	MD5 values of requested targets.
SimpleInfo		Unsigned int	N		Indicates whether to obtain simplified information, which contains only Id , GrpId , and Name of a target. The options include

					1 (yes) and 0 (no). If this parameter is not transferred, non-simplified information is obtained by default.
WithImage		Unsigned int	N		Indicates whether to obtain images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, the images are not obtained by default.
WithFeature		Unsigned int	N		Indicates whether to obtain features. The options include 1 (yes) and 0 (no). If this parameter is not transferred, the features are not obtained by default.
NeedMD5		Unsigned int	N		Indicates whether MD5 values are required. This parameter is transferred even in non-simplified mode. If it is not transferred, MD5 values are not required by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	

Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of targets. (If the value is less than that of Count in the request, the actual number of targets has decreased.)
FaceInfo		JSON array		For details, see Table 15-5.

15.4.7.1 Example and Comment

This API is used to query target information in a list based on the target ID or MD5 value. The MD5 value is preferentially used for query. That is, if you transfer both **FacesId** and **FacesMD5**, only **FacesMD5** will be used.

Request example:

POST/API/AI/AddedFaces/GetById?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{  
  
  "data": {
```

```
"MsgId": null,  
  
"FacesId": [7], // Unique ID of a target.  
  
"SimpleInfo": 0,  
  
"WithImage": 0,  
  
"WithFeature": 0,  
  
"NeedMD5": 0 // Indicates whether to return the MD5 value of an image. This parameter  
is invalid when SimpleInfo is set to 0.  
  
}  
  
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 2340

```
{  
  
  "data": {  
  
    "Result": 0,  
  
    "Count": 1,  
  
    "FaceInfo": [  
  
      {  
  
        "Id": 7,  
  
        "GrpId": 2,  
  
        "Name": "cxl",  
  
        "MD5": "8F0C85295FB19F36426246EC0F66ADC3",  
  
        "Sex": 1,  
  
        "Age": 32,  
  
      }  
    ]  
  }  
}
```

```
        "ModifyCnt": 4,

        "Country": "zg",

        "NativePlace": "",

        "IdCode": "",

        "Job": "",

        "Phone": "",

        "Email": "",

        "Domicile": "",

        "Remark": "12345",

        "EnableChnAlarm": []

    }

]

}
```

15.4.8 Adding More Images for an Existing Target

API Description	
Function	This API is used to add more images for an existing target.
API URL	/API/AI/ExtraFaces/Add
Request Method	POST
Message ID	
Description	The FacelId and Image parameters must be transferred to invoke this API.

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
ExtFaceInfo		JSON array	Y		For details, see Table 15-8.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
data	Data structure.	Object	Y		
Elements in the data object					
Result	Operation result.	JSON array	[0,0,...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of elements in the ExtFaceInfo array.	
Id		JSON array	[1,2,...N]	IDs of additional target images.	

Table 15-8 Elements in the ExtFaceInfo array

Elements in the ExtFaceInfo array				
		JSON object		(The listed parameters vary depending on the API.) Elements in a single object are as follows.

Elements in a single object				
Id		Long long		ID of an additional target image. This parameter is not required for invoking this API.
FaceId		Long long		ID of the target for which an image is to be added.
Image		String	Base64-encoded image data	Additional target image. The image data is encoded using Base64.
Feature		String	Base64-encoded feature	Feature in an additional target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version. Return this parameter together with the feature.

15.4.9 Deleting Additional Target Images

API Description					
Function	This API is used to delete additional target images.				
API URL	/API/AI/ExtraFaces/Remove				
Request Method	POST				
Message ID					
Description	To delete additional target images, you only need to set Id and use default values for other parameters.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	Object	Y		
Elements in the data object					
ExtFaceInfo		JSON array	Y		Elements in the array are as follows.
Elements in the ExtFaceInfo array					
		JSON object			Elements in a single object are as follows.
Elements in a single object					
Id		Long			ID of an additional target image.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0,...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of elements in the ExtFaceInfo array.	
Id		JSON array	[1,2,...N]	IDs of additional target images.	

15.4.10 Obtaining Information About Additional Target Images

API Description					
Function	This API is used to obtain information about additional target images.				
API URL	/API/AI/ExtraFaces/Get				
Request Method	POST				
Message ID					
Description	Since there are only a few additional images of a target, you can obtain their information at a time. Therefore, no API for obtaining the information in pagination mode is provided, and all additional targets are returned on only one page.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
FaceId		Long long	Y		ID of the target to which an additional target image belongs.
WithImage		Unsigned int	N		Indicates whether to obtain images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, images are not obtained by default.
WithFeature		Unsigned int	N		Indicates whether to obtain features. The options include 1 (yes) and 0 (no). If this parameter is not

					transferred, features are not obtained by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of elements in the ExtFaceInfo array.	
ExtFaceInfo		JSON array		For details, see Table 15-8.	

15.4.11 Obtaining Information About Additional Target Images Based on the Target ID

API Description					
Function	This API is used to obtain information about additional target images based on the target ID.				
API URL	/API/AI/ExtraFaces/GetById				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data	Mandat	Value	Server

		Type	ory (Y/N)	Range	Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Id		Long	Y		ID of an additional target image. Do not set this parameter to the ID of the existing target for which the image is added.
WithImage		Unsigned int	N		Indicates whether to obtain images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, images are not obtained by default.
WithFeature		Unsigned int	N		Indicates whether to obtain features. The options include 1 (yes) and 0 (no). If this parameter is not transferred, features are not obtained by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details,	

				see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of elements in the ExtFaceInfo array.
ExtFaceInfo		JSON array		For details, see Table 15-8.

15.4.12 Searching for or Matching Target Snapshots

API Description						
Function	This API is used to search for or match target snapshots.					
API URL	/API/AI/SnapedFaces/Search					
Request Method	POST					
Message ID						
Description	To invoke this API to match a target snapshot, you only need to transfer the target ID (specified by Id), which is the unique ID of a common target or a target in a target list, MD5 value (specified by MD5) of the target image, or features (specified by Feature) of the target. The priorities of the three parameters are as follows: Feature > MD5 > Id To reduce the memory overhead for transfer and processing, transfer only the required parameters based on the application scenario. If the number of targets to be matched is 0, target snapshot search, instead of target match, will be performed. Note that if you want to transfer an external image to this API, invoke the /API/AI/CompareFaces/Add API first to add the external image to the list of common targets. After the image is successfully added, its Id and MD5 parameters are returned and then can be used to search for and match target snapshots.					
Input Parameters						
Parameter	Description	Data Type	Mandatory	Value Range	Server Verification	

			(Y/N)		Rule
data	Data structure.	Object	Y		
Elements in the data object					
StartTime		String	Y	Format: <i>yyyy-mm-dd hh:mm:ss</i> , for example, 2018-10-20 00:00:00	Search start time.
EndTime		String	Y	Format: <i>yyyy-mm-dd hh:mm:ss</i> , for example, 2018-10-28 23:59:59	Search end time.
Chn		JSON array	N	[0, 1, 2, ..., N]	Channel to be searched. All channels are searched if this parameter is not transferred.
AlarmGroup		JSON array	N	[1, 2, ..., N]	ID of the target list to be searched. If this parameter is not transferred, all target lists will be searched.
Similarity		Int	N		Similarity threshold for match, which ranges from 0 to 100. If the similarity between an input target snapshot and that in the list is

					lower than the threshold, the target snapshot in the list will be filtered out. If the match is not required, set this parameter to -1 . If the value of FacelInfo does not contain any target snapshots to be matched, this parameter is invalid.
Engine		Unsigned int	N		Search engine. The options include 0 and 1. These two engines are provided to work around the impact caused by the overlap between the search process and operation logic. For example, if you open the dialog box for selecting targets to be searched when a list of match results are already displayed, the cached match results will be overwritten by the search result and become invalid. The engines are independent of each other in terms of data and cache and are allocated by the client. For example, engine

					0 can be used to implement temporary target selection tasks such as target selection in the dialog box; engine 1 can be used to display search or match results. If this parameter is not transferred, 0 is used by default.
FaceInfo		JSON array	N		Information about input target snapshots to be matched. For details, see Table 15-9. (Note that you can transfer only one of Id, MD5, and Feature based on the application scenario. Generally, Id is preferred.)
Sex		JSON array	N	0–2	Search by gender. The options are constants in the Gender_e enum.
Age		JSON array	N	0–7	Search by age group. The options include under 18, 18–25, 26–30, 31–35, 36–40, 41–50, over 51, and unknown.
MouthMaskType		JSON array	N	0–2	Indicates whether to search for targets wearing a mask. The

					options are constants in the MouthMask_e enum.
HairType		JSON array	N	0–2	Search by hairstyle. The options are constants in the HairType_e enum.
GlassesType		JSON array	N	0–3	Indicates whether to search for targets wearing glasses. The options are constants in the GlassType_e enum.
HatType		JSON array	N	0–2	Indicates whether to search for targets wearing a hat. The options are constants in the HatState_e enum.
SortType		Unsigned int	N		Sorting type. The options include 0 (ascending) and 1 (descending). If this parameter is not transferred, value 1 is used by default. For details, see AIResultSortType .
EventType		JSON array	N	0–4	Event alarm type. The options include 0 (target recognition alarm), 1 (high-frequency stranger detection alarm), 2

					(high-frequency snapshot alarm for listed persons), 3 (other alarms), and 4 (mask detection alarm). If this parameter is not transferred, all alarm types will be used for search.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of images that are found or matched.	

Table 15-9 Elements in the FaceInfo array

Elements in the FaceInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Id		Long long		64-bit signed target ID, which uniquely identifies a target.

Feature		String	Base64-encoded feature	Feature in a target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version. Return this parameter together with the feature.
MD5		String		MD5 value of an image.

15.4.13 Obtaining Results of Target Snapshot Search or Match in Pagination Mode

API Description					
Function	This API is used to obtain results of target snapshot search or match in pagination mode.				
API URL	/API/AI/SnapedFaces/GetByIndex				
Request Method	POST				
Message ID					
Description	This API needs to be invoked after /API/AI/SnapedFaces/Search is invoked. When SimpleInfo is set to 0: 1. If WithFacelImage, WithBodyImage, or WithBackgroud is not set to 0, information about a maximum of 30 targets can be obtained. 2. If WithFacelImage, WithBodyImage, and WithBackgroud are all set to 0 while WithFeature is not set to 0, information about a maximum of 500 targets can be obtained.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					

Engine		Unsigned int	N		Search engine. The options include 0 and 1 . This parameter is used together with the search API. If this parameter is not transferred, 0 is used by default.
StartIndex		Unsigned int	N	0–Max (The maximum value is 99999 for an HWT-NVR800 with only one disk tray and 199999 for an HWT-NVR800 with two or more disk trays.)	Start index of a request. For example, if the index ranges from 0 to 19 for the first request and from 20 to 39 for the second request, the values of this parameter are 0 and 20 respectively. If this parameter is not transferred, 0 is used by default.
Count		Unsigned int	N		Number of requested targets. If this parameter is not transferred, the first target image is returned by default.
SimpleInfo		Unsigned int	N		Indicates whether to obtain simplified or complete information. Simplified information contains at most MatchedId , Similarity , and UId . If no target is

					matched, only UUID is contained. If this parameter is not transferred, value 0 (indicating that simplified information is obtained) is used by default. In simplified mode, the WithFacelImage , WithBackground , and WithFeature parameters are invalid.
WithFacelImage		Unsigned int	N		Indicates whether to obtain target images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithBackground		Unsigned int	N		Indicates whether to obtain background images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithBodyImage		Unsigned int	N		Indicates whether to obtain person images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default. Note that if no

					person images are available, the parameter indicating person images will not be returned even if you set this parameter to 1 .
WithFeature		Unsigned int	N		Indicates whether to obtain features. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
TotalCount		Int		Total number of captured targets that are found or successfully matched.
Count		Int		Total number of captured targets returned in the current response.
SnapedFaceInfo		JSON array		For details, see Table 15-10.

Table 15-10 Elements in the **SnapedFaceInfo** array

Elements in the SnapedFaceInfo array				
		JSON object		(Parameters in the object vary depending on the API.) Elements in a single object are as follows.
Elements in a single object				
UUIId		String		Unique ID of a captured target, which can be used for obtaining details about the target.
MatchedFaceId		Int		ID of a matched target. If no target is matched, this parameter is not returned.
MatchedMD5		String		MD5 value of a matched target. If no target is matched, this parameter is not returned.
Chn		Int		Channel where a target is captured.
Similarity		float		Target similarity. For image-based search, this parameter specifies the similarity between the uploaded image and the required target image in lists. For search based on snapshot records, this parameter specifies the similarity between the target image captured during real-time target recognition and the required target image in lists.
StartTime		String	Fixed format: <i>yyyy-mm-dd</i>	Start time.

			<i>hh:mm:ss</i> , for example, 2020-09-21 21:59:35	
EndTime		String	Fixed format: <i>yyyy-mm-dd</i> <i>hh:mm:ss</i> , for example, 2020-09-21 21:59:35	End time.
FacelImage		String	Base64-enco ded image data	Target snapshot (encoded using Base64).
Background		String	Base64-enco ded image data	Captured background image (encoded using Base64).
BodyImage		String	Base64-enco ded image data	Person snapshot (encoded using Base64).
Feature		String	Base64-enco ded image data	Feature in the captured target image (encoded using Base64).
FtVersion		Int		Feature version.
SnapId		Int		Target trajectory ID of a camera, which is of no use to the HWT-NVR800.
Gender		Int		Gender. The options are constants in the Gender_e enum.
Age		Int		Age, which ranges from 1 to 100 on the camera side.
Expression		Int		Facial expression. The options are constants in the FaceExpression_e enum.
Hair		Int		Hairstyle. The options are constants in the HairType_e enum.
Hat		Int		Indicates whether a target wears a hat. The options are

				constants in the HatState_e enum.
GlassType		Int		Indicates whether a target wears glasses. The options are constants in the GlassType_e enum.
MouthMask		Int		Indicates whether a target wears a mask. The options are constants in the MouthMask_e enum.
Mustache		Int		Indicates whether a target wears a beard. The options are constants in the Mustache_e enum.
AlarmFlags		Int	4-bit value, with each bit (from right to left) indicating the following: Bit 0: common snapshot Bit 1: alarm-triggered snapshot Bit 2: mask alarm snapshot Bit 3: high-frequency stranger alarm snapshot Bit 4: high-frequency alarm-triggered snapshot for listed persons	Alarm type. The options are constants in the AIAlarmType_e enum. Multiple alarm types can be specified.
AppearCnt		Int		Number of times that a target appears. The maximum value is 21 . That is, if a target appears for more than

				21 times, the value of this parameter is still 21 .
RepeatersId		String		ID of a regular customer, which is unique for regular customer counting. This parameter is generated when AppearCnt is not set to 0 .
FaceRect		JSON object		Position of the captured target in the background image. For details, see Table 15-11.

Table 15-11 Elements in the FaceRect object

Elements in the FaceRect object				
Note: Use relative coordinates (10000 x 10000) in the background image.				
X		Int	0–10,000	X coordinate of the upper left corner of the target position.
Y		Int	0–10,000	Y coordinate of the upper left corner of the target position.
Width		Int	0–10,000	Target position width.
Height		Int	0–10,000	Target position height.

15.4.14 Obtaining Information About Captured Targets Based on the UUID

API Description	
Function	This API is used to obtain information about captured targets based on the UUID.
API URL	/API/AI/SnapedFaces/GetById
Request Method	POST

Message ID					
Description	This API is used to obtain information about captured targets based on the UUID, without searching for targets in the snapshot list. 1. If WithFacelImage, WithBodyImage, or WithBackgroud is not set to 0, information about a maximum of 30 targets can be obtained. 2. If WithFacelImage, WithBodyImage, and WithBackgroud are all set to 0 while WithFeature is not set to 0, information about a maximum of 500 targets can be obtained. 3. If snapshots, background images, and features are not required, information about a maximum of 10,000 targets can be obtained at a time.				
Input Parameters					
Parameter	Description	Data Type	Mandat ory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigne d int	N		Search engine. The options include 0 and 1 . This parameter is used together with the search API. If this parameter is not transferred, 0 is used by default.
UUIds		JSON array	N	Referen ce: "nQkAB oxluT8E BAEA", "nQkAB oxluT8E BBEX"	Unique ID of a captured target. If this parameter is not transferred, information about the target cannot be obtained.
WithFacelImage		Unsigne d int	N		Indicates whether to obtain target images. The options include 1 (yes) and 0 (no). If this

					parameter is not transferred, 1 is used by default.
WithBackgroud		Unsigned int	N		Indicates whether to obtain background images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithBodyImage		Unsigned int	N		Indicates whether to obtain person images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default. Note that if no person images are available, the parameter indicating person images will not be returned even if you set this parameter to 1 .
WithFeature		Unsigned int	N		Indicates whether to obtain features. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful,	

				AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Total number of captured targets returned in the current response.
SnapedFaceInfo		JSON array		For details, see Table 15-10.

15.4.15 Stopping Searching for Targets in the Snapshot List

API Description					
Function	This API is used to stop searching for targets in the snapshot list.				
API URL	/API/AI/SnapedFaces/StopSearch				
Request Method	POST				
Message ID					
Description	This API is used to stop searching for targets in the snapshot list. The client invokes this API when it determines that the search is no longer required, for example, when the target selection dialog box or the target match page is closed, or when the search takes a long time and needs to be canceled. Then, the device releases all caches of the specified search engine to alleviate pressure on the device.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
Engine		JSON array	Y	[0,1...N]	Engines whose caches are to be released.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	

15.4.16 Adding Target Images for Match

API Description	
Function	This API is used to add target images for match.
API URL	/API/AI/CompareFaces/Add
Request Method	POST
Message ID	
Description	This API is used to add target images for match. Target images found on the hard disk, or selected from a USB flash drive or the local computer must be added to a list of common targets before they can be used for match. The request message is similar to that of adding targets. The difference is, only the Image parameter is mandatory (set Feature if any) in the request and you do not need to set the GrpId parameter. The added target images are stored in the internal list. You can obtain the internal group ID by invoking the API in section 15.3.4 "Querying Target List Parameters", and then query the target information in the internal list by invoking the APIs in sections 15.4.5 15.4.6

"Obtaining Target Information from a List in Pagination Mode." Note that if the information about the target images to be matched is the same as those in the existing list, the target images will be moved to the list that records matched targets.					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
WithImage		Unsigned int	N		Indicates whether to return images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithFeature		Unsigned int	N		Indicates whether to return features extracted from images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
FaceInfo		JSON array			List of targets to be added. For details, see Table 15-12.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N]	Target adding result. If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are	

				returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Number of elements in the FaceInfo array.
FaceInfo		JSON array	The Image and Feature parameters are empty if WithImage is set to 0 in the request.	For details, see Table 15-12.

Table 15-12 Elements in the FaceInfo array

Elements in the FaceInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Image		String	Base64-encoded image data, which is mandatory	Main image of a target in the list, which is Base64-encoded. This parameter must be transferred for adding a target.
Feature		String	Base64-encoded feature	Feature in a target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version. Return this parameter together with the feature.

15.4.16.2 Example

Supported image specifications are as follows:

Format: JPG (recommended), PNG, and BMP.

Resolution: ranging from 80 x 80 to 1200 x 1200. The recommended value is 150 x 200. If the image resolution exceeds this recommended value, you are advised to scale down the image before invoking related APIs. Otherwise, the device load increases, but the target recognition effect will not be improved.

Size: no more than 500 KB. It is recommended that the image size be less than or equal to 50 KB. If the image quality is acceptable, a smaller image size is preferred. Generally, the size of a 150 x 200 JPG ID photo is about 7 KB. If the size exceeds 50 KB, scale down the image to within 150 x 200 and encode the image into a JPG image with 80% quality. Then, you can invoke related APIs for target recognition. Otherwise, the network transmission efficiency deteriorates and the device consumes more memory, but the target recognition effect will not be improved.

Request example:

```
POST/API/AI/CompareFaces/Add?2020-10-17%2014:21:33 HTTP/1.1
```

```
Content-Type: application/json; charset=UTF-8
```

```
Accept: application/json, text/javascript, */*; q=0.01
```

```
X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdc81b08
```

```
Content-Length: 2438
```

```
Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc
```

```
{
```

```
  "data": {
```

```
    "WithImage": 0, // Indicates whether to return images. The options include 1 (yes) and 0 (no).
```

```
    "WithFeature": 0, // Indicates whether to return features extracted from images. The options include 1 (yes) and 0 (no).
```

```
    "FaceInfo": [
```

```
      {
```

```
        "Image": "base64(imgData)", // Target image to be imported.
```

```
        "Feature": "base64(feature)", //: Features in the image to be imported. If no feature is available, this parameter will not be specified and its value will be automatically calculated by the HWT-NVR800.
```

```
    "FtVersion": 0      // Feature version. Return this parameter together with the feature.

  },

  ...

]

}

}
```

After a target is successfully added, all target information parameters will be returned, with most of them being empty. Important information is stored in parameters like **Id**, **MD5**, **GrpId**, **Image**, and **Feature**. The response format is as follows:

```
HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 658

{
  "data": {
    "Result": [
      0
    ],
    "Count": 1,
    "FaceInfo": [
      {
        "Id": 652,
        "GrpId": 1,
        "Name": "",
        "MD5": "102819BE9B2C3BEFE3DA2FB245D2909D",
        "Sex": 3,
        "Age": 255,
        "ModifyCnt": 1,
        "Country": "",
        "NativePlace": ""
      }
    ]
  }
}
```

```

        "IdCode": "",
        "Job": "",
        "Phone": "",
        "Email": "",
        "Domicile": "",
        "Remark": "",
        "EnableChnAlarm": []
    }
}
}
}

```

15.4.17 Searching for Target Statistics

API Description					
Function	This API is used to search for target statistics.				
API URL	/API/AI/FaceStatistics/Search				
Request Method	POST				
Message ID					
Description	This API is used to search for target statistics and display search results on the client.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . If this parameter is not transferred, 0 is

					used by default.
StartTime		String	Y	Format: yyyy-m m-dd hh:mm: ss, for exampl e, 2018-10 -20 00:00:0 0	Start time.
EndTime		String	Y	Format: yyyy-m m-dd hh:mm: ss, for exampl e, 2018-10 -28 23:59:5 9	End time.
Chn		JSON array	Y	[0,1,2,... N]	Channel to be searched. If no channel is available or this parameter is not transferred, no channel will be found.
Group		JSON array	N	[1,2,3... N]	ID of the target list to be searched. If this parameter is left empty, all target lists (including deleted ones) will be searched.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details,	

				see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Actual number of target statistics records.

15.4.17.1 Example

Request example:

POST/API/AI/FaceStatistics/Search?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "Engine": 0,
    "StartTime": "2020-08-18 00:00:00",
    "EndTime": "2020-08-18 23:59:59",
    "Chn": [0, 1, 2, 3, 4, 5, 6, 7, 8]
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 396

```
{
  "data": {
    "Result": 0,
    "Count": 329
  }
}
```

15.4.18 Obtaining Target Statistics

API Description					
Function	This API is used to obtain target statistics.				
API URL	/API/AI/FaceStatistics/Get				
Request Method	POST				
Message ID					
Description	You can invoke this API only after the API in 15.4.18 is invoked.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.

StartIndex		Int	Y		Start index of a request. For example, if the index ranges from 0 to 9,999 for the first request and from 10,000 to 19,999 for the second request, the values of this parameter are 0 and 10000 respectively.
Count		Int	N		Number of target statistics records to be returned. If this parameter is not transferred, one target statistics record is returned by default.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Total number of target records in this response.
Statistics		JSON array		For details, see Table 15-13.

Table 15-13 Elements in the Statistics array

Elements in the Statistics array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Group		Int		ID of the list to which a target belongs.
Time		Unsigned long long		Unix timestamp, for example, 1597737970 . You need to convert the timestamp to the UTC time, and then display the local time according to the UTC time.
Chn		Int		Channel to which a target belongs.

15.4.19 Regular Customer Recognition

Note: To implement this function, use both the existing APIs and following new APIs:

- (1) **/API/AI/SnapedFaces/Search**: searches for all targets that meet specified criteria.
- (2) **/API/AI/SnapedFeaturesId/Get**: obtains all IDs and feature IDs that meet specified criteria.
- (3) **/API/AI/FilterSnapedFaces/Get**: obtains the filtered results. This API is used in a cyclical manner until all targets are filtered.
- (4) **/API/AI/MatchAddedFaces/Get**: obtains the identity information about all regular customers.
- (5) Summarize identity information by combining the information about the same regular customer.

15.4.19.1 Obtaining Target IDs and Feature IDs

API Description	
Function	This API is used to obtain target IDs and feature IDs.
API URL	/API/AI/SnapedFeaturesId/Get
Request Method	POST
Message ID	

Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine, which should be the same as that of /API/All/Snaped Faces/Search . The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.
StartIndex		Unsigned int	N		Start index of a request. For example, if the index ranges from 0 to 9,999 for the first request and from 10,000 to 19,999 for the second request, the values of this parameter are 0 and 10000 respectively. If this parameter is not transferred, 0 is used by default.
Count		Unsigned int	N		Number of requested records. If there are just several IDs to be queried, request them at a time. If this parameter is not transferred, the

					UUID of a target snapshot is returned by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
UUIds		JSON array	Reference: "nQkABoxluT8EBAEA", "nQkABoxluT8EBBEX"	Unique IDs of captured target snapshots.	

15.4.19.2 Filtering Captured Targets

API Description					
Function	This API is used to filter captured targets.				
API URL	/API/AI/FilterSnapedFaces/Get				
Request Method	POST				
Message ID					
Description	This API is used to filter targets in the specified set and return the IDs of targets meeting conditions in Filter .				
Input Parameters					
Parameter	Description	Data	Mandat	Value	Server

		Type	ory (Y/N)	Range	Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine, which should be the same as that of /API/Al/Snaped Faces/Search . The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.
MinInterval		Unsigned int	N	0–65535	Minimum interval between two snapshots which belong to the same target, in seconds. When an interval is shorter than the value of this parameter, the target will be counted. In the response, bValid indicates whether a target will be counted and returned. If this parameter is not transferred, 0 is used by default.
Similarity		Unsigned int	N		Similarity used for filtering. If this parameter is not transferred, 70 is used by default.
Filter		String	N		Desired captured target. Data in UUIDSet

					is filtered to find the target which is similar to that specified in Filter . If this parameter is not transferred, no filtered data will be obtained.
UUIdSet		JSON array	N	Reference: "nQkABoxluT8EBAEA", "nQkABoxluT8ESCEA"	Set to be filtered, which is an array of UUIDs. A UUID is a unique ID of a captured target. If this parameter is not transferred, no filtered data will be obtained.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of filtered targets.	
MatchedIds		JSON array	Reference: "nQkABoxluT8EBAEA", "nQkABoxluT8EBBEX"	Unique IDs of captured targets.	
Counting		JSON array	[1, 0, 0, 1, 0, 1, ..., N]	Indicates whether a target will be counted. When an interval between two snapshots (belonging to the same target) is shorter than the	

				minimum, the value of this parameter is 0 , which indicates the target will not be counted. When an interval is greater than the minimum, the value of this parameter is 1 , which indicates the target will be counted.
--	--	--	--	--

15.4.19.3 Obtaining the Identity Information About Regular Customers

API Description					
Function	This API is used to obtain the identity information about regular customers.				
API URL	/API/AI/MatchAddedFaces/Get				
Request Method	POST				
Message ID					
Description	This API is used to filter targets in the specified set and return the IDs of targets meeting the filtering conditions.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine, which should be the same as that of /API/AI/Snaped Faces/Search . The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.

Similarity		Unsigned int	N		Similarity used for filtering. If this parameter is not transferred, 70 is used by default.
GrpIds		JSON array	N	[1,2,3...N]	Target lists in the base library, which are specified to recognize identities of regular customers. If the parameter is not transferred or is left empty, all lists (excluding the internal list) will be used. Generally, you do not need to transfer this parameter.
UUIs		JSON array	N	Reference: "nQkABoxluT8EBAEA", "nQkABoxluT8ESCEA"	UUIs of regular customers. If this parameter is not transferred, no data will be obtained.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
reason		String		This parameter is returned when a request error occurs.

Facelds		JSON array	[3,-1,2,-1...N]	Targets found in the base library. Each element in the array is of the signed long long type and matches with those in UUI ds in the request. Value -1 indicates that a target has no match in the base library.
---------	--	------------	-----------------	--

15.4.20 Target Attendance

API Description					
Function	This API is used for target attendance.				
API URL	/API/AI/AddedFaces/GetId				
Request Method	POST				
Message ID					
Description	This API is used for target attendance. Similar to the target search function, target attendance allows you to search for captured targets which resemble the specified target, and perform classification and display statistics according to the specified clock-in/out time. To quickly obtain all targets in a specified list, you can use all target IDs in the lists obtained by using this API.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
GrpId		JSON array	N	[1,2,3... N]	Lists whose target IDs are to be obtained.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation	Int		If the operation is	

	result.			successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Facelds		JSON array	[1,2,3...N]	Unique IDs of targets in a list, which are of the signed long long type. This parameter can be used to obtain all information about a target.

15.4.21 Extracting Target Features

API Description					
Function	This API is used to extract target features.				
API URL	/API/AI/ImagesFeature/Get				
Request Method	POST				
Message ID					
Description	The extracted features and feature versions can be used as input parameters for adding a target, adding a target to be matched, adding more images for an existing target, modifying target information, and checking whether a target has its match in the target list. With this API, you are free from the bother to repeatedly extract target features when invoking APIs for the preceding scenarios, thus saving your time. Note that this API can extract features from a maximum of 10 target images at a time.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
ImagesInfo		JSON array	Y		Target image information. For details, see Table 15-14.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	JSON array	[0,0...N]	If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String array		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
FeaturesInfo		JSON array		Information about features extracted from target images. For details, see Table 15-15.	

Table 15-14 Elements in the ImagesInfo array

Elements in the ImagesInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Image		String	Base64-encoded image data	Target image, which is Base64-encoded.

Table 15-15 Elements in the FeaturesInfo array

Elements in the FeaturesInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Feature		String	Base64-encoded feature	Feature in a target image, which is Base64-encoded.
FtVersion		Unsigned int		Feature version.

15.4.21.2 Example

Supported image specifications are as follows:

Format: JPG (recommended), PNG, and BMP.

Resolution: ranging from 80 x 80 to 1200 x 1200. The recommended value is 150 x 200. If the image resolution exceeds this recommended value, you are advised to scale down the image before invoking related APIs. Otherwise, the device load increases, but the target recognition effect will not be improved.

Size: no more than 500 KB. It is recommended that the image size be less than or equal to 50 KB. If the image quality is acceptable, a smaller image size is preferred. Generally, the size of a 150 x 200 JPG ID photo is about 7 KB. If the size exceeds 50 KB, scale down the image to 150 x 200 or less and encode the image into a JPG image with 80% quality. Then, you can invoke related APIs for target recognition. Otherwise, the network transmission efficiency deteriorates and the device consumes more memory, but the target recognition effect will not be improved.

Request example:

POST https://172.16.10.24/API/AI/ImagesFeature/Get HTTP/1.1

Cookie: session=10e7a139a44a97693861b4f07e2f04da94c4468130ff54051096a09f1895230b

X-csrfToken: e926722f80cb87a9769bf6ce27087b15afd694509a934048543ad45008c14f0e

Content-Type: text/plain

User-Agent: PostmanRuntime/7.26.8

Accept: */*

Cache-Control: no-cache

Postman-Token: 7b9d0bc8-31d6-49a1-8722-6ec6f8c84c22

Host: 172.16.10.24

Accept-Encoding: gzip, deflate, br

Connection: keep-alive

Content-Length: 73896

```
{
  "data": {
    "ImagesInfo": [
      {
        "Image": "base64(imgData)" // Data of a target image from which features are to be extracted.
      },
      {
        "Image": "base64(imgData)"
      }
    ]
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Length: 8499

Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors 'self'

Content-Type: application/json

Strict-Transport-Security: max-age=315360000; includeSubDomains; preload

X-Content-Type-Options: nosniff

X-Frame-Options: SAMEORIGIN

X-XSS-Protection: 1; mode=block

server: server

```
{
  "data": {
    "Result": [
      0,
      0
    ],
    "FeaturesInfo": [
```

```

    {
        "Feature": "base64data", // Feature in a target image.
        "FtVersion": 12591105 // Feature version.
    }
    {
        "Feature": "base64(imgData)",
        "FtVersion": 12591105
    }

```

15.4.22 Matching a Target with Those in the Target List

API Description					
Function	This API is used to match a target with those in the target list.				
API URL	/API/AI/MatchGroupFaces/Get				
Request Method	POST				
Message ID					
Description	If a target image transferred by the client has its match in the target list, this API returns the ID of the matched target. Otherwise, -1 will be returned. For details about the matched target in the target list, see 15.4.7 . A maximum of 10 target images can be transferred for match at a time.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Engine to be used. If this parameter is not transferred, engine 0 is used by default.
Similarity		Unsigned int	N		Similarity. If this parameter is not transferred, 85

					is used by default.
FaceInfo		JSON array	Y		Target image information. For details, see Table 15-16.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
MatchInfo		Object array		Each element in this array is of the object type and specifies a target match result. For details, see Table 15-17.	

Table 15-16 Elements in the FaceInfo array

Elements in the FaceInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Image		String	Base64-encoded image data	Target image, which is Base64-encoded.
Feature		String	Base64-encoded feature	Feature in a target image, which is

				Base64-encoded.
FtVersion		Unsigned int		Feature version. Feature match is valid only when it is performed on features of the same version.

Table 15-17 Elements in the MatchInfo array

Elements in the MatchInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
FaceId		Long		ID of a matched target in the target list. If no target is matched, the value is -1 .
Similarity		Float		Similarity between the transferred target and the matched target in the target list. If no target in the target list is matched, the value is 0.0 .

15.4.22.2 Example

Supported image specifications are as follows:

Format: JPG (recommended), PNG, and BMP.

Resolution: ranging from 80 x 80 to 1200 x 1200. The recommended value is 150 x 200. If the image resolution exceeds this recommended value, you are advised to scale down the image before invoking related APIs. Otherwise, the device load increases, but the target recognition effect will not be improved.

Size: no more than 500 KB. It is recommended that the image size be less than or equal to 50 KB. If the image quality is acceptable, a smaller image size is preferred. Generally, the size of a 150 x 200 JPG ID photo is about 7 KB. If the size exceeds 50 KB, scale down the image to 150 x 200 or less and encode the image into a JPG image with 80% quality. Then, you can invoke related APIs for target recognition. Otherwise, the network transmission efficiency deteriorates and the device consumes more memory, but the target recognition effect will not be improved.

Request example:

POST https://172.16.10.24/API/AI/MatchGroupFaces/Get HTTP/1.1

Cookie: session=10e7a139a44a97693861b4f07e2f04da94c4468130ff54051096a09f1895230b

X-csrfToken: e926722f80cb87a9769bf6ce27087b15afd694509a934048543ad45008c14f0e

Content-Type: text/plain

User-Agent: PostmanRuntime/7.26.8

Accept: */*

Cache-Control: no-cache

Postman-Token: 7b9d0bc8-31d6-49a1-8722-6ec6f8c84c22

Host: 172.16.10.24

Accept-Encoding: gzip, deflate, br

Connection: keep-alive

Content-Length: 103896

```
{
  "data": {
    "Engine":0,
    "Similarity":85,
    "FaceInfo":[
      {
        "Image":"base64(imgData)", // Data of a target image to be matched.
        "Feature":"base64(imgData)", // Feature in a target image. For details, see 15.4.21 .
        "FtVersion":12591105 // Target feature version. For details, see 15.4.21 .
      },
      {
        "Image":"base64(imgData)", // Data of a target image to be matched.
        "Feature":"base64(imgData)", // Feature in a target image. For details, see 15.4.21 .
        "FtVersion":12591105 // Target feature version. For details, see 15.4.21 .
      }
    ]
  }
}
```


Response example:

```
HTTP/1.1 200 OK
Content-Length: 8499
Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors 'self'
Content-Type: application/json
Strict-Transport-Security: max-age=315360000; includeSubDomains; preload
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
X-XSS-Protection: 1; mode=block
server: server
{
  "data": {
    "Result": 0,
    "Facelds": [
      1,
      -1
    ]
  }
}
```

15.4.23 Searching for Target Reports

API Description	
Function	This API is used to search for target reports.
API URL	/API/AI/FaceReport/Get
Request Method	POST
Message ID	
Description	This API returns to the client the numbers of targets counted by age group and gender. If you want to obtain the numbers counted by age group and gender in customer flow analysis, set IsFlowStatistic to true. Then AgeGenderInfo will be returned.
Input Parameters	

Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.
Date		String	N	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-20 00:00:00	Period in which data to be searched is generated. If StatisticstType is set to day , week , month , quarter , or year , data generated in the current day, week, month, quarter, or year will be searched. If this parameter is not transferred, no valid data can be found.
Chn		JSON array	Y	[0,1,2,...N]	Channel to be searched. The maximum element in the array cannot exceed the number of HWT-NVR800 channels.
StatisticsType	Statistics collection type.	String	Y	Value options: day , week , month , quarter , and year	Data can be collected by day, week, month, quarter, or year.
IsFlowStatistic	Indicates	Unsigned	N		The options

	whether to request the numbers counted by age group and gender in customer flow analysis.	int			include 0 (no, indicating that target statistics are returned) and 1 (yes, indicating that target statistics are not returned).
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
AgeInfo		Int array	Example: [0,0,0,0]	Numbers of targets collected by age group (unit: person). This array consists of four elements of the int type. The first to fourth elements indicate the numbers of targets under the age of 18, in the age group of 18 to 35, in the age group of 36 to 59, and at or above the age of 60, respectively.	
GenderInfo		Int array	Example: [0,0]	Numbers of targets collected by gender (unit: person). This array consists of two elements of the int type. The first and second elements	

				indicate the numbers of male targets and female targets, respectively.
AgeGenderInfo		Object array		Numbers of targets counted by age group and gender in customer flow analysis. The array consists of four elements of the object type. The first to fourth elements indicate the numbers of male targets and female targets under the age of 18, in the age group of 18 to 35, in the age group of 36 to 59, and at or above the age of 60, respectively. For details, see Table 15-18.

Table 15-18 Elements in the AgeGenderInfo array

Elements in the AgeGenderInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Male		Int		Total number of male targets in an age group.
Female		Int		Total number of female targets in an age group.

15.4.24 Searching for Regular Customers in Real Time

Note: To implement this function, use both the existing APIs and following new APIs:

- (1) Use **/API/AI/RealtimeRepeaters/Search** to search for basic information about regular customers by specified search criteria.

(2) Use basic information about the regular customers obtained in step 1 as the input parameter and use **/API/AI/RealtimeRepeaters/Get** to obtain the UUIDs of captured targets of these regular customers.

(3) Use **/API/AI/SnapedFaces/GetById** to obtain details about the regular customers based on the UUIDs.

15.4.24.1 Searching for Regular Customer Data

API Description					
Function	This API is used to search for regular customer data.				
API URL	/API/AI/RealtimeRepeaters/Search				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
StartTime		String	Y	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-20 00:00:00	Search start time.
EndTime		String	Y	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-28 23:59:59	Search end time.
Group		Int array	Y	[1,2,...N]	ID of the list containing the regular customers to be searched for.
MinAppearCount		Unsigned int	N	2–21	Minimum number of appearance times. If this parameter is not

					transferred, 2 is used by default.
MaxAppearCount		Unsigned int	N	2–21	Minimum number of appearance times. If this parameter is not transferred, 21 is used by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
RepeatersInfo		Object array		Basic information about regular customers. For details, see Table 15-19.	

Table 15-19 Elements in the RepeatersInfo array

Elements in the RepeatersInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
RepeaterUUID		String		UUID of a regular customer.
FaceId		Int		ID of a matched target in the target list. If no target is matched, -1 is

				used by default.
AppearInfo		Object array		Number of times that a regular customer appears.
Elements in the AppearInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Chn		Int		Channel where a regular customer appearing at the current time is captured.
Time		Unsigned int		Unix timestamp when a regular customer appears.

15.4.24.2 Obtaining Regular Customer Data

API Description					
Function	This API is used to obtain regular customer data.				
API URL	/API/AI/RealtimeRepeaters/Get				
Request Method	POST				
Message ID					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
RequestOne		Unsigned int	N	0–1	Indicates whether to request the cover image of regular customers. If this parameter is not transferred or is

					set to 1 , the UUID of the target corresponding to the cover image is requested. If this parameter is set to 0 , all target UUIDs of regular customers are requested.
RepeatersInfo		Object array	Y		Basic information about regular customers. For details, see Table 15-20.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
RepeatersInfo		Object array		UUIDs of captured targets of regular customers. For details, see Table 15-21.	

Table 15-20 Elements in the RepeatersInfo array

Elements in the RepeatersInfo array				
		JSON object		Elements in a single object are as follows.

Elements in a single object				
RepeaterUUId		String		UUID of a regular customer.
AppearInfo		Object array		Number of times that a regular customer appears.
Elements in the AppearInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
Chn		Int		Channel where a regular customer appearing at the current time is captured. The value starts from 0 and is obtained using the API in 15.4.24.1. The maximum value must be less than the maximum number of HWT-NVR800 channels. If this parameter is transferred, use its original value.
Time		Unsigned int		Unix timestamp when a regular customer appears. This parameter is set to its original value obtained using the API in 15.4.24.1.

Table 15-21 Elements in the RepeatersInfo array

Elements in the RepeatersInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
RepeaterUUId		String		UUID of a regular

				customer.
AppearUUIDs		String array		UUID of a captured target in a single element. The array size is the same as the size of AppearInfo . If no target is found, the UUID is an empty string, which indicates that the captured target does not exist.

15.5 Obtaining Head Counting Statistics

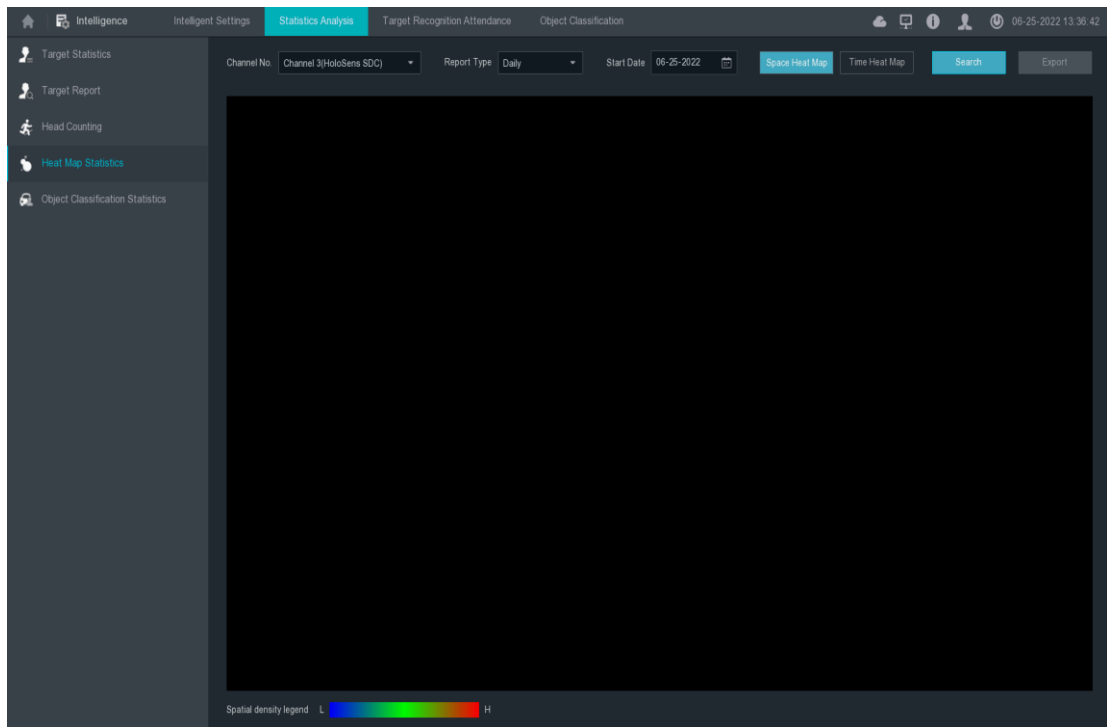
API Description					
Function	This API is used to obtain head counting statistics.				
API URL	/API/AI/CCStatistics/Search				
Request Method	POST				
Message ID					
Description	This API is used to obtain the head counting statistics collected by a camera.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
chn	Channel ID list.	Int/Int array	N	[0,1,2...N]	The number of elements in the array cannot exceed the maximum number of channels. The channel ID starts from 0. The data type of this parameter was changed from int to int

					array on January 5, 2021 to support search by multiple channels. In addition, it is compatible with the single-channel search mode in earlier versions.
date	Period in which data to be searched is generated.	String	N	Format: yyyy-mm-dd hh:mm:ss, for example, 2020-08-14 00:00:00	If reportType is set to day , week , month , quarter , or year , data generated in the current day, week, month, quarter, or year will be searched. If this parameter is not transferred, no valid data can be found.
ReportType	Statistics collection type.	String	N	Value options: day , week , month , quarter , and year	Data can be collected by day, week, month, quarter, or year. If this parameter is not transferred, no valid data can be found.
DetectType	Detection type, which specifies whether to count the number of persons entering or exiting a detection area.	String	N	Value options: enter , exit , and all	If this parameter is not transferred, no valid data can be found.
Output Parameters					
Parameter	Descriptio	Data	Value Range	Remarks	

	n	Type		
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description .	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		JSON array	[0, 1, 292, 33... N]	Numbers of persons counted in different time segments. For example, if ReportType is set to day , there will be 24 elements in the array and each element indicates the number of persons counted in each hour in one day. Similarly, if ReportType is set to week , there will be seven elements in this array and each element indicates the number of persons counted on each day in one week. The same rule applies to search by month and year. If DetectType is set to all , this parameter indicates the number of persons entering the detection area.
CountExt		JSON array	[0, 1, 292, 33... N]	Numbers of persons exiting the detection area in different time segments. If DetectType is set to all , this parameter and

15.6 Obtaining Heat Map Statistics

This API is used to obtain heat map statistics.



API Description

Function	This API is used to obtain heat map statistics.
API URL	/API/AI/HeatMapStatistics/Search
Request Method	POST
Message ID	
Description	This API is used to obtain the heat map statistics collected by a camera. The output parameter Image of this API indicates the spatial intensity distribution and will be obtained for drawing a spatial heat map. Its value is a 352 x 288 two-dimensional array, which represents the original data of the spatial heat map obtained by the SDC. Each element in the array is a UCHAR variable in either of the following ranges: 0–12 (zero heat; transparent; recommended Alpha = 0) and 13–255 (color adjusted from left to right on the color scale; translucent; recommended Alpha = 160). To draw a heat map, perform the following steps:

	1. Invoke this API to obtain the two-dimensional grayscale array (specified by Image) of the heat map, and invoke the API in 12 to obtain the background. 2. Convert the two-dimensional grayscale array into a pixel map by following the preceding step. 3. Draw the background in a specified area. 4. Draw the pixel map obtained in step 2 in the background area.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
chn	Channel ID list.	Int	Y	[0,1,2...N]	The number of elements in the array cannot exceed the maximum number of channels. The channel ID starts from 0.
date	Period in which data to be searched is generated.	String	N	Format: yyyy-mm-dd hh:mm:ss, for example, 2020-08-14 00:00:00	If ReportType is set to day , week , or month , data generated in the current day, week, or month will be searched. If this parameter is not transferred, no valid data can be found.
ReportType	Statistics collection type.	String	N	Value options: day , week , and month	Data can be collected by day, week, or month. If this parameter is not transferred, no valid data can be found.
getSpaceMap	Indicates	Int	N	0 or 1	The options

	whether to search for the temporal heat map or spatial heat map.				include 0 (temporal heat map) and 1 (spatial heat map). If this parameter is not transferred, the temporal heat map is searched by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
HeatMapWidth		Int		Heat map width. This parameter is returned when getSpaceMap is set to 1 .	
HeatMapHeight		Int		Heat map height. This parameter is returned when getSpaceMap is set to 1 .	
Image		String	Base64-encoded data.	Two-dimensional grayscale array of the heat map. This parameter is returned when getSpaceMap is set to 1 .	
TimeHeatMap		JSON array	[0,1,292,33...N]	Statistical results generated in different time segments. For example, if ReportType is set to	

				day, there will be 24 elements in the array and each element indicates the statistical result in each hour in one day. Similarly, if ReportType is set to week, there will be seven elements in this array and each element indicates the statistical result on each day in one week. The same rule applies to search by month. Each element indicates temporal heat map statistics with no specific unit, representing the activeness in a heat map area within a specified period.
--	--	--	--	--

15.6.1 Example

Request example for obtaining temporal heat map statistics:

POST/API/AI/HeatMapStatistics/Search?2020-10-17%2014:21:33 HTTP/1.1

Content-Type: application/json; charset=UTF-8

Accept: application/json, text/javascript, */*; q=0.01

X-csrfToken: 14559384d58d55d9d80bf4baf048684f366c77905d665d99ba1bde2cdcf81b08

Content-Length: 2438

Cookie: session=cc3ec99c6f1295489c86f8842e1dd719c194c637af5b965a23932c41188a3acc

```
{
  "data": {
    "chn": 2,
    "date": "2020-09-25 15:06:30",
    "ReportType": "day",
    "getSpaceMap": 0
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 516

```
{"data":{"Result":0,"TimeHeatMap":[77,0,45,0,45,0,0,0,29301,14958,30069,25705,14394,26214,25185,14436,11572,13413,14137,13357,25904,11619,12856,25954]}}
```

15.7 Object Classification Operations

15.7.1 Searching for Objects

API Description					
Function	This API is used to search for objects.				
API URL	/API/AI/SnapedObjects/Search				
Request Method	POST				
Message ID					
Description	If ACPT_Vehicle is not transferred in the Type array, the LicensePlate , VehicleBrand , VehicleType , VehicleColor , PlateType , and PlateColor parameters are invalid.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
StartTime		String	Y	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-20 00:00:00	Search start time.
EndTime		String	Y	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-28 23:59:59	Search end time.

Chn		JSON array	N	[0, 1, 2, ...N]	Channel to be searched. All channels are searched if this parameter is not transferred.
SortType		Unsigned int	N	0 or 1	Sort type. The options include 0 (ascending order) and 1 (descending order). The default value is 0 .
Type		JSON array	Y	[1, 2, 7]	Search type, which corresponds to constants in the AI Cam Pic Type enum. Among these constants, only ACPT_HumanBody , ACPT_Vehicle , and ACPT_NonVehicle are valid.
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . These two engines are provided to work around the impact caused by the overlap between the search process and operation logic. For example, if you open the dialog box for selecting targets to be searched when a list of match results are already displayed, the cached match results will be overwritten by the search result and

					become invalid. The engines are independent of each other in terms of data and cache and are allocated by the client. For example, engine 0 can be used to implement temporary target selection tasks such as target selection in the dialog box; engine 1 can be used to display search or match results. If this parameter is not transferred, 0 is used by default.
LicensePlate		String	N	Maximum length: 32 bytes If Unknown is transferred, an empty string is returned, indicating that the license plate is unknown.	License plate characters. If the parameter is not transferred or set to an empty string, search by license plate is not supported.
VehicleBrand		Unsigned int	N		Vehicle make, which corresponds to constants in the PU_VEHICLE_BRAND_E enum. The maximum value of this parameter corresponds to PU_LPR_VB_ALFAROMEIO. If this parameter is not transferred, search by vehicle make is not supported.

VehicleType		JSON array	N	0–35 and 120	Vehicle type, which corresponds to constants in the PU_VEHICLE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_VT_MOTOR . If this parameter is not transferred, search by vehicle type is not supported.
VehicleColor		JSON array	N	0–14	Vehicle color, which corresponds to constants in the PU_VEHICLE_COLOR_E enum. If this parameter is not transferred, search by vehicle color is not supported.
PlateType		JSON array	N	0–20	License plate type, which corresponds to constants in the PU_PLATE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_LPT_AIRPORT . If this parameter is not transferred, search by license plate type is not supported.
PlateColor		JSON array	N	0–15	License plate color, which corresponds to constants in the PU_PLATE_COLOR_E enum. If this parameter is not transferred,

					search by license plate color is not supported.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
Count		Int		Number of objects found.	

15.7.1.1 Example and Comment

If **ACPT_Vehicle** is not transferred in the **Type** array, you do not need to transfer other parameters related to vehicle search. The request example is as follows:

POST https://172.16.10.24/API/AI/SnapedObjects/Search HTTP/1.

Cookie: session=7af0e6045fecc2c0a283f9af6c9c22cfb60562a0d3f80e10af235367b79e999d

X-csrfToken: 377963fcd497efe28401ec316fb484747113a43edfe3ffd773c35fc3837e3b33

Content-Type: text/plain

User-Agent: PostmanRuntime/7.26.8

Accept: */*

Cache-Control: no-cache

Postman-Token: 020e03f3-bc47-48cd-89cd-cf8e0fb6635c

Host: 172.16.10.24

Accept-Encoding: gzip, deflate, br

Connection: keep-alive

Content-Length: 238

```
{
  "data": {
    "Engine": 1,
    "StartTime": "2020-11-24 00:00:00",
    "EndTime": "2020-11-24 23:59:59",
    "Type": [2],
    "LicensePlate": "*A716CF",
    "VehicleType": [],
    "PlateType": []
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Length: 31

Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors 'self'

Content-Type: application/json

Strict-Transport-Security: max-age=315360000; includeSubDomains; preload

X-Content-Type-Options: nosniff

X-Frame-Options: SAMEORIGIN

X-XSS-Protection: 1; mode=block

server: server

```
{"data":{"Result":0,"Count":1}}
```

15.7.2 Obtaining Search Results in Pagination Mode

API Description	
Function	This API is used to obtain search results in pagination mode.

API URL	/API/AI/SnapedObjects/GetByIndex				
Request Method	POST				
Message ID					
Description	You can invoke this API to obtain search results in pagination mode after /API/AI/SnapedObjects/Search is invoked. Vehicle information can be obtained only when ACPT_Vehicle is transferred in the Type array. The vehicle information includes the license plate characters, vehicle make, vehicle model, vehicle make ID, vehicle model ID, vehicle model year, vehicle type, vehicle color, license plate type, license plate color, vehicle position, and license plate position. When SimpleInfo is set to 0: If WithObjectImage or WithBackground is not set to 0, a maximum of 30 snapshots and 30 background images can be obtained.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . This parameter is used together with the search API. If this parameter is not transferred, 0 is used by default.
StartIndex		Unsigned int	N	0–Max (The maximum value is 99999 for an HWT-NVR800 with only one disk tray and	Start index of a request. For example, if the index ranges from 0 to 19 for the first request and from 20 to 39 for the second request, the values of this parameter are 0 and 20 respectively. If this parameter

				199999 for an HWT-NVR800 with two or more disk trays.)	is not transferred, the default value 0 is used.
Count		Unsigned int	N		Number of requested objects. If this parameter is not transferred, the first object image is returned by default.
SimpleInfo		Unsigned int	N		Indicates whether to obtain simplified information or complete information. The options include 0 (simplified information) and 1 (complete information). Simplified information contains at most Type and UUid . If this parameter is not transferred, 0 is used by default.
WithObjectImage		Unsigned int	N		Indicates whether to obtain snapshots. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithBackground		Unsigned int	N		Indicates whether to obtain background images. The options include

					1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
TotalCount		Int		Total number of objects found.	
Count		Int		Total number of captured objects returned in the current response.	
SnapedObjInfo		JSON array		For details, see Table 15-22.	

Table 15-22 Elements in the SnapedObjInfo array

Elements in the SnapedObjInfo array				
		JSON object		Elements in a single object are as follows.
Elements in a single object				
UUIId		String		Unique ID of a captured object, which can be used for obtaining details about

				the object.
Type		Int	1, 2, and 7	Type of a captured object, which corresponds to constants in the AI CamPicType enum. Among these constants, only ACPT_HumanBody , ACPT_Vehicle , and ACPT_NonVehicle are valid.
Chn		Int		Capture channel.
StartTime		String	Fixed format: yyyy-mm-dd hh:mm:ss, for example, 2020-09-21 21:59:35	Start time.
EndTime		String	Fixed format: yyyy-mm-dd hh:mm:ss, for example, 2020-09-21 21:59:35	End time.
ObjectImage		String	Base64-encoded image data	Captured image (encoded using Base64) of a person , vehicle , or non-motorized vehicle .
Background		String	Base64-encoded image data	Captured background image (encoded using Base64).
SnapId		Int		Target trajectory ID of a camera, which is of no use to the HWT-NVR800.
LicensePlate		String		License plate characters. An empty string indicates an unknown license plate.
CarPreBrand		String		Vehicle make. An empty string indicates an unknown vehicle make.
CarPreBrandId		Unsigned int	0–0xffffffff	Vehicle make ID, which is returned by

				the camera. For details, see the type table provided by the camera.
CarSubBrand		String		Vehicle model. An empty string indicates an unknown vehicle model.
CarSubBrandId		Unsigned int	0–0xffffffff	Vehicle model ID, which is returned by the camera. For details, see the type table provided by the camera.
CarYearBrand		String		Vehicle model year. An empty string indicates an unknown model year.
CarType		Int	0–35 and 120	Vehicle type, which corresponds to constants in the PU_VEHICLE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_VT_MOTOR .
CarColor		Int	0–14	Vehicle color, which corresponds to constants in the PU_VEHICLE_COLOR_E enum.
PlateType		Int	0–20	License plate type, which corresponds to constants in the PU_PLATE_TYPE_E enum. The maximum value of this parameter corresponds to PU_VLPR_LPT_AIRPORT .
PlateColor		Int	0–15	License plate color, which corresponds to constants in the PU_PLATE_COLOR_E enum.
ObjPos		JSON		Position of a captured vehicle in the

		object		background image. For details, see Table 15-23.
PlatePos		JSON object		Position of a captured license plate in the background image. For details, see Table 15-23.
HumanImagePos		JSON object		Position of a captured person in the snapshot. This parameter is returned when Type is set to the value corresponding to persons. For details, see Table 15-23.
HumanBgPos		JSON object		Position of a captured person in the background image. This parameter is returned when Type is set to the value corresponding to persons. For details, see Table 15-23.
NonVehicleBgPos		JSON object		Position of a non-motorized vehicle in the background image. This parameter is returned when Type is set to the value corresponding to non-motorized vehicles. For details, see Table 15-23.

Table 15-23 Elements in the ObjPos, PlatePos, HumanImagePos, HumanBgPos, and NonVehicleBgPos objects

Elements in the ObjPos , PlatePos , HumanImagePos , HumanBgPos , and NonVehicleBgPos objects				
Note: Use relative coordinates (10000 x 10000) in the background image. If X , Y , Width , and Height are all set to 0 , the position is unknown.				
X		Int	0–10,000	X coordinate of the upper left corner of the position.

Y		Int	0–10,000	Y coordinate of the upper left corner of the position.
Width		Int	0–10,000	Position width.
Height		Int	0–10,000	Position height.

15.7.2.1 Example

Request example:

POST https://172.16.10.24/API/AI/ShapeObjects/GetByIndex HTTP/1.1

Content-Type: text/plain

Cookie: session=7af0e6045fecc2c0a283f9af6c9c22cfb60562a0d3f80e10af235367b79e999d

X-csrfToken: 377963fcd497efe28401ec316fb484747113a43edfe3ffd773c35fc3837e3b33

User-Agent: PostmanRuntime/7.26.8

Accept: */*

Cache-Control: no-cache

Postman-Token: 703733dd-a0f8-4c8a-ae24-0082e18eb4e3

Host: 172.16.10.24

Accept-Encoding: gzip, deflate, br

Connection: keep-alive

Content-Length: 146

```
{
  "data": {
    "Engine": 1,
    "StartIndex": 0,
    "Count": 1,
    "SimpleInfo": 0,
    "WithObjectImage": 0,
    "WithBackground": 0
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Length: 455

Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors 'self'

Content-Type: application/json

Strict-Transport-Security: max-age=315360000; includeSubDomains; preload

X-Content-Type-Options: nosniff

X-Frame-Options: SAMEORIGIN

X-XSS-Protection: 1; mode=block

server: server

```
{
  "data": {
    "Result": 0,
    "TotalCount": 1,
    "Count": 1,
    "SnapedObjInfo": [
      {
        "UUId": "ZyMAADDknz8AAAEA",
        "Type": 2,
        "Chn": 0,
        "StartTime": "2020-11-24 08:47:38",
        "EndTime": "2020-11-24 08:47:48",
        "SnapId": 2,
        "LicensePlate": "*A716CF",
        "CarPreBrand": "Ford",
        "CarSubBrand": "Focus",
        "CarYearBrand": "2011_2012",
        "CarType": 1,
        "CarColor": 1,
        "PlateType": 1,
        "PlateColor": 1,

```

```

    "ObjPos": {
        "X": 6078,
        "Y": 3861,
        "Width": 3020,
        "Height": 5305
    },
    "PlatePos": {
        "X": 7333,
        "Y": 7796,
        "Width": 916,
        "Height": 518
    }
}
]
}
}

```

15.7.3 Obtaining Information About Captured Objects by UUID

API Description					
Function	This API is used to obtain information about captured objects by UUID.				
API URL	/API/AI/SnapedObjects/GetById				
Request Method	POST				
Message ID					
Description	This API is used to obtain information about captured objects directly by UUID without search.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . This parameter is used together with the search API. If this parameter is not transferred, 0 is used by default.
UUIds		JSON array	Y	Reference: "nQkABoxluT8EBAEA", "nQkABoxluT8EBBEX"	Unique ID of a captured object whose information is to be obtained.
WithObjectImage		Unsigned int	N		Indicates whether to obtain target images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
WithBackground		Unsigned int	N		Indicates whether to obtain background images. The options include 1 (yes) and 0 (no). If this parameter is not transferred, 0 is used by default.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise,	

				other results are returned. For details, see section 15.1 "Operation Results."
ResultReason	Operation result description	String		This parameter is returned when no request errors occur on the client.
reason		String		This parameter is returned when a request error occurs.
Count		Int		Total number of captured objects returned in the current response.
SnapedObjInfo		JSON array		For details, see Table 15-22.

15.7.4 Searching for Object Statistics

API Description					
Function	This API is used to search for object statistics.				
API URL	/API/AI/ObjectStatistics/Get				
Request Method	POST				
Message ID					
Description	This API is used to search for object statistics and display the search results on the client.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . If this parameter is not transferred, 0 is

					used by default.
Date		String	N	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-2 0 00:00:00	Period in which data to be searched is generated. If StatisticstType is set to day , week , month , quarter , or year , data generated in the current day, week, month, quarter, or year will be searched. If this parameter is not transferred, no valid data can be found.
Chn		JSON array	Y	[0,1,2,...N]	Channel to be searched. The maximum element in the array cannot exceed the number of HWT-NVR800 channels.
StatisticsType	Statistics collection type.	String	Y	Value options: day , week , month , quarter , and year	Data can be collected by day, week, month, quarter, or year.
Output Parameters					
Parameter	Descriptio n	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description	String		This parameter is returned when no request errors occur	

	.			on the client.
reason		String		This parameter is returned when a request error occurs.
Count		JSON array	Example (with StatisticsType set to quarter): [[0,0,0],[7944,7460,18036],[33,49,729]]	Statistical results in different time segments. For example, if StatisticsType is set to day , there will be 24 elements in the array and each element indicates the statistical result in each hour in one day. Similarly, if StatisticsType is set to week , there will be seven elements in this array and each element indicates the statistical result on each day in one week. The same rule applies to search by month and year. Each element is of the JSON array type. For details, see Table 15-24.
TotalCount		JSON array	Example: [0,0,0]	Numbers of objects found. The value consists of three elements of the int type. The first to third elements indicate the number of motor vehicles, non-motorized vehicles, and pedestrians, respectively.

Table 15-24 Elements in the Count array

Elements in the Count array				
		JSON array	Example: [0,0,0]	The value consists of three elements of the int type. The first to

				third elements indicate the number of motor vehicles, non-motorized vehicles, and pedestrians, respectively.
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15.7.4.1 Example

Request example:

POST https://172.16.10.24/API/AI/SnapedObjects/Search HTTP/1.1

Cookie: session=7af0e6045fecc2c0a283f9af6c9c22cfb60562a0d3f80e10af235367b79e999d

X-csrfToken: 377963fcd497efe28401ec316fb484747113a43edfe3ffd773c35fc3837e3b33

Content-Type: text/plain

User-Agent: PostmanRuntime/7.26.8

Accept: */*

Cache-Control: no-cache

Postman-Token: 020e03f3-bc47-48cd-89cd-cf8e0fb6635c

Host: 172.16.10.24

Accept-Encoding: gzip, deflate, br

Connection: keep-alive

Content-Length: 140

```
{
  "data": {
    "Engine": 0,
    "Date": "2020-12-07 00:00:00",
    "StatisticsType": "day",
    "Chn": [0]
  }
}
```

Response example:

HTTP/1.1 200 OK

Content-Length: 455

```

Content-Security-Policy: default-src 'self'; object-src 'self'; script-src 'self'; frame-ancestors
'self'

Content-Type: application/json

Strict-Transport-Security: max-age=315360000; includeSubDomains; preload

X-Content-Type-Options: nosniff

X-Frame-Options: SAMEORIGIN

X-XSS-Protection: 1; mode=block

server: server

{
  "data":{
    "Result":0,
    "Count":[[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,2,0],[32,48,37],[0,0,0],[0,0,0],[0,
0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0],[0,0,0]],
    "TotalCount":[32,50,37]
  }
}

```

15.7.5 Stopping Searching for Objects

API Description					
Function	This API is used to stop searching for objects.				
API URL	/API/AI/SnapedObjects/StopSearch				
Request Method	POST				
Message ID					
Description	This API is used to stop searching for objects. The client invokes this API when it determines that the search is no longer required, for example, when the search takes too long and needs to be canceled. Then, the device releases all caches of the specified search engine. In this way, pressure on the device is reduced.				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		

Elements in the data object					
Engine		JSON array	Y	[0,1...N]	Engines whose caches are to be released.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description.	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	

15.8 Customer Flow Analysis

Customer flow analysis is implemented using different APIs to display information like customer flow, customer density, and personal information.

This function consists of the following modules:

1. Heat map

This module reuses the heat map statistics API to display a spatial heat map.

2. Age and gender

This module reuses the API for searching for target reports and transfers the **IsFlowStatistic** parameter (set to **true**) through the API to obtain statistics on male targets and female targets in a specific age group.

3. Information about recent customers

This module uses the API for querying event alarms to obtain information about recent customers, that is, data carried by the **face_alarm** parameter in the alarms.

4. Customer type

This module uses the API for obtaining the customer type to obtain customer information.

5. Customer flow in a specific time segment

This module reuses the API for obtaining head counting statistics to calculate customer flow in a specific time segment and display the customer flow. The customer flow is calculated as the number of persons entering an area, the number of persons leaving an area, or the sum of the two values.

6. Store visit rate

This module reuses the API for obtaining head counting statistics to calculate the store visit rate. The formula for calculating the store visit rate is as follows: Store visit rate = Number of persons entering the store calculated by the in-store channel/Number of persons near the store entrance calculated by the out-store channel x 100%

15.8.1 Obtaining the Customer Type

API Description					
Function	This API is used to obtain the customer type for customer flow analysis.				
API URL	/API/All/FaceCustomerStatistics/Get				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	Object	Y		
Elements in the data object					
Engine		Unsigned int	N		Search engine. The options include 0 and 1 . If this parameter is not transferred, 0 is used by default.
Date		String	N	Format: yyyy-mm-dd hh:mm:ss, for example, 2018-10-20 00:00:00	Period in which data to be searched is generated. If StatisticstType is set to day , week , or month , data

					generated in the current day, week, or month will be searched. If this parameter is not transferred, no valid data can be found.
StatisticsType	Statistics collection type.	String	Y	Value options: day , week , and month	Data can be collected by day, week, or month.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
Result	Operation result.	Int		If the operation is successful, AORT_SUCCESS is returned. Otherwise, other results are returned. For details, see section 15.1 "Operation Results."	
ResultReason	Operation result description	String		This parameter is returned when no request errors occur on the client.	
reason		String		This parameter is returned when a request error occurs.	
FaceCustomerInfo		Object array		Customer type in customer flow analysis. Each element in this array is an object indicating the number of customers whose targets are in different target lists. For details, see Table 15-25.	

Table 15-25 Elements in the FaceCustomerInfo arrayElements in the **FaceCustomerInfo** array

		JSON object		Elements in a single object are as follows.
Elements in a single object				
GrpId		Int		Target list ID corresponding to the current element.
Num		Int		Number of recognized targets in the target list corresponding to the current element. Note that this parameter is not available for the stranger list.
HighFreqStrangers		Int		Number of frequently detected strangers. This parameter is available only for the stranger list.
LowFreqStrangers		Int		Number of less frequently detected strangers. This parameter is available only for the stranger list.

16 Intelligent Service Configuration

16.1 Intelligent Capabilities

16.1.1 Obtaining the Intelligent Capability Set of a Channel

API Description					
Function	This API is used to obtain the intelligent capability set of a channel.				
API URL	/API/Intelligent/Ability/Get				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory	Value Range	Server Verification Rule

			(Y/N)		
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
ability		string array	"PerimeterZone", "PerimeterLine", "AreaEnter", "AreaExit", "Loitering", "FastMove", "CrossCounting", "QueueDetect", "CrowdDensity", "HeatMap", "FaceDetection",		PerimeterZone : intrusion detection. PerimeterLine : tripwire crossing detection. AreaEnter : area entry detection. AreaExit : area exit detection. Loitering : loitering detection. FastMove : fast movement detection. CrossCounting :

			"MaskAlarm", "FaceRecognition", "SoundDetect", "LensBlocking", "MotionDetect", "ThirdPartyAlgorithm", "FaceAttribute", "ItemLeave", "ItemMove", "ElectroMobile", "VehicleDetection", "VehicleParameter", "VehicleIllegalPark", "AudioOutAlarm", "LightAlarm", "SoundAbnormalAlarm", "LeaveDetect"	crossing-line head counting. QueueDetect : queue length detection. CrowdDensity : crowd density detection. HeatMap : heat map. FaceDetection : target detection. MaskAlarm : mask detection. FaceRecognition : target recognition. SoundDetect : audio diagnosis. LensBlocking : lens blocking detection. MotionDetect : motion detection. ThirdPartyAlgorithm : third-party algorithm. FaceAttribute : target attribute recognition. ItemLeave : abandoned object detection. ItemMove : removed object detection. ElectroMobile : electric bike detection. VehicleDetection : object classification. VehicleParameter : motor vehicle detection. VehicleIllegalPark : parking violation detection. AudioOutAlarm : audible alarm. LightAlarm : visual alarm. SoundAbnormalAlarm : abnormal sound alarm. LeaveDetect : off-duty detection.
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16.2 Behavior Analysis

16.2.1 Obtaining Values of Behavior Analysis Parameters

API Description					
Function	This API is used to obtain values of behavior analysis parameters.				
API URL	Intrusion detection: /API/Intelligent/PerimeterZone/Get Tripwire crossing detection: /API/Intelligent/PerimeterLine/Get Area entry detection: /API/Intelligent/AreaEnter/Get Area exit detection: /API/Intelligent/AreaExit/Get Loitering detection: /API/Intelligent/Loitering/Get Fast movement detection: /API/Intelligent/FastMovement/Get				
Request Method	POST				
Message ID					
Description	Behavior analysis: intrusion detection, tripwire crossing detection, area entry/exit detection, loitering detection, and fast movement detection URL used to obtain the parameter value range /API/Intelligent/PerimeterZone/Range /API/Intelligent/PerimeterLine/Range /API/Intelligent/AreaEnter/Range /API/Intelligent/AreaExit/Range /API/Intelligent/Loitering/Range /API/Intelligent/FastMovement/Range (Used to obtain the parameter value range only. The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.

page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
detect_type		Array	"Pedestrian", "Vehicle", "NonVehicle"		Type of the object to be detected. (ChannelConfig)
area_list		JSON Array	JSON object		Area list. For details, see Table 16-1. This parameter is valid only when page_type is set to ChannelConfig , and not required for tripwire crossing detection.
line_list		JSON array	JSON object		Detection line list. For details, see Table 16-3. This parameter is valid only when page_type is set to ChannelConfig and dedicated for tripwire

				crossing detection.
sensitivity		Int	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
tolerance_time		Int	1–180	Alarm tolerance time, in seconds. This parameter is valid only when page_type is set to ChannelConfig and dedicated for loitering detection.
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clock In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified	Audio file. (ChannelConfig)

			Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels.)
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)

record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking upon alarm occurrence is enabled for a channel. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

Table 16-1 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
index		Int	1–4	Area index.
enable		Bool	Bool	Indicates whether a detection area is enabled.
point_num		Int	0–10	Number of vertices in an area. (This parameter is not required during setting.)
point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-2.

Table 16-2 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.
y		Int	0–576	Y coordinate.

Table 16-3 Elements in the line_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single detection line object.
Elements in a single detection line object				
index		Int	1–4	Area index.
enable		Bool	Bool	Indicates whether a detection area is enabled.
rule_type		String	"A->B", "B->A", "A<->B"	Crossing direction.
x1		Int	0–704	X1 coordinate.
y1		Int	0–576	Y1 coordinate.
x2		Int	0–704	X2 coordinate.
y2		Int	0–576	Y2 coordinate.

16.2.2 Setting Behavior Analysis Parameters

API Description	
Function	This API is used to set behavior analysis parameters.
API URL	Intrusion detection: /API/Intelligent/PerimeterZone/Set Tripwire crossing detection: /API/Intelligent/PerimeterLine/Set Area entry detection: /API/Intelligent/AreaEnter/Set Area exit detection: /API/Intelligent/AreaExit/Set

	Loitering detection: /API/Intelligent/Loitering/Set Fast movement detection: /API/Intelligent/FastMovement/Set				
Request Method	POST				
Message ID					
Description	Behavior analysis: intrusion detection, tripwire crossing detection, area entry/exit detection, loitering detection, and fast movement detection				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
detect_type		Array	N	"Pedestrian", "Vehicle", "NonVehicle"	Type of the object to be detected. (ChannelConfig)
area_list		JSON array	N	JSON object	Area list. For details, see Table 16-1. This parameter is valid only when page_type is set to ChannelConfig , and not required for tripwire crossing

					detection.
line_list		JSON array	N	JSON object	Detection line list. For details, see Table 16-3. This parameter is valid only when page_type is set to ChannelConfig and dedicated for tripwire crossing detection.
sensitivity		Int	N	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	N	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
tolerance_time		Int	N	1–180	Alarm tolerance time, in seconds. This parameter is valid only when page_type is set to ChannelConfig and dedicated for loitering detection.
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav",	Audio file. (ChannelConfig)

				"Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon

					alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable alarm-linked recording for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking upon alarm occurrence for a channel. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.3 Target Capture

16.3.1 Obtaining Values of Target Capture Parameters

API Description					
Function	This API is used to obtain values of target capture parameters.				
API URL	/API/Intelligent/PerimeterFace/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/PerimeterFace/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline",		Channel status (online or offline). This parameter is

			"Online"	available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool		Indicates whether target capture is enabled.
face_attr_switch		Bool		Indicates whether target attribute recognition is enabled.
face_detect_mode		Int	0–2	Detection mode. The options include 0 (target detection), 1 (person detection), and 2 (both).
alarm_interval		Int	1–10	Alarm interval, in seconds.
face_sensitivity		Int	1–5	Target detection sensitivity.
body_sensitivity		Int	1–5	Person detection sensitivity.
min_pixel		Int	26–399	Minimum resolution of a target that can be detected in 1080p video images, in pixels.
max_pixel		Int	27–400	Maximum resolution of a target that can be detected in 1080p video images, in pixels.
face_exposure		Bool		Indicates whether target exposure is enabled.
roi_meter_level		Int	1–100	Target exposure sensitivity.
face_cap_mode		Int	0–2	Capture mode. The options include 0 (default), 1 (quick capture), and 2 (scheduled capture).
face_cap_sensitivity		Int	1–10	Target capture sensitivity (in quick capture mode).
face_exposure_time		Int	1–60	Target exposure duration, in minutes.
time_capture		Int	3–300	Scheduled capture interval, in seconds.
send_face_img		Bool		Indicates whether target images are sent.

send_body_img		Bool		Indicates whether person images are sent.
send_panorama		Bool		Indicates whether full images are sent.
face_enhancement		Bool		Indicates whether image enhancement is enabled.
face_img_quality		Int	1–5	Quality of target images.
body_img_quality		Int	1–5	Quality of full-body images.
panorama_img_quality		Int	1–5	Full image quality.
send_face_num		Int	1–3	Number of sent target images.
send_body_num		Int	1–3	Number of sent person images.
area_list		JSON array	JSON object. A maximum of two elements are contained in this list, one is the detection area, and the other is the exclude area.	Area list. For details, see Table 16-1.

16.3.2 Setting Target Capture Parameters

API Description					
Function	This API is used to set target capture parameters.				
API URL	/API/Intelligent/PerimeterFace/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					

channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N		Indicates whether to enable target capture.
face_attr_switch		Bool	N		Indicates whether to enable target attribute recognition.
face_detect_mode		Int	N	0–2	Detection mode. The options include 0 (target detection), 1 (person detection), and 2 (both).
alarm_interval		Int	N	1–10	Alarm interval, in seconds.
face_sensitivity		Int	N	1–5	Target detection sensitivity.
body_sensitivity		Int	N	1–5	Person detection sensitivity.
min_pixel		Int	N	26–399	Minimum resolution of a target that can be detected in 1080p video images, in pixels.
max_pixel		Int	N	27–400	Maximum resolution of a target that can be detected in 1080p video images, in pixels.
face_exposure		Bool	N		Indicates whether to enable target exposure.
roi_meter_level		Int	N	1–100	Target exposure sensitivity.
face_cap_mode		Int	N	0–2	Capture mode. The options include 0 (default), 1 (quick capture), and 2 (scheduled capture).
face_cap_sensitivity		Int	N	1–10	Target capture sensitivity (in quick capture mode).
face_exposure_time		Int	N	1–60	Target exposure duration, in minutes.

time_capture		Int	N	3–300	Scheduled capture interval, in seconds.
send_face_img		Bool	N		Indicates whether to send target images.
send_body_img		Bool	N		Indicates whether to send person images.
send_panorama		Bool	N		Indicates whether to send full images.
face_enhancement		Bool	N		Indicates whether to enable image enhancement.
face_img_quality		Int	N	1–5	Quality of target images.
body_img_quality		Int	N	1–5	Quality of person images.
panorama_img_quality		Int	N	1–5	Full image quality.
send_face_num		Int	N	1–3	Number of sent target images.
send_body_num		Int	N	1–3	Number of sent person images.
area_list		JSON array	N	JSON object. A maximum of two elements are contained in this list, one is the detection area, and the other is the exclude area.	Area list. For details, see Table 16-1.

Output Parameters

Parameter	Description	Data Type	Value Range	Remarks
need_chk_face_quick_cmp		JSON array	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	Trustlist-based fast match must be set in the camera web system. Note that this function needs to be enabled in fast capture mode and disabled it in optimal capture mode. Incorrect setting will cause failure to use the target recognition function on a camera. (This parameter is returned only in this case.)
result		String	Maximum length: 256 bytes	If the operation is successful, success is

				returned. If the operation fails, other values (causes of failure) are returned.
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16.4 Algorithm Mode Switchover

16.4.1 Obtaining the Camera Algorithm Mode

API Description					
Function	This API is used to obtain the camera algorithm mode.				
API URL	/API/Intelligent/CameraModeConfig/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/CameraModeConfig/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON			Elements in the channel

		object		object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
camera_mode		String	"FaceCardMode" "MicroCardMode" "MixedWalkMode" "BehaviorAnalyseMode"	Algorithm mode: target checkpoint, micro checkpoint, full intelligence, and behavior analysis.

16.4.2 Setting the Camera Algorithm Mode

API Description					
Function	This API is used to set the camera algorithm mode.				
API URL	/API/Intelligent/CameraModeConfig/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.

Elements in the channel object					
camera_mode		String	Y	"FaceCardMode" "MicroCardMode" "MixedWalkMode" "BehaviorAnalyseMode"	Algorithm mode: target checkpoint, micro checkpoint, full intelligence, and behavior analysis.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.5 Global Behavior Analysis Configuration

16.5.1 Obtaining Values of Global Behavior Analysis Parameters

API Description					
Function	This API is used to obtain values of global behavior analysis parameters.				
API URL	/API/Intelligent/IGTAlarmConfig/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/IGTAlarmConfig/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number	The value must be in the format of CHn . The value of n cannot exceed the

				of channels varies depending on the capabilities of the device.	maximum number of channels of a device.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool			Indicates whether behavior analysis is enabled.
send_metadata_s witch		Bool			Indicates whether metadata is sent.
shadow_removal_ mode		String	"WeakShadow" "NormalShadow" "StrongShadow"		Shadow removal mode. The options include WeakShadow , NormalShadow , and StrongShadow .
detection_sensitiv ty		String	"LowSensitivity" "MiddleSensitivity" "HighSensitivity" "LowerSensitivity" "LowestSensitivity"		Target detection sensitivity. The options include LowSensitivity , MiddleSensitivity , HighSensitivity , LowerSensitivity , and LowestSensitivity .
background_upda te_rate		String	"Low", "Middle", "High"		Background update rate. The options include Low , Middle , and High .
min_target_size		JSON object			Minimum object resolution. For details, see Table

				16-4.
max_target_size		JSON object		Maximum object resolution. For details, see Table 16-4.
perimeter_enhance_switch		Bool		Indicates whether enhanced perimeter detection is enabled. (This parameter is available only when the cameras support this function.)
alert_inhibition_switch		Bool		Indicates whether alarm suppression is enabled. (This parameter is available only when the cameras support this function.)
alarm_pic_aiInfo_switch		Bool		Indicates whether information about intelligent analysis results is superimposed on the images that is captured upon alarm generation. (This parameter is available only when the cameras support this function.)

Table 16-4 Elements in the area_list array

Element in a single area object				
width		Int	5–200	Width.
height		Int	5–200	Height.

16.5.2 Setting Global Behavior Analysis Parameters

API Description	
Function	This API is used to set global behavior analysis parameters.
API URL	/API/Intelligent/IGTAlarmConfig/Set
Request Method	POST
Message ID	

Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N		Indicates whether to enable behavior analysis.
send_metadata_switch		Bool	N		Indicates whether to send metadata.
shadow_removal_mode		String	N	"WeakShadow" "NormalShadow" "StrongShadow"	Shadow removal mode. The options include WeakShadow , NormalShadow , and StrongShadow .
detection_sensitivity		String	N	"LowSensitivity" "MiddleSensitivity" "HighSensitivity" "LowerSensitivity" "LowestSensitivity"	Target detection sensitivity. The options include LowSensitivity , MiddleSensitivity , HighSensitivity , LowerSensitivity , and LowestSensitivity .
background_update_rate		String	N	"Low", "Middle", "High"	Background update rate. The options include Low , Middle , and High .
min_target_size		JSON object	N		Minimum object resolution. For details, see Table 16-4.
max_target_size		JSON object	N		Maximum object resolution. For details, see

					Table 16-4.
perimeter_enhance_switch		Bool	N		Indicates whether to enable enhanced perimeter detection. (This parameter is configurable only when the cameras support this function.)
alert_inhibition_switch		Bool	N		Indicates whether to enable alarm suppression. (This parameter is configurable only when the cameras support this function.)
alarm_pic_aiInfo_switch		Bool	N		Indicates whether to superimpose information about intelligent analysis results on the images that is captured upon alarm generation. (This parameter is configurable only when the cameras support this function.)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.	

16.6 Lens Blocking Detection

16.6.1 Obtaining Values of Lens Blocking Detection Parameters

API Description	
Function	This API is used to obtain values of lens blocking detection parameters.
API URL	/API/Intelligent/OcclusionDetection/Get
Request Method	POST
Message ID	
Description	URL used to obtain the parameter value range API used to obtain the parameter value range:

		/API/Intelligent/OcclusionDetection/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)			
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in a data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool		Indicates whether lens blocking detection is

				enabled. (ChannelConfig)
sensitivity		Int	1–5	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clock In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Audio file. (ChannelConfig)
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds.

				(ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking upon alarm occurrence is enabled for a channel. (AlarmConfig)

show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)
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16.6.2 Setting Lens Blocking Detection Parameters

API Description					
Function	This API is used to set lens blocking detection parameters.				
API URL	/API/Intelligent/OcclusionDetection/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON	N		Elements in the channel object are as

		object			follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable lens blocking detection. (ChannelConfig)
sensitivity		Int	N	1–5	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Audio file. (ChannelConfig)
audio_playback_ti		Int	N	1–50	Number of times an audio file is played.

mes					(ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence

					for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking upon alarm occurrence for a channel. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.7 Mask Detection

16.7.1 Obtaining Values of Mask Detection Parameters

API Description					
Function	This API is used to obtain values of mask detection parameters.				
API URL	/API/Intelligent/MaskAlarm/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/MaskAlarm/Range				
Input Parameters					
Parameter	Descri ption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data structu re.	object	Y		

Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
channel_max		Int		Total number of channels of the current device. This parameter is available only when the parameter value range is obtained.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.	
alarm_type		String	"None", "Exist", "Close"	Alarm type. The options include None (generated when a person not wearing a mask is detected), Exist (generated when a person wearing a mask is detected), and Close (disabled).	
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds.	
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2"	Alarm output channel. Alarm output can be triggered by the local system or camera (up to two channels).	

			The maximum number of alarm output sources varies depending on the capabilities of a device.	
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds.
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel.
record_channel		Array	CH1 to CHn	Alarm output channel. Indicates whether the channel alarm linkage is enabled.
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds.
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled.
show_message		Bool	Bool	Indicates whether the alarm ID is displayed.
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence.
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode.

16.7.2 Setting Mask Detection Parameters

API Description					
Function		This API is used to set mask detection parameters.			
API URL		/API/Intelligent/MaskAlarm/Set			
Request Method		POST			
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
alarm_type		String	N	"None", "Exist", "Close"	Alarm type.
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. The values are fixed options.
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1".. "IP_CHn->1" "IP_CH1->2".. "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. The values are fixed options.
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. The values are fixed options.
record_enable		Bool	N	Bool	Bool
record_channel		Array	N	CH1 to CHn	
post_recording		String		"30", "60", "120", "300"	Alarm output duration, in seconds. The values are fixed options.

snap_enable		Bool		Bool	Indicates whether to enable snapshot taking.
show_message		Bool		Bool	
send_email		Bool		Bool	Bool
full_screen		Bool		Bool	Bool
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.	

16.8 Target Recognition Model

16.8.1 Obtaining Values of Target Recognition Model Parameters

API Description					
Function	This API is used to obtain values of target recognition model parameters.				
API URL	/API/Intelligent/FaceRecognition/Get				
Request Method	POST				
Message ID					
Description	The current_module_version parameter specifies the unique version of the target recognition model that is currently used by the HWT-NVR800. You need to set this parameter to a specified value of FR_module_version, which refers to the version of the target recognition model that is currently used by cameras connected to the HWT-NVR800. Cameras using the specified model version can work. Cameras that use other versions of target recognition models will be invalid. After the parameters are set, you need to invoke this API again to obtain parameter values because the obtained parameter values may be different from the specified ones due to parameter setting validity verification. API used to obtain the parameter value range: /API/Intelligent/FaceRecognition/Range				
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
current_module_version		UINT	0–0xffffffff		Version of the target recognition model that is currently used.
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool			Indicates whether target recognition is enabled.
FR_module_version		UINT	0–0xffffffff		Version of the target recognition model, in UINT format. This parameter is read-only and cannot be transferred during parameter settings.
face_recognize_module		String	Maximum length: 48 bytes		Version of the target recognition model, in string format. This parameter is

				read-only and cannot be transferred during parameter setting.
face_detect_module		String	Maximum length: 48 bytes	Version of the target detection model, in string format. This parameter is read-only and cannot be transferred during parameter setting.
is_support_synergy		Bool		Indicates whether feature-based collaboration is supported. Only when this parameter is set to true can the feature_usage_mode parameter be available for a channel. This parameter is read-only.
feature_usage_mode		Int	0–1	Feature-based collaboration mode. The options include 0 (preferred mode) and 1 (access control mode). This parameter is available only when is_support_synergy is set to true .
face_usage_mode		Int	0–2	Mode in which the target recognition model is used. The options include 0 (preferred), 1 (fast), and 2 (scheduled). This parameter is read-only and available only when is_support_synergy is set to false .

16.8.2 Setting Target Recognition Model Parameters

API Description	
Function	This API is used to set target recognition model parameters.
API URL	/API/Intelligent/FaceRecognition/Set
Request Method	POST
Message ID	
Description	

Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
current_module_version		UINT	N	0–0xffffffff	Version of the target recognition model that is currently used.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N		Indicates whether to enable target recognition.
feature_usage_mode		Int	N		Feature-based collaboration mode. The options include 0 (preferred mode) and 1 (access control mode). This parameter is available only when is_support_synergy is set to true .
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
need_chk_face_quick_cmp		JSON array	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.		Trustlist-based fast match must be set in the camera web system. Note that this function needs to be enabled in fast capture mode and disabled it in optimal capture mode. Incorrect setting will cause failure to use the target

				recognition function on a camera. (This parameter is returned only in this case.)
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.9 Heat Map

16.9.1 Obtaining Values of Heat Map Parameters

API Description					
Function	This API is used to obtain values of heat map parameters.				
API URL	/API/Intelligent/HeatMapConfig/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/HeatMapConfig/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in a data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks

channel_info		JSON object		JSON information about the elements in the object is as follows.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool		Indicates whether to heat map is enabled.
meta_data_switch		Bool		Indicates whether metadata is sent.
area_list		JSON array	JSON object	Area list. For details, see Table 16-5.
alarm_plan_list		JSON array	A maximum seven elements are contained in the array, and each of them corresponds to a day in the week. The number of elements is determined by the number of requested elements. You can modify the parameters of an element as required.	Alert plan list. For details, see Table 16-7.

Table 16-5 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
index		Int	1–8	Area index.
point_num		Int	0–10	Number of vertices in an area. (This parameter is not required during setting.)

point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-6.
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Table 16-6 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.
y		Int	0–576	Y coordinate.

Table 16-7 Elements in the alarm_plan_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON array	A maximum of eight elements (JSON objects) can be contained in the array. That is, a maximum of eight time segments can be set for a day.	Single area object.
Elements in a single area object				
start_time		UINT	0–86400	Start time, which is the number of seconds from 00:00.
end_time		UINT	0–86400	End time, which is the number of seconds from 00:00.

16.9.2 Setting Heat Map Parameters

API Description	
Function	This API is used to set heat map parameters.
API URL	/API/Intelligent/HeatMapConfig/Set
Request Method	POST

Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable heat map.
meta_data_switch		Bool	N	Bool	Indicates whether to send metadata.
area_list		JSON array	N		Area list. For details, see Table 16-5.
alarm_plan_list		JSON array	N		Alert plan list. For details, see Table 16-7.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.10 Audio Diagnosis

16.10.1 Obtaining Values of Audio Diagnosis Parameters

API Description					
Function	This API is used to obtain values of audio diagnosis parameters.				
API URL	/API/Intelligent/SoundDetection/Get				
Request Method	POST				
Message ID					
Remarks	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/SoundDetection/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.

Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
exception_switch		Bool	Bool	Indicates whether audio anomaly detection is enabled. (ChannelConfig)
exception_sensitivity		String	"Low", "Middle", "High"	Audio anomaly detection sensitivity. (ChannelConfig)
exception_interval		Int	1–100	Interval between alarms generated upon detecting audio anomalies, in seconds. (ChannelConfig)
exception_audio_enable		Bool	Bool	Indicates whether an audio file is displayed upon audio anomaly occurrence. (ChannelConfig)
exception_audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not	Alarm audio file to be played upon audio anomaly occurrence. (ChannelConfig)

			Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
exception_audio_playback_times		Int	1–50	Number of times an audio file is played upon audio anomaly occurrence. (ChannelConfig)
exception_white_light_illuminator_enable		Bool	Bool	Indicates whether to enable visual alarms (light flashing) upon audio anomaly occurrence. (ChannelConfig)
exception_flash_duration		Int	2–60	Duration of light flashing after audio anomaly occurrence, in seconds. (ChannelConfig)
exception_white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Frequency of alarm flashing after audio anomaly occurrence. (ChannelConfig)
rise_switch		Bool	Bool	Indicates whether sudden volume increase detection is enabled. (ChannelConfig)
rise_sensitivity		Int	1–100	Sensitivity of sudden volume increase detection. (ChannelConfig)
rise_alarm_interval		Int	1–100	Interval between sudden volume increase alarms, in seconds. (ChannelConfig)
rise_audio_enable		Bool	Bool	Indicates whether an audio file is played when sudden volume increase is detected. (ChannelConfig)
rise_audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav",	Alarm audio file to be played when sudden volume increase is detected. (ChannelConfig)

			"Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
rise_audio_playback_times		Int	1–50	Number of times an audio file is played after sudden volume increase is detected. (ChannelConfig)
rise_white_light_illuminator_enable		Bool	Bool	Indicates whether light flashing is enabled after sudden volume decrease is detected. (ChannelConfig)
rise_flash_duration		Int	2–60	Duration of light flashing after sudden volume increase is detected, in seconds. (ChannelConfig)
rise_white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Frequency of light flashing after sudden volume increase is detected. (ChannelConfig)
decline_switch		Bool	Bool	Indicates whether sudden volume decrease detection is enabled. (ChannelConfig)
decline_sensitivity		Int	1–100	Sensitivity of sudden volume decrease detection. (ChannelConfig)

decline_alarm_interval		Int	1–100	Interval between alarms after sudden volume decrease is detected, in seconds. (ChannelConfig)
decline_audio_enable		Bool	Bool	Indicates whether an audio file is played when sudden volume decrease is detected. (ChannelConfig)
decline_audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Alarm audio file to be played when sudden volume decrease is detected. (ChannelConfig)
decline_audio_playback_times		Int	1–50	Number of times an audio file is played after sudden volume decrease is detected. (ChannelConfig)
decline_white_light_illuminator_enable		Bool	Bool	Indicates whether light flashing is enabled after sudden volume decrease is detected. (ChannelConfig)
decline_flash_duration		Int	2–60	Duration of light flashing after sudden volume decrease is detected, in

				seconds. (ChannelConfig)
decline_white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Frequency of light flashing after sudden volume decrease is detected. (ChannelConfig)
exception_alarm_plan_list		JSON array	JSON object	List of alert plans for audio anomaly detection. For details, see Table 6-1. (ChannelConfig)
rise_alarm_plan_list		JSON array	JSON object	List of alert plans for sudden volume increase detection. For details, see Table 6-1. (ChannelConfig)
decline_alarm_plan_list		JSON array	JSON object	List of alert plans for sudden volume decrease detection. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a

				channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking upon alarm occurrence is enabled for a channel. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan (related to audio anomalies, sudden volume increase, or sudden volume decrease) starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

16.10.2 Setting Audio Diagnosis Parameters

API Description					
Function	This API is used to set audio diagnosis parameters.				
API URL	/API/Intelligent/SoundDetection/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct	object	Y		

	ure.				
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
exception_switch		Bool		Bool	Indicates whether to enable audio anomaly detection. (ChannelConfig)
exception_sensitivity		String		"Low", "Middle", "High"	Audio anomaly detection sensitivity. (ChannelConfig)
exception_interval		Int	N	1–100	Interval between alarms generated upon detecting audio anomalies, in seconds. (ChannelConfig)
exception_audio_enable		Bool	N	Bool	Indicates whether an audio file is displayed upon audio anomaly occurrence. (ChannelConfig)
exception_audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park	Alarm audio file to be played upon audio anomaly occurrence. (ChannelConfig)

				Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
exception_audio_playback_times		Int	N	1–50	Number of times an audio file is played upon audio anomaly occurrence. (ChannelConfig)
exception_white_light_illuminator_enable		Bool	N	Bool	Indicates whether the visual alarm (light flashing) upon audio anomaly occurrence is enabled. (ChannelConfig)
exception_flash_duration		Int	N	2–60	Duration of light flashing after audio anomaly occurrence, in seconds. (ChannelConfig)
exception_white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Frequency of alarm flashing after audio anomaly occurrence. (ChannelConfig)
rise_switch		Bool		Bool	Indicates whether to enable sudden volume increase detection. (ChannelConfig)
rise_sensitivity		Int		1–100	Sensitivity of sudden volume increase detection. (ChannelConfig)

rise_alarm_interval		Int	N	1–100	Interval between sudden volume increase alarms, in seconds. (ChannelConfig)
rise_audio_enable		Bool	N	Bool	Indicates whether to play an audio file when sudden volume increase is detected. (ChannelConfig)
rise_audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Alarm audio file to be played when sudden volume increase is detected. (ChannelConfig)
rise_audio_playback_times		Int	N	1–50	Number of times an audio file is played after sudden volume increase is detected. (ChannelConfig)
rise_white_light_ill		Bool	N	Bool	Indicates whether to

uminator_enable					enable light flashing after sudden volume decrease is detected. (ChannelConfig)
rise_flash_duration		Int	N	2–60	Duration of light flashing after sudden volume increase is detected, in seconds. (ChannelConfig)
rise_white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Frequency of light flashing after sudden volume increase is detected. (ChannelConfig)
decline_switch		Bool		Bool	Indicates whether to enable sudden volume decrease detection. (ChannelConfig)
decline_sensitivity		Int		1–100	Sensitivity of sudden volume decrease detection. (ChannelConfig)
decline_alarm_interval		Int	N	1–100	Interval between alarms after sudden volume decrease is detected, in seconds. (ChannelConfig)
decline_audio_enable		Bool	N	Bool	Indicates whether to play an audio file when sudden volume decrease is detected. (ChannelConfig)
decline_audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav",	Alarm audio file to be played when sudden volume decrease is detected. (ChannelConfig)

				"Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
decline_audio_playback_times		Int	N	1–50	Number of times an audio file is played after sudden volume decrease is detected. (ChannelConfig)
decline_white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable light flashing after sudden volume decrease is detected. (ChannelConfig)
decline_flash_duration		Int	N	2–60	Duration of light flashing after sudden volume decrease is detected, in seconds. (ChannelConfig)
decline_white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Frequency of light flashing after sudden volume decrease is detected. (ChannelConfig)
exception_alarm_plan_list		JSON array	N	JSON object	List of alert plans for audio anomaly detection. For details, see Table 6-1. (ChannelConfig)
rise_alarm_plan_list		JSON array	N	JSON object	List of alert plans for sudden volume increase detection. For details, see Table 6-1. (ChannelConfig)
decline_alarm_plan_list		JSON array	N	JSON object	List of alert plans for sudden volume decrease detection. For details, see Table 6-1.

					(ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking upon alarm occurrence for a channel. (AlarmConfig)
Output Parameters					
Parameter	Desc ription	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is

				returned. If the operation fails, other values (causes of failure) are returned.
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16.11 Crossing-Line Head Counting

16.11.1 Obtaining Values of Crossing-Line Head Counting Parameters

API Description					
Function	This API is used to obtain values of crossing-line head counting parameters.				
API URL	/API/Intelligent/CrossCounting/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/CrossCounting/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr	Data	Value Range		Remarks

	Option	Type		
channel_info		JSON object		JSON information about the elements in the object is as follows.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool	Indicates whether crossing-line head counting is enabled. (ChannelConfig)
line_list		JSON array	JSON object	Detection line list. For details, see Table 16-8. (ChannelConfig)
area_list		JSON array	JSON object	Area list. For details, see Table 16-9. (ChannelConfig)
per_report_switch		Bool	Bool	Indicates whether people count is periodically reported. (ChannelConfig)
report_interval		Int	1–600	Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_human_number		Int	1–65536	Alarm count. (ChannelConfig)
clear_at_zero		Bool	Bool	Indicates whether data is cleared at 00:00. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav",	Audio file. (ChannelConfig)

			"Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)

show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

Table 16-8 Elements in the line_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single detection line object.
Elements in a single detection line object				
rule_type		String	"A->B", "B->A"	Crossing direction.
x1		Int	0–704	X1 coordinate.
y1		Int	0–576	Y1 coordinate.
x2		Int	0–704	X2 coordinate.
y2		Int	0–576	Y2 coordinate.

Table 16-9 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
point_num		Int	4	Number of vertices in an area. (This parameter is not required during setting.)
point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-10.

Table 16-10 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.

y		Int	0–576	Y coordinate.
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16.11.2 Setting Crossing-Line Head Counting Parameters

API Description					
Function	This API is used to set crossing-line head counting parameters.				
API URL	/API/Intelligent/CrossCounting/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in a CH1-CHn object					
switch		Bool	N	Bool	Indicates whether to enable crossing-line head counting. (ChannelConfig)
line_list		JSON array		JSON object	Detection line list. (ChannelConfig) For details, see Table 16-8.
area_list		JSON		JSON object	Area list. For details, see

		array			Table 16-9. (ChannelConfig)
per_report_switch		Bool		Bool	Indicates whether to periodically report data. (ChannelConfig)
report_interval		Int		1–600	Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int		1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_human_number		Int		1–65536	Alarm count. (ChannelConfig)
clear_at_zero		Bool		Bool	Indicates whether to clear data at 00:00. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort	Audio file. (ChannelConfig)

				Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).

latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.11.3 Clearing Crossing-Line Head Counting Data

API Description					
Function	This API is used to clear crossing-line head counting data.				
API URL	/API/Intelligent/CrossCounting/Clear				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		

Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.12 Queue Length Detection

16.12.1 Obtaining Values of Queue Length Detection Parameters

API Description					
Function	This API is used to obtain values of queue length detection parameters.				
API URL	/API/Intelligent/QueueDetect/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/QueueDetect/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					

channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool		Indicates whether queue length detection is enabled. (ChannelConfig)
area_list		JSON array	JSON object		Area list. For details, see Table 16-9. (ChannelConfig)
per_report_switch		Bool	Bool		Indicates whether data is periodically reported. (ChannelConfig)
report_interval		Int	1–600		Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int	1–256		Alarm interval, in seconds. (ChannelConfig)

stay_time		Int	1-3600	Duration of servicing a person, in seconds. (ChannelConfig)
alarm_threshold		Int	1–1000	Alarm threshold, in minutes. (ChannelConfig)
sensitivity		Int	1–5	Detection sensitivity. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Audio file. (ChannelConfig)
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in

				seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed.

				The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)
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16.12.2 Setting Queue Length Detection Parameters

API Description					
Function	This API is used to set queue length detection parameters.				
API URL	/API/Intelligent/QueueDetect/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to

					enable queue length detection. (ChannelConfig)
area_list		JSON array		JSON object	Area list. For details, see Table 16-9. (ChannelConfig)
per_report_switch		Bool		Bool	Indicates whether to periodically report data. (ChannelConfig)
report_interval		Int		1–600	Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int		1–256	Alarm interval, in seconds. (ChannelConfig)
stay_time		Int		1-3600	Duration of servicing a person, in seconds. (ChannelConfig)
alarm_threshold		Int		1–1000	Alarm threshold, in minutes. (ChannelConfig)
sensitivity		Int		1–5	Detection sensitivity. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav",	Audio file. (ChannelConfig)

				"Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1"	Alarm output channel. (AlarmConfig) Alarm output can be

				"IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Description
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.13 Crowd Density Detection

16.13.1 Obtaining Values of Crowd Density Detection Parameters

API Description	
Function	This API is used to obtain values of crowd density detection parameters.
API URL	/API/Intelligent/CrowdDensity/Get
Request Method	POST
Message ID	
Description	URL used to obtain the parameter value range

	API used to obtain the parameter value range: /API/Intelligent/CrowdDensity/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool		Indicates whether crowd

				density detection is enabled. (ChannelConfig)
area_list		JSON array	JSON object	Area list. For details, see Table 16-9. (ChannelConfig)
per_report_switch		Bool	Bool	Indicates whether data is periodically reported. (ChannelConfig)
report_interval		Int	1–600	Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_threshold		Int	1–50	Alarm threshold, in number of persons. (ChannelConfig)
sensitivity		Int	1–5	Detection sensitivity. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav",	Audio file. (ChannelConfig)

			"Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)

record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

16.13.2 Setting Crowd Density Detection Parameters

API Description					
Function	This API is used to set crowd density detection parameters.				
API URL	/API/Intelligent/CrowdDensity/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.

channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable crowd density detection. (ChannelConfig)
area_list		JSON array		JSON object	Area list. For details, see Table 16-9. (ChannelConfig)
per_report_switch		Bool		Bool	Indicates whether to periodically report data. (ChannelConfig)
report_interval		Int		1–600	Data reporting interval, in seconds. (ChannelConfig)
alarm_interval		Int		1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_threshold		Int		1–50	Alarm threshold, in number of persons. (ChannelConfig)
sensitivity		Int		1–5	Detection sensitivity. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav",	Audio file. (ChannelConfig)

				"Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)

send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.14 Third-Party Algorithm Alarms

16.14.1 Obtaining Values of Third-Party Algorithm Alarm Parameters

API Description	
Function	This API is used to obtain values of third-party algorithm alarm parameters.

API URL	/API/Intelligent/ThirdPartyAlgorithm/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/ThirdPartyAlgorithm/Range				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
channel_info		JSON object		JSON information about the elements in the object is as follows.	
channel_max		Int		Total number of channels of the current device. This parameter is available only when the parameter value range is obtained.	
Elements in the channel_info object					
CH1-CHn		JSON object		Elements in the channel object are as follows.	
Elements in the channel object					
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for	

				digital channels. Note: This parameter is unavailable when the channel is online.
alarm_type_list		JSON array	Each element in the array is of the JSON object type, which identifies an algorithm with its status (enabled or disabled).	This parameter is used only to obtain third-party algorithm parameters and cannot be transferred in the request. For details, see Table 16-11.
current_alarm_type		String	Maximum length: 63 bytes	Current algorithm type. You can set the value to one of the algorithms types in alarm_type_list .
current_switch_state		Bool		Indicates whether to enable the current algorithm. You can obtain the value but cannot change it.
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds.
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds.
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel.
record_channel		Array	CH1 to CHn	Recording channel. Indicates whether the channel alarm linkage is enabled.
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds.
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled.
show_message		Bool	Bool	Indicates whether the alarm ID is displayed.
send_email		Bool	Bool	Indicates whether emails are sent

				upon alarm occurrence.
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode.

Table 16-11 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Elements in a single object are as follows.
Elements in a single vertex object				
type_name		String	Maximum length: 63 bytes	Algorithm type name.
type_switch_state		Bool		Indicates whether the specified type of algorithms is enabled.

16.14.2 Setting Third-Party Algorithm Alarm Parameters

API Description					
Function	This API is used to set third-party algorithm alarm parameters.				
API URL	/API/Intelligent/ThirdPartyAlgorithm/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON	Y		JSON information about the elements in

		object			the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
current_alarm_type		String	N	Maximum length: 63 bytes	Type of the current third-party algorithm.
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. The values are fixed options.
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1".. "IP_CHn->1" "IP_CH1->2".. "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. The values are fixed options.
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. The values are fixed options.
record_enable		Bool	N	Bool	Bool
record_channel		Array	N	CH1 to CHn	
post_recording		String		"30", "60", "120", "300"	Video recording delay, in seconds. The values are fixed options.
snap_enable		Bool		Bool	Indicates whether to enable snapshot taking.
show_message		Bool		Bool	
send_email		Bool		Bool	Bool
full_screen		Bool		Bool	Bool
Output Parameters					
Parameter	Desc	Data Type	Value Range	Remarks	

	ri tio n			
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.15 Regular Customer Detection

16.15.1 Obtaining Values of Regular Customer Detection Parameters

API Description					
Function	This API is used to obtain values of regular customer detection parameters.				
API URL	/API/Intelligent/RepeatVisitorAlarm/Get				
Request Method	POST				
Message ID					
Description	API used to obtain the parameter value range: /API/Intelligent/RepeatVisitorAlarm/Range				
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
alarm_switch		Bool	Bool		Indicates whether the alarm function is enabled.
frequency_interval		Int	1-86400		Interval between alarm sounds, in seconds.
frequency_threshold_value		Int	2-20		Alarm frequency threshold.
cycle		Int	1-90		Alarm statistical period, in days.
similarity		Int	1-100		Target match rate.
max_statistics		Int	1-xxx		Maximum number of persons that can be detected. The value varies depending on the number of disk trays. HWT-NVR800 with one disk tray:

				10000 HWT-NVR800 with 2 disk trays: 15000 HWT-NVR800 with 4, 8, or 16 disk trays: 50000
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds.
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds.
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel.
alarm_channel		Array	CH1 to CHn	Alarm output channel. Indicates whether the channel alarm linkage is enabled.
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds.
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled.
show_message		Bool	Bool	Indicates whether the alarm ID is displayed.
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence.
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode.
age_filter_switch		Bool	Bool	Indicates whether alarms generated when minors are detected are filtered out.
alarm_min_age		Int	1–18	Minimum age of regular customers that will trigger alarms.
group_info_list		JSON array	Each element in the array is of the object type, which contains	Target lists. For details, see Table 16-12.

			the information about a target list.	
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Table 16-12 Elements in a single object

Elements in a single object				
name		String	Maximum length: 127 bytes	Target list name. (This parameter is invalid when you set regular customer detection parameters.)
group_id		UINT	1–0xffffffff	Target list ID, which is mandatory when you set target list parameters. You must select an ID from the list you have obtained during the settings. Otherwise, a message indicating that the ID is invalid will be returned. The value uniquely identifies a target list.
is_selected		Bool	Bool	Indicates whether the target list is used, which is mandatory when you set target list parameters. The options include true (yes) and false (no). (You can set the parameter to true for a maximum of 16 lists, that is, a maximum of 16 target lists can be used.)

16.15.2 Setting Regular Customer Detection Parameters

API Description	
Function	This API is used to set regular customer detection parameters.
API URL	/API/Intelligent/RepeatVisitorAlarm/Set
Request Method	POST

Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
alarm_switch		Bool	N	Bool	Indicates whether to enable the alarm function.
frequency_interval		Int	N	1–86400	Interval between alarm sounds, in seconds.
frequency_threshold_value		Int	N	2–20	Alarm frequency threshold.
cycle		Int	N	1–90	Alarm statistical period, in days.
similarity		Int	N	1–100	Target match rate.
max_statistics		Int	N	1–xxx	Maximum number of persons that can be detected. The value varies depending on the number of disk trays. HWT-NVR800 with one disk tray: 10000 HWT-NVR800 with 2 disk trays: 15000 HWT-NVR800 with 4, 8, or 16 disk trays : 50000
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. The values are fixed options.
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1".. "IP_CHn->1" "IP_CH1->2".. "IP_CHn->2" The maximum number of	Alarm output channel. The values are fixed options.

				alarm output sources varies depending on the capabilities of a device.	
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. The values are fixed options.
record_enable		Bool	N	Bool	Bool
alarm_channel		Array	N	CH1 to CHn	
post_recording		String		"30", "60", "120", "300"	Video recording delay, in seconds. The values are fixed options.
snap_enable		Bool		Bool	Indicates whether to enable snapshot taking.
show_message		Bool		Bool	
send_email		Bool		Bool	Bool
full_screen		Bool		Bool	Bool
age_filter_switch		Bool		Bool	Indicates whether to filter out alarms generated when minors are detected.
alarm_min_age		Int		1–18	Minimum age of regular customers that will trigger alarms.
group_info_list		JSON array		Each element in the array is of the object type, which contains the information about a target list.	Target lists. For details, see Table 16-12.
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.	

16.16 Object Classification

16.16.1 Obtaining Values of Object Classification Parameters

API Description					
Function	This API is used to obtain values of object classification parameters.				
API URL	/API/Intelligent/VehicleDetect/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/VehicleDetect/Range				
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					

status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool	Indicates whether object classification is enabled.
area_list		JSON array	The parameter is a JSON object. A maximum of one detection area can be drawn.	Area list. For details, see Table 16-13.
bright_compensation_switch		Bool	Bool	Indicates whether license plate brightness compensation is enabled.
compensation_level		Int	1–100	License plate brightness compensation level.
priority_province_or_city		String	None or the single-character abbreviations of Chinese provinces and municipalities.	Province or municipalities to be preferentially detected. The options include fixed and only one value can be transferred.

Table 16-13 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
area_name		String	Maximum length: 32 bytes	Detection area name.
point_num		Int		Number of vertices in an area. (This parameter is not required during setting.)
point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-14.

Table 16-14 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.
y		Int	0–576	Y coordinate.

16.16.2 Setting Object Classification Parameters

API Description					
Function	This API is used to set object classification parameters.				
API URL	/API/Intelligent/VehicleDetect/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable object

					classification.
area_list		JSON array	N	The parameter is a JSON object. A maximum of one detection area can be drawn.	Area list. For details, see Table 16-13.
bright_compensation_switch		Bool	N	Bool	Indicates whether to enable license plate brightness compensation.
compensation_level		Int	N	1–100	License plate brightness compensation level. (This parameter is valid only when bright_compensation_switch is set to true .)
priority_province_or_city		String	N	None or the single-character abbreviations of Chinese provinces and municipalities.	Province or municipalities to be preferentially detected. The options include fixed and only one value can be transferred.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.17 Motor Vehicle Detection

16.17.1 Obtaining Values of Motor Vehicle Detection Parameters

API Description	
Function	This API is used to obtain values of motor vehicle detection parameters.
API URL	/API/Intelligent/VehicleParameter/Get
Request Method	POST
Message ID	
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/VehicleParameter/Range

Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool		Indicates whether motor vehicle detection is enabled.
area_list		JSON array	The parameter is a JSON object. A maximum of one detection area can be drawn.		Area list. For details, see Table 16-13.
bright_compensati on_switch		Bool	Bool		Indicates whether license plate brightness compensation is enabled.

compensation_level		Int	1–100	License plate brightness compensation level.
priority_province_or_city		String	None or the single-character abbreviations of Chinese provinces and municipalities.	Province or municipalities to be preferentially detected. The options include fixed and only one value can be transferred.

16.17.2 Setting Motor Vehicle Detection Parameters

API Description					
Function	This API is used to set motor vehicle detection parameters.				
API URL	/API/Intelligent/VehicleParameter/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable motor vehicle detection.
area_list		JSON array		The parameter is a JSON object. A maximum of one	Area list. For details, see Table 16-13.

				detection area can be drawn.	
bright_compensation_switch		Bool		Bool	Indicates whether to enable license plate brightness compensation.
compensation_level		Int		1–100	License plate brightness compensation level. (This parameter is valid only when bright_compensation_switch is set to true .)
priority_province_or_city		String		None or the single-character abbreviations of Chinese provinces and municipalities.	Province or municipalities to be preferentially detected. The options include fixed and only one value can be transferred.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.18 Parking Violation Detection

16.18.1 Obtaining Values of Parking Violation Detection Parameters

API Description	
Function	This API is used to obtain values of parking violation detection parameters.
API URL	/API/Intelligent/VehicleIllegalAlarm/Get
Request Method	POST
Message ID	
Description	URL used to obtain the parameter value range API used to obtain the parameter value range: /API/Intelligent/VehicleIllegalAlarm/Range (The required parameter value range can be obtained only when the page_type parameter is transferred.)
Input Parameters	

Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
switch		Bool	Bool		Indicates whether parking violation detection is enabled. (ChannelConfig)
area_list		JSON array	The parameter is a JSON object. A maximum of four detection area can be drawn.		Area list. For details, see Table 16-15. (ChannelConfig)

sensitivity		Int	1–100	Detection sensitivity. (ChannelConfig)
illegal_parking_trigger_time		Int	10–600	Parking duration that triggers parking violation detection, in seconds. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)

show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)
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Table 16-15 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
index		Int	1–4	Area index.
point_num		Int	0–10	Number of vertices in an area. (This parameter is not required during setting.)
point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-16.

Table 16-16 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.
y		Int	0–576	Y coordinate.

16.18.2 Setting Parking Violation Detection Parameters

API Description					
Function	This API is used to set parking violation detection parameters.				
API URL	/API/Intelligent/VehicleIllegalAlarm/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
switch		Bool	N	Bool	Indicates whether to enable parking violation detection. (ChannelConfig)
area_list		JSON array		The parameter is a JSON object. A maximum of four detection area can be drawn.	Area list. For details, see Table 16-15. (ChannelConfig)
sensitivity		Int		1–100	Detection sensitivity. (ChannelConfig)

illegal_parking_trigger_time		Int		10–600	Parking duration that triggers parking violation detection, in seconds. (ChannelConfig)
alarm_interval		Int		1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0","1","3","5","10","20","40","60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30","60","120","300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking.

				(AlarmConfig)
Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.19 Abandoned Object Detection

16.19.1 Obtaining Values of Abandoned Object Detection Parameters

API Description					
Function	This API is used to obtain values of abandoned object detection parameters.				
API URL	/API/Intelligent/ItemLeave/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range /API/Intelligent/ItemLeave/Range (Used to obtain the parameter value range only. The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.

page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
area_list		JSON array	JSON object		Area list. For details, see Table 16-1. (ChannelConfig)
sensitivity		Int	1–100		Detection sensitivity. (ChannelConfig)
alarm_interval		Int	1–256		Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	"Center", "Bottom", "Top"		Recognition mode. (ChannelConfig)
tolerance_time		Int	10–300		Alarm tolerance time, in seconds. (ChannelConfig)
audio_enable		Bool	Bool		Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav",		Audio file. (ChannelConfig)

			"Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clock In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed.

				(AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

16.19.2 Setting Abandoned Object Detection Parameters

API Description	
Function	This API is used to set abandoned object detection parameters.

API URL	/API/Intelligent/ItemLeave/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
area_list		JSON array	N	JSON object	Area list. For details, see Table 16-1. (ChannelConfig)
sensitivity		Int	N	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	N	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
tolerance_time		Int	N	10–300	Alarm tolerance time, in seconds. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms.

					(ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Audio file. (ChannelConfig)
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium",	Light flashing frequency. (ChannelConfig)

				"High", "Steady on"	
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)

Output Parameters				
Parameter	Description	Data Type	Value Range	Remarks
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.20 Removed Object Detection

16.20.1 Obtaining Values of Removed Object Detection Parameters

API Description					
Function	This API is used to obtain values of removed object detection parameters.				
API URL	/API/Intelligent/ItemMove/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range /API/Intelligent/ItemMove/Range (Used to obtain the parameter value range only. The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig",	This parameter is used to distinguish the channel

				"AlarmConfig"	configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
area_list		JSON array	JSON object		Area list. For details, see Table 16-1. (ChannelConfig)
sensitivity		Int	1–100		Detection sensitivity. (ChannelConfig)
alarm_interval		Int	1–256		Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	"Center", "Bottom", "Top"		Recognition mode. (ChannelConfig)
audio_enable		Bool	Bool		Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav",		Audio file. (ChannelConfig)

			"Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)

send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

16.20.2 Setting Removed Object Detection Parameters

API Description	
Function	This API is used to set removed object detection parameters.
API URL	/API/Intelligent/ItemMove/Set
Request Method	POST
Message ID	

Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.
Elements in the channel object					
area_list		JSON array	N	JSON object	Area list. For details, see Table 16-1. (ChannelConfig)
sensitivity		Int	N	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	N	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of	Audio file. (ChannelConfig)

				People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clock In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)

show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range	Remarks	
result		String	Maximum length: 256 bytes	If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.	

16.21 Electric Bike Detection

16.21.1 Obtaining Values of Electric Bike Detection Parameters

API Description					
Function	This API is used to obtain values of electric bike detection parameters.				
API URL	/API/Intelligent/ElectroMobile/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range /API/Intelligent/ElectroMobile/Range (Used to obtain the parameter value range only. The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Descr iption	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object

				is as follows.
Elements in the channel_info object				
CH1-CHn		JSON object		Elements in the channel object are as follows.
Elements in the channel object				
status		String	"Offline", "Online"	Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
area_list		JSON array	JSON object	Area list. For details, see Table 16-17. (ChannelConfig)
sensitivity		Int	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav",	Audio file. (ChannelConfig)

			"Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled. (ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).

			a device.	
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)
show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)

Table 16-17 Elements in the area_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single area object.
Elements in a single area object				
enable		Bool	Bool	Indicates whether a detection area is enabled.
point_num		Int	0–10	Number of vertices in an area. (This parameter is not required during setting.)
point_list		JSON array	JSON object	Vertex list of a detection area. For details, see Table 16-18.

Table 16-18 Elements in the point_list array

Parameter	Description	Data Type	Value Range	Remarks
		JSON object		Single vertex object.
Elements in a single vertex object				
x		Int	0–704	X coordinate.
y		Int	0–576	Y coordinate.

16.21.2 Setting Electric Bike Detection Parameters

API Description					
Function	This API is used to set electric bike detection parameters.				
API URL	/API/Intelligent/ElectroMobile/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Descr iption	Data Type	Mand atory (Y/N)	Value Range	Server Verification Rule
data	Data struct ure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.

Elements in the channel object					
area_list		JSON array	N	JSON object	Area list. For details, see Table 16-17. (ChannelConfig)
sensitivity		Int	N	1–100	Detection sensitivity. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
recognize_mode		String	N	"Center", "Bottom", "Top"	Recognition mode. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clock In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in	Audio file. (ChannelConfig)

				Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to

					enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

16.22 Off-Duty Detection

16.22.1 Obtaining Values of Off-Duty Detection Parameters

API Description					
Function	This API is used to obtain values of off-duty detection parameters.				
API URL	/API/Intelligent/LeaveDetect/Get				
Request Method	POST				
Message ID					
Description	URL used to obtain the parameter value range /API/Intelligent/LeaveDetect/Range (Used to obtain the parameter value range only. The required parameter value range can be obtained only when the page_type parameter is transferred.)				
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule

data	Data structure.	object	Y		
Elements in the data object					
channel		JSON array	N	CH1 to CHn The maximum number of channels varies depending on the capabilities of the device.	The value must be in the format of CHn . The value of n cannot exceed the maximum number of channels of a device.
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
Output Parameters					
Parameter	Description	Data Type	Value Range		Remarks
channel_info		JSON object			JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object			Elements in the channel object are as follows.
Elements in the channel object					
status		String	"Offline", "Online"		Channel status (online or offline). This parameter is available only for digital channels. Note: This parameter is unavailable when the channel is online.
area_list		JSON array	JSON object		Area list. For details, see Table 16-9. (ChannelConfig)
switch		Bool	Bool		Indicates whether off-duty detection is enabled. (ChannelConfig)
alarm_interval		Int	1–256		Alarm interval, in seconds. (ChannelConfig)

alarm_osd_enable		Bool	Bool	Indicates whether metadata is superimposed on snapshots taken upon alarm occurrence. (ChannelConfig)
on_duty_num		Int	1–5	Number of on-duty persons. (ChannelConfig)
leave_time		Int	1–600	Off-duty duration, in seconds. (ChannelConfig)
audio_enable		Bool	Bool	Indicates whether the audible alarm is enabled. (ChannelConfig)
audio_file		String	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav", "Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	Audio file. (ChannelConfig)
audio_playback_times		Int	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	Bool	Indicates whether the visual alarm (light flashing) is enabled.

				(ChannelConfig)
flash_duration		Int	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	Bool	Indicates whether the alarm ID is displayed. (AlarmConfig)
full_screen		Bool	Bool	Indicates whether alarms are displayed in full screen mode. (AlarmConfig)
send_email		Bool	Bool	Indicates whether emails are sent upon alarm occurrence. (AlarmConfig)
alarm_out		Array	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2" The maximum number of alarm output sources varies depending on the capabilities of a device.	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to two channels).
latch_time		String	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	Bool	Indicates whether video recording upon alarm occurrence is enabled for a channel. (AlarmConfig)
record_channel		Array	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	Bool	Indicates whether snapshot taking is enabled. (AlarmConfig)

show_start_date		Bool	This parameter is read-only and does not need to be transferred.	Indicates whether the date is displayed when the alert plan starts to be executed. The options include true (yes) and false (no). If this parameter is set to false , start_date becomes invalid and its original value will be transferred when you set the alert plan. (ChannelConfig)
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16.22.2 Setting Off-Duty Detection Parameters

API Description					
Function	This API is used to set off-duty detection parameters.				
API URL	/API/Intelligent/LeaveDetect/Set				
Request Method	POST				
Message ID					
Description					
Input Parameters					
Parameter	Description	Data Type	Mandatory (Y/N)	Value Range	Server Verification Rule
data	Data structure.	object	Y		
Elements in the data object					
page_type		String	Y	"ChannelConfig", "AlarmConfig"	This parameter is used to distinguish the channel configuration page from the alarm configuration page.
channel_info		JSON object	Y		JSON information about the elements in the object is as follows.
Elements in the channel_info object					
CH1-CHn		JSON object	N		Elements in the channel object are as follows.

Elements in the channel object					
area_list		JSON array	N	JSON object	Area list. For details, see Table 16-9. (ChannelConfig)
switch		Bool	N	Bool	Indicates whether to enable off-duty detection. (ChannelConfig)
alarm_interval		Int	N	1–256	Alarm interval, in seconds. (ChannelConfig)
alarm_osd_enable		Bool	N	Bool	Indicates whether to superimpose metadata on snapshots taken upon alarm occurrence. (ChannelConfig)
on_duty_num		Int	N	1–5	Number of on-duty persons. (ChannelConfig)
leave_time		Int	N	1–600	Off-duty duration, in seconds. (ChannelConfig)
audio_enable		Bool	N	Bool	Indicates whether to enable audible alarms. (ChannelConfig)
audio_file		String	N	"Alarm Generated.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Leave the Protected Area.wav", "Please Wear a Mask.wav", "Alarm Sound.wav", "Do Not Park Here.wav", "Excessive Number of People.wav", "Please Wear a Mask.wav", "Please Leave the Protected Area.wav", "Alarm Generated.wav", "Electric Bike Entry Not Allowed.wav",	Audio file. (ChannelConfig)

				"Welcome.wav", "Clocked In.wav", "Please Sort Waste.wav", "Off-Duty Conduct.wav", "Deep Water.wav", "Waste Sorting in Specified Period.wav"	
audio_playback_times		Int	N	1–50	Number of times an audio file is played. (ChannelConfig)
white_light_illuminator_enable		Bool	N	Bool	Indicates whether to enable visual alarms (light flashing). (ChannelConfig)
flash_duration		Int	N	2–60	Light flashing duration, in seconds. (ChannelConfig)
white_light_illuminator		String	N	"Low", "Medium", "High", "Steady on"	Light flashing frequency. (ChannelConfig)
alarm_plan_list		JSON array	N	JSON object	Alert plan list. For details, see Table 6-1. (ChannelConfig)
buzzer		String	N	"0", "1", "3", "5", "10", "20", "40", "60"	Beep duration of the buzzer, in seconds. (AlarmConfig)
show_message		Bool	N	Bool	Indicates whether to display the alarm ID. (AlarmConfig)
full_screen		Bool	N	Bool	Indicates whether to display alarms in full screen mode. (AlarmConfig)
send_email		Bool	N	Bool	Indicates whether to send emails upon alarm occurrence. (AlarmConfig)
alarm_out		Array	N	"Local->1"... "Local->n" "IP_CH1->1"... "IP_CHn->1" "IP_CH1->2"... "IP_CHn->2"	Alarm output channel. (AlarmConfig) Alarm output can be triggered by the local system or camera (up to

				The maximum number of alarm output sources varies depending on the capabilities of a device.	two channels).
latch_time		String	N	"10", "20", "40", "60"	Alarm output duration, in seconds. (AlarmConfig)
record_enable		Bool	N	Bool	Indicates whether to enable video recording upon alarm occurrence for a channel. (AlarmConfig)
record_channel		Array	N	CH1 to CHn	Recording channel. (AlarmConfig)
post_recording		String	N	"30", "60", "120", "300"	Video recording delay, in seconds. (AlarmConfig)
snap_enable		Bool	N	Bool	Indicates whether to enable snapshot taking. (AlarmConfig)
Output Parameters					
Parameter	Description	Data Type	Value Range		Description
result		String	Maximum length: 256 bytes		If the operation is successful, success is returned. If the operation fails, other values (causes of failure) are returned.

17 Search in Broadcast Mode

API Description	
Function	This API is used to search for HWT-NVR800s in a LAN.
Request Method	Search message (MsgHead) sent to port 9333 in the broadcast address 255.255.255.255.
MsgHead	<pre> Typedefstruct { unsignedchar version: 2 bit; /* Message header version. The value is fixed to 0.*/ unsignedchar extsize: 6 bit; /* Extension size of the message header. The value is fixed to 0. */ unsignedchar reserved; unsignedshort ackflag; /* Ignored. */ </pre>

	<pre> unsignedshort msgtype; /* Message type. The value is fixed to 12. */ unsignedshort msgsubtype; /* Message sub-type, which can be ignored. */ unsignedint msgsize; /* Size of the data following the message header. The value is fixed to 0. */ unsignedint seq; /* Message sequence number, which uniquely identifies a message at a time. It is created by the requester and the original value of this parameter is returned in the response. */ }MsgHead; </pre> The size of MsgHead is 16 bytes.
MsgBody	<pre> typedefstruct { char ip[16]; //IP charreserve[104]; unsignedint ieport; /* Internet Explorer port. */ char reserv1[40]; charDevVer[36]; /* Device version. */ charreserve1[40]; char macaddr[36]; /* Device MAC address. */ charreserve2[4]; charreserved[170]; } MsgBody; </pre> The size of MsgBody is 448 bytes.
Description	The response message is a combination of MsgHead and MsgBody .

18 FAQs

18.1 Login

Q: When I invoke **/API/Web/Login** and **/API/Login/Range** on Postman, the error code indicating "Error:Self signed certificate in certificate chain" is displayed. What should I do?

A: Disable the SSL verification function in settings on Postman.

Why is no value returned when the **/API/Web/Login** API is invoked?

A: The login follows standard digest authentication process. If the API is invoked for the first time, **401** is returned. After that, digest authentication is required.

Q: The error code **first_login** is returned after the login API is invoked, asking me to change the password for the **apiadmin** user before login. However, I have changed the password to a new one for the user name on my device. Why do I have to change the password again after invoking the API for login?

A: The **apiadmin** user is available only on the internal platform. On the LDU, you can only use the **admin** user to register an **apiadmin** user or set the password for it. Note that after you set a

password for a common user, for example, **apiadmin** using the **admin** user on the LDU, the next login of the common user will be considered as the first login. This is because the **admin** user cannot manage the passwords of common users.

Q: When I invoke **/API/Login/TransKey/Get**, why does the request fail no matter whether I set the data type to **x-www-form-urlencoded** or **JSON**?

A: Set the data type to **JSON** in the request when invoking the API and store both request and response data in the **data** parameter. For details about the format, see section 1.3 "Request Description."

Q: Why is **no_permission** returned after **/API/Login/TransKey/Get** is invoked?

A: This API is used only for the first login. If you have set a password for the **admin** account, use **/API/Maintenance/TransKey/Get** instead.

Q: Why **login_failed_or_block** is returned upon login although my user name and password are correct? Will the login to the LDU fail if I have used the same user name to log in to the LDU or camera web system?

A: Check whether the login with this account has failed or the login has failed for more than five consecutive times. This error code is returned either when the authentication fails or when your account is locked for 3 minutes due to more than five login failures. Besides, an account can be used for login to the LDU and API invoking at the same time.

Q: How can I encrypt a password? How can I decode a hex string?

A: RSA encryption is used together with Optimal Asymmetric Encryption Padding (OAEP). For example, **Int RSA_public_encrypt(Int flen, const unsigned char *from, unsigned char *to, RSA *rsa, Int padding)**. For details, see section 11.1 "Login."

Q: Is there any process provided for first login and password change upon first login?

A: Yes. For details, see section 11.1 "Login."

FAQs

Q: Why is it not feasible to use a developed web application to send requests to the device through HTTP or HTTPS APIs?

A: Due to the cross-domain resource sharing (CORS) policy of the device server, no cross-domain request can be made on the browser. For details about CORS, visit <https://developer.mozilla.org/en-US/docs/Web/HTTP/CORS>. Only the web page program that uses the same origin policy as the device server can access the device. For details about Same-origin policy, https://developer.mozilla.org/en-US/docs/Web/Security/Same-origin_policy. You are advised to use other methods, instead of web application, to send requests through HTTP or HTTPS APIs.

18.2 Cameras

Q: How can I query the list of cameras connected to an HWT-NVR800, connect cameras to the HWT-NVR800, delete cameras from the HWT-NVR800?

A: API used to query the list of cameras connected to an HWT-NVR800: **/API/ChannelConfig/IPChannel/Get**. API used to query the list of cameras connected to an HWT-NVR800 in broadcast mode: **/API/ChannelConfig/RemoteDev/Search**, which returns the list of cameras on the LAN.

API used to connect cameras to an HWT-NVR800: **/API/ChannelConfig/IPChannel/Set**, which requires you to set **operation_type** to **AddOrEditChannel**. If you know the camera information such as the IP address, you can directly use this API to connect cameras.

API used to delete cameras from the HWT-NVR800: **/API/ChannelConfig/IPChannel/Set**, (which requires you to set **remove_ipc**).

18.3 Alarm and Target Recognition Results

Q: Why no alarm data is returned for all channels when I invoke **/API/Event/Check**, even though alarm information and images are displayed in the HWT-NVR800?

A: For the first invoking, the server always returns **reader_id**, **lapnumber**, and **sequence**. Values of the three parameters must be carried in subsequent requests. Otherwise, the subsequent requests will always be treated as the first one.

Q: Do I need to invoke an API to query alarms of each HWT-NVR800 channel? Is there any callback mechanism to push alarm data in real time?

A: For the first invoking, the current alarm information is returned. After that, your invoking will always be treated as the first one if the parameters **reader_id**, **lapnumber**, and **sequence** are not set. Currently, no callback mechanism is available. You can obtain alarms only by invoking the API.

Q: How should I set **sequence** and **lap_number** to obtain all alarm information?

A: After the values of **reader_id**, **lapnumber**, and **sequence** are returned by the server upon your first invoking, use the values of these three parameters in your next request. Then, you will obtain all alarm information after your next request is successful. Among the three parameters, each session requires **reader_id**. The **lapnumber** and **sequence** parameter indicate indexes of the system alarm list. In general cases, after your first request is sent, **reader_id** remains valid as long as you send another request within 30s. If it expires, you need to send another request to obtain a new **reader_id**. The validity of **reader_id** is maintained in the current session if requests are sent constantly.

Q: Why cannot I receive the latest alarms after I send the first request and then the second one strictly following the instructions?

A: According to the request and the returned data, the possible causes are as follows:

If the value of **reader_id** in the request is different from that in the response, the server did not receive **reader_id**. If **reader_id** was received, the server either returned correct alarm events or an invalid value of **reader_id**.

Q: Why does the server not receive **reader_id** sent by the client?

A: Further analyses show that the data in the client request is not placed in **data**.

Q: What is the alarm/event interconnection process?

A: For details, see section 12.1.2.

Q: What are the APIs for managing a target list?

A:

API used to add a target list: **/API/AI/FDGroup/Add**

API used to delete a target list: **/API/AI/FDGroup/Remove**

API used to query a target list: **/API/AI/FDGroup/Get**

Note that **/API/AI/FDGroup/Get** is used to query the target lists only, not the listed target records.

Q: What are the APIs for managing target records?

A:

API used to add target information: **/API/AI/Faces/Add**

API used to delete target information: **/API/AI/Faces/Remove**

API used to change the list to which target records belong: **/API/AI/FDGroup/Change**

API used to query target records: **/API/AI/AddedFaces/GetByIndex**

Note: The **/API/AI/AddedFaces/GetByIndex** API can only be used to obtain target records in a certain list. If a large number of target records are to be obtained, you have to obtain them for multiple times. Before using this API to obtain the target records in pagination mode, you need to use **/API/AI/AddedFaces/Search** to search for the total number of target records.

Q: What are the APIs for reporting target recognition results and querying target capture data? How can I set target capture parameters?

A:

API used to set target capture parameters: **/API/Intelligent/PerimeterFace/Set**

API used to report target recognition results: **/API/Event/Check** (that same as that used to report alarm results). For details, see the API description and examples in section 14 "Event."

To implement target recognition, set **subscribe_ai_metadata** to **true**. If target capture is implemented, the server returns **face_alarm**, which contains data including captured target images, background images, countenances, and hairstyles.

Q: What are the APIs for querying target snapshots?

Target snapshots are stored on the server. You can invoke the following APIs to query target snapshots. Note that you need to first query the total number of snapshots, and then obtain snapshots.

API used to query the total number of snapshots: **/API/AI/SnapedFaces/Search**

API used to obtain the target snapshots in pagination mode: **/API/AI/SnapedFaces/GetByIndex**

API used to obtain the target snapshots by target ID: **/API/AI/SnapedFaces/GetById**

Q: What APIs should I use to allow the platform to deliver target lists to an HWT-NVR800 for target recognition and related functions and the HWT-NVR800 to report target recognition results to the platform?

A:

Target list delivery:

A target list is a list of targets, for example, blocklisted or trustlisted targets. A target list contains information like list ID, list name, strategy (trusted or blocked), match rate, alarm configuration, and target images. Operations on target records are separate from those on target lists. Therefore, target records and target lists are delivered through different APIs. The delivery of a target list or a target record covers operations like adding, removing, modifying, and querying a target list or a target record. Note that only the **admin** and **apiadmin** users have the permission to perform these operations. For details, see section 15 "AI."

The following table describes APIs for managing target lists.

Function	API
Adding target lists	/API/AI/FDGroup/Add
Modifying alarm configuration for target lists	/API/AI/FDGroup/Modify
Deleting target lists	/API/AI/FDGroup/Remove
Querying target lists	/API/AI/FDGroup/Get

The following table describes APIs for managing target records.

Function	API
Adding target records to a target list	/API/AI/Faces/Add
Modifying target records	/API/AI/Faces/Modify
Deleting target records from a target list	/API/AI/Faces/Remove
Changing the list to which a target record belongs	/API/AI/FDGroup/Change
Querying target records in a list (querying the total number of targets in the list, then, querying targets in pagination mode or by ID)	/API/AI/AddedFaces/Search /API/AI/AddedFaces/GetByIndex /API/AI/AddedFaces/GetById

For details about other APIs, see section 15 "AI."

18.4 RTSP

Q: How can I play RTSP streams returned by **/API/Preview/GetRtspUrl/Get** without entering the user name and password?

A: Check whether the URL returned by the API contains a token. If so, you do not have to enter the user name or password. Note that this token is valid for 30s and can be used only once.

Q: Are the user name and password for obtaining RTSP streams different from those for logging in to the HWT-NVR800?

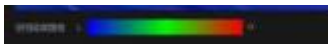
A: No. They are the same.

Q: How can I obtain the RTSP video playback URL that supports both live video viewing and video playback? Does **/API/Playback/PlaybacRtspkUrl/Get** support live video viewing from any camera connected to the HWT-NVR800?

A: The **/API/Playback/PlaybacRtspkUrl/Get** API is used to obtain the video playback URL. You can invoke **/API/Preview/GetRtspUrl/Get** to obtain the live video viewing URL. For details, see section 5.2 "RTSP Preview Stream Parameters."

18.5 Heat Map Statistics

Q: What is the color range of the scale ?



A: The functions of a scale are similar to those of Photoshop. The range of each color on the scale does not need to be too precise. You can use the scale to extract colors without knowing the specific color values.

Q: Does a scale present only colors?

A: Yes.

Q: What does the number representing grayscale in **Image** returned by the API refer to?

A: The value **13** indicates the leftmost color on the scale, and **255** the rightmost color on the scale. Each value between 13 and 255 indicates a color on the scale in a linear way. In this way, you do not need to pay attention to specific color values. When the scale changes, the style of the heat map will change accordingly, without requiring you to map values to colors on the new scale. In addition, there is no negative color value. Therefore, after the grayscale data is decoded using Base64, you need to define a container of the **unsigned char** type, instead of a **signed char** one, to store the data.